# ETC LED Test Results

LM79 and InSitu Test

10/01/2016

# Table of contents


**SECTION 1 -** Source Four LED Series 2
- Lustr
- Tungsten HD
- Daylight HD

**SECTION 2 -** Source Four LED
- Lustr+
- Studio HD
- Tungsten
- Daylight

**SECTION 3 -** Desire D22
- Lustr+
- Studio HD
- Tungsten
- Daylight

**SECTION 4 -** Desire D40
- Lustr+
- Vivid
- Studio HD
- Tungsten
- Daylight

**SECTION 5 -** Desire D60
- Lustr+
- Vivid
- Studio HD
- Tungsten
- Daylight

**SECTION 6 -** Selador Classic Series
- Lustr 11"
- Lustr 21"
- Vivid-R 11"
- Paletta 11"
- Pearl 11"

**SECTION 7 -** ColorSource Family
- PAR
- PAR (Deep Blue)
- Spot
- Spot (Deep Blue)
- Linear 1

**SECTION 8 -** Source 4WRD
- 80 CRI
- 90 CRI

**SECTION 9 -** Irideon
- FPZ

# SECTION 1 - Source Four LED Series 2

Lustr
Tungsten HD
Daylight HD

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101507248 Date: March 5, 2014

REPORT NO. 101507248CRT-001

TEST OF ONE LED STAGE LIGHT

MODEL NO. SOURCE FOUR LED SERIES 2 LUSTR
LED MODEL NO. LUXEON REBEL AND REBEL ES

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500504417.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number Source Four LED Series 2 Lustr. The sample was received by Intertek on Feb. 12, 2014, in undamaged condition and one sample was tested as received. The sample designation was CRT1402121259-001.

DATES OF TESTS: February 26, 2014 through March 3, 2014.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | Source Four LED Series 2 Lustr |
| Description: | LED Stage Light |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 5847 | 6244 |
| Total Power (W) | 161.5 | 161.5 |
| Luminaire Efficacy (LPW) | 36.2 | 38.66 |

| Criteria | Result |
|---|---|
| Power Factor | 0.984 |
| Current ATHD % | 4.77 |
| Correlated Color Temperature (CCT - K) | 3958 |
| Color Rendering Index (CRI - Ra) | 90.4 |
| Color Rendering Index (CRI - R9) | 51.5 |
| DUV | 0.003 |
| Chromaticity Coordinate (x) | 0.384 |
| Chromaticity Coordinate (y) | 0.385 |
| Chromaticity Coordinate (u') | 0.224 |
| Chromaticity Coordinate (v') | 0.506 |
| Maximum In-Situ Source Temperature Point (°C) - Lime | 52.2 |
| Maximum In-Situ Source Temperature Point (°C) - Red | 69.6 |
| Maximum In-Situ Source Temperature Point (°C) - Amber | 67.2 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Sorenson DC Power Supply | XFR 150-8 | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | N/A | 09/18/10 | 100 hrs of use |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/27/13 | 11/27/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 02/24/14 | 03/24/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| ExTech Hygro Thermometer | 445703 | T1357 | 11/25/13 | 11/25/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| Mitutoyo Digital Level/Protractor | Pro 3600 950-316 | N1390 | 12/11/13 | 12/11/14 |
| Cole Parmer Hygro-thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

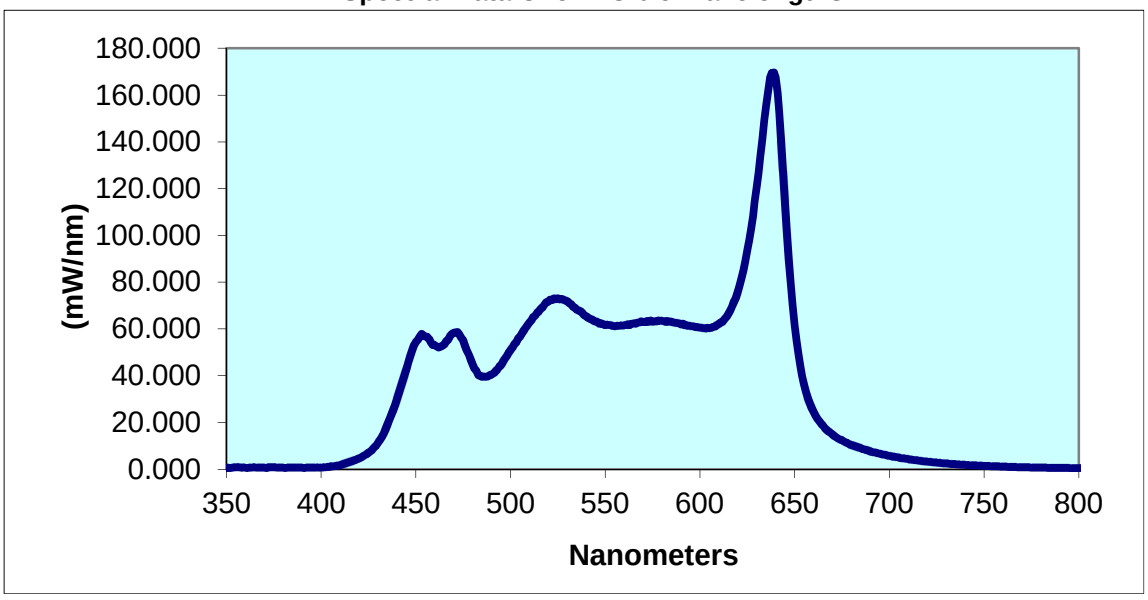
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1402121259-001 | UP | 120.0 | 1368 | 161.5 | 0.984 | 4.77 | 5847 | 36.2 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3958 | 90.4 | 51.5 | 0.003 | 0.384 | 0.385 | 0.224 | 0.506 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.335 | 440 | 29.920 | 530 | 71.830 | 620 | 75.770 | 710 | 4.313 |
| 355 | 0.731 | 445 | 42.980 | 535 | 68.290 | 625 | 93.680 | 715 | 3.740 |
| 360 | 0.718 | 450 | 54.280 | 540 | 65.350 | 630 | 120.500 | 720 | 3.227 |
| 365 | 0.857 | 455 | 56.710 | 545 | 63.190 | 635 | 155.700 | 725 | 2.805 |
| 370 | 0.626 | 460 | 53.090 | 550 | 61.750 | 640 | 167.200 | 730 | 2.408 |
| 375 | 0.771 | 465 | 53.280 | 555 | 61.150 | 645 | 114.200 | 735 | 2.119 |
| 380 | 0.623 | 470 | 58.280 | 560 | 61.570 | 650 | 61.460 | 740 | 1.842 |
| 385 | 0.706 | 475 | 55.130 | 565 | 62.430 | 655 | 36.320 | 745 | 1.574 |
| 390 | 0.586 | 480 | 44.880 | 570 | 62.990 | 660 | 24.570 | 750 | 1.371 |
| 395 | 0.604 | 485 | 39.520 | 575 | 63.030 | 665 | 18.550 | 755 | 1.208 |
| 400 | 0.702 | 490 | 40.540 | 580 | 63.460 | 670 | 14.930 | 760 | 1.068 |
| 405 | 1.141 | 495 | 44.500 | 585 | 62.990 | 675 | 12.400 | 765 | 0.932 |
| 410 | 1.815 | 500 | 50.910 | 590 | 61.930 | 680 | 10.400 | 770 | 0.818 |
| 415 | 3.148 | 505 | 56.790 | 595 | 61.230 | 685 | 8.933 | 775 | 0.709 |
| 420 | 4.688 | 510 | 62.630 | 600 | 60.620 | 690 | 7.682 | 780 | 0.646 |
| 425 | 7.069 | 515 | 67.590 | 605 | 60.310 | 695 | 6.641 | | |
| 430 | 11.260 | 520 | 71.710 | 610 | 62.040 | 700 | 5.728 | | |
| 435 | 19.050 | 525 | 73.000 | 615 | 66.410 | 705 | 4.975 | | |

**Spectral Data Over Visible Wavelengths**



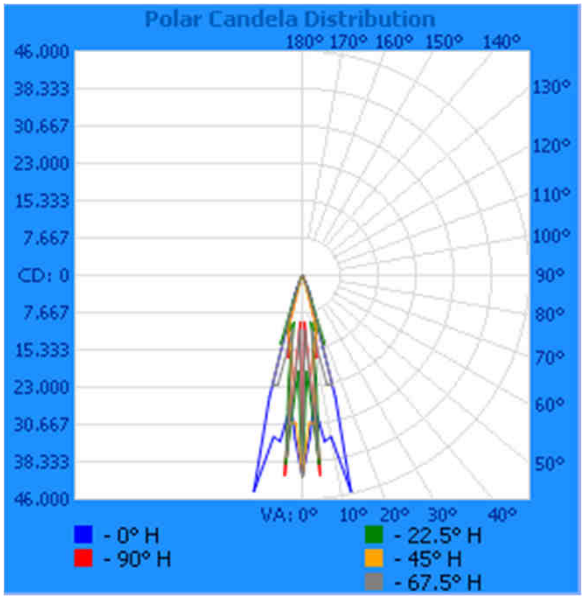
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1402121259-001 | UP | 120.1 | 1364 | 161.5 | 0.986 | 6244 | 38.66 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 41296 | 41296 | 41296 | 41296 | 41296 |
| 5 | 26542 | 38913 | 30644 | 37479 | 41288 |
| 10 | 33692 | 9776 | 14337 | 11383 | 17190 |
| 15 | 25810 | 10002 | 10261 | 23428 | 10579 |
| 20 | 4788 | 5248 | 2490 | 2954 | 4844 |
| 25 | 1258 | 629 | 681 | 1744 | 581 |
| 30 | 76 | 514 | 432 | 171 | 191 |
| 35 | 6 | 9 | 13 | 63 | 33 |
| 40 | 0 | 0 | 0 | 6 | 31 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

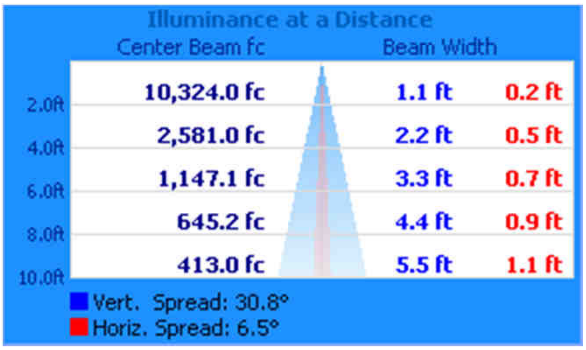


RESULTS OF TEST (cont'd)

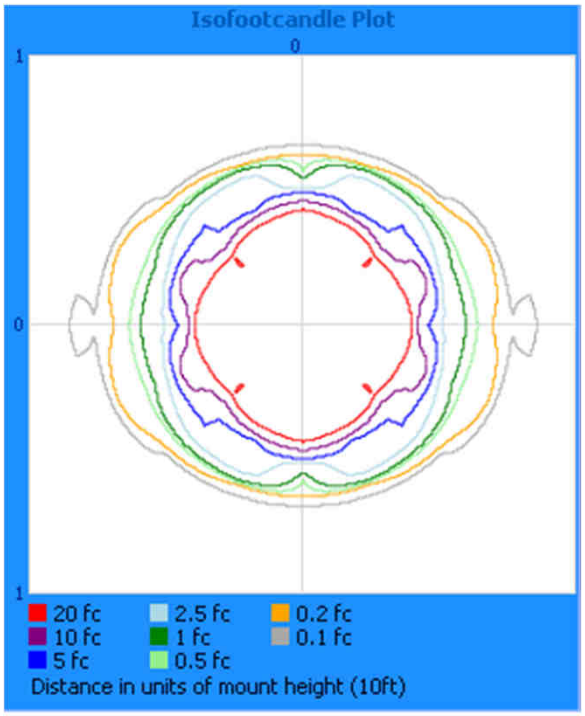
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 6208 | 99.4 |
| 0-40 | 6243 | 100.0 |
| 0-60 | 6244 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 6244 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 6244 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2196 | 35.2 |
| 10-20 | 3443 | 55.1 |
| 20-30 | 568.8 | 9.1 |
| 30-40 | 35.7 | 0.6 |
| 40-50 | 0.5 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied LED Documentation:

# LUXEON® Rebel and LUXEON® Rebel ES

## Electrical Characteristics for LUXEON Rebel and LUXEON Rebel ES Colors at Test Current and Temperature [1]

Table 3.

| Color | Part Number | Forward Voltage $V_f$ (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_F / \Delta T_J$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|---|
| | | Minimum | Typical | Maximum | | |
| Lime | LXML-PX02 | 2.60 | 2.75 | 3.00 | -2.0 to -4.0 | 6 |
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB02<br>LXML-PB01 | 2.50<br>2.55 | 2.95<br>2.95 | 3.50<br>3.51 | - 2.0 to - 4.0<br>- 2.0 to - 4.0 | 6<br>10 |
| Royal Blue | LXML-PR02<br>LXML-PR01 | 2.50<br>2.55 | 2.90<br>2.95 | 3.50<br>3.51 | - 2.0 to - 4.0<br>- 2.0 to - 4.0 | 6<br>10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red-Orange | LXM2-PH01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

## Absolute Maximum Ratings

Table 4.

| Parameter | Green/Cyan/Blue/ Royal Blue | ES Royal Blue/ES Blue/Lime | Red/Deep Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |
| Operating Case Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 120°C | -40°C - 110°C |
| Storage Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 | 3 | 3 | 3 |
| Autoclave Conditions | 121°C at 2 ATM<br>100% Relative Humidity for 96 Hours Maximum | | | |
| Reverse Voltage (Vr) | LUXEON Rebel Color Portfolio LEDs are not designed to be driven in reverse bias. | | | |

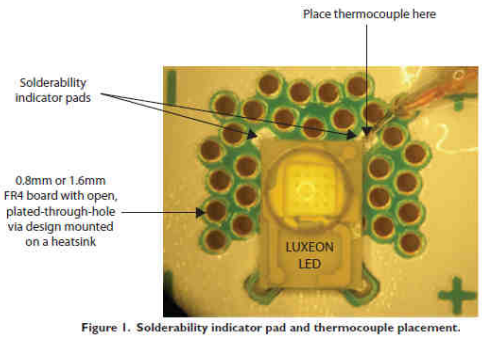
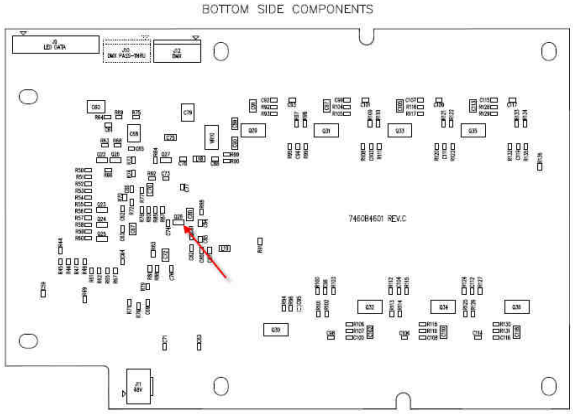



Figure 1. Solderability indicator pad and thermocouple placement.

RESULTS OF TEST (cont'd)

Manufacturer Supplied Driver Documentation:




Note from Client:
Temperature should be measured on the bottom side of the control card, 7460B5601, at location Q28.
Client requested 100 ºC maximum driver temperature.

In-Situ Maximum Measured LED Source Temperature

Lime
Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3V
Measured LED Current = 586mA
Calculated LED Wattage = Vf x Measured LED Current = 1.758W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 139.5°C

Red (LMX2)
Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 8°C/W
Maximum Forward Voltage (Vf) from LED specification = 2.8V
Measured LED Current = 585mA
Calculated LED Wattage = Vf x Measured LED Current = 1.638W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 121.9°C

PC Amber
Maximum Junction Temperature from LED specification (Tj) = 130°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 580mA
Calculated LED Wattage = Vf x Measured LED Current = 2.03W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 109.7°C

RESULTS OF TEST (cont'd)

Maximum Measured Manufacturer Designated Source Temperature

Lime

| Sample No. | Maximum Measured LED Source Temperature (°C) | Location | Maximum Rated LED Source Temperature (°C) |
|---|---|---|---|
| CRT1402121259-001 | 52.2 | Per diagram | 139.5 |

Red (LMF2)

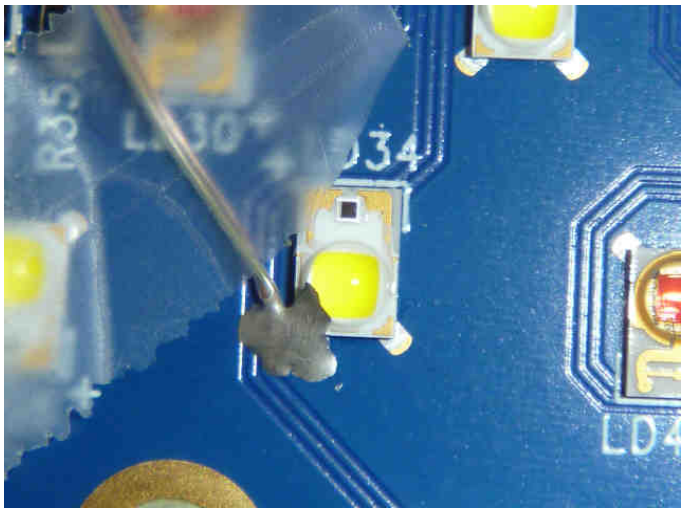
| Sample No. | Maximum Measured LED Source Temperature (°C) | Location | Maximum Rated LED Source Temperature (°C) |
|---|---|---|---|
| CRT1402121259-001 | 69.6 | Per diagram | 121.9 |

PC Amber

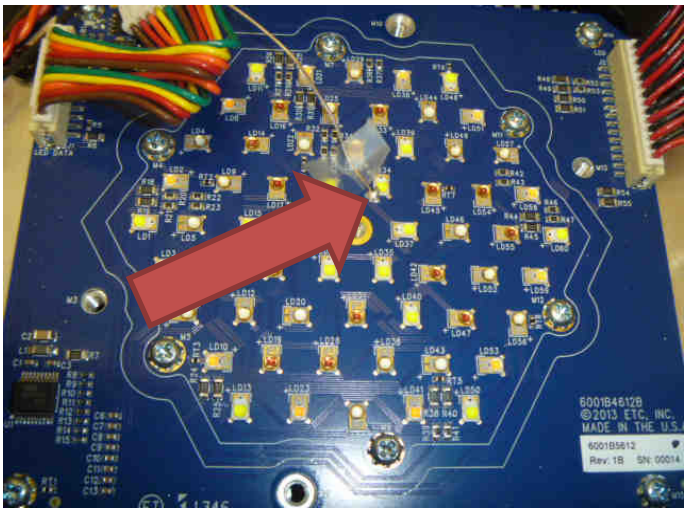
| Sample No. | Maximum Measured LED Source Temperature (°C) | Location | Maximum Rated LED Source Temperature (°C) |
|---|---|---|---|
| CRT1402121259-001 | 67.2 | Per diagram | 109.7 |

| Sample No. | Maximum Measured Driver Source Temperature (°C) | Location | Maximum Rated Driver Source Temperature (°C) |
|---|---|---|---|
| CRT1402121259-001 | 62.2 | Per diagram | 100 |

In-Situ Picture – Ts - Lime

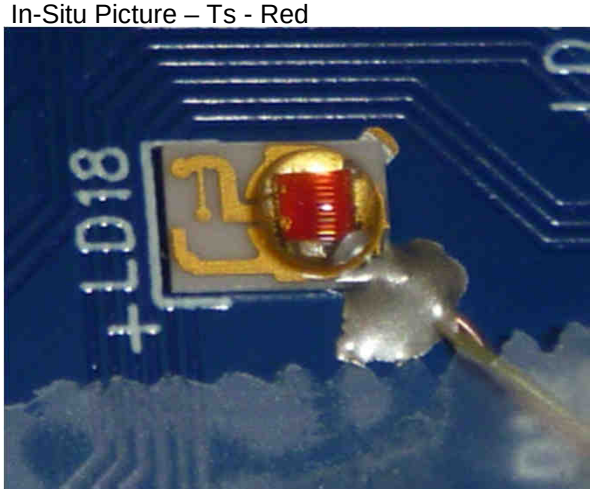


In-Situ Picture – Ts location - Lime

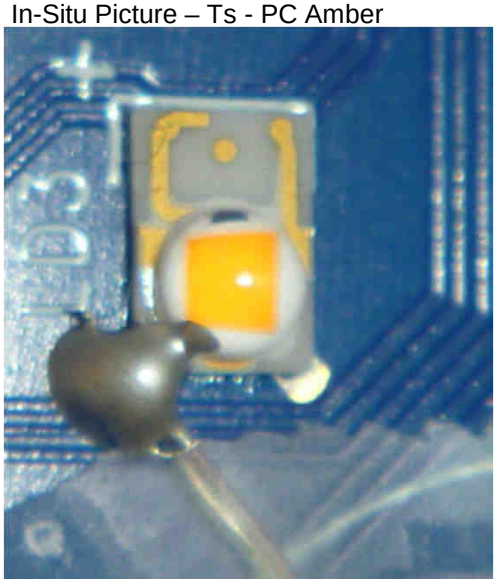


RESULTS OF TEST (cont'd)

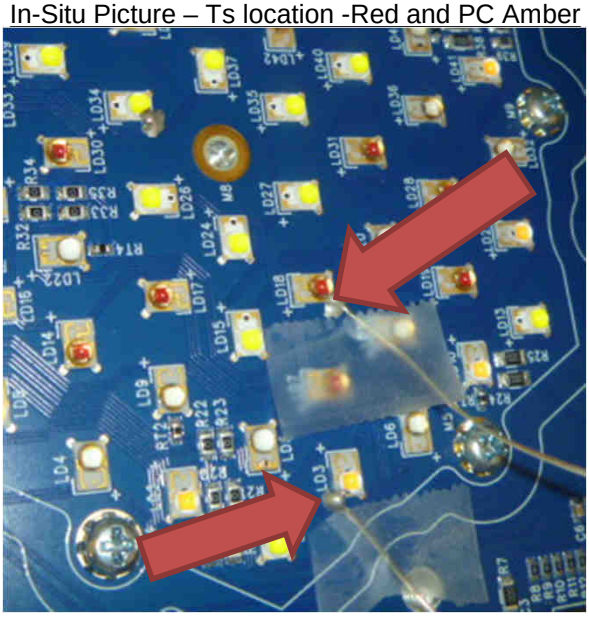
In-Situ Picture – Ts - Red



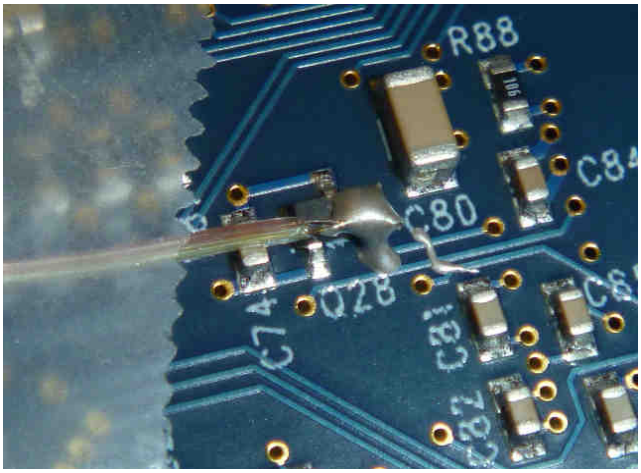

In-Situ Picture – Ts - PC Amber



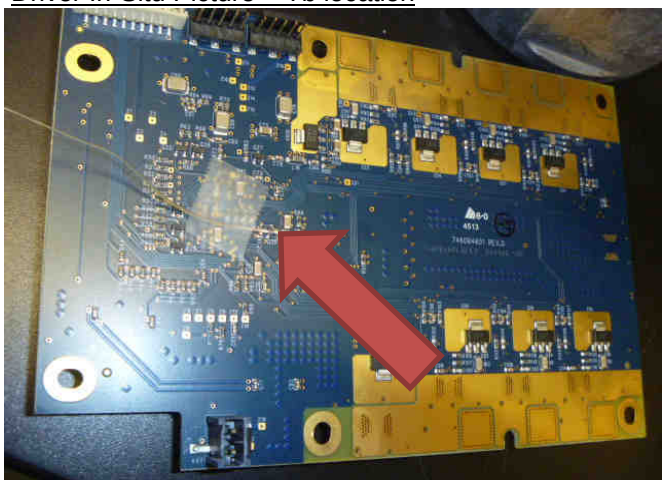
In-Situ Picture – Ts location -Red and PC Amber



Driver In-Situ Picture – Tb




Driver In-Situ Picture – Tb location



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Ryan Siddon
Engineer
Lighting Division

Report Reviewed By:

Peter Leshkiv
Engineering Supervisor
Acoustical/Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101902506

Date: November 21, 2014
Revision Date: November 30, 2014

REPORT NO. 101902506CRT-001

TEST OF ONE LED PROJECTION SPOT LIGHT

MODEL NO. S4LED SERIES 2 TUNGSTEN HD
DRIVER MODEL NO. ELECTRONIC THEATRE CONTROLS 7460B5609
LED MODEL NO. OSRAM OSLON LUW CQAR-NQNS-MMMW-1
LED MODEL NO. PHILIPS LUXEON REBEL ES COLOR

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC.
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

REVISION NOTE: Samples were retested with attachment lens provided by client

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500547958.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one prototype sample of model number S4LED Series 2 Tungsten HD. The sample was received by Intertek on November 12, 2014, in undamaged condition and one sample was tested as received. The sample designation was CRT1411121056-001.

DATES OF TESTS: November 20, 2014 through November 26, 2014

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | S4LED Series 2 Tungsten HD |
| Description: | LED Projection Spot Light |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 7993 | 7759 |
| Total Power (W) | 199.5 | 206.8 |
| Lumen Efficacy (LPW) | 40.07 | 37.52 |

| Criteria | Result |
|---|---|
| Power Factor | 0.993 |
| Current ATHD % | 3.69 |
| Correlated Color Temperature (CCT - K) | 3058 |
| Color Rendering Index (CRI - Ra) | 84.6 |
| Color Rendering Index (CRI - R9) | 51.4 |
| DUV | 0.014 |
| Chromaticity Coordinate (x) | 0.414 |
| Chromaticity Coordinate (y) | 0.363 |
| Chromaticity Coordinate (u') | 0.254 |
| Chromaticity Coordinate (v') | 0.500 |

EQUIPMENT LIST

| | Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|---|
| | Yokogawa Power Analyzer | WT1600 | E474 | 03/07/14 | 03/07/15 |
| | LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| | Fluke Temperature Meter | 53 II | T1318 | 03/21/14 | 03/21/15 |
| | Elgar Power Supply | CW1251 | --- | VBU | VBU |
| | Extech Hygro-Thermometer | 445703 | T1366 | 11/27/13 | 11/27/14 |
| | SORENSEN POWER SUPPLY | XFR 150-8 | --- | VBU | VBU |
| | NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| | LSI High Speed Mirror Goniometer | 6440 | --- | 11/19/14 | 12/19/14 |
| | Elgar Power Supply | CW1251 | --- | VBU | VBU |
| | Yokogawa Power Analyzer | WT210 | E464 | 04/17/14 | 04/17/15 |
| * | ExTech Hygro Thermometer | 445703 | T1357 | 11/25/13 | 11/25/14 |
| | Fisher Scientific | 14-649-9 | N1405 | 08/25/14 | 08/25/15 |
| | M-D Building Products | Smart Tool | L112 | 03/14/14 | 03/15/15 |

* Used Before Cal Due Date

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

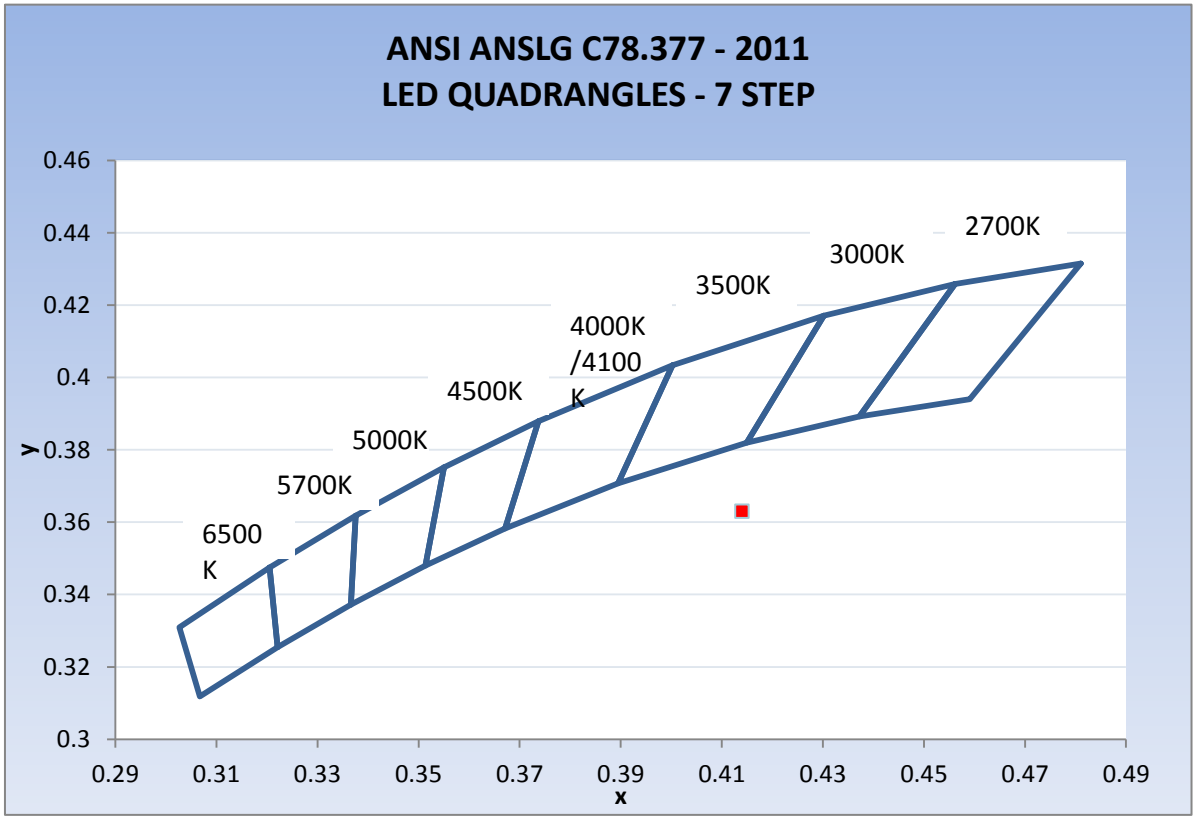
RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1411121056-001 | UP | 120.0 | 1674 | 199.5 | 0.993 | 3.69 | 7993 | 40.07 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3058 | 84.6 | 51.4 | 0.014 | 0.414 | 0.363 | 0.254 | 0.500 |

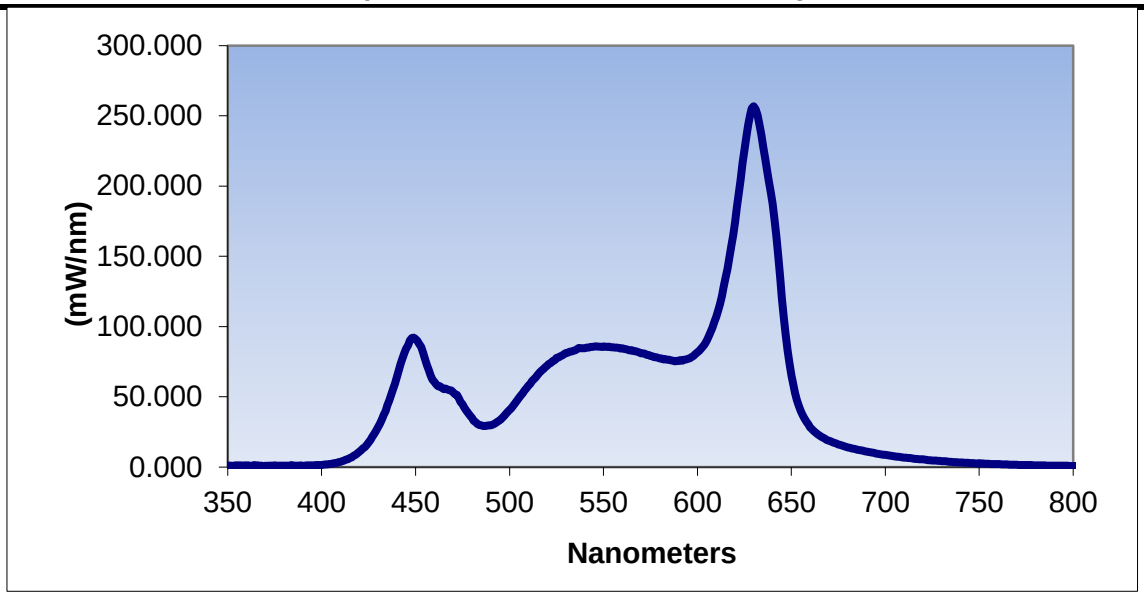
**ANSI C78.377 SSL Chromaticity (2011 Version)**



Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 1.262 | 440 | 64.018 | 530 | 81.102 | 620 | 174.532 | 710 | 6.759 |
| 355 | 1.088 | 445 | 84.343 | 535 | 83.279 | 625 | 225.337 | 715 | 6.012 |
| 360 | 1.085 | 450 | 91.073 | 540 | 84.482 | 630 | 256.829 | 720 | 5.443 |
| 365 | 1.227 | 455 | 76.768 | 545 | 85.790 | 635 | 227.556 | 725 | 4.662 |
| 370 | 0.939 | 460 | 60.686 | 550 | 85.645 | 640 | 187.716 | 730 | 4.136 |
| 375 | 1.145 | 465 | 55.470 | 555 | 85.195 | 645 | 120.376 | 735 | 3.699 |
| 380 | 0.887 | 470 | 53.055 | 560 | 84.615 | 650 | 65.242 | 740 | 3.300 |
| 385 | 1.024 | 475 | 44.653 | 565 | 83.124 | 655 | 40.023 | 745 | 2.871 |
| 390 | 0.809 | 480 | 34.907 | 570 | 80.988 | 660 | 28.230 | 750 | 2.548 |
| 395 | 1.174 | 485 | 29.656 | 575 | 78.992 | 665 | 22.347 | 755 | 2.267 |
| 400 | 1.530 | 490 | 29.691 | 580 | 77.340 | 670 | 18.674 | 760 | 1.996 |
| 405 | 2.210 | 495 | 33.553 | 585 | 76.260 | 675 | 16.059 | 765 | 1.827 |
| 410 | 3.772 | 500 | 40.607 | 590 | 75.361 | 680 | 13.984 | 770 | 1.553 |
| 415 | 6.376 | 505 | 48.962 | 595 | 76.970 | 685 | 12.419 | 775 | 1.409 |
| 420 | 10.700 | 510 | 57.909 | 600 | 81.576 | 690 | 11.002 | 780 | 1.251 |
| 425 | 17.582 | 515 | 65.641 | 605 | 90.192 | 695 | 9.714 | | |
| 430 | 28.268 | 520 | 72.131 | 610 | 106.973 | 700 | 8.647 | | |
| 435 | 43.856 | 525 | 77.633 | 615 | 134.588 | 705 | 7.623 | | |

**Spectral Data Over Visible Wavelengths**



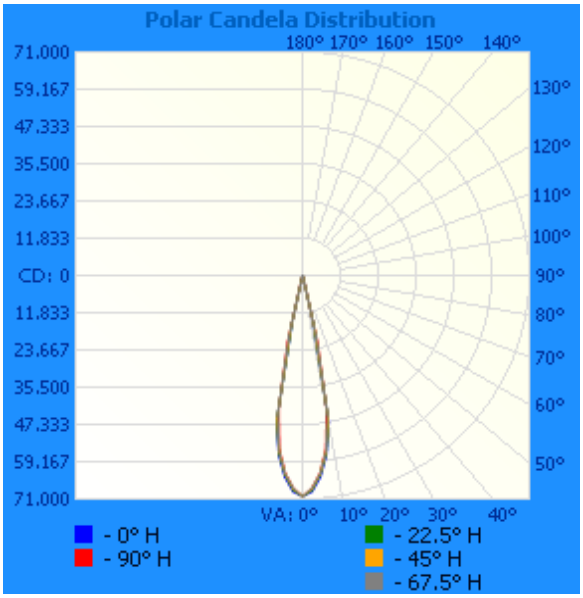
RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1411121056-001 | UP | 120.0 | 1736 | 206.8 | 0.993 | 7759 | 37.52 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 70118 | 70118 | 70118 | 70118 | 70118 |
| 5 | 64199 | 63440 | 63155 | 63004 | 62903 |
| 10 | 46888 | 46493 | 46045 | 44914 | 43127 |
| 15 | 460 | 482 | 635 | 878 | 1510 |
| 20 | 98 | 95 | 96 | 98 | 106 |
| 25 | 30 | 28 | 28 | 30 | 30 |
| 30 | 0 | 0 | 0 | 1 | 2 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

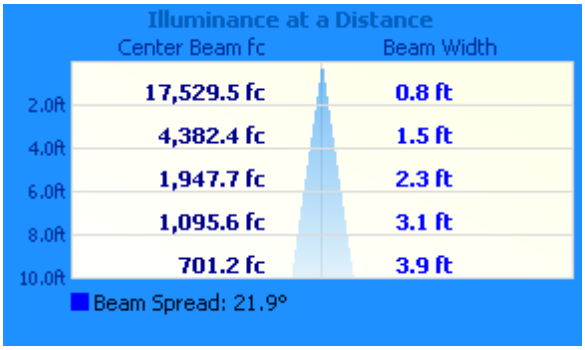


RESULTS OF TEST (cont'd)

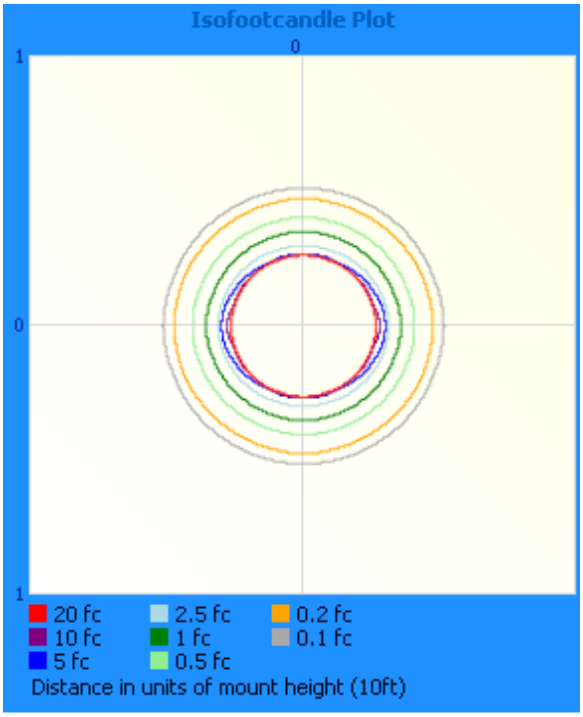
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 7759 | 100.0 |
| 0-40 | 7759 | 100.0 |
| 0-60 | 7759 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 7759 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 7759 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 5467 | 70.5 |
| 10-20 | 2277 | 29.3 |
| 20-30 | 15.2 | 0.2 |
| 30-40 | 0.0 | 0.0 |
| 40-50 | 0.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

Product Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Ryan Siddon
Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101902506 Date: November 30, 2014

REPORT NO. 101902506CRT-003

TEST OF ONE LED PROJECTION SPOT LIGHT

MODEL NO. S4LED SERIES 2 TUNGSTEN HD
DRIVER MODEL NO. ELECTRONIC THEATRE CONTROLS 7460B5609
LED MODEL NO. OSRAM OSLON LUW CQAR-NQNS-MMMW-1
LED MODEL NO. PHILIPS LUXEON REBEL ES COLOR

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC.
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

TEST: Electrical and ISTMT test.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500547958.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

UL 1598-2009: Underwriters Laboratories Inc. Standard for Safety - Luminaires.

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4LED SERIES 2 TUNGSTEN HD. The sample was received by Intertek on November 12, 2014, in undamaged condition and one sample was tested as received. The sample designation was CRT1411121056-001.

DATES OF TESTS: November 26, 2014

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | S4LED SERIES 2 TUNGSTEN HD |
| Description: | LED PROJECTION SPOT LIGHT |

| Criteria | Result |
|---|---|
| Maximum In-Situ Source Temperature Point (°C) LUXEON REBEL BLUE | 76.6 |
| Maximum In-Situ Source Temperature Point (°C) LUXEON REBEL RED | 75.8 |
| Maximum In-Situ Source Temperature Point (°C) OSLON MINT | 77.0 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Extech Hygro-Thermometer | 445703 | T1355 | 12/30/13 | 12/30/14 |
| Fluke Multimeter | 87 | E259 | 03/20/14 | 03/20/15 |
| Fluke Temperature Meter | 53 II | N1324 | 03/21/14 | 03/21/15 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

## LUXEON Rebel and LUXEON Rebel ES Colors at 350 mA or 700 mA, Test Temperature [1]

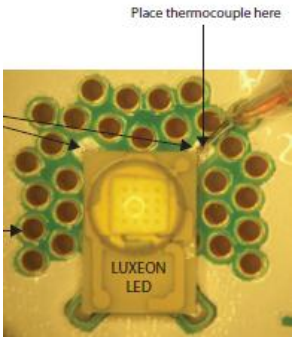



Table 2.

| Color | Dominant Wavelength [2] $\lambda_D$, or Peak Wavelength [3] $\lambda_P$ | | | Typical Spectral Half-width [5] (nm) $\Delta\lambda_{1/2}$ | Typical Temperature Coefficient of Dominant or Peak Wavelength (nm/°C) $\Delta\lambda_D / \Delta T_J$ | Typical Total Included Angle [5] (degrees) $\theta_{0.90V}$ | Typical Viewing Angle [6] (degrees) 2θ 1/2 |
|---|---|---|---|---|---|---|---|
| | Minimum | Typical | Maximum | | | | |
| Lime [7] [9] | 566.0 nm | 567.5 nm | 569.0 nm | 100 | 0.01 | 160 | 125 |
| Green [7] | 520.0 nm | 530.0 nm | 540.0 nm | 30 | 0.05 | 160 | 125 |
| Cyan [7] | 490.0 nm | 505.0 nm | 515.0 nm | 30 | 0.04 | 160 | 125 |
| Blue [7] | 460.0 nm | 470.0 nm | 485.0 nm | 20 | 0.05 | 160 | 125 |
| Royal Blue [3] [7] | 440.0 nm | 447.5 nm | 460.0 nm | 20 | 0.04 | 160 | 125 |
| Red [8] | 620.0 nm | 627.0 nm | 645.0 nm | 20 | 0.05 | 160 | 125 |
| Deep Red [3] [8] | 650.0 nm | 655.0 nm | 670.0 nm | 20 | 0.05 | 160 | 125 |
| Red-Orange [8] | 610.0 nm | 617.0 nm | 620.0 nm | 20 | 0.08 | 160 | 125 |
| Amber [8] | 584.5 nm | 590.0 nm | 594.5 nm | 20 | 0.10 | 160 | 125 |
| PC Amber [7] [9] | 587.8 nm | 591.0 nm | 592.0 nm | 80 | -0.01 | 160 | 120 |

## Electrical Characteristics for LUXEON Rebel and LUXEON Rebel ES Colors at Test Current and Temperature [1]

Table 3.

| Color | Part Number | Forward Voltage $V_f$ (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_F / \Delta T_J$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|---|
| | | Minimum | Typical | Maximum | | |
| Lime | LXML-PX02 | 2.60 | 2.75 | 3.00 | -2.0 to -4.0 | 6 |
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB02 | 2.50 | 2.95 | 3.50 | - 2.0 to - 4.0 | 6 |
| | LXML-PB01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Royal Blue | LXML-PR02 | 2.50 | 2.90 | 3.50 | - 2.0 to - 4.0 | 6 |
| | LXML-PR01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red-Orange | LXM2-PH01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

## Absolute Maximum Ratings

Table 4.

| Parameter | Green/Cyan/Blue/ Royal Blue | ES Royal Blue/ES Blue/Lime | Red/Deep Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |
| Operating Case Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 120°C | -40°C - 110°C |
| Storage Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 | 3 | 3 | 3 |
| Autoclave Conditions | 121°C at 2 ATM 100% Relative Humidity for 96 Hours Maximum | | | |
| Reverse Voltage (Vr) | LUXEON Rebel Color Portfolio LEDs are not designed to be driven in reverse bias. | | | |

RESULTS OF TEST

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

## OSLON Square®
## LUW CQAR (EQW)

**Maximum Ratings**
**Grenzwerte**

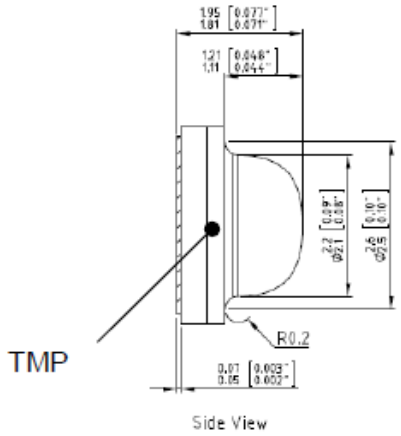
| Parameter<br>Bezeichnung | Symbol<br>Symbol | Values<br>Werte | Unit<br>Einheit |
|---|---|---|---|
| Forward current<br>Durchlassstrom<br>($T_S$ = 25 °C) | $I_F$ | 200 ... 1500 | mA |
| Junction temperature absolute *<br>Sperrschichttemperatur absolut * | $T_{j, abs}$ | 150 | °C |
| Junction temperature<br>Sperrschichttemperatur | $T_j$ | 125 | °C |

**Characteristics** ($T_S$ = 25 °C; $I_F$ = 700 mA)
**Kennwerte**

| Parameter<br>Bezeichnung | | Symbol<br>Symbol | Values<br>Werte | Unit<br>Einheit |
|---|---|---|---|---|
| Real thermal resistance junction / solder point<br>6) *page 24*<br>Realer Wärmewiderstand Sperrschicht / Lötpad<br>6) *Seite 24* | (typ.)<br>(max.) | $R_{th\ JS\ real}$<br>$R_{th\ JS\ real}$ | 3.8<br>5 | K/W<br>K/W |

**Forward Voltage Groups** 4) *page 24*
**Durchlassspannungsgruppen** 4) *Seite 24*

| Group<br>Gruppe | (min.) $V_F$ [V] | (max.) $V_F$ [V] |
|---|---|---|
| L1 | 2.80 | 2.90 |
| L2 | 2.90 | 3.00 |
| M1 | 3.00 | 3.10 |
| MX | 3.10 | 3.25 |



RESULTS OF TEST

LED: PHILIPS LUXEON REBEL BLUE
Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 920mA
Calculated LED Wattage = Vf x Measured LED Current = 3.22W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 130.7

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1411121056-001 | 76.6 | Per diagram | 130.7 |

LED: PHILIPS LUXEON REBEL RED
Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 8°C/W
Maximum Forward Voltage (Vf) from LED specification = 2.8V
Measured LED Current = 611mA
Calculated LED Wattage = Vf x Measured LED Current = 1.711W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 121.3

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1411121056-001 | 75.8 | Per diagram | 121.3 |

LED: OSRAM OSLON SQ MINT
Maximum Junction Temperature from LED specification (Tj) = 125°C
Thermal Resistance Formula from LED specification = 5°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.25V
Measured LED Current = 1445mA
Calculated LED Wattage = Vf x Measured LED Current = 4.696W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 101.5

Maximum Measured Manufacturer Designated Source Temperature

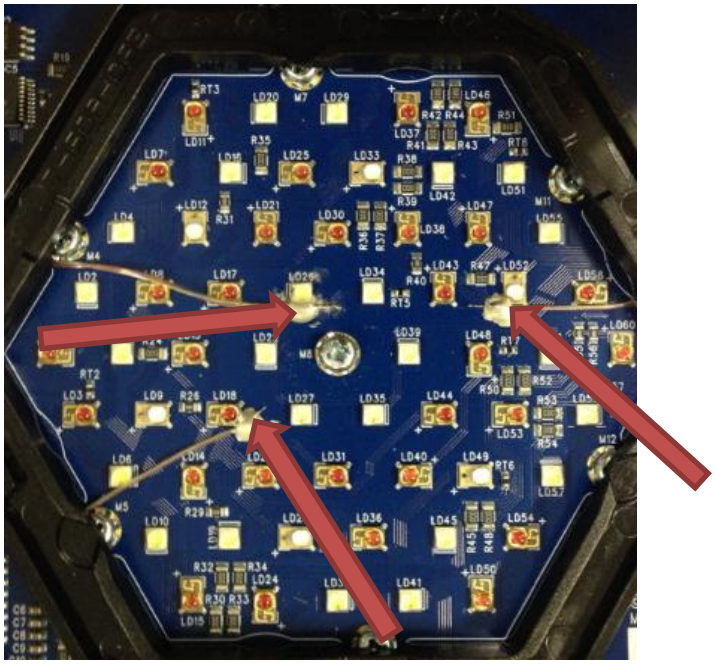
| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1411121056-001 | 77.0 | Per diagram | 101.5 |

RESULTS OF TEST

In-Situ Maximum Measured LED Source Temperature

In-Situ Picture – Ts

In-Situ Picture – Ts location



| INSITU LEDs |
| --- |
| LED26 - OSRAM OSLON SQ MINT<br>LED18 - PHILIPS LUXEON REBEL<br>LED52 - PHILIPS LUXEON REBEL BLUE |

Product Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Ryan Siddon
Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101902506

Date: November 21, 2014
Revision Date: November 30, 2014

REPORT NO. 101902506CRT-002

TEST OF ONE LED PROJECTION SPOT LIGHT

MODEL NO. S4LED SERIES 2 DAYLIGHT HD
DRIVER MODEL NO. ELECTRONIC THEATRE CONTROLS 7460B5609
LED MODEL NO. OSRAM OSLON LUW CQAR-NQNS-MMMW-1
LED MODEL NO. PHILIPS LUXEON REBEL ES COLOR

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC.
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

REVISION NOTE: Samples were retested with attachment lens provided by client

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500547958.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one prototype sample of model number S4LED Series 2 Daylight HD. The sample was received by Intertek on November 12, 2014, in undamaged condition and one sample was tested as received. The sample designation was CRT1411121056-002.

DATES OF TESTS: November 20, 2014 through November 26, 2014

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | S4LED Series 2 Daylight HD |
| Description: | LED Projection Spot Light |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 9084 | 8604 |
| Total Power (W) | 248.9 | 248.0 |
| Lumen Efficacy (LPW) | 36.5 | 34.69 |

| Criteria | Result |
|---|---|
| Power Factor | 0.993 |
| Current ATHD % | 4.10 |
| Correlated Color Temperature (CCT - K) | 5641 |
| Color Rendering Index (CRI - Ra) | 84.5 |
| Color Rendering Index (CRI - R9) | 25.4 |
| DUV | 0.009 |
| Chromaticity Coordinate (x) | 0.330 |
| Chromaticity Coordinate (y) | 0.321 |
| Chromaticity Coordinate (u') | 0.213 |
| Chromaticity Coordinate (v') | 0.466 |

EQUIPMENT LIST

| | Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|---|
| | Yokogawa Power Analyzer | WT1600 | E474 | 03/07/14 | 03/07/15 |
| | LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| | Fluke Temperature Meter | 53 II | T1318 | 03/21/14 | 03/21/15 |
| | Elgar Power Supply | CW1251 | --- | VBU | VBU |
| | Extech Hygro-Thermometer | 445703 | T1366 | 11/27/13 | 11/27/14 |
| | SORENSEN POWER SUPPLY | XFR 150-8 | --- | VBU | VBU |
| | NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| | LSI High Speed Mirror Goniometer | 6440 | --- | 11/19/14 | 12/19/14 |
| | Elgar Power Supply | CW1251 | --- | VBU | VBU |
| | Yokogawa Power Analyzer | WT210 | E464 | 04/17/14 | 04/17/15 |
| * | ExTech Hygro Thermometer | 445703 | T1357 | 11/25/13 | 11/25/14 |
| | Fisher Scientific | 14-649-9 | N1405 | 08/25/14 | 08/25/15 |
| | M-D Building Products | Smart Tool | L112 | 03/14/14 | 03/15/15 |

* Used Before Cal Due Date

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

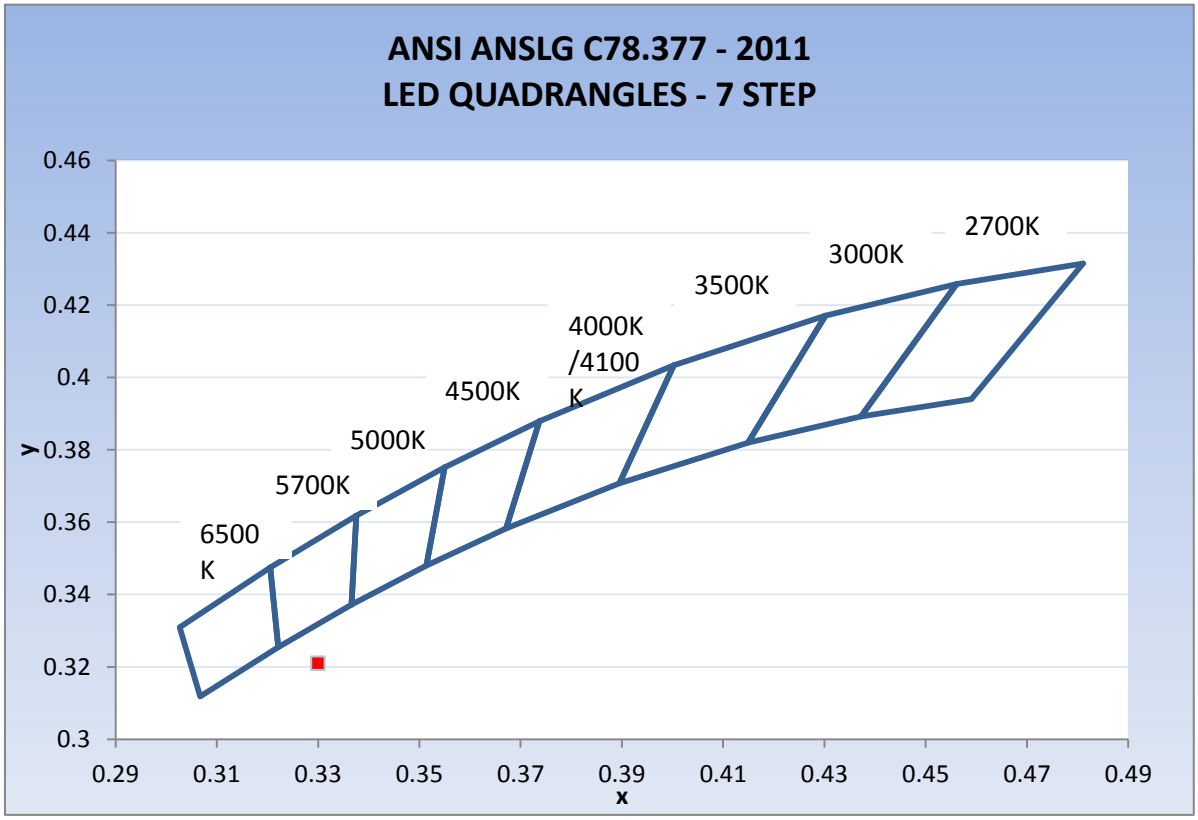
RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1411121056-002 | UP | 120.0 | 2088 | 248.9 | 0.993 | 4.10 | 9084 | 36.5 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 5641 | 84.5 | 25.4 | 0.009 | 0.330 | 0.321 | 0.213 | 0.466 |

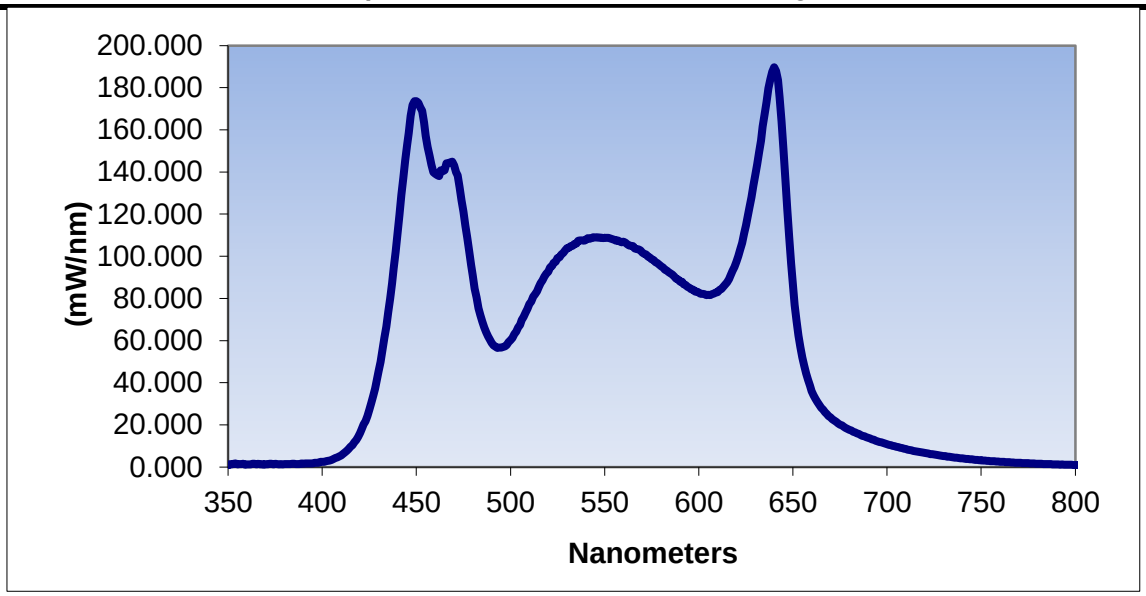
**ANSI C78.377 SSL Chromaticity (2011 Version)**



Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.829 | 440 | 110.803 | 530 | 103.941 | 620 | 97.374 | 710 | 8.566 |
| 355 | 1.322 | 445 | 152.855 | 535 | 106.077 | 625 | 114.627 | 715 | 7.526 |
| 360 | 1.287 | 450 | 173.600 | 540 | 107.674 | 630 | 138.470 | 720 | 6.697 |
| 365 | 1.318 | 455 | 157.133 | 545 | 108.879 | 635 | 167.279 | 725 | 5.900 |
| 370 | 1.390 | 460 | 139.193 | 550 | 108.738 | 640 | 189.708 | 730 | 5.257 |
| 375 | 1.503 | 465 | 140.835 | 555 | 107.828 | 645 | 150.898 | 735 | 4.613 |
| 380 | 1.399 | 470 | 143.135 | 560 | 106.938 | 650 | 85.783 | 740 | 4.087 |
| 385 | 1.522 | 475 | 121.251 | 565 | 104.804 | 655 | 51.938 | 745 | 3.631 |
| 390 | 1.491 | 480 | 90.952 | 570 | 101.819 | 660 | 36.096 | 750 | 3.195 |
| 395 | 1.766 | 485 | 68.987 | 575 | 98.517 | 665 | 28.118 | 755 | 2.832 |
| 400 | 2.437 | 490 | 58.929 | 580 | 95.307 | 670 | 23.543 | 760 | 2.535 |
| 405 | 3.384 | 495 | 56.601 | 585 | 91.843 | 675 | 20.041 | 765 | 2.242 |
| 410 | 5.572 | 500 | 60.190 | 590 | 87.912 | 680 | 17.641 | 770 | 1.963 |
| 415 | 9.544 | 505 | 67.488 | 595 | 85.002 | 685 | 15.711 | 775 | 1.785 |
| 420 | 16.122 | 510 | 77.279 | 600 | 82.863 | 690 | 13.815 | 780 | 1.575 |
| 425 | 27.185 | 515 | 85.202 | 605 | 81.704 | 695 | 12.166 | | |
| 430 | 45.533 | 520 | 92.599 | 610 | 83.008 | 700 | 10.819 | | |
| 435 | 73.539 | 525 | 99.186 | 615 | 87.722 | 705 | 9.631 | | |

**Spectral Data Over Visible Wavelengths**



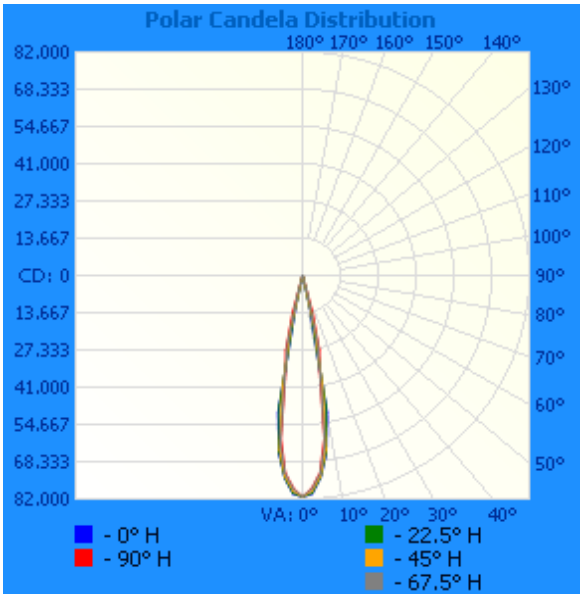
RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1411121056-002 | UP | 120.0 | 2079 | 248.0 | 0.994 | 8604 | 34.69 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 81136 | 81136 | 81136 | 81136 | 81136 |
| 5 | 74457 | 74298 | 73846 | 72789 | 72510 |
| 10 | 51056 | 48319 | 42044 | 36538 | 35530 |
| 15 | 258 | 453 | 1975 | 6299 | 13861 |
| 20 | 72 | 75 | 85 | 95 | 113 |
| 25 | 21 | 23 | 23 | 26 | 31 |
| 30 | 0 | 0 | 1 | 4 | 8 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

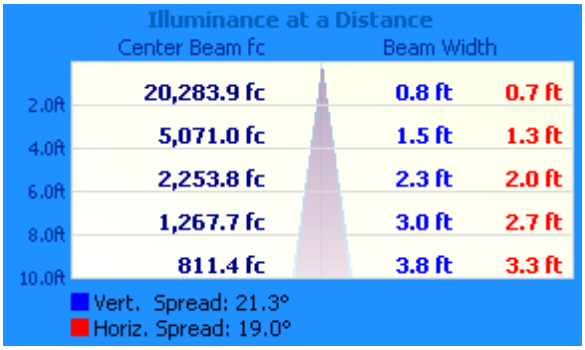


RESULTS OF TEST (cont'd)

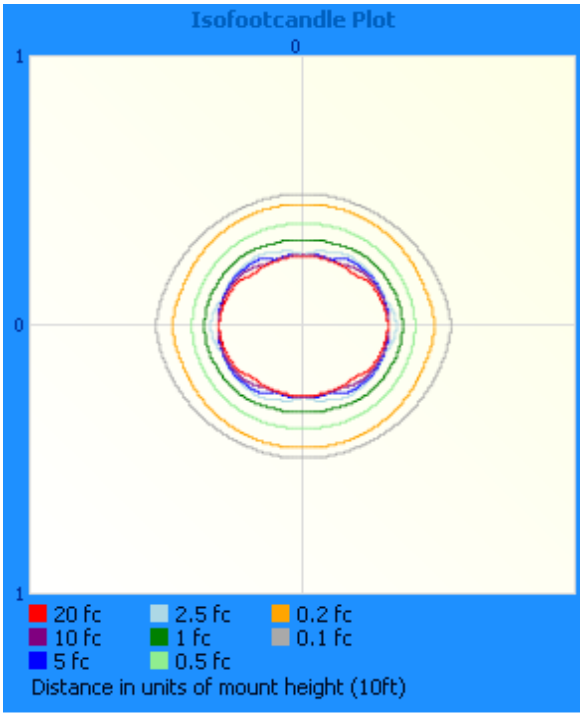
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 8604 | 100.0 |
| 0-40 | 8604 | 100.0 |
| 0-60 | 8604 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 8604 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 8604 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 6050 | 70.3 |
| 10-20 | 2540 | 29.5 |
| 20-30 | 13.5 | 0.2 |
| 30-40 | 0.2 | 0.0 |
| 40-50 | 0.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

Product Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Ryan Siddon
Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101902506 Date: November 30, 2014

REPORT NO. 101902506CRT-004

TEST OF ONE LED PROJECTION SPOT LIGHT

MODEL NO. S4LED SERIES 2 DAYLIGHT HD
DRIVER MODEL NO. ELECTRONIC THEATRE CONTROLS 7460B5609
LED MODEL NO. OSRAM OSLON LUW CQAR-NQNS-MMMW-1
LED MODEL NO. PHILIPS LUXEON REBEL ES COLOR

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC.
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

TEST: Electrical and ISTMT test.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500547958.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

UL 1598-2009: Underwriters Laboratories Inc. Standard for Safety - Luminaires.

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4LED SERIES 2 DAYLIGHT HD. The sample was received by Intertek on November 12, 2014, in undamaged condition and one sample was tested as received. The sample designation was CRT1411121056-002.

DATES OF TESTS: November 26, 2014

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | S4LED SERIES 2 DAYLIGHT HD |
| Description: | LED PROJECTION SPOT LIGHT |

| Criteria | Result |
|---|---|
| Maximum In-Situ Source Temperature Point (°C) LUXEON REBEL BLUE | 87.8 |
| Maximum In-Situ Source Temperature Point (°C) LUXEON REBEL RED | 85.9 |
| Maximum In-Situ Source Temperature Point (°C) OSLON MINT | 85.6 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Extech Hygro-Thermometer | 445703 | T1355 | 12/30/13 | 12/30/14 |
| Fluke Multimeter | 87 | E259 | 03/20/14 | 03/20/15 |
| Fluke Temperature Meter | 53 II | N1324 | 03/21/14 | 03/21/15 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

## LUXEON Rebel and LUXEON Rebel ES Colors at 350 mA or 700 mA, Test Temperature [1]

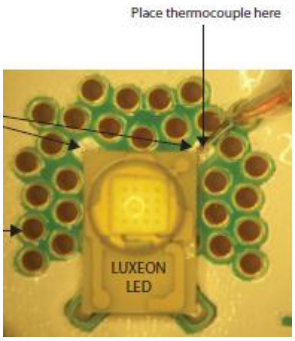



Table 2.

| Color | Dominant Wavelength [2] $\lambda_D$, or Peak Wavelength [3] $\lambda_P$ | | | Typical Spectral Half-width [5] (nm) $\Delta\lambda_{1/2}$ | Typical Temperature Coefficient of Dominant or Peak Wavelength (nm/°C) $\Delta\lambda_D / \Delta T_J$ | Typical Total Included Angle [5] (degrees) $\theta_{0.90V}$ | Typical Viewing Angle [6] (degrees) 2θ 1/2 |
|---|---|---|---|---|---|---|---|
| | Minimum | Typical | Maximum | | | | |
| Lime [7] [9] | 566.0 nm | 567.5 nm | 569.0 nm | 100 | 0.01 | 160 | 125 |
| Green [7] | 520.0 nm | 530.0 nm | 540.0 nm | 30 | 0.05 | 160 | 125 |
| Cyan [7] | 490.0 nm | 505.0 nm | 515.0 nm | 30 | 0.04 | 160 | 125 |
| Blue [7] | 460.0 nm | 470.0 nm | 485.0 nm | 20 | 0.05 | 160 | 125 |
| Royal Blue [3] [7] | 440.0 nm | 447.5 nm | 460.0 nm | 20 | 0.04 | 160 | 125 |
| Red [8] | 620.0 nm | 627.0 nm | 645.0 nm | 20 | 0.05 | 160 | 125 |
| Deep Red [3] [8] | 650.0 nm | 655.0 nm | 670.0 nm | 20 | 0.05 | 160 | 125 |
| Red-Orange [8] | 610.0 nm | 617.0 nm | 620.0 nm | 20 | 0.08 | 160 | 125 |
| Amber [8] | 584.5 nm | 590.0 nm | 594.5 nm | 20 | 0.10 | 160 | 125 |
| PC Amber [7] [9] | 587.8 nm | 591.0 nm | 592.0 nm | 80 | -0.01 | 160 | 120 |

## Electrical Characteristics for LUXEON Rebel and LUXEON Rebel ES Colors at Test Current and Temperature [1]

Table 3.

| Color | Part Number | Forward Voltage $V_f$ (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_F / \Delta T_J$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|---|
| | | Minimum | Typical | Maximum | | |
| Lime | LXML-PX02 | 2.60 | 2.75 | 3.00 | -2.0 to -4.0 | 6 |
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB02 | 2.50 | 2.95 | 3.50 | - 2.0 to - 4.0 | 6 |
| | LXML-PB01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Royal Blue | LXML-PR02 | 2.50 | 2.90 | 3.50 | - 2.0 to - 4.0 | 6 |
| | LXML-PR01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red-Orange | LXM2-PH01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

## Absolute Maximum Ratings

Table 4.

| Parameter | Green/Cyan/Blue/ Royal Blue | ES Royal Blue/ES Blue/Lime | Red/Deep Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |
| Operating Case Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 120°C | -40°C - 110°C |
| Storage Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 | 3 | 3 | 3 |
| Autoclave Conditions | 121°C at 2 ATM 100% Relative Humidity for 96 Hours Maximum | | | |
| Reverse Voltage (Vr) | LUXEON Rebel Color Portfolio LEDs are not designed to be driven in reverse bias. | | | |

RESULTS OF TEST

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

# OSLON Square®

# LUW CQAR (EQW)

**Maximum Ratings**
**Grenzwerte**

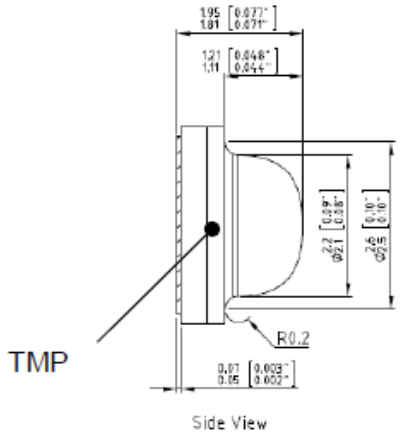
| Parameter<br>Bezeichnung | Symbol<br>Symbol | Values<br>Werte | Unit<br>Einheit |
|---|---|---|---|
| Forward current<br>Durchlassstrom<br>($T_S$ = 25 °C) | $I_F$ | 200 ... 1500 | mA |
| Junction temperature absolute *<br>Sperrschichttemperatur absolut * | $T_{j, abs}$ | 150 | °C |
| Junction temperature<br>Sperrschichttemperatur | $T_j$ | 125 | °C |

**Characteristics** ($T_S$ = 25 °C; $I_F$ = 700 mA)
**Kennwerte**

| Parameter<br>Bezeichnung | | Symbol<br>Symbol | Values<br>Werte | Unit<br>Einheit |
|---|---|---|---|---|
| Real thermal resistance junction / solder point<br>6) *page 24*<br>Realer Wärmewiderstand Sperrschicht / Lötpad<br>6) *Seite 24* | (typ.)<br>(max.) | $R_{th\ JS\ real}$<br>$R_{th\ JS\ real}$ | 3.8<br>5 | K/W<br>K/W |

**Forward Voltage Groups** 4) *page 24*
**Durchlassspannungsgruppen** 4) *Seite 24*

| Group<br>Gruppe | (min.) $V_F$ [V] | (max.) $V_F$ [V] |
|---|---|---|
| L1 | 2.80 | 2.90 |
| L2 | 2.90 | 3.00 |
| M1 | 3.00 | 3.10 |
| MX | 3.10 | 3.25 |



RESULTS OF TEST

LED: PHILIPS LUXEON REBEL BLUE
Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 967mA
Calculated LED Wattage = Vf x Measured LED Current = 3.384W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 129.7°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1411121056-002 | 87.8 | Per diagram | 129.7 |

LED: PHILIPS LUXEON REBEL RED
Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 8°C/W
Maximum Forward Voltage (Vf) from LED specification = 2.8V
Measured LED Current = 260mA
Calculated LED Wattage = Vf x Measured LED Current = 0.728W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 129.2°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1411121056-002 | 85.9 | Per diagram | 129.2 |

LED: OSRAM OSLON SQ MINT
Maximum Junction Temperature from LED specification (Tj) = 125°C
Thermal Resistance Formula from LED specification = 5°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.25V
Measured LED Current = 1410mA
Calculated LED Wattage = Vf x Measured LED Current = 4.582W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 102.1°C

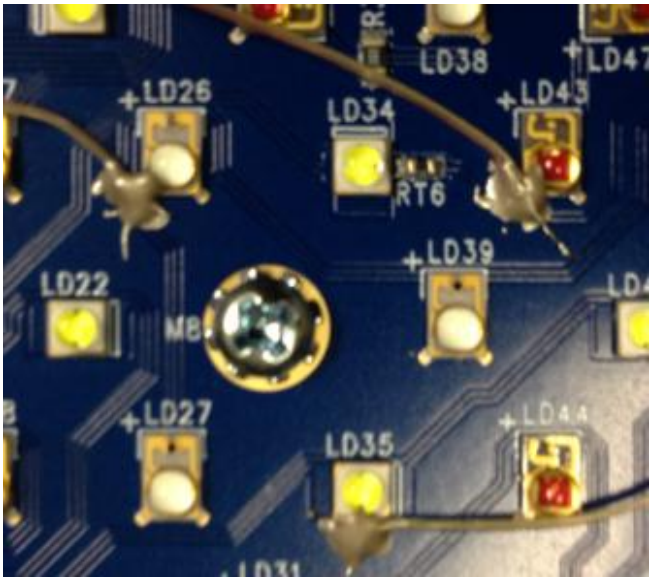
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1411121056-002 | 85.6 | Per diagram | 102.1 |

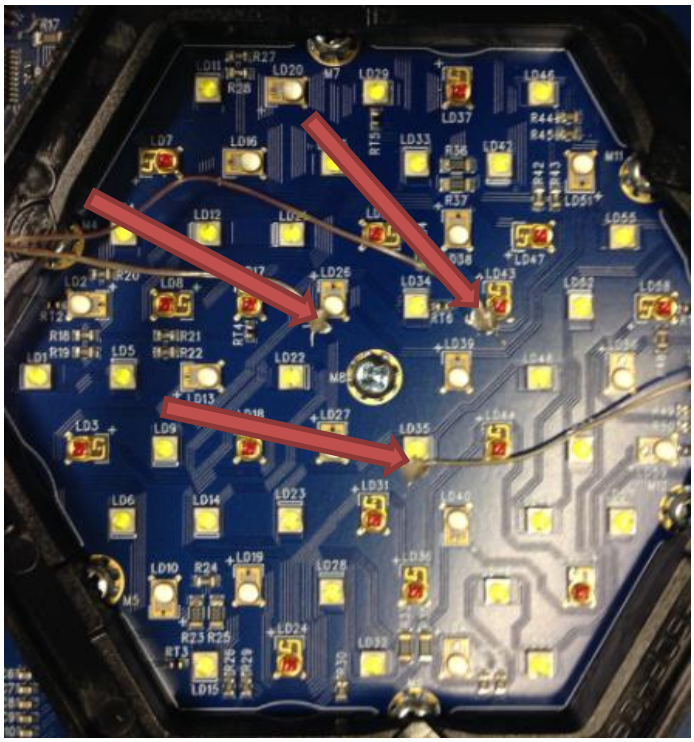
RESULTS OF TEST

In-Situ Maximum Measured LED Source Temperature

In-Situ Picture – Ts




In-Situ Picture – Ts location



| INSITU LEDs |
| --- |
| LED35 - OSRAM OSLON SQ MINT<br>LED43 - PHILIPS LUXEON REBEL<br>LED26 - PHILIPS LUXEON REBEL BLUE |

Product Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Ryan Siddon
Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

# SECTION 2 - Source Four LED

Lustr+
Studio HD
Tungsten
Daylight

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: May 23, 2013

REPORT NO. 101158242CRT-002

TEST OF ONE LED PROFILE

FIXTURE MODEL NO. S4 LUSTR+

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79: 2008 Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products

ANSI ANSLG C38.377: 2012 Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4 LUSTR+. The sample was received by Intertek on May 6, 2013, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-003.

DATES OF TESTS: May 15, 2013 through May 22, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | S4 LUSTR+ |
| Description: | LED PROFILE |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 3724 | 3704 |
| Total Power (W) | 122.7 | 122.1 |
| Luminaire Efficacy (LPW) | 30.35 | 30.33 |

| Criteria | Result |
|---|---|
| Power Factor | 0.989 |
| Current ATHD (%) | 5.53 |
| Correlated Color Temperature (CCT - K) | 5543 |
| Color Rendering Index (CRI) – Ra | 62.2 |
| Color Rendering Index (CRI) - R9 | -74.5 |
| Duv | 0.027 |
| Chromaticity Coordinate (x) | 0.331 |
| Chromaticity Coordinate (y) | 0.293 |
| Chromaticity Coordinate (u') | 0.226 |
| Chromaticity Coordinate (v') | 0.451 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 05/20/13 | 06/20/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | --- | N1132 | 04/22/13 | 04/22/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Yokogawa Power Analyzer | WT1600 | E462 | 07/06/12 | 07/06/13 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | NA | NA | NA |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Sorensen DC Power Supply | DLM150-20E | N/A | N/A | N/A |
| FLUKE | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

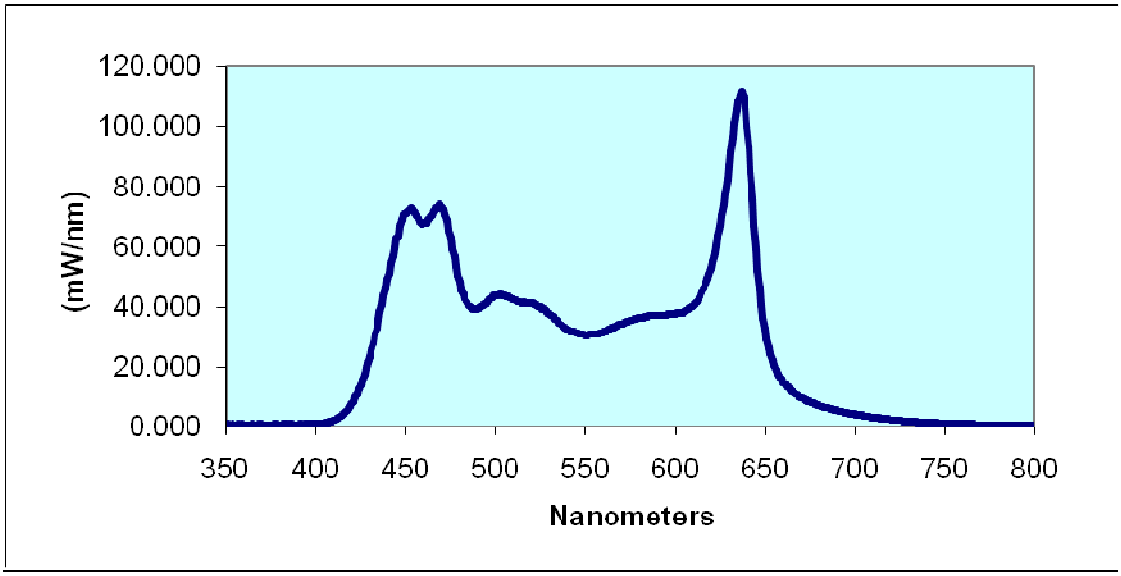
The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.799 | 460 | 67.806 | 570 | 34.004 | 680 | 7.042 |
| 355 | 0.525 | 465 | 70.360 | 575 | 35.244 | 685 | 6.055 |
| 360 | 0.620 | 470 | 73.215 | 580 | 36.125 | 690 | 5.252 |
| 365 | 0.652 | 475 | 62.632 | 585 | 36.767 | 695 | 4.537 |
| 370 | 0.651 | 480 | 48.124 | 590 | 37.031 | 700 | 3.972 |
| 375 | 0.616 | 485 | 40.432 | 595 | 37.040 | 705 | 3.439 |
| 380 | 0.550 | 490 | 39.240 | 600 | 37.556 | 710 | 2.985 |
| 385 | 0.576 | 495 | 41.090 | 605 | 38.280 | 715 | 2.600 |
| 390 | 0.620 | 500 | 43.612 | 610 | 40.255 | 720 | 2.285 |
| 395 | 0.711 | 505 | 43.486 | 615 | 44.732 | 725 | 1.947 |
| 400 | 0.806 | 510 | 42.431 | 620 | 52.909 | 730 | 1.688 |
| 405 | 1.145 | 515 | 41.319 | 625 | 66.875 | 735 | 1.474 |
| 410 | 1.949 | 520 | 41.035 | 630 | 86.688 | 740 | 1.294 |
| 415 | 3.848 | 525 | 39.917 | 635 | 108.281 | 745 | 1.093 |
| 420 | 7.620 | 530 | 37.586 | 640 | 100.082 | 750 | 0.999 |
| 425 | 13.850 | 535 | 34.522 | 645 | 58.941 | 755 | 0.870 |
| 430 | 23.396 | 540 | 32.367 | 650 | 32.404 | 760 | 0.738 |
| 435 | 35.641 | 545 | 31.080 | 655 | 20.270 | 765 | 0.000 |
| 440 | 48.897 | 550 | 30.438 | 660 | 14.716 | 770 | 0.581 |
| 445 | 62.526 | 555 | 30.738 | 665 | 11.716 | 775 | 0.521 |
| 450 | 70.845 | 560 | 31.539 | 670 | 9.721 | 780 | 0.445 |
| 455 | 71.081 | 565 | 32.777 | 675 | 8.162 | | |

**Sample No. CRT1304251519-003**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25°C – Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-003 | UP | 120.0 | 1033 | 122.7 | 0.989 | 5.53 | 3724 | 30.35 |

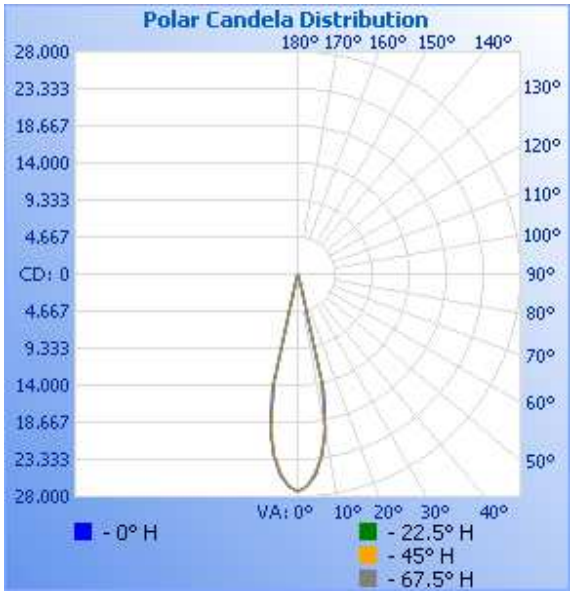
| Intertek Sample No. | Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-003 | 5543 | 62.2 | -74.5 | 0.027 | 0.331 | 0.293 | 0.226 | 0.451 |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1304251519-003 | UP | 120.1 | 1027 | 122.1 | 0.991 | 3704 | 30.33 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 27360 | 27360 | 27360 | 27360 | 27360 |
| 5 | 25396 | 25109 | 25046 | 25242 | 25323 |
| 10 | 19936 | 19796 | 19406 | 19995 | 20003 |
| 15 | 1184 | 1280 | 1577 | 2347 | 2404 |
| 20 | 56 | 57 | 57 | 57 | 56 |
| 25 | 21 | 19 | 20 | 20 | 22 |
| 30 | 5 | 4 | 4 | 3 | 5 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

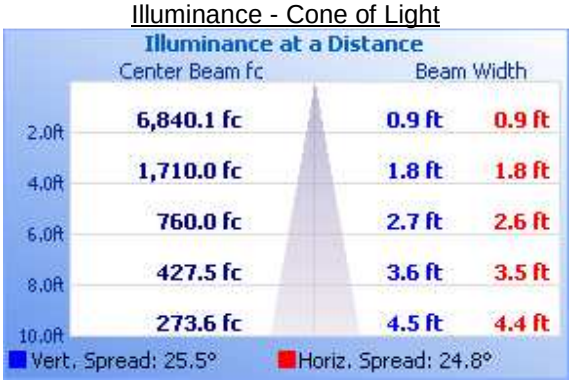


RESULTS OF TESTS (cont'd)

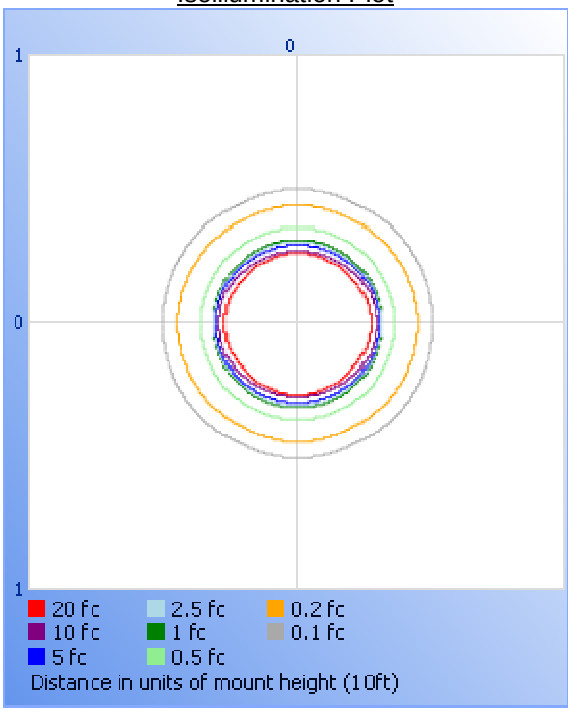
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3704 | 100.0 |
| 0-40 | 3704 | 100.0 |
| 0-60 | 3704 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 3704 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3704 | 100.0 |

Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

David Ellis
Senior Project Engineer
Lighting Division

Attachment: None

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: July 30, 2013

REPORT NO. 101158242CRT-010

TEST OF ONE LED PROFILE LIGHT

FIXTURE MODEL NO. S4 LUSTR+
LED MODEL NO. LUXEON REBEL ES

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and LED In-Situ tests.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

Energy Star Version 1.2 (2012): Program Requirements for Luminaires

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4 LUSTR+. The sample was received by Intertek on May 6, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-003.

DATES OF TESTS: July 29, 2013

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | S4 LUSTR+ |
| Description: | LED PROFILE LIGHT |

| Criteria | Result |
|---|---|
| Maximum In-Situ Source Temperature Point – LXML-PM01 GREEN (°C) | 72.4 |
| Maximum In-Situ Source Temperature Point – LXML-PE01 CYAN (°C) | 71.2 |
| Maximum In-Situ Source Temperature Point – LXML-PB01 BLUE (°C) | 74.5 |
| Maximum In-Situ Source Temperature Point – LXML-PR01 ROYAL BLUE (°C) | 73.8 |
| Maximum In-Situ Source Temperature Point – LXM2-PL01 PC AMBER (°C) | 74.9 |
| Maximum In-Situ Source Temperature Point – LXM2-PD01 RED LOW VF (°C) | 77.0 |
| Maximum In-Situ Source Temperature Point – LXML-PWN1 WHITE (°C) | 73.2 |
| Maximum Power Supply Case Temperature | 48.9 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Fluke Multimeter | PM2525 | M127 | 10/17/12 | 10/17/13 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

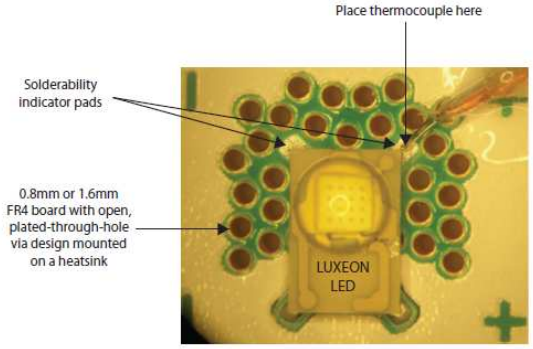
In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Agilent 34970A Data Logger. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TESTS

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: LUXEON REBEL ES




## Electrical Characteristics at 350 mA for LUXEON Rebel color, Thermal Pad Temperature = 25°C

**Table 3.**

| Color | Part Number | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|---|
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Royal-Blue | LXML-PR01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red-Orange | LXM2-PH01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

## Absolute Maximum Ratings

**Table 5.**

| Parameter | Green/Cyan/ Blue/Royal Blue | LUXEON Rebel ES Royal Blue/ES Blue | Red/Deep-Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |

RESULTS OF TESTS (cont'd)

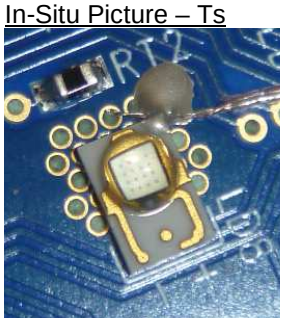
In-Situ Maximum Measured LED Source Temperature - LXML-PM01 GREEN

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 554 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.945W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 131°C

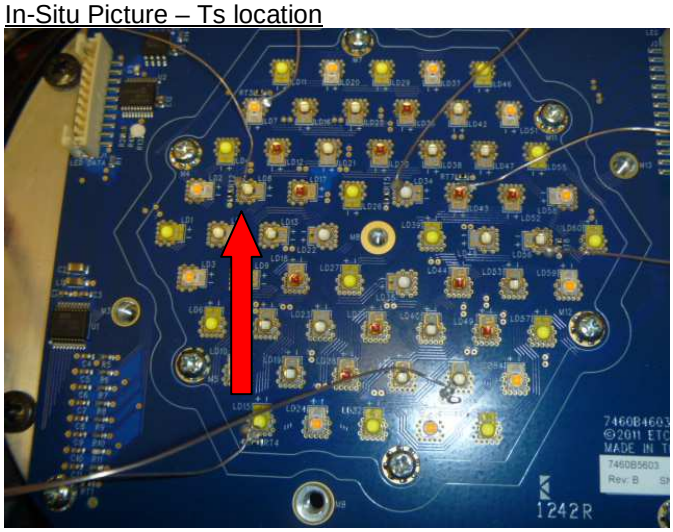
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-003 | 72.4 | Per diagram above | 131 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

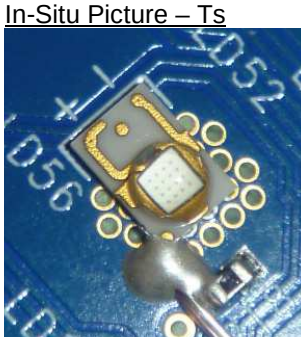
In-Situ Maximum Measured LED Source Temperature - LXML-PE01 CYAN

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 557 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.955W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 130°C

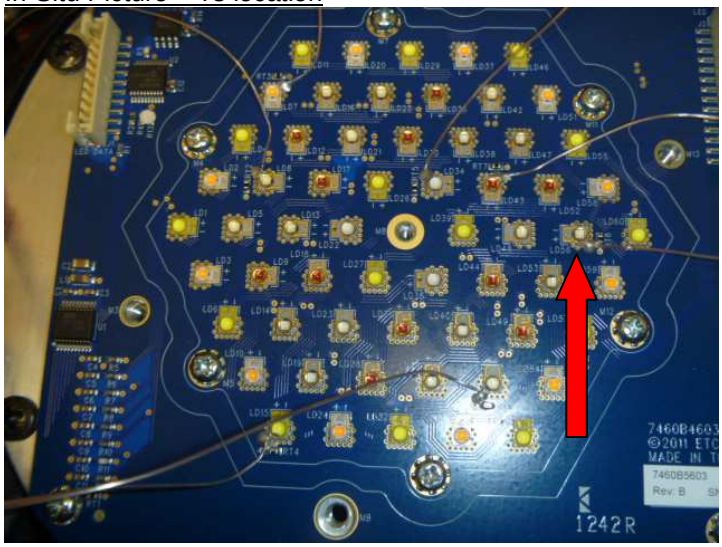
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-003 | 71.2 | Per diagram above | 130 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

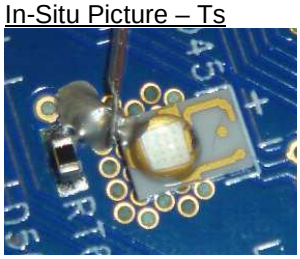
In-Situ Maximum Measured LED Source Temperature - LXML-PB01 BLUE

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 555 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.948W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 131°C

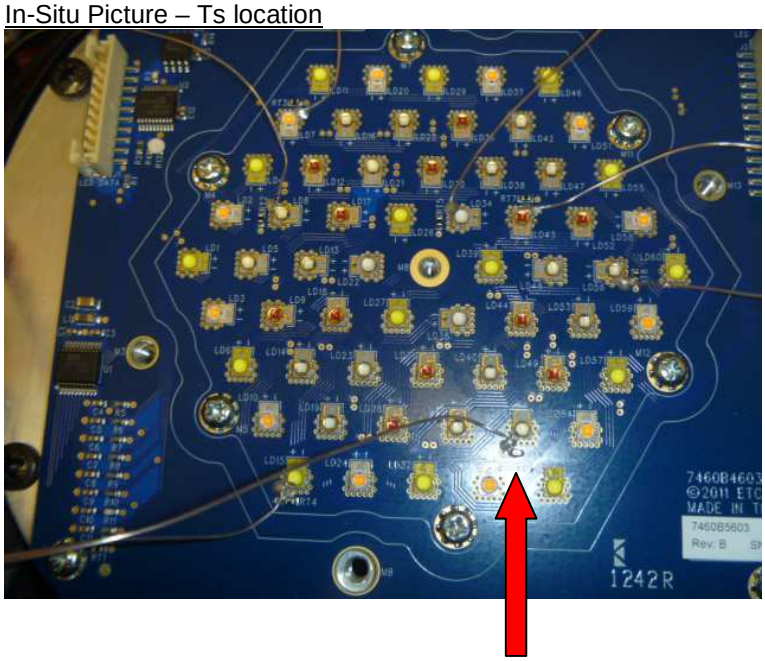
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-003 | 74.5 | Per diagram above | 131 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

In-Situ Maximum Measured LED Source Temperature - LXML-PR01 ROYAL BLUE

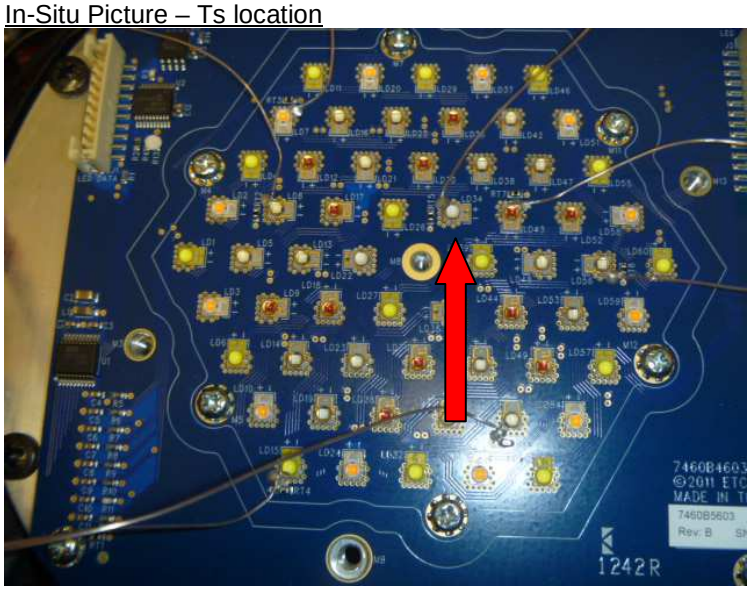
Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 553 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.941W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 131°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-003 | 73.8 | Per diagram above | 131 |

In-Situ Picture – Ts

In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

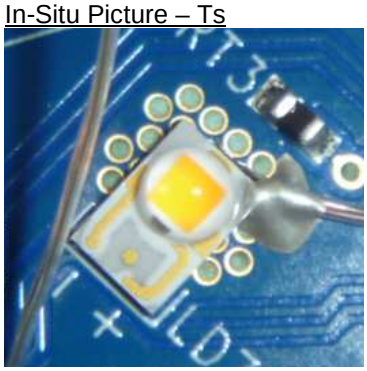
In-Situ Maximum Measured LED Source Temperature - LXM2-PL01 PC AMBER

Maximum Junction Temperature from LED specification (Tj) = 130°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 557 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.955W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 110°C

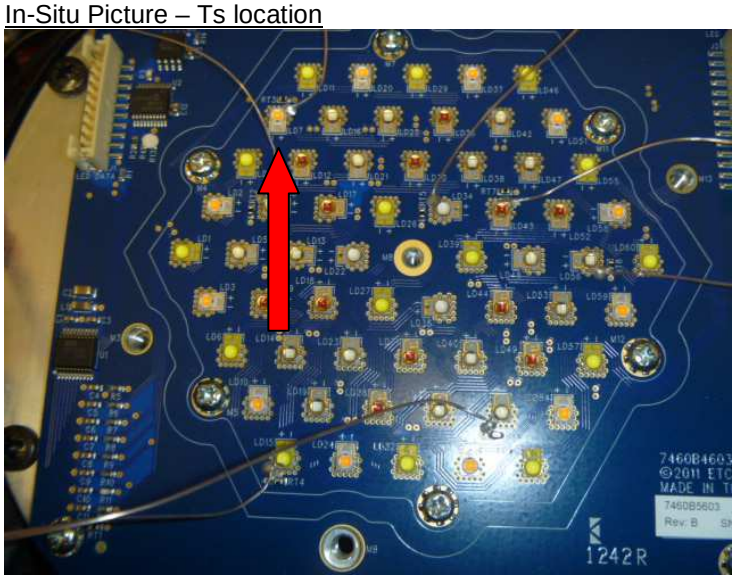
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-003 | 74.9 | Per diagram above | 110 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

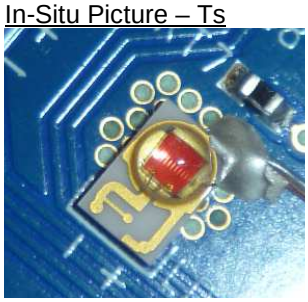
In-Situ Maximum Measured LED Source Temperature - LXM2-PD01 RED  LOW VF

Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 8°C/W
Maximum Forward Voltage (Vf) from LED specification = 2.8V
Measured LED Current = 560 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.568W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 122°C

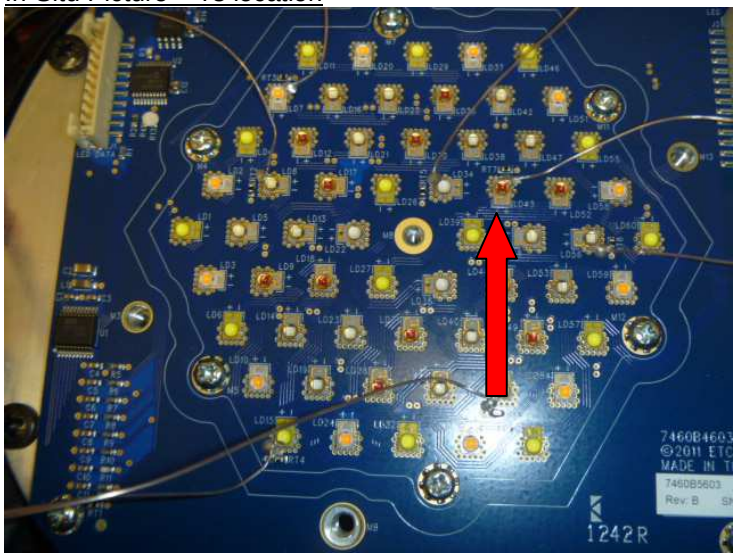
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-003 | 77.0 | Per diagram above | 122 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

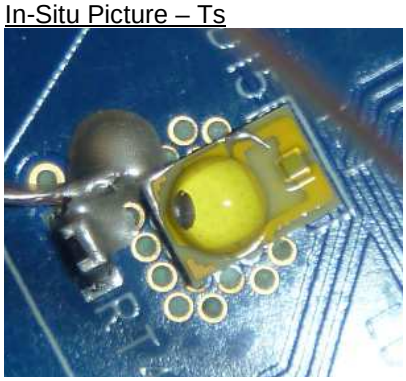
In-Situ Maximum Measured LED Source Temperature - LXML-PWN1 WHITE

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 548 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.918W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 138°C

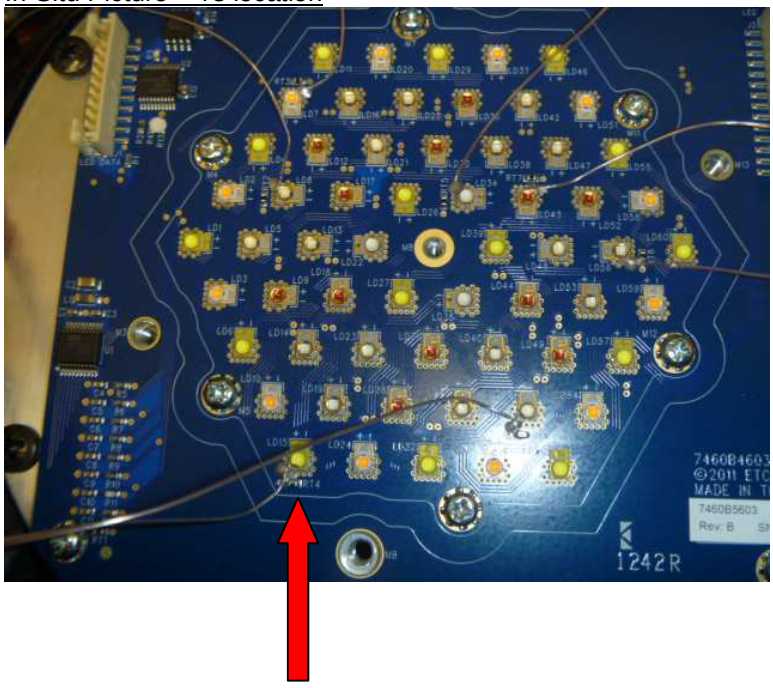
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-003 | 73.2 | Per diagram above | 138 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



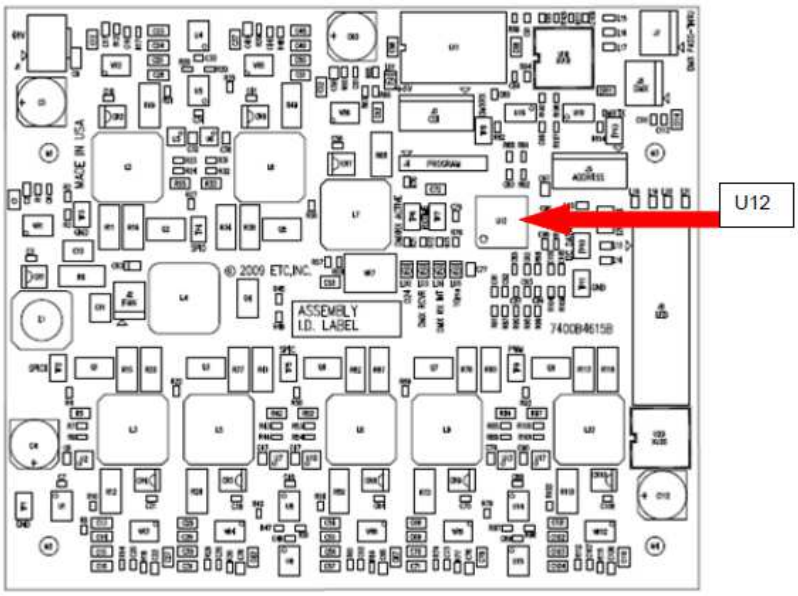
RESULTS OF TESTS (cont'd)

Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:
Driver identified as: Model No. 7400B5615

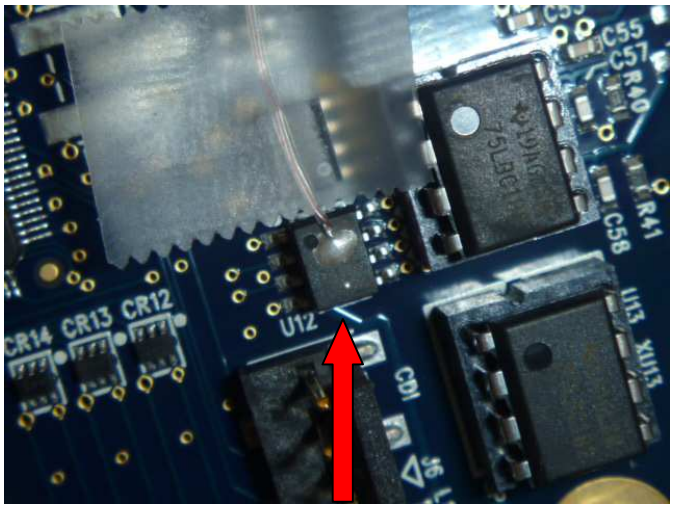
Manufacturer Supplied Documentation:
Driver identified as: 7400B5615, Rev. F

For Selador Classic, measure the temperature at U12.

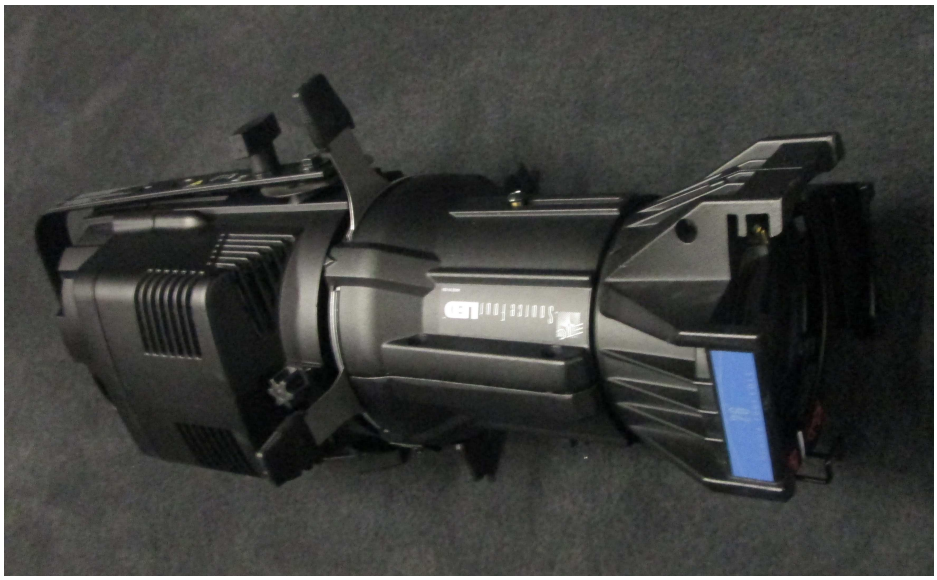



Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Measured Power Supply Case Temperature (°C) | Location | Maximum Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1304251519-003 | 48.9 | U12 | 100 |



Picture (not to scale)



CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

Joseph Schledorn
Project Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 29, 2013

REPORT NO. 101414036CRT-005

TEST OF ONE S4LED STUDIOHD ETL LTENG W/ BARRL (BLK)

MODEL NO. S4LED STUDIOHD
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4LED STUDIOHD. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-005.

DATES OF TESTS: November 21, 2013 through November 26, 2013.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | S4LED STUDIOHD |
| Description: | S4LED STUDIOHD ETL LTENG W/ BARRL (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 5510 | 5718 |
| Total Power (W) | 109.1 | 109.3 |
| Luminaire Efficacy (LPW) | 50.5 | 52.31 |

| Criteria | Result |
|---|---|
| Power Factor | 0.985 |
| Current ATHD % | 6.35 |
| Correlated Color Temperature (CCT - K) | 4018 |
| Color Rendering Index (CRI - Ra) | 89.6 |
| Color Rendering Index (CRI - R9) | 65.2 |
| DUV | 0.014 |
| Chromaticity Coordinate (x) | 0.371 |
| Chromaticity Coordinate (y) | 0.343 |
| Chromaticity Coordinate (u') | 0.233 |
| Chromaticity Coordinate (v') | 0.484 |
| Maximum In-Situ Source Temperature Point (°C) | 79.6 |
| Maximum In-Situ Driver Case Temperature (°C) | 60.6 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Fluke Multimeter | 45 | M133 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

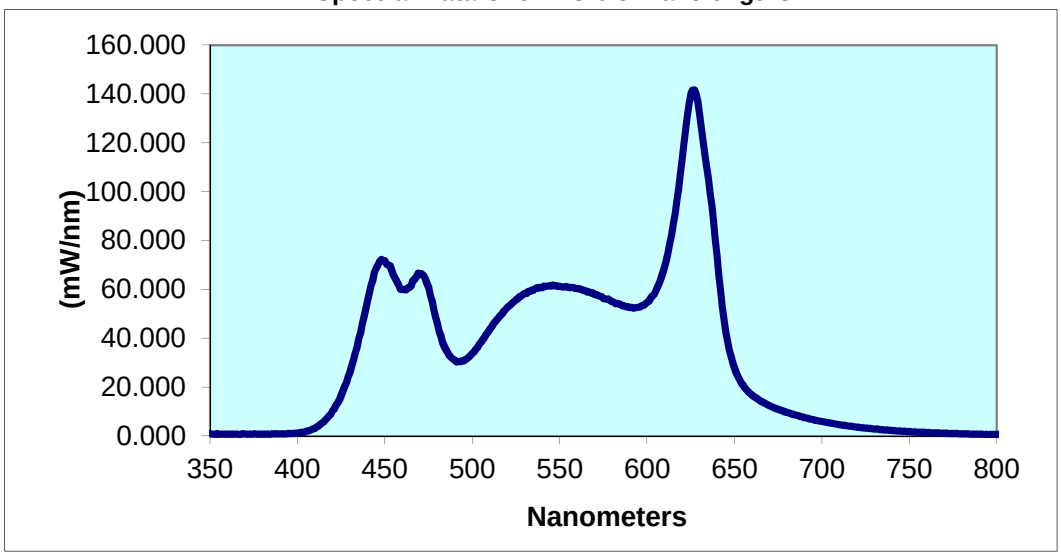
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-005 | UP | 120.0 | 922.3 | 109.1 | 0.985 | 6.35 | 5510 | 50.5 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 4018 | 89.6 | 65.2 | 0.014 | 0.371 | 0.343 | 0.233 | 0.484 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.979 | 440 | 54.97 | 530 | 58.28 | 620 | 113.6 | 710 | 4.606 |
| 355 | 0.722 | 445 | 68.63 | 535 | 59.65 | 625 | 139.9 | 715 | 4.129 |
| 360 | 0.737 | 450 | 71.65 | 540 | 60.66 | 630 | 131.7 | 720 | 3.59 |
| 365 | 0.767 | 455 | 65.73 | 545 | 61.31 | 635 | 105.1 | 725 | 3.216 |
| 370 | 0.793 | 460 | 59.92 | 550 | 61.18 | 640 | 73.54 | 730 | 2.796 |
| 375 | 0.753 | 465 | 61.31 | 555 | 60.77 | 645 | 42.54 | 735 | 2.469 |
| 380 | 0.652 | 470 | 66.3 | 560 | 60.3 | 650 | 27.56 | 740 | 2.223 |
| 385 | 0.782 | 475 | 60.73 | 565 | 59.45 | 655 | 20.55 | 745 | 1.937 |
| 390 | 0.83 | 480 | 46.02 | 570 | 57.87 | 660 | 16.73 | 750 | 1.728 |
| 395 | 0.914 | 485 | 35.29 | 575 | 56.19 | 665 | 14.23 | 755 | 1.502 |
| 400 | 1.141 | 490 | 30.89 | 580 | 55.09 | 670 | 12.39 | 760 | 1.362 |
| 405 | 1.849 | 495 | 30.75 | 585 | 53.72 | 675 | 10.87 | 765 | 1.183 |
| 410 | 3.181 | 500 | 33.89 | 590 | 52.53 | 680 | 9.546 | 770 | 1.073 |
| 415 | 5.788 | 505 | 38.3 | 595 | 52.74 | 685 | 8.494 | 775 | 0.934 |
| 420 | 10.2 | 510 | 43.86 | 600 | 54.64 | 690 | 7.525 | 780 | 0.829 |
| 425 | 16.52 | 515 | 48.41 | 605 | 59.56 | 695 | 6.661 | | |
| 430 | 26.08 | 520 | 52.54 | 610 | 69.39 | 700 | 5.902 | | |
| 435 | 39.74 | 525 | 55.9 | 615 | 86.78 | 705 | 5.231 | | |

**Spectral Data Over Visible Wavelengths**



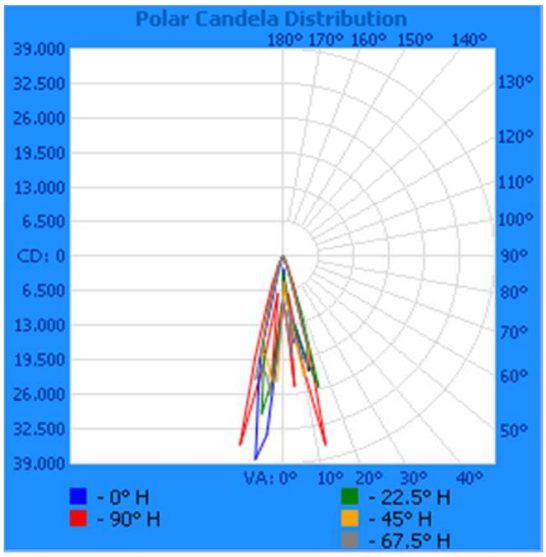
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-005 | UP | 120.0 | 922.1 | 109.3 | 0.988 | 5718 | 52.31 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 8297 | 8297 | 8297 | 8297 | 8297 |
| 5 | 7799 | 5750 | 8357 | 18889 | 24545 |
| 10 | 14040 | 13260 | 22891 | 18836 | 20994 |
| 15 | 17069 | 25405 | 18552 | 17019 | 19740 |
| 20 | 2130 | 2930 | 2944 | 2885 | 5593 |
| 25 | 249 | 768 | 1012 | 299 | 2443 |
| 30 | 939 | 462 | 326 | 330 | 136 |
| 35 | 42 | 26 | 16 | 10 | 6 |
| 40 | 13 | 3 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

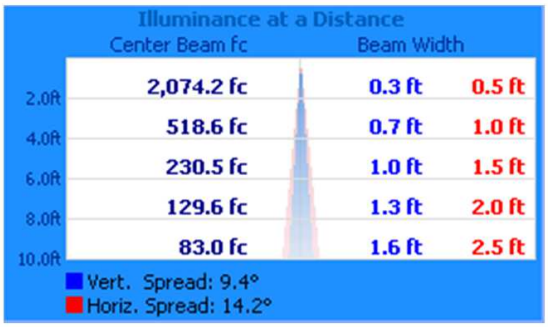


RESULTS OF TEST (cont'd)

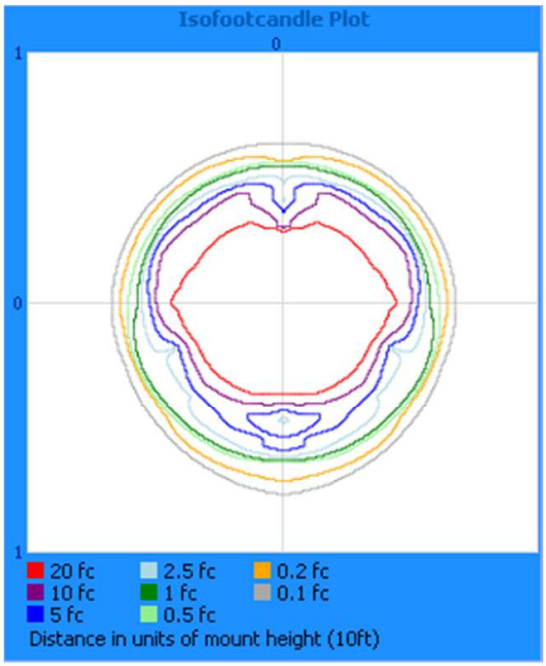
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 5692 | 99.5 |
| 0-40 | 5717 | 100.0 |
| 0-60 | 5718 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 5718 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 5718 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 1695 | 29.6 |
| 10-20 | 3445 | 60.3 |
| 20-30 | 552.3 | 9.7 |
| 30-40 | 25.6 | 0.4 |
| 40-50 | 0.2 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as:

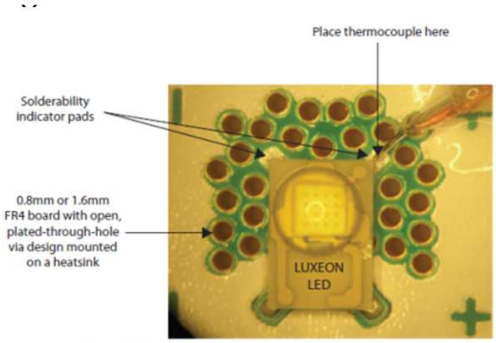


Figure 1. Solderability indicator pad and thermocouple placement.

| Forward Voltage $V_f$ [1] (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) | Typical Thermal Resistance Junction to Thermal Pad (°C/W) |
|---|---|---|---|---|
| Min. | Typ. | Max. | $\Delta V_f / \Delta T_J$ | $R\theta_{J-C}$ |
| 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |

| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

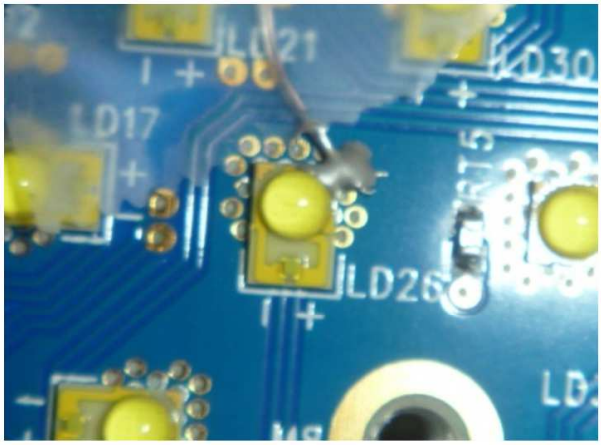
Green LXML-PM01

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 583mA
Calculated LED Wattage = Vf x Measured LED Current = 2.046W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 129.5°C

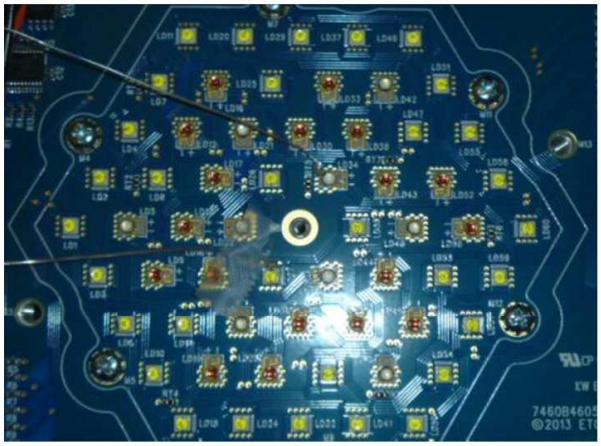
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-005 | 79.6 | Per diagram | 129.5 |

In-Situ Picture – Ts



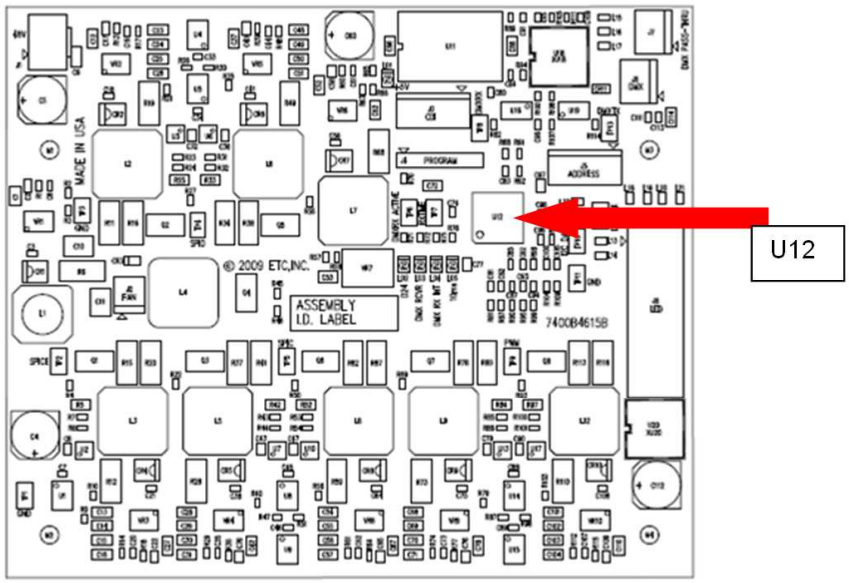

In-Situ Picture – Ts location



RESULTS OF TEST (cont'd)

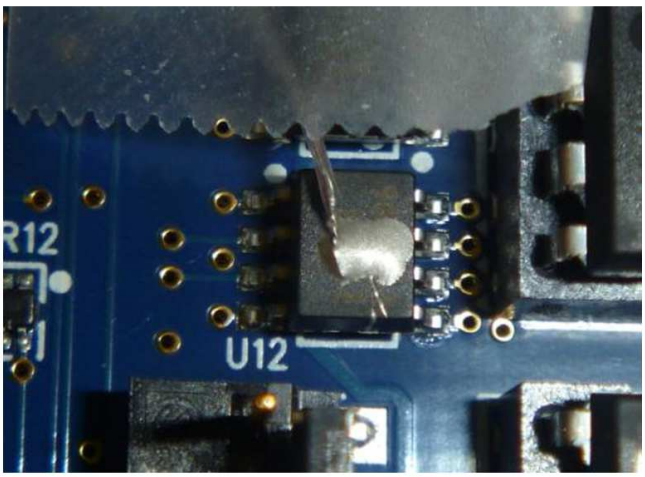
Maximum Measured Power Supply Case Temperature

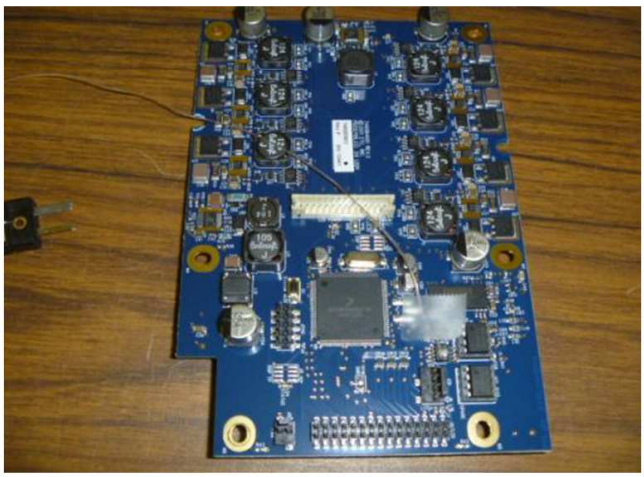
Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-005 | 60.6 | Per diagram | 100 |

Power Supply Picture – Tc location



PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: May 23, 2013

REPORT NO. 101158242CRT-004

TEST OF ONE LED PROFILE

FIXTURE MODEL NO. S4 STUDIO TUNGSTEN

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79: 2008 Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products

ANSI ANSLG C38.377: 2012 Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4 STUDIO TUNGSTEN. The sample was received by Intertek on May 6, 2013, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-006.

DATES OF TESTS: May 15, 2013 through May 22, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | S4 STUDIO TUNGSTEN |
| Description: | LED PROFILE |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 5479 | 5487 |
| Total Power (W) | 151.9 | 151.3 |
| Luminaire Efficacy (LPW) | 36.08 | 36.27 |

| Criteria | Result |
|---|---|
| Power Factor | 0.992 |
| Current ATHD (%) | 5.36 |
| Correlated Color Temperature (CCT - K) | 3050 |
| Color Rendering Index (CRI) – Ra | 83.2 |
| Color Rendering Index (CRI) - R9 | 16.0 |
| Duv | 0.001 |
| Chromaticity Coordinate (x) | 0.435 |
| Chromaticity Coordinate (y) | 0.406 |
| Chromaticity Coordinate (u') | 0.248 |
| Chromaticity Coordinate (v') | 0.522 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 05/20/13 | 06/20/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | --- | N1132 | 04/22/13 | 04/22/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Yokogawa Power Analyzer | WT1600 | E462 | 07/06/12 | 07/06/13 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | NA | NA | NA |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Sorensen DC Power Supply | DLM150-20E | N/A | N/A | N/A |
| FLUKE | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

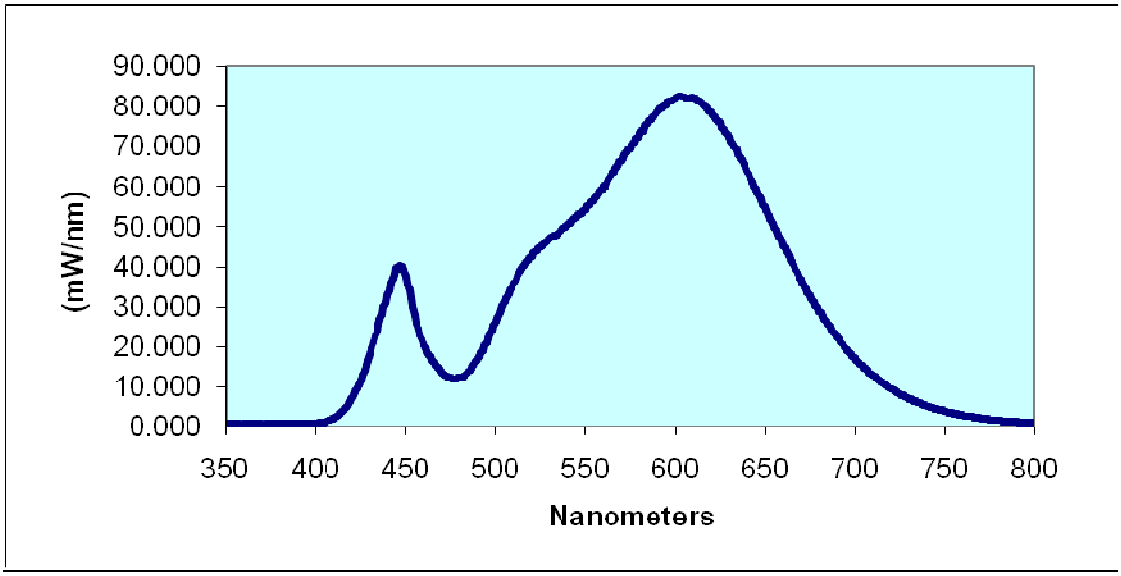
The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.620 | 460 | 20.569 | 570 | 66.386 | 680 | 28.804 |
| 355 | 0.522 | 465 | 16.591 | 575 | 69.591 | 685 | 25.451 |
| 360 | 0.585 | 470 | 13.692 | 580 | 73.058 | 690 | 22.379 |
| 365 | 0.518 | 475 | 12.052 | 585 | 76.303 | 695 | 19.445 |
| 370 | 0.579 | 480 | 12.151 | 590 | 78.370 | 700 | 16.995 |
| 375 | 0.602 | 485 | 13.738 | 595 | 80.349 | 705 | 14.705 |
| 380 | 0.516 | 490 | 16.620 | 600 | 82.045 | 710 | 12.742 |
| 385 | 0.583 | 495 | 20.953 | 605 | 82.221 | 715 | 10.993 |
| 390 | 0.500 | 500 | 26.187 | 610 | 81.797 | 720 | 9.484 |
| 395 | 0.612 | 505 | 31.059 | 615 | 80.555 | 725 | 8.182 |
| 400 | 0.758 | 510 | 35.752 | 620 | 78.520 | 730 | 6.954 |
| 405 | 1.106 | 515 | 39.518 | 625 | 75.846 | 735 | 5.992 |
| 410 | 1.965 | 520 | 42.378 | 630 | 72.102 | 740 | 5.116 |
| 415 | 3.797 | 525 | 44.999 | 635 | 67.905 | 745 | 4.434 |
| 420 | 6.960 | 530 | 47.009 | 640 | 63.765 | 750 | 3.758 |
| 425 | 11.540 | 535 | 48.433 | 645 | 59.247 | 755 | 3.224 |
| 430 | 18.120 | 540 | 50.117 | 650 | 54.678 | 760 | 2.807 |
| 435 | 25.516 | 545 | 52.224 | 655 | 49.848 | 765 | 0.000 |
| 440 | 32.652 | 550 | 54.311 | 660 | 45.295 | 770 | 2.030 |
| 445 | 39.500 | 555 | 56.996 | 665 | 40.976 | 775 | 1.727 |
| 450 | 37.328 | 560 | 59.992 | 670 | 36.720 | 780 | 1.487 |
| 455 | 27.756 | 565 | 63.072 | 675 | 32.664 | | |

**Sample No. CRT1304251519-006**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25°C – Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-006 | UP | 120.0 | 1274 | 151.9 | 0.992 | 5.36 | 5479 | 36.08 |

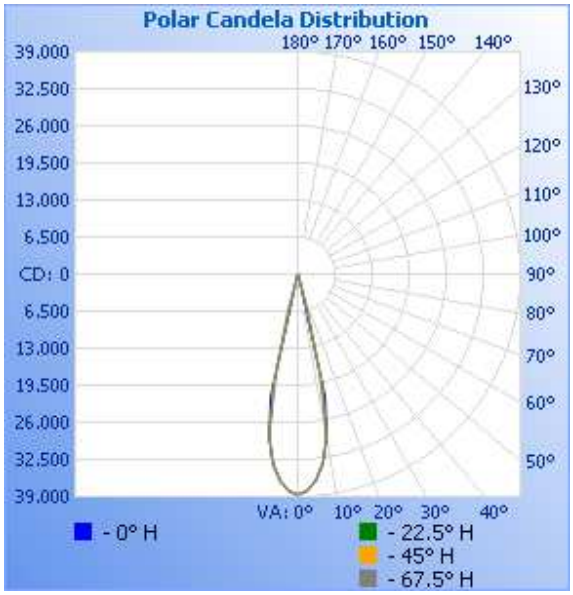
| Intertek Sample No. | Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-006 | 3050 | 83.2 | 16.0 | 0.001 | 0.435 | 0.406 | 0.248 | 0.522 |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1304251519-006 | UP | 120.0 | 1269 | 151.3 | 0.993 | 5487 | 36.27 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 38675 | 38675 | 38675 | 38675 | 38675 |
| 5 | 36308 | 36269 | 36252 | 36256 | 36109 |
| 10 | 28996 | 28960 | 28889 | 28952 | 28917 |
| 15 | 1633 | 2956 | 5528 | 6888 | 7621 |
| 20 | 102 | 100 | 102 | 101 | 103 |
| 25 | 40 | 40 | 41 | 39 | 41 |
| 30 | 14 | 14 | 13 | 12 | 12 |
| 35 | 2 | 1 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

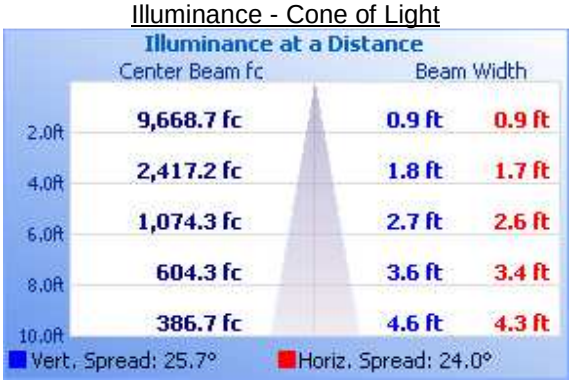


RESULTS OF TESTS (cont'd)

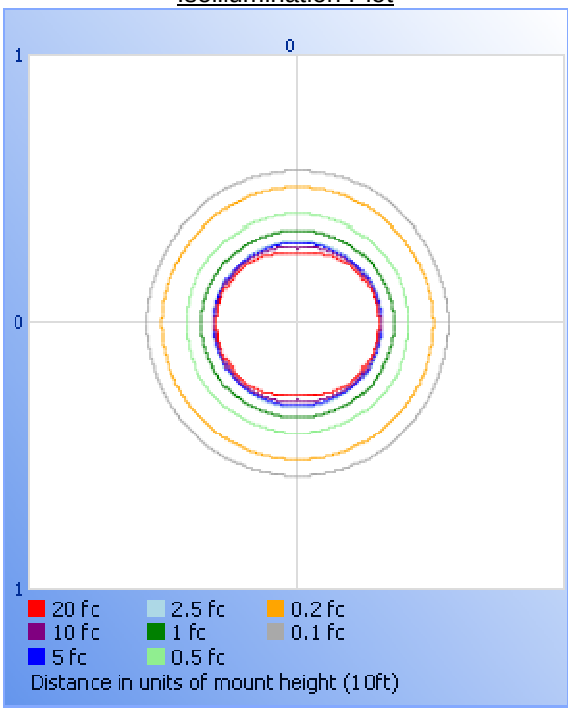
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light



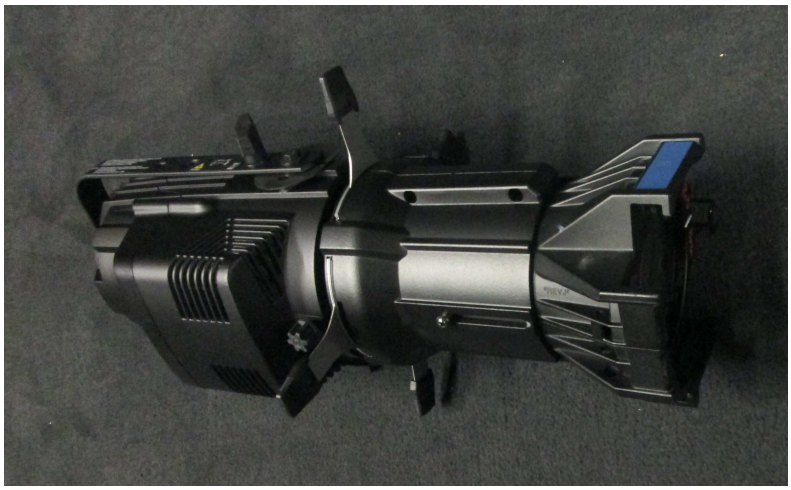

Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 5485 | 100.0 |
| 0-40 | 5487 | 100.0 |
| 0-60 | 5487 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 5487 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 5487 | 100.0 |

Picture (not to scale)



CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

David Ellis
Senior Project Engineer
Lighting Division

Attachment: None

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: July 30, 2013

REPORT NO. 101158242CRT-009

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. S4 STUDIO TUNGSTEN
LED MODEL NO. LUXEON REBEL ES

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and LED In-Situ tests.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

Energy Star Version 1.2 (2012): Program Requirements for Luminaires

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4 STUDIO TUNGSTEN. The sample was received by Intertek on May 6, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-006.

DATES OF TESTS: July 29, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | S4 STUDIO TUNGSTEN |
| Description: | LED WASH LIGHT |

| Criteria | Result |
|---|---|
| Maximum In-Situ Source Temperature Point (°C) | 78.9 |
| Maximum Power Supply Case Temperature (°C) | 65.8 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Fluke Multimeter | PM2525 | M127 | 10/17/12 | 10/17/13 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

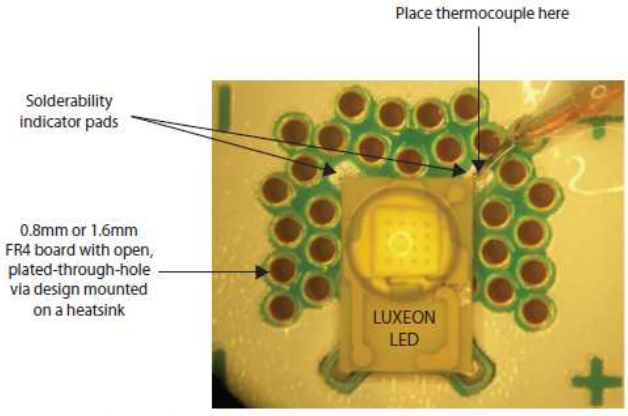
In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Agilent 34970A Data Logger. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TESTS

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: LUXEON REBEL ES




| Color | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|
| Neutral-White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |
| Cool-White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |

| Parameter | Cool-White / Neutral-White |
|---|---|
| DC Forward Current (mA) | 1000 mA |
| Peak Pulsed Forward Current (mA) [2] | 1200 mA |
| Average Forward Current (mA) | 1000 mA |
| ESD Sensitivity | < 8000V Human Body Model (HBM)<br>Class 3B JESD22-A114-E |
| LED Junction Temperature[3] | 150°C |
| Operating Case Temperature at 350 mA | -40°C - 135°C |
| Storage Temperature | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 |
| Autoclave Conditions | 121°C at 2 ATM<br>100% Relative Humidity for 96 Hours Maximum |
| Reverse Voltage (Vr) | LUXEON Rebel ES LEDs are not designed to be driven in reverse bias |

RESULTS OF TESTS (cont'd)

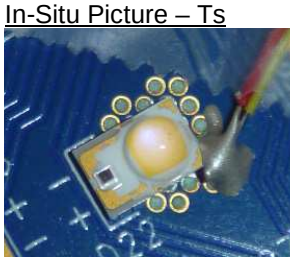
In-Situ Maximum Measured LED Source Temperature

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5 V
Measured LED Current = 553 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.9355W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 138°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-006 | 78.9 | Per diagram above | 138 |

In-Situ Picture – Ts



In-Situ Picture – Ts location

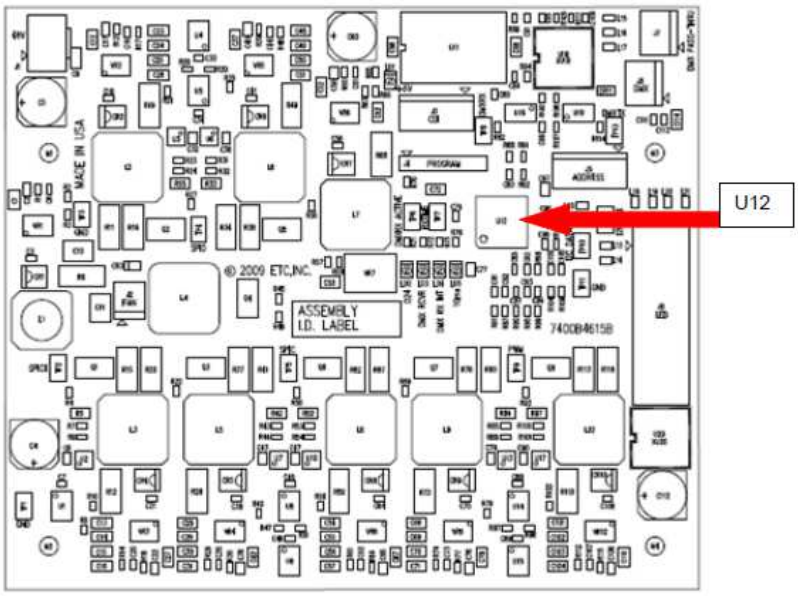
RESULTS OF TESTS (cont'd)

Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:
Driver identified as: Model No. 7400B5615

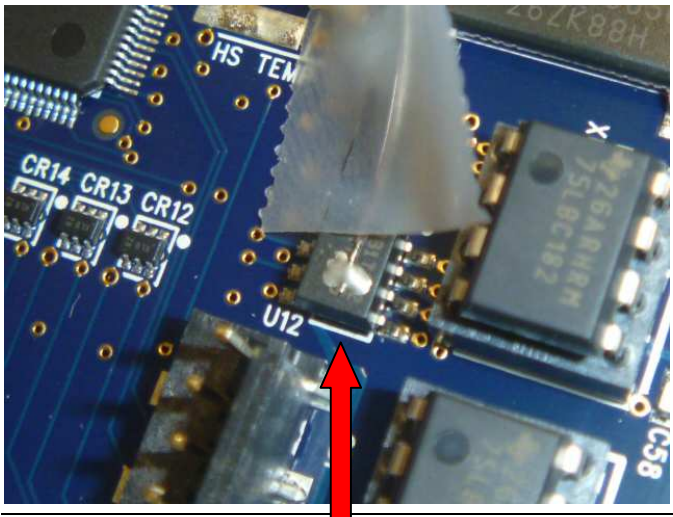
Manufacturer Supplied Documentation:
Driver identified as: 7400B5615, Rev. F

For Selador Classic, measure the temperature at U12.

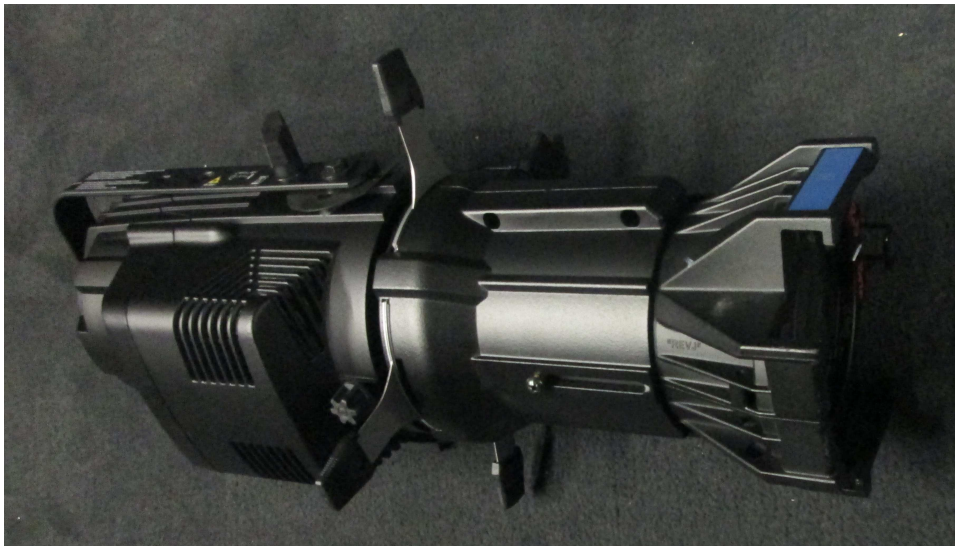



Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Measured Power Supply Case Temperature (°C) | Location | Maximum Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1304251519-006 | 65.8 | U12 | 100 |



Picture (not to scale)



CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

Joseph Schledorn
Project Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 29, 2013

REPORT NO. 101414036CRT-004

TEST OF ONE S4LED DAYLIGHT ETL LTENG W/ BARRL (BLK)

MODEL NO. S4LED DAYLIGHT
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4LED DAYLIGHT. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-004.

DATES OF TESTS: November 21, 2013 through November 25, 2013.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | S4LED DAYLIGHT |
| Description: | S4LED DAYLIGHT ETL LTENG W/ BARRL (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 7363 | 7522 |
| Total Power (W) | 145.4 | 145.6 |
| Luminaire Efficacy (LPW) | 50.64 | 51.66 |

| Criteria | Result |
|---|---|
| Power Factor | 0.991 |
| Current ATHD % | 5.18 |
| Correlated Color Temperature (CCT - K) | 5417 |
| Color Rendering Index (CRI - Ra) | 69.5 |
| Color Rendering Index (CRI - R9) | -13.5 |
| DUV | 0.008 |
| Chromaticity Coordinate (x) | 0.335 |
| Chromaticity Coordinate (y) | 0.359 |
| Chromaticity Coordinate (u') | 0.202 |
| Chromaticity Coordinate (v') | 0.487 |
| Maximum In-Situ Source Temperature Point (°C) | 81.5 |
| Maximum In-Situ Driver Case Temperature (°C) | 52.8 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Fluke Multimeter | 45 | M133 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1355 | 11/26/12 | 11/26/13 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

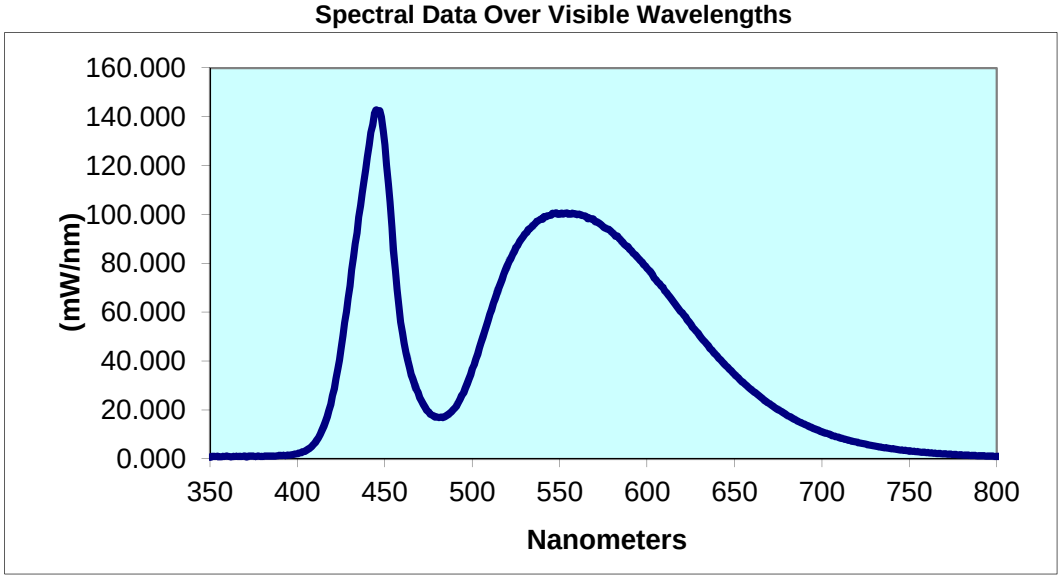
Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-004 | UP | 120.0 | 1222 | 145.4 | 0.991 | 5.18 | 7363 | 50.64 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 5417 | 69.5 | -13.5 | 0.008 | 0.335 | 0.359 | 0.202 | 0.487 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.675 | 440 | 123.5 | 530 | 92.01 | 620 | 59.76 | 710 | 8.702 |
| 355 | 0.846 | 445 | 142.9 | 535 | 95.66 | 625 | 55.01 | 715 | 7.664 |
| 360 | 1.062 | 450 | 128.6 | 540 | 97.99 | 630 | 50.52 | 720 | 6.794 |
| 365 | 1.014 | 455 | 85.48 | 545 | 99.71 | 635 | 46.18 | 725 | 6.011 |
| 370 | 1.048 | 460 | 51.5 | 550 | 100.2 | 640 | 42.13 | 730 | 5.237 |
| 375 | 1.054 | 465 | 34.49 | 555 | 100.2 | 645 | 38.16 | 735 | 4.658 |
| 380 | 0.914 | 470 | 24.95 | 560 | 100.1 | 650 | 34.46 | 740 | 4.096 |
| 385 | 1.131 | 475 | 19.1 | 565 | 99.29 | 655 | 31.17 | 745 | 3.623 |
| 390 | 1.322 | 480 | 17 | 570 | 97.32 | 660 | 28.1 | 750 | 3.21 |
| 395 | 1.5 | 485 | 17.58 | 575 | 94.88 | 665 | 25.2 | 755 | 2.851 |
| 400 | 2.116 | 490 | 20.91 | 580 | 92.72 | 670 | 22.56 | 760 | 2.514 |
| 405 | 3.456 | 495 | 27.08 | 585 | 89.58 | 675 | 20.04 | 765 | 2.24 |
| 410 | 6.516 | 500 | 36.8 | 590 | 85.76 | 680 | 17.85 | 770 | 1.971 |
| 415 | 13.22 | 505 | 47.42 | 595 | 82.11 | 685 | 15.87 | 775 | 1.737 |
| 420 | 25.87 | 510 | 59.22 | 600 | 77.98 | 690 | 14.14 | 780 | 1.542 |
| 425 | 45.06 | 515 | 69.73 | 605 | 73.25 | 695 | 12.51 | | |
| 430 | 70.84 | 520 | 79.04 | 610 | 68.77 | 700 | 11.06 | | |
| 435 | 99.01 | 525 | 86.66 | 615 | 64.38 | 705 | 9.803 | | |



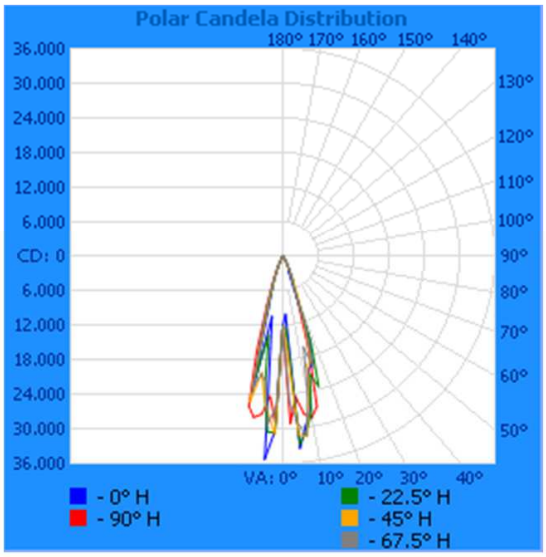
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-004 | UP | 120.0 | 1223 | 145.6 | 0.992 | 7522 | 51.66 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 12182 | 12182 | 12182 | 12182 | 12182 |
| 5 | 33484 | 32846 | 31122 | 29099 | 24506 |
| 10 | 26950 | 27931 | 25253 | 24231 | 28425 |
| 15 | 19212 | 23416 | 18431 | 20092 | 17228 |
| 20 | 2760 | 4930 | 6954 | 4860 | 5452 |
| 25 | 2697 | 2052 | 1993 | 1739 | 2177 |
| 30 | 1382 | 951 | 496 | 556 | 121 |
| 35 | 361 | 172 | 113 | 65 | 20 |
| 40 | 45 | 32 | 15 | 7 | 2 |
| 45 | 18 | 6 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

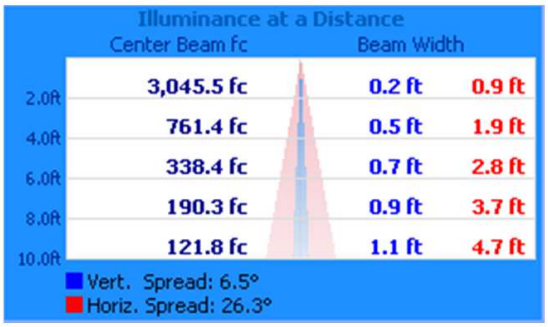


RESULTS OF TEST (cont'd)

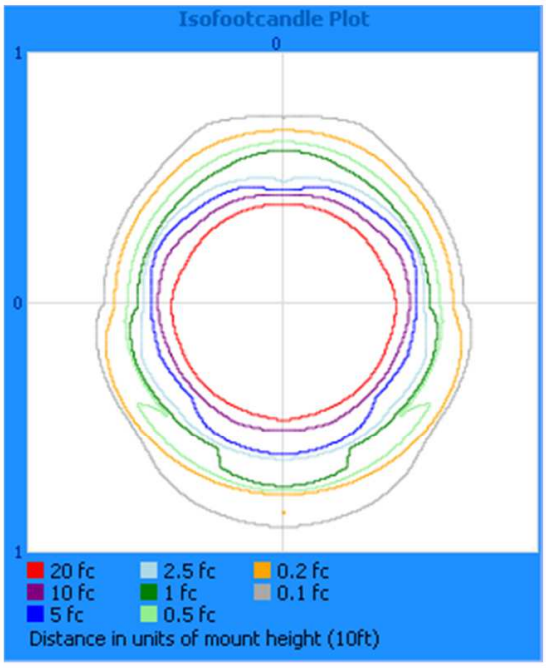
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 7461 | 99.2 |
| 0-40 | 7520 | 100.0 |
| 0-60 | 7522 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 7522 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 7522 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2482 | 33.0 |
| 10-20 | 4097 | 54.5 |
| 20-30 | 882.5 | 11.7 |
| 30-40 | 59.5 | 0.8 |
| 40-50 | 2.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as: Luxeon Rebel

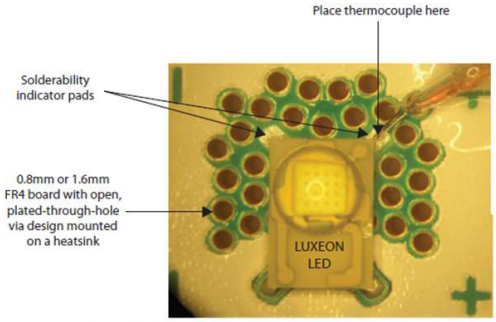


Figure 1. Solderability indicator pad and thermocouple placement.

| Nominal ANSI CCT | Forward Voltage $V_f$ [1] (V) Min. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|
| 2700K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 3000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 4000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 5000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |

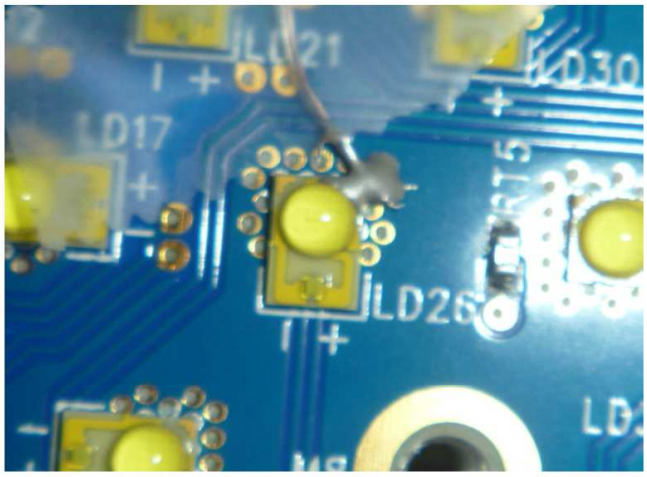
| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 592mA
Calculated LED Wattage = Vf x Measured LED Current = 2.072W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 137.6°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-004 | 81.5 | Per diagram | 137.6 |

In-Situ Picture – Ts

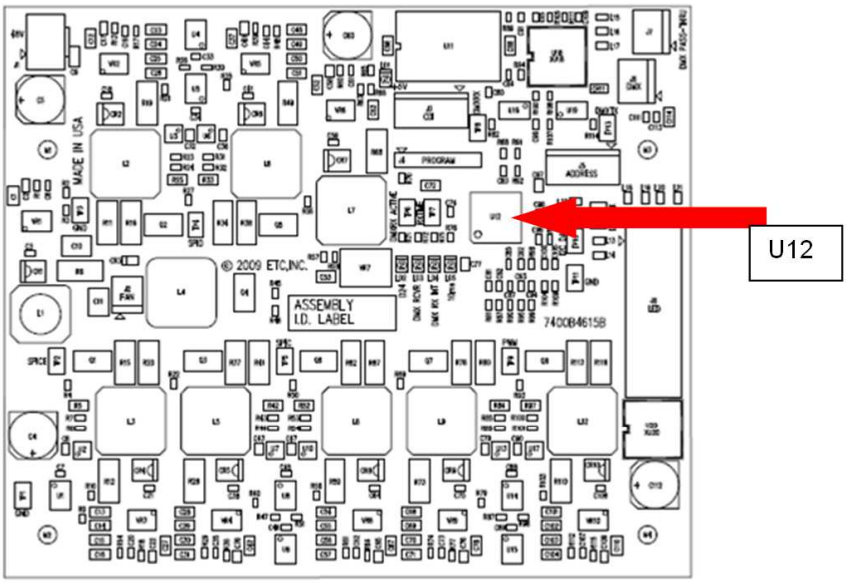



In-Situ Picture – Ts location

RESULTS OF TEST (cont'd)

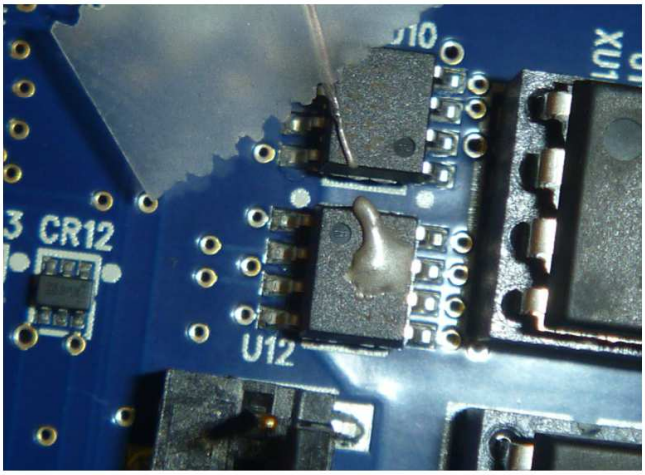
Maximum Measured Power Supply Case Temperature

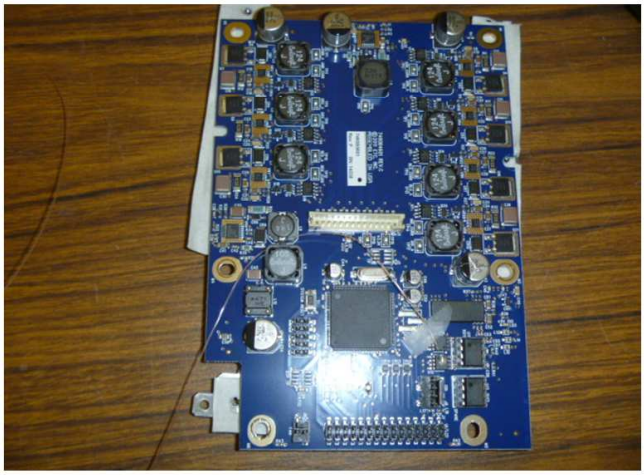
Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-004 | 52.8 | Per diagram | 100 |

Power Supply Picture – Tc location



PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Attachment: None

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

# SECTION 3 - Desire D22

Lustr+
Studio HD
Tungsten
Daylight

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: October 7 , 2013

REPORT NO. 101158242CRT-013

TEST OF ONE STAGE LIGHT

MODEL NO. D22 LUSTR+
DRIVER MODEL NO. 7410B4605
LED MODEL NO. LUXEON REBEL ES

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Version 1.2 (2012): Program Requirements for Luminaires (Light Fixtures)

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D22 LUSTR+. The sample was received by Intertek on September 6, 2013, in undamaged condition and one sample was tested as received. The sample designation was 1309060840-002.

DATES OF TESTS: September 30, 2013 through October 3, 201

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D22 LUSTR+ |
| Description: | STAGE LIGHT |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 3671 | 3482 |
| Total Power (W) | 51.70 | 50.77 |
| Luminaire Efficacy (LPW) | 71.01 | 68.58 |

| Criteria | Result |
|---|---|
| Power Factor | 0.982 |
| Current ATHD % | 6.05 |
| Correlated Color Temperature (CCT - K) | 5375 |
| Color Rendering Index (CRI - Ra) | 68.7 |
| Color Rendering Index (CRI - R9) | -11.5 |
| DUV | 0.007 |
| Chromaticity Coordinate (x) | 0.336 |
| Chromaticity Coordinate (y) | 0.358 |
| Chromaticity Coordinate (u') | 0.203 |
| Chromaticity Coordinate (v') | 0.486 |
| Maximum In-Situ Source Temperature Point (°C) | 67.5 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Fluke Multimeter | 45 | M133 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 09/20/13 | 10/20/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | --- | N1132 | 04/22/13 | 04/22/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Fluke Multimeter | PM2525 | M127 | 10/17/12 | 10/17/13 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

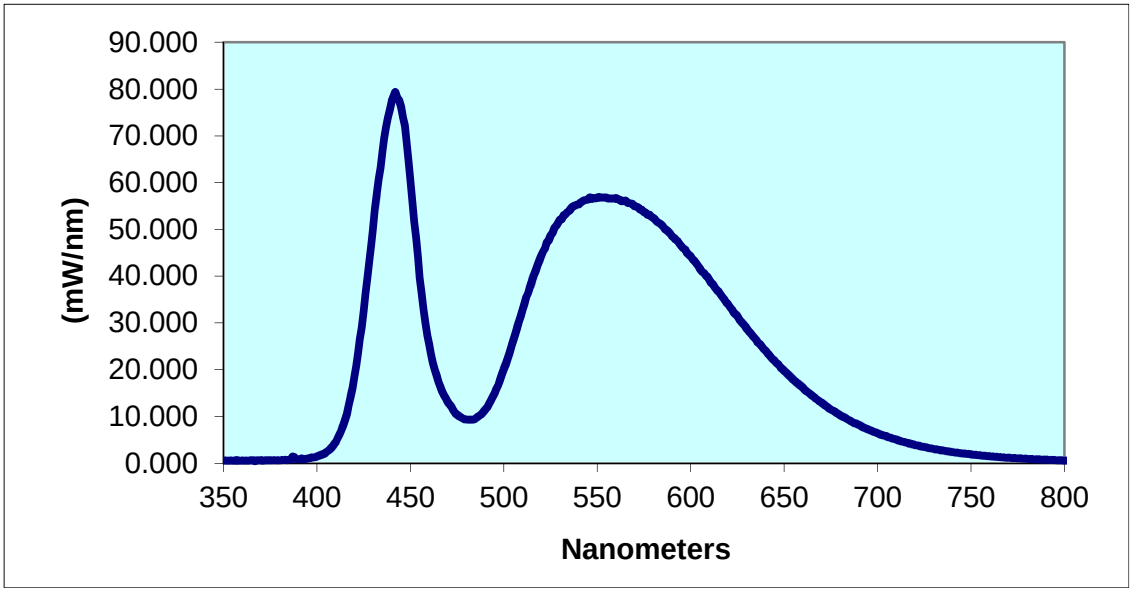
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| 1309060840-002 | UP | 120.0 | 438.5 | 51.70 | 0.982 | 6.05 | 3671 | 71.01 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 5375 | 68.7 | -11.5 | 0.007 | 0.336 | 0.358 | 0.203 | 0.486 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.52 | 440 | 77.42 | 530 | 52.06 | 620 | 33.99 | 710 | 5.03 |
| 355 | 0.54 | 445 | 76.23 | 535 | 53.95 | 625 | 31.45 | 715 | 4.42 |
| 360 | 0.60 | 450 | 60.57 | 540 | 55.37 | 630 | 28.82 | 720 | 3.91 |
| 365 | 0.58 | 455 | 39.67 | 545 | 56.36 | 635 | 26.32 | 725 | 3.46 |
| 370 | 0.65 | 460 | 25.38 | 550 | 56.79 | 640 | 24.04 | 730 | 3.04 |
| 375 | 0.60 | 465 | 17.38 | 555 | 56.75 | 645 | 21.88 | 735 | 2.69 |
| 380 | 0.57 | 470 | 13.06 | 560 | 56.71 | 650 | 19.79 | 740 | 2.37 |
| 385 | 0.70 | 475 | 10.40 | 565 | 56.17 | 655 | 17.86 | 745 | 2.12 |
| 390 | 0.78 | 480 | 9.35 | 570 | 54.91 | 660 | 16.12 | 750 | 1.90 |
| 395 | 1.03 | 485 | 9.65 | 575 | 53.67 | 665 | 14.45 | 755 | 1.66 |
| 400 | 1.45 | 490 | 11.41 | 580 | 52.49 | 670 | 12.98 | 760 | 1.49 |
| 405 | 2.38 | 495 | 14.92 | 585 | 50.70 | 675 | 11.51 | 765 | 1.30 |
| 410 | 4.51 | 500 | 20.12 | 590 | 48.47 | 680 | 10.26 | 770 | 1.17 |
| 415 | 9.30 | 505 | 26.16 | 595 | 46.45 | 685 | 9.161 | 775 | 1.03 |
| 420 | 18.48 | 510 | 32.88 | 600 | 44.27 | 690 | 8.179 | 780 | 0.92 |
| 425 | 32.44 | 515 | 39.01 | 605 | 41.62 | 695 | 7.243 | | |
| 430 | 50.09 | 520 | 44.41 | 610 | 39.12 | 700 | 6.429 | | |
| 435 | 66.79 | 525 | 48.66 | 615 | 36.63 | 705 | 5.698 | | |

**Spectral Data Over Visible Wavelengths**



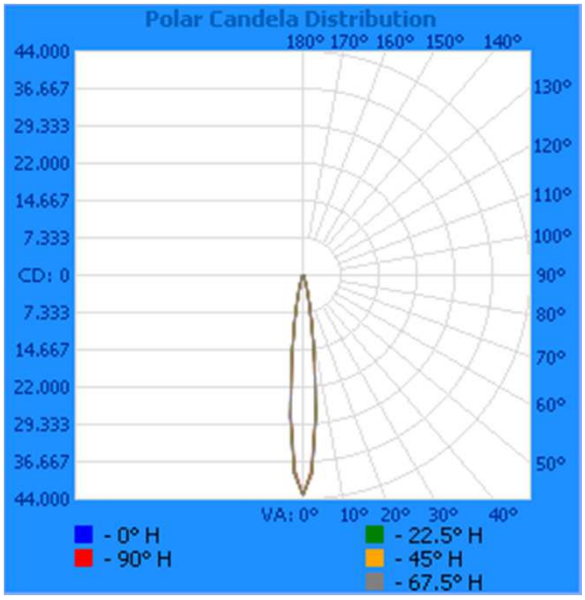
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| 1309060840-002 | Select One | 120.0 | 429.9 | 50.77 | 0.984 | 3482 | 68.58 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 43220 | 43220 | 43220 | 43220 | 43220 |
| 5 | 28176 | 27875 | 27829 | 27720 | 27476 |
| 10 | 9919 | 9864 | 9876 | 9719 | 9738 |
| 15 | 3146 | 3191 | 3179 | 3092 | 3095 |
| 20 | 1188 | 1206 | 1207 | 1192 | 1176 |
| 25 | 492 | 497 | 495 | 490 | 489 |
| 30 | 253 | 254 | 249 | 249 | 248 |
| 35 | 158 | 158 | 154 | 155 | 152 |
| 40 | 110 | 106 | 109 | 107 | 101 |
| 45 | 77 | 76 | 78 | 78 | 73 |
| 50 | 50 | 52 | 54 | 52 | 51 |
| 55 | 30 | 30 | 33 | 33 | 30 |
| 60 | 15 | 16 | 16 | 17 | 15 |
| 65 | 4 | 3 | 4 | 6 | 5 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

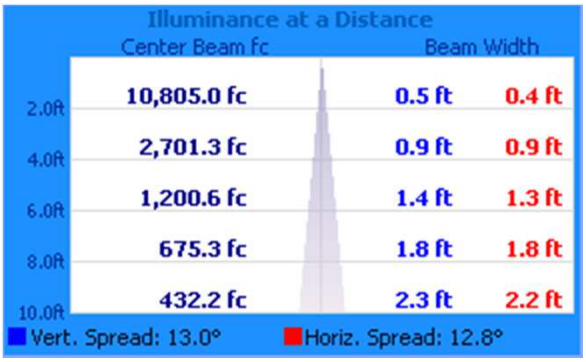


RESULTS OF TEST (cont'd)

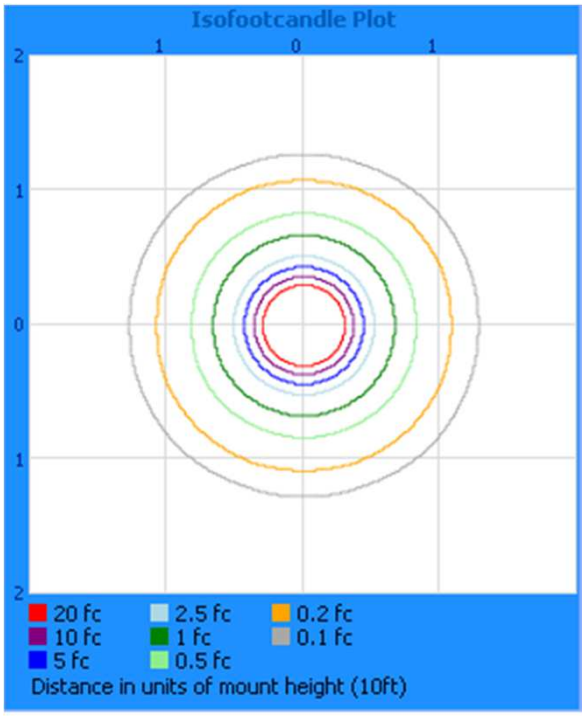
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot

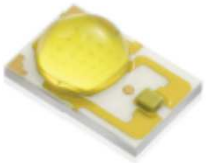



Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3287 | 94.4 |
| 0-40 | 3388 | 97.3 |
| 0-60 | 3477 | 99.8 |
| 60-90 | 5.6 | 0.2 |
| 0-90 | 3482 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3482 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2022 | 58.1 |
| 10-20 | 1010 | 29.0 |
| 20-30 | 255.2 | 7.3 |
| 30-40 | 101.3 | 2.9 |
| 40-50 | 59.4 | 1.7 |
| 50-60 | 28.8 | 0.8 |
| 60-70 | 5.6 | 0.2 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |



RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: Luxeom Rebel ES

Electrical Characteristics at 700 mA for LUXEON Rebel ES
Thermal Pad Temperature = 25°C

Table 4.

| Color | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{JC}$ |
|---|---|---|---|---|---|
| Neutral-White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |
| Cool-White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |

Absolute Maximum Ratings

Table 6.

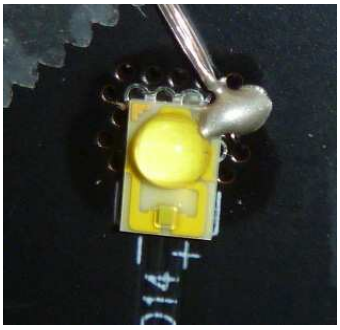
| Parameter | Cool-White / Neutral-White |
|---|---|
| DC Forward Current (mA) | 1000 mA |
| Peak Pulsed Forward Current (mA) [2] | 1200 mA |
| Average Forward Current (mA) | 1000 mA |
| ESD Sensitivity | < 8000V Human Body Model (HBM)<br>Class 3B JESD22-A114-E |
| LED Junction Temperature [1] | 150°C |
| Operating Case Temperature at 350 mA | -40°C - 135°C |
| Storage Temperature | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 |
| Autoclave Conditions | 121°C at 2 ATM<br>100% Relative Humidity for 96 Hours Maximum |
| Reverse Voltage (VR) | LUXEON Rebel ES LEDs are not designed to be driven in reverse bias |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.45V
Measured LED Current = 587mA
Calculated LED Wattage = Vf x Measured LED Current = 2.025W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 137.8°C

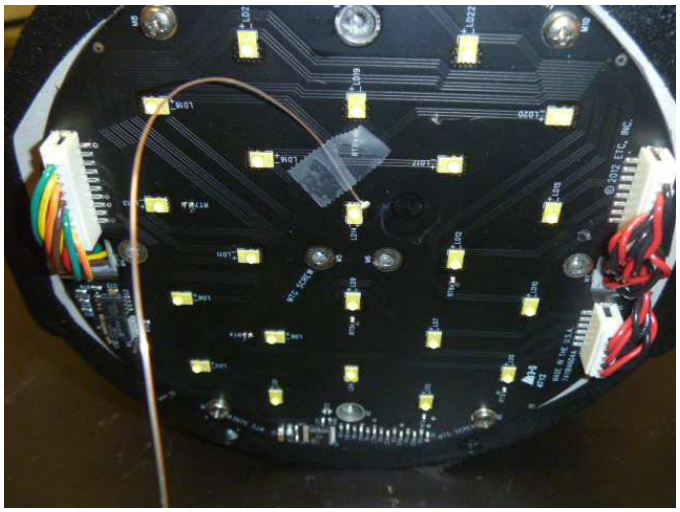
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| 1309060840-002 | 67.5 | Per diagram | 137.8 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Jacki Swiernik
Staff Engineer
Lighting Division

Report Reviewed By:

David Ellis
Senior Project Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 29, 2013

REPORT NO. 101414036CRT-001

TEST OF ONE D22 STUDIO HD PORTABLE LED ETL FIX (BLK)

MODEL NO. D22 STUDIO HD
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D22 STUDIO HD. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-001.

DATES OF TESTS: November 19, 2013 through November 25, 2013.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D22 STUDIO HD |
| Description: | D22 STUDIO HD PORTABLE LED ETL FIX (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 1822 | 1761 |
| Total Power (W) | 47.53 | 47.79 |
| Luminaire Efficacy (LPW) | 38.33 | 36.85 |

| Criteria | Result |
|---|---|
| Power Factor | 0.978 |
| Current ATHD % | 5.93 |
| Correlated Color Temperature (CCT - K) | 4468 |
| Color Rendering Index (CRI - Ra) | 85.8 |
| Color Rendering Index (CRI - R9) | 78.5 |
| DUV | 0.003 |
| Chromaticity Coordinate (x) | 0.361 |
| Chromaticity Coordinate (y) | 0.359 |
| Chromaticity Coordinate (u') | 0.219 |
| Chromaticity Coordinate (v') | 0.490 |
| Maximum In-Situ Source Temperature Point (°C) | 73.3 |
| Maximum In-Situ Driver Case Temperature (°C) | 73.5 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Fluke Multimeter | 45 | M133 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 10/21/13 | 11/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole-Palmer Hygro Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1355 | 11/26/12 | 11/26/13 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

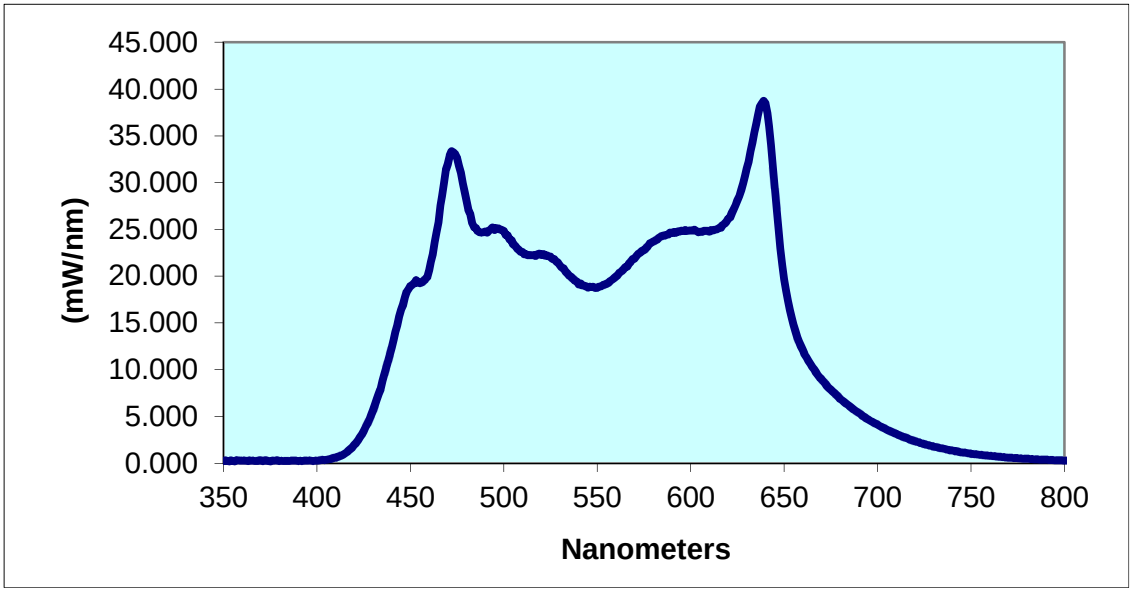
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-001 | UP | 120.0 | 404.7 | 47.53 | 0.978 | 5.93 | 1822 | 38.33 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 4468 | 85.8 | 78.5 | 0.003 | 0.361 | 0.359 | 0.219 | 0.490 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.216 | 440 | 12.38 | 530 | 21.18 | 620 | 26.15 | 710 | 3.13 |
| 355 | 0.216 | 445 | 16.41 | 535 | 20.03 | 625 | 28.13 | 715 | 2.728 |
| 360 | 0.25 | 450 | 18.94 | 540 | 19.11 | 630 | 31.51 | 720 | 2.368 |
| 365 | 0.269 | 455 | 19.25 | 545 | 18.78 | 635 | 36.31 | 725 | 2.053 |
| 370 | 0.309 | 460 | 20.64 | 550 | 18.78 | 640 | 38.44 | 730 | 1.782 |
| 375 | 0.209 | 465 | 25.76 | 555 | 19.2 | 645 | 29.37 | 735 | 1.551 |
| 380 | 0.226 | 470 | 32 | 560 | 19.97 | 650 | 19.66 | 740 | 1.356 |
| 385 | 0.239 | 475 | 32.61 | 565 | 20.98 | 655 | 14.83 | 745 | 1.167 |
| 390 | 0.231 | 480 | 28.17 | 570 | 21.88 | 660 | 12.05 | 750 | 1.024 |
| 395 | 0.266 | 485 | 25.14 | 575 | 22.78 | 665 | 10.31 | 755 | 0.887 |
| 400 | 0.277 | 490 | 24.76 | 580 | 23.7 | 670 | 8.991 | 760 | 0.792 |
| 405 | 0.358 | 495 | 24.94 | 585 | 24.33 | 675 | 7.853 | 765 | 0.687 |
| 410 | 0.59 | 500 | 24.83 | 590 | 24.61 | 680 | 6.876 | 770 | 0.587 |
| 415 | 1.045 | 505 | 23.44 | 595 | 24.87 | 685 | 6.086 | 775 | 0.531 |
| 420 | 1.966 | 510 | 22.57 | 600 | 24.85 | 690 | 5.349 | 780 | 0.455 |
| 425 | 3.421 | 515 | 22.22 | 605 | 24.73 | 695 | 4.676 | | |
| 430 | 5.605 | 520 | 22.28 | 610 | 24.78 | 700 | 4.129 | | |
| 435 | 8.811 | 525 | 22.09 | 615 | 25.13 | 705 | 3.57 | | |

**Spectral Data Over Visible Wavelengths**



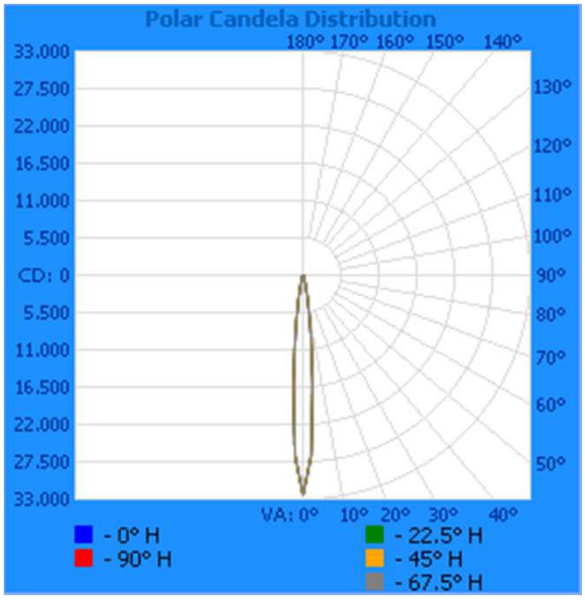
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-001 | UP | 120.0 | 405.7 | 47.79 | 0.982 | 1761 | 36.85 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 32341 | 32341 | 32341 | 32341 | 32341 |
| 5 | 16099 | 16192 | 16153 | 16105 | 16324 |
| 10 | 4346 | 4433 | 4527 | 4466 | 4418 |
| 15 | 1217 | 1233 | 1266 | 1246 | 1230 |
| 20 | 490 | 487 | 486 | 491 | 492 |
| 25 | 208 | 203 | 198 | 203 | 205 |
| 30 | 106 | 102 | 99 | 102 | 105 |
| 35 | 65 | 66 | 62 | 63 | 60 |
| 40 | 40 | 43 | 43 | 47 | 37 |
| 45 | 31 | 32 | 28 | 29 | 29 |
| 50 | 22 | 20 | 20 | 19 | 20 |
| 55 | 10 | 10 | 10 | 11 | 12 |
| 60 | 1 | 1 | 3 | 4 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

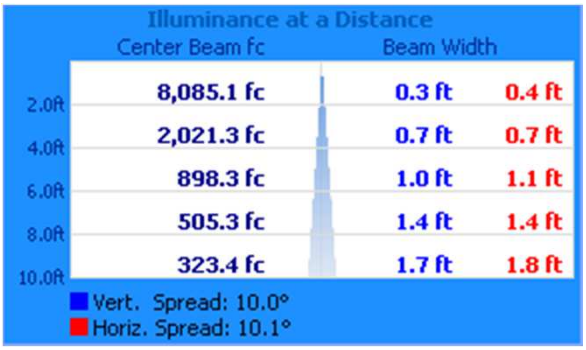


RESULTS OF TEST (cont'd)

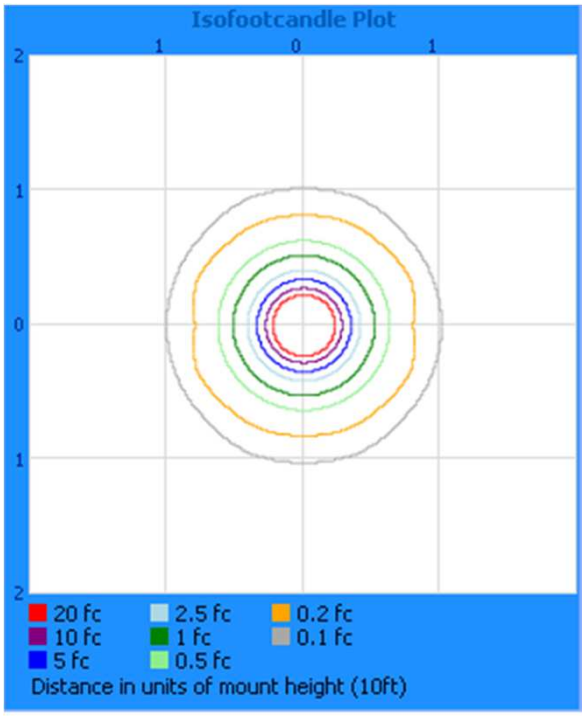
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 1686 | 95.8 |
| 0-40 | 1727 | 98.1 |
| 0-60 | 1760 | 100.0 |
| 60-90 | 0.4 | 0.0 |
| 0-90 | 1761 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 1761 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 1161 | 66.0 |
| 10-20 | 421.0 | 23.9 |
| 20-30 | 104.3 | 5.9 |
| 30-40 | 40.7 | 2.3 |
| 40-50 | 23.4 | 1.3 |
| 50-60 | 9.7 | 0.6 |
| 60-70 | 0.4 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as:

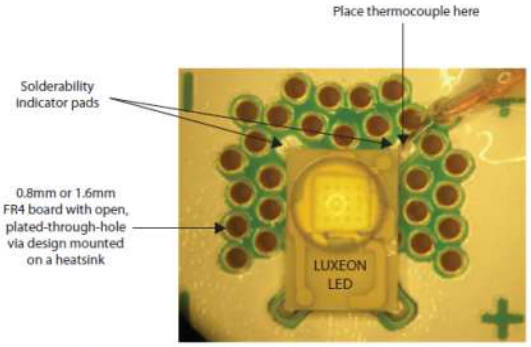


Figure 1. Solderability indicator pad and thermocouple placement.

| Forward Voltage $V_f$ [1] (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) | Typical Thermal Resistance Junction to Thermal Pad (°C/W) |
|---|---|---|---|---|
| Min. | Typ. | Max. | $\Delta V_f / \Delta T_j$ | $R\theta_{j-c}$ |
| 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |

| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

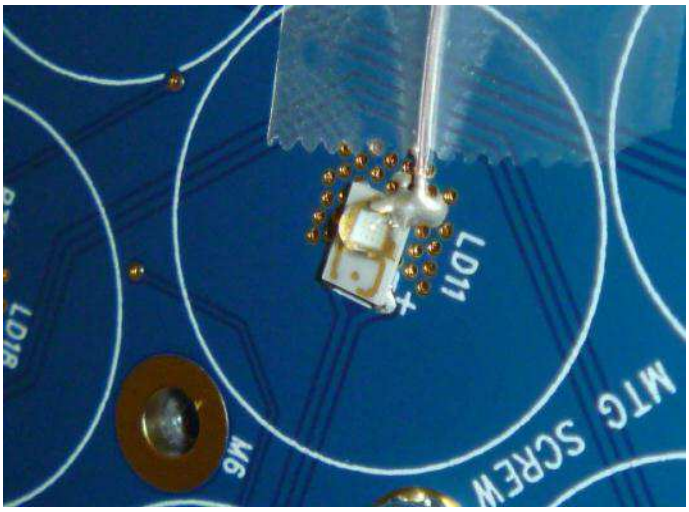
Green LXML-PM01

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.3V
Measured LED Current = 589mA
Calculated LED Wattage = Vf x Measured LED Current = 1.944W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 130.6°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-001 | 73.3 | Per diagram | 130.6 |

In-Situ Picture – Ts

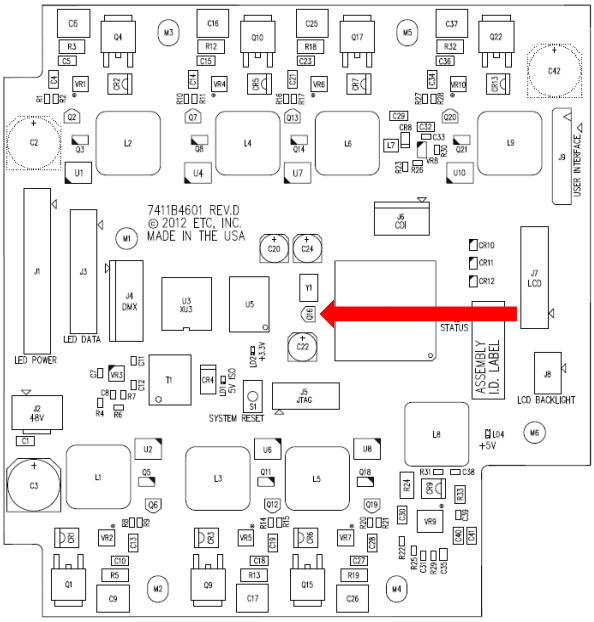



In-Situ Picture – Ts location

RESULTS OF TEST (cont'd)

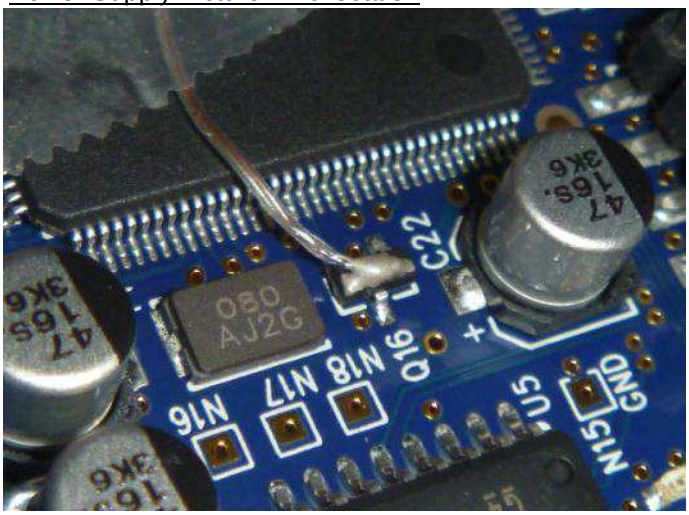
Maximum Measured Power Supply Case Temperature

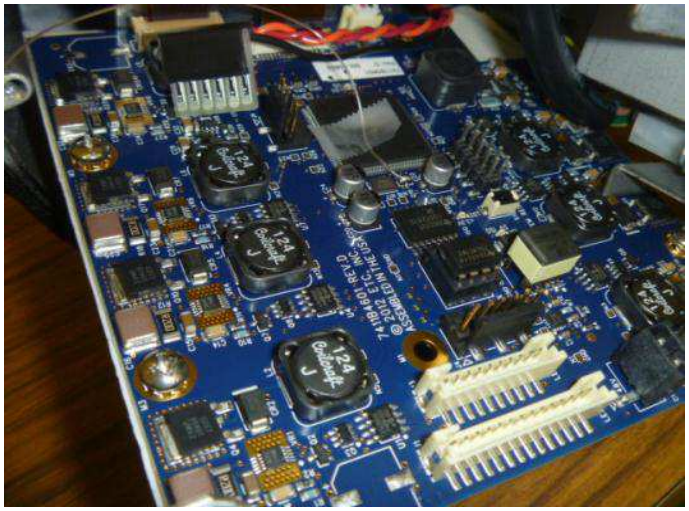
Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-001 | 73.5 | Per diagram | 100 |

Power Supply Picture – Tc location



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: October 7, 2013

REPORT NO. 101158242CRT-014

TEST OF ONE STAGE LUMINAIRE

MODEL NO. D22 TUNGSTEN
DRIVER MODEL NO. 7410B4605
LED MODEL NO. LUXEON REBEL ES

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Version 1.2 (2012): Program Requirements for Luminaires (Light Fixtures)

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D22 TUNGSTEN. The sample was received by Intertek on September 6, 2013, in undamaged condition and one sample was tested as received. The sample designation was 1309060840-001.

DATES OF TESTS: September 30, 2013 through October 4, 201

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D22 TUNGSTEN |
| Description: | STAGE LUMINAIRE |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 2572 | 2427 |
| Total Power (W) | 52.34 | 51.17 |
| Luminaire Efficacy (LPW) | 49.14 | 47.43 |

| Criteria | Result |
|---|---|
| Power Factor | 0.969 |
| Current ATHD % | 7.83 |
| Correlated Color Temperature (CCT - K) | 3131 |
| Color Rendering Index (CRI - Ra) | 86.2 |
| Color Rendering Index (CRI - R9) | 26.3 |
| DUV | 0.000 |
| Chromaticity Coordinate (x) | 0.428 |
| Chromaticity Coordinate (y) | 0.401 |
| Chromaticity Coordinate (u') | 0.246 |
| Chromaticity Coordinate (v') | 0.519 |
| Maximum In-Situ Source Temperature Point (°C) | 63.5 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Fluke Multimeter | 45 | M133 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 09/20/13 | 10/20/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | --- | N1132 | 04/22/13 | 04/22/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Fluke Multimeter | PM2525 | M127 | 10/17/12 | 10/17/13 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

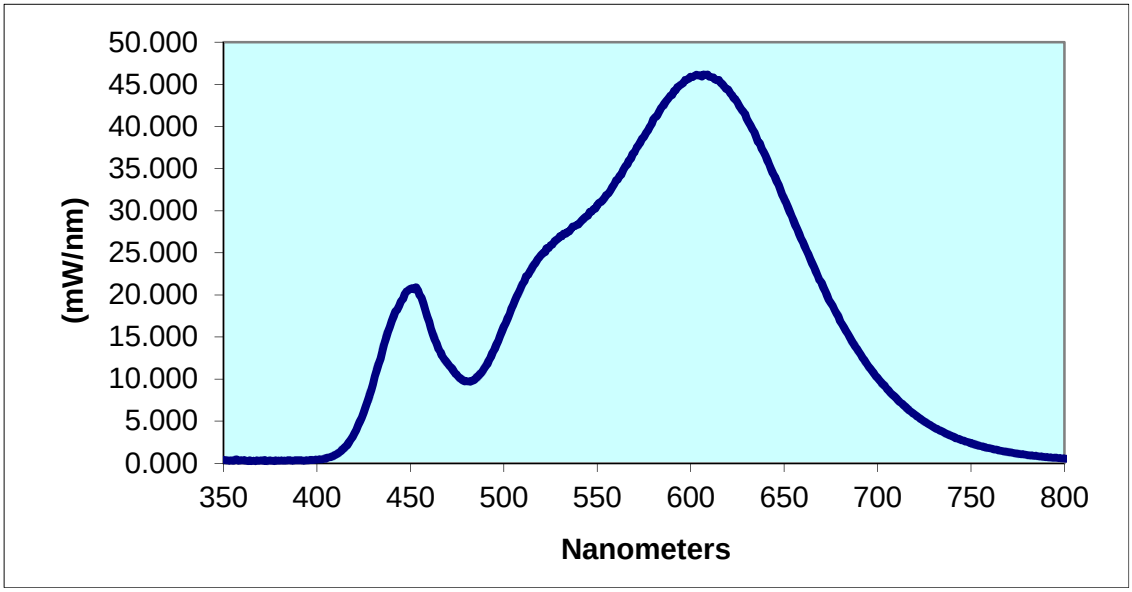
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| 1309060840-001 | UP | 120.0 | 449.9 | 52.34 | 0.969 | 7.83 | 2572 | 49.14 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3131 | 86.2 | 26.3 | 0.000 | 0.428 | 0.401 | 0.246 | 0.519 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.40 | 440 | 16.89 | 530 | 26.95 | 620 | 44.38 | 710 | 7.71 |
| 355 | 0.28 | 445 | 19.22 | 535 | 27.55 | 625 | 42.87 | 715 | 6.68 |
| 360 | 0.33 | 450 | 20.72 | 540 | 28.40 | 630 | 40.95 | 720 | 5.78 |
| 365 | 0.27 | 455 | 19.95 | 545 | 29.43 | 635 | 38.72 | 725 | 4.97 |
| 370 | 0.34 | 460 | 16.80 | 550 | 30.62 | 640 | 36.51 | 730 | 4.28 |
| 375 | 0.34 | 465 | 13.55 | 555 | 31.88 | 645 | 34.03 | 735 | 3.70 |
| 380 | 0.28 | 470 | 11.81 | 560 | 33.60 | 650 | 31.45 | 740 | 3.18 |
| 385 | 0.34 | 475 | 10.47 | 565 | 35.28 | 655 | 28.83 | 745 | 2.75 |
| 390 | 0.32 | 480 | 9.77 | 570 | 36.98 | 660 | 26.24 | 750 | 2.38 |
| 395 | 0.35 | 485 | 10.11 | 575 | 38.76 | 665 | 23.71 | 755 | 2.04 |
| 400 | 0.41 | 490 | 11.43 | 580 | 40.80 | 670 | 21.41 | 760 | 1.79 |
| 405 | 0.59 | 495 | 13.52 | 585 | 42.40 | 675 | 19.05 | 765 | 1.52 |
| 410 | 1.02 | 500 | 16.24 | 590 | 43.73 | 680 | 16.86 | 770 | 1.32 |
| 415 | 1.91 | 505 | 18.77 | 595 | 45.01 | 685 | 14.98 | 775 | 1.14 |
| 420 | 3.54 | 510 | 21.32 | 600 | 45.91 | 690 | 13.23 | 780 | 0.98 |
| 425 | 6.00 | 515 | 23.18 | 605 | 46.05 | 695 | 11.55 | | |
| 430 | 9.36 | 520 | 24.75 | 610 | 45.91 | 700 | 10.15 | | |
| 435 | 13.39 | 525 | 25.89 | 615 | 45.52 | 705 | 8.85 | | |

**Spectral Data Over Visible Wavelengths**



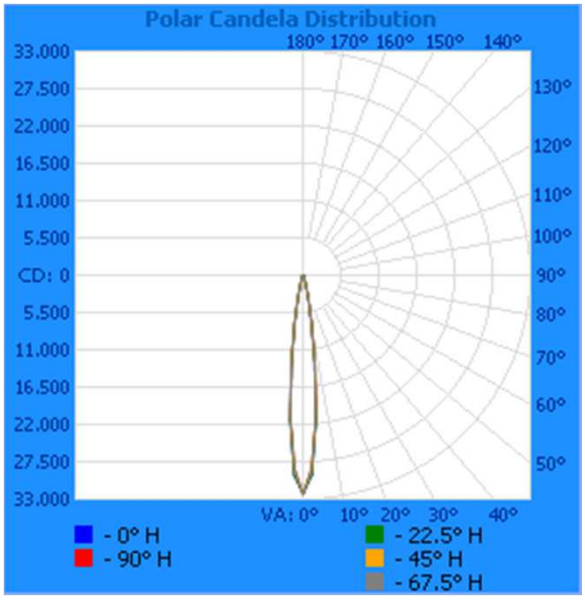
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| 1309060840-001 | UP | 120.0 | 439.1 | 51.17 | 0.971 | 2427 | 47.43 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 32212 | 32212 | 32212 | 32212 | 32212 |
| 5 | 21318 | 21443 | 20998 | 21008 | 21183 |
| 10 | 6750 | 6917 | 6972 | 7121 | 7254 |
| 15 | 1845 | 1874 | 1902 | 1920 | 1945 |
| 20 | 693 | 686 | 690 | 707 | 713 |
| 25 | 291 | 287 | 287 | 300 | 306 |
| 30 | 139 | 138 | 139 | 146 | 144 |
| 35 | 83 | 83 | 83 | 86 | 86 |
| 40 | 60 | 58 | 58 | 62 | 56 |
| 45 | 44 | 42 | 43 | 44 | 40 |
| 50 | 30 | 31 | 32 | 32 | 30 |
| 55 | 21 | 18 | 19 | 20 | 18 |
| 60 | 10 | 11 | 10 | 12 | 9 |
| 65 | 2 | 0 | 3 | 2 | 2 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

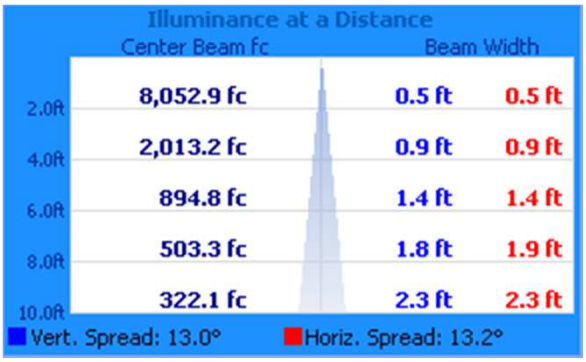


RESULTS OF TEST (cont'd)

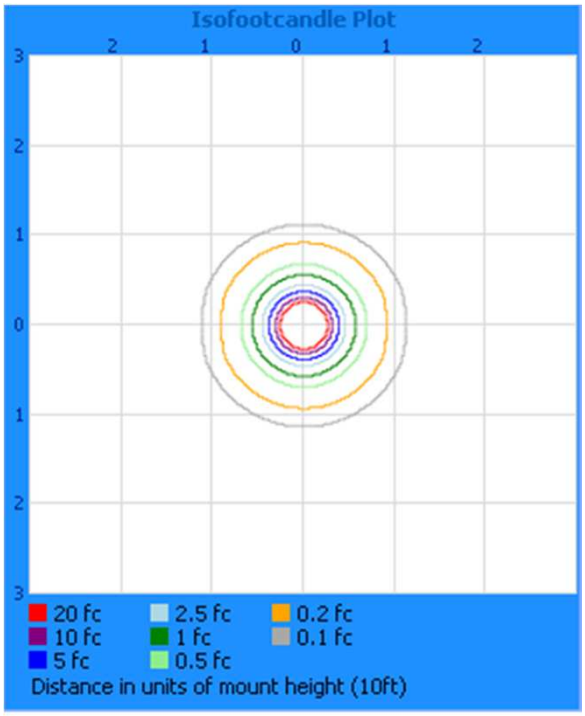
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 2317 | 95.5 |
| 0-40 | 2372 | 97.8 |
| 0-60 | 2424 | 99.9 |
| 60-90 | 3.1 | 0.1 |
| 0-90 | 2427 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 2427 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 1520 | 62.6 |
| 10-20 | 647.4 | 26.7 |
| 20-30 | 149.6 | 6.2 |
| 30-40 | 55.6 | 2.3 |
| 40-50 | 33.8 | 1.4 |
| 50-60 | 17.7 | 0.7 |
| 60-70 | 3.1 | 0.1 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: Luxeom Rebel ES

## Electrical Characteristics at 700 mA for LUXEON Rebel ES
## Thermal Pad Temperature = 25°C

Table 4.

| Color | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|---|
| Neutral White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |
| Cool White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |

## Absolute Maximum Ratings

Table 6.

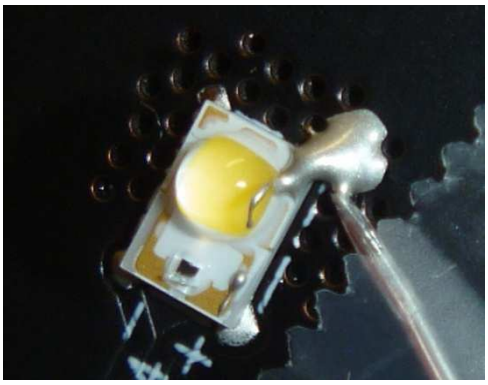
| Parameter | Cool White / Neutral White |
|---|---|
| DC Forward Current (mA) | 1000 mA |
| Peak Pulsed Forward Current (mA) [2] | 1200 mA |
| Average Forward Current (mA) | 1000 mA |
| ESD Sensitivity | < 8000V Human Body Model (HBM)<br>Class 3B JESD22-A114-E |
| LED Junction Temperature [1] | 150°C |
| Operating Case Temperature at 350 mA | -40°C - 135°C |
| Storage Temperature | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 |
| Autoclave Conditions | 121°C at 2 ATM<br>100% Relative Humidity for 96 Hours Maximum |
| Reverse Voltage (VR) | LUXEON Rebel ES LEDs are not designed to be driven in reverse bias |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 590mA
Calculated LED Wattage = Vf x Measured LED Current = 2.065W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 137.6°C

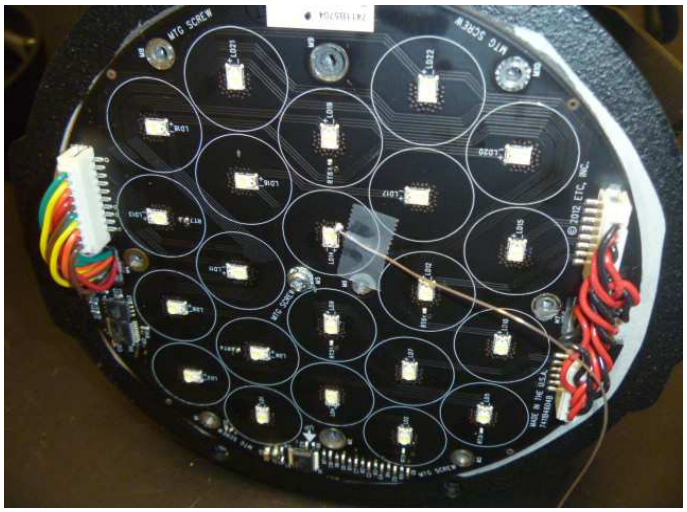
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| 1309060840-001 | 63.5 | Per diagram | 137.6 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Jacki Swiernik
Staff Engineer
Lighting Division

Report Reviewed By:

David Ellis
Senior Project Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 29, 2013

REPORT NO. 101414036CRT-002

TEST OF ONE D22 DAYLIGHT PORTABLE LED ETL FIX (BLK)

MODEL NO. D22 DAYLIGHT
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D22 DAYLIGHT. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-002.

DATES OF TESTS: November 20, 2013 through November 25, 2013.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D22 DAYLIGHT |
| Description: | D22 DAYLIGHT PORTABLE LED ETL FIX (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 3540 | 3510 |
| Total Power (W) | 53.32 | 53.44 |
| Luminaire Efficacy (LPW) | 66.39 | 65.68 |

| Criteria | Result |
|---|---|
| Power Factor | 0.982 |
| Current ATHD % | 6.12 |
| Correlated Color Temperature (CCT - K) | 5556 |
| Color Rendering Index (CRI - Ra) | 72.6 |
| Color Rendering Index (CRI - R9) | -6.2 |
| DUV | 0.007 |
| Chromaticity Coordinate (x) | 0.331 |
| Chromaticity Coordinate (y) | 0.354 |
| Chromaticity Coordinate (u') | 0.201 |
| Chromaticity Coordinate (v') | 0.484 |
| Maximum In-Situ Source Temperature Point (°C) | 73.0 |
| Maximum In-Situ Driver Case Temperature (°C) | 71.6 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Fluke Multimeter | 45 | M133 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Cole Palmer Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 10/21/13 | 11/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole-Palmer Hygro Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1355 | 11/26/12 | 11/26/13 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

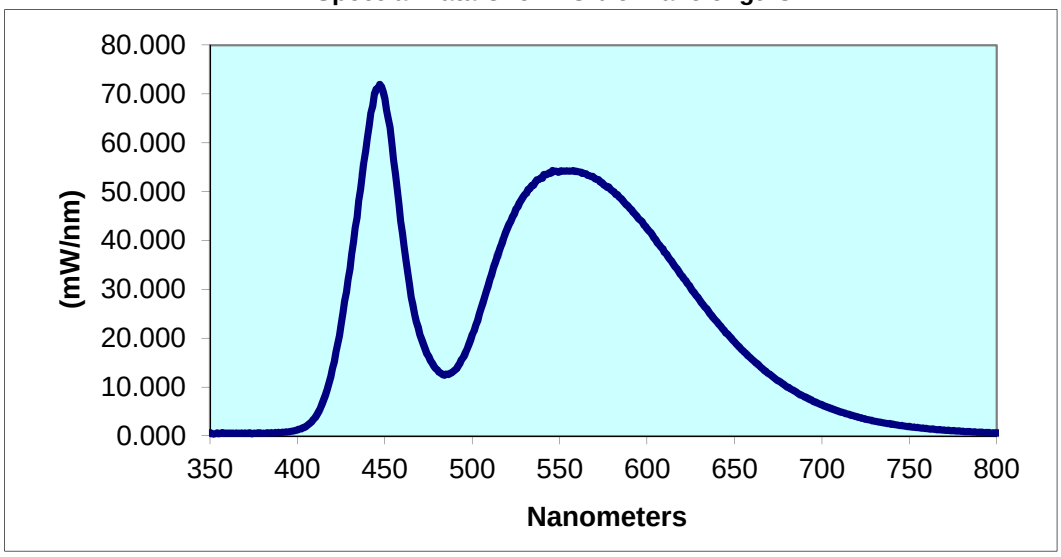
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-002 | UP | 120.0 | 452.1 | 53.32 | 0.982 | 6.12 | 3540 | 66.39 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 5556 | 72.6 | -6.2 | 0.007 | 0.331 | 0.354 | 0.201 | 0.484 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.67 | 440 | 60.93 | 530 | 49.43 | 620 | 32.71 | 710 | 5.004 |
| 355 | 0.452 | 445 | 70.99 | 535 | 51.35 | 625 | 30.24 | 715 | 4.45 |
| 360 | 0.535 | 450 | 69.02 | 540 | 52.78 | 630 | 27.89 | 720 | 3.912 |
| 365 | 0.596 | 455 | 56.48 | 545 | 53.81 | 635 | 25.5 | 725 | 3.448 |
| 370 | 0.618 | 460 | 41.44 | 550 | 54.11 | 640 | 23.33 | 730 | 3.039 |
| 375 | 0.56 | 465 | 28.36 | 555 | 54.23 | 645 | 21.14 | 735 | 2.704 |
| 380 | 0.546 | 470 | 20.81 | 560 | 54.12 | 650 | 19.2 | 740 | 2.397 |
| 385 | 0.691 | 475 | 16.25 | 565 | 53.66 | 655 | 17.39 | 745 | 2.12 |
| 390 | 0.688 | 480 | 13.49 | 570 | 52.69 | 660 | 15.66 | 750 | 1.885 |
| 395 | 0.876 | 485 | 12.63 | 575 | 51.42 | 665 | 14.06 | 755 | 1.688 |
| 400 | 1.238 | 490 | 13.54 | 580 | 50.2 | 670 | 12.67 | 760 | 1.49 |
| 405 | 1.994 | 495 | 16.2 | 585 | 48.62 | 675 | 11.3 | 765 | 1.334 |
| 410 | 3.756 | 500 | 20.76 | 590 | 46.57 | 680 | 10.06 | 770 | 1.181 |
| 415 | 7.455 | 505 | 26.04 | 595 | 44.52 | 685 | 9.026 | 775 | 1.034 |
| 420 | 13.76 | 510 | 32.07 | 600 | 42.42 | 690 | 8.02 | 780 | 0.927 |
| 425 | 22.76 | 515 | 37.46 | 605 | 40.07 | 695 | 7.118 | | |
| 430 | 34.21 | 520 | 42.42 | 610 | 37.59 | 700 | 6.379 | | |
| 435 | 48.11 | 525 | 46.52 | 615 | 35.21 | 705 | 5.632 | | |

**Spectral Data Over Visible Wavelengths**



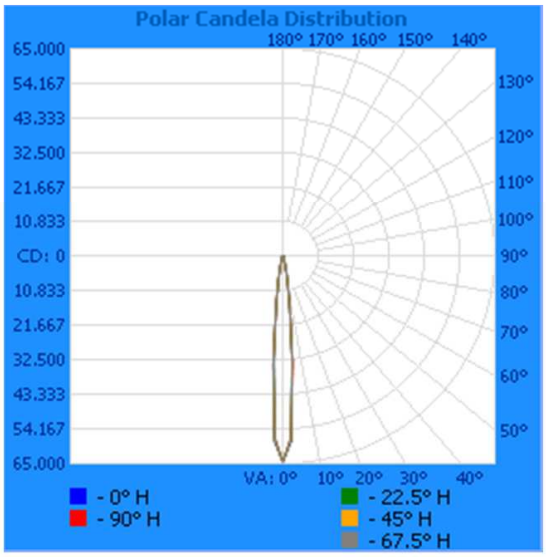
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-002 | UP | 120.0 | 451.9 | 53.44 | 0.985 | 3510 | 65.68 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 64464 | 64464 | 64464 | 64464 | 64464 |
| 5 | 32815 | 32518 | 33183 | 33655 | 34204 |
| 10 | 8318 | 8246 | 8339 | 8383 | 8367 |
| 15 | 2206 | 2324 | 2533 | 2438 | 2253 |
| 20 | 899 | 904 | 952 | 932 | 882 |
| 25 | 306 | 313 | 325 | 312 | 315 |
| 30 | 200 | 210 | 207 | 187 | 202 |
| 35 | 177 | 198 | 193 | 179 | 185 |
| 40 | 93 | 88 | 87 | 128 | 80 |
| 45 | 65 | 56 | 58 | 64 | 52 |
| 50 | 43 | 45 | 48 | 44 | 44 |
| 55 | 30 | 28 | 32 | 30 | 28 |
| 60 | 15 | 14 | 13 | 17 | 14 |
| 65 | 4 | 5 | 4 | 6 | 4 |
| 70 | 0 | 0 | 0 | 1 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

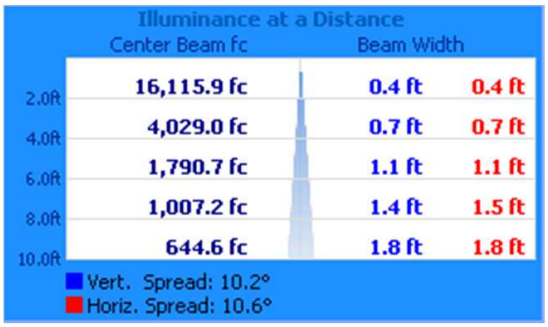


RESULTS OF TEST (cont'd)

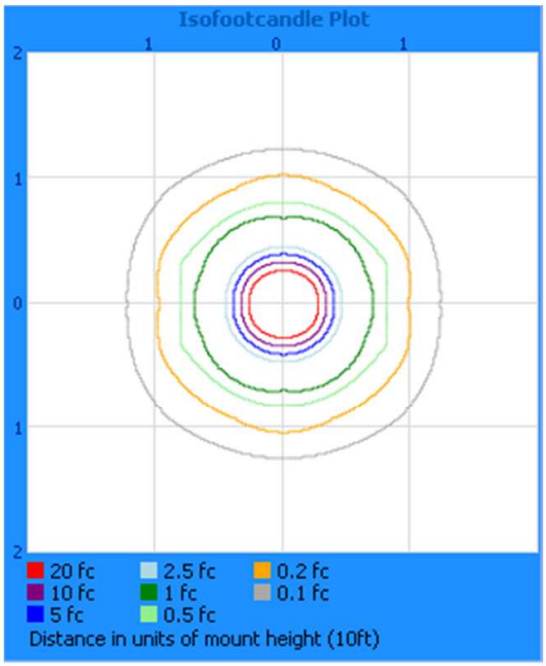
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3321 | 94.6 |
| 0-40 | 3428 | 97.7 |
| 0-60 | 3504 | 99.8 |
| 60-90 | 5.7 | 0.2 |
| 0-90 | 3510 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3510 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2339 | 66.6 |
| 10-20 | 801.7 | 22.8 |
| 20-30 | 180.4 | 5.1 |
| 30-40 | 106.8 | 3.0 |
| 40-50 | 49.5 | 1.4 |
| 50-60 | 26.5 | 0.8 |
| 60-70 | 5.6 | 0.2 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as: Luxeon Rebel A

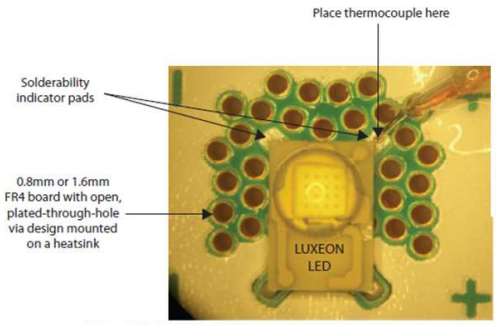


Figure 1. Solderability indicator pad and thermocouple placement.

| Nominal ANSI CCT | Forward Voltage $V_f$ [1] (V) Min. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|
| 2700K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 3000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 4000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 5000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |

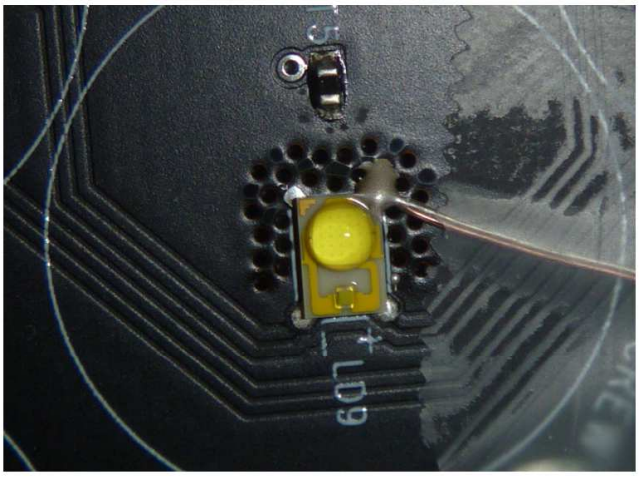
| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 585mA
Calculated LED Wattage = Vf x Measured LED Current = 2.047W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 137.7°C

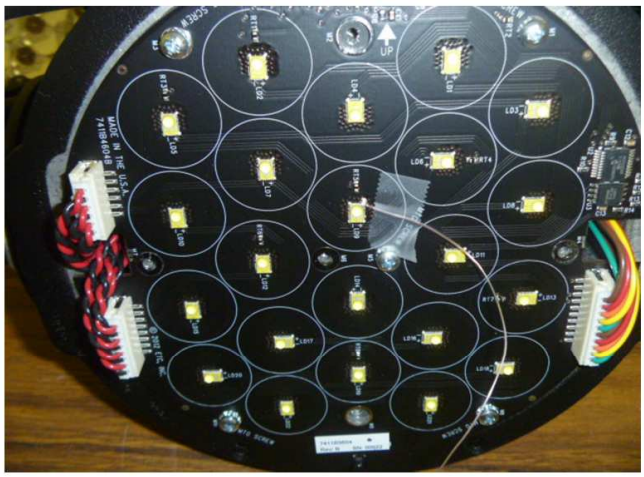
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-002 | 73.0 | Per diagram | 137.7 |

In-Situ Picture – Ts



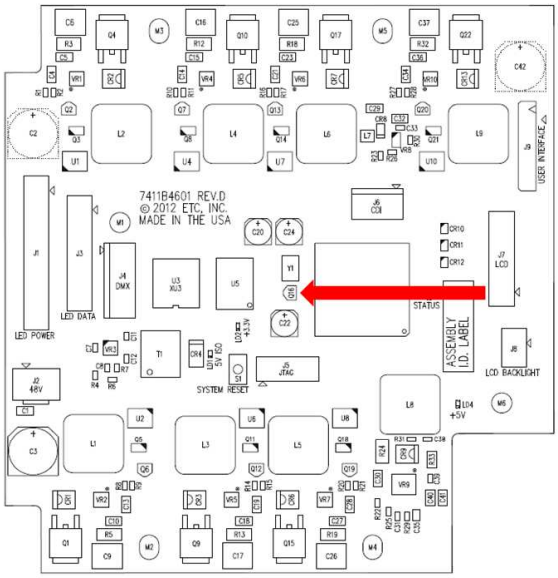
In-Situ Picture – Ts location



RESULTS OF TEST (cont'd)

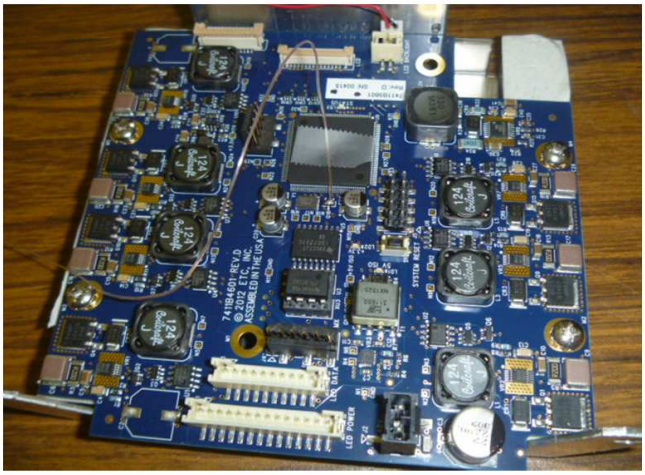
Maximum Measured Power Supply Case Temperature

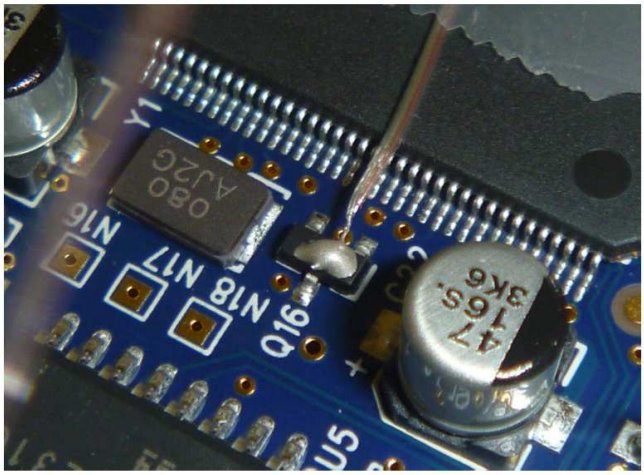
Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-001 | 71.6 | Per diagram | 100 |

Power Supply Picture – Tc location





PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

# SECTION 4 - Desire D40

Lustr+
Vivid
Studio HD
Tungsten
Daylight

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: May 23, 2013

REPORT NO. 101158242CRT-005

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. D40 LUSTR+

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79: 2008 Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products

ANSI ANSLG C38.377: 2012 Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D40 LUSTR+. The sample was received by Intertek on May 6, 2013, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-007.

DATES OF TESTS: May 15, 2013 through May 22, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | D40 LUSTR+ |
| Description: | LED WASH LIGHT |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 3303 | 3324 |
| Total Power (W) | 94.24 | 93.79 |
| Luminaire Efficacy (LPW) | 35.05 | 35.44 |

| Criteria | Result |
|---|---|
| Power Factor | 0.996 |
| Current ATHD (%) | 3.16 |
| Correlated Color Temperature (CCT - K) | 6139 |
| Color Rendering Index (CRI) – Ra | 64.5 |
| Color Rendering Index (CRI) - R9 | -60.9 |
| Duv | 0.029 |
| Chromaticity Coordinate (x) | 0.324 |
| Chromaticity Coordinate (y) | 0.284 |
| Chromaticity Coordinate (u') | 0.225 |
| Chromaticity Coordinate (v') | 0.443 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 05/20/13 | 06/20/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | --- | N1132 | 04/22/13 | 04/22/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Yokogawa Power Analyzer | WT1600 | E462 | 07/06/12 | 07/06/13 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | NA | NA | NA |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Sorensen DC Power Supply | DLM150-20E | N/A | N/A | N/A |
| FLUKE | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

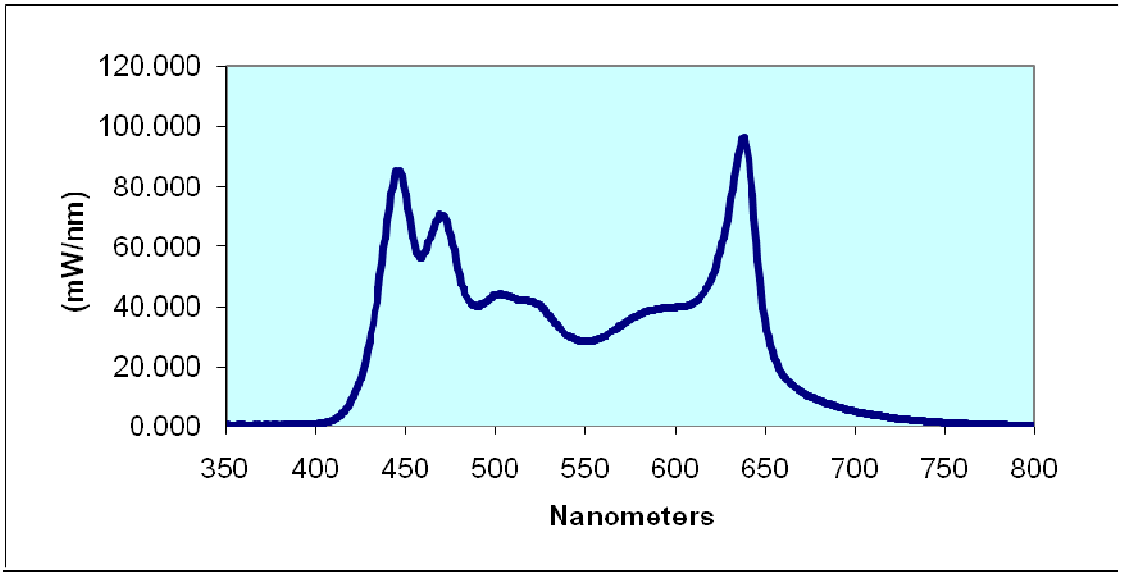
The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.770 | 460 | 57.199 | 570 | 33.728 | 680 | 8.589 |
| 355 | 0.587 | 465 | 64.405 | 575 | 35.622 | 685 | 7.525 |
| 360 | 0.676 | 470 | 70.295 | 580 | 37.315 | 690 | 6.574 |
| 365 | 0.634 | 475 | 63.499 | 585 | 38.571 | 695 | 5.739 |
| 370 | 0.597 | 480 | 50.295 | 590 | 38.916 | 700 | 5.059 |
| 375 | 0.647 | 485 | 41.893 | 595 | 39.307 | 705 | 4.438 |
| 380 | 0.613 | 490 | 39.954 | 600 | 39.648 | 710 | 3.922 |
| 385 | 0.681 | 495 | 41.373 | 605 | 39.913 | 715 | 3.429 |
| 390 | 0.714 | 500 | 43.460 | 610 | 41.108 | 720 | 3.045 |
| 395 | 0.785 | 505 | 43.612 | 615 | 43.643 | 725 | 2.629 |
| 400 | 0.953 | 510 | 42.726 | 620 | 48.890 | 730 | 2.340 |
| 405 | 1.363 | 515 | 41.992 | 625 | 58.167 | 735 | 2.038 |
| 410 | 2.281 | 520 | 41.418 | 630 | 72.279 | 740 | 1.790 |
| 415 | 4.468 | 525 | 40.109 | 635 | 90.276 | 745 | 1.613 |
| 420 | 8.659 | 530 | 37.048 | 640 | 93.514 | 750 | 1.407 |
| 425 | 16.052 | 535 | 33.183 | 645 | 62.005 | 755 | 1.255 |
| 430 | 28.282 | 540 | 30.390 | 650 | 35.558 | 760 | 1.107 |
| 435 | 45.917 | 545 | 28.705 | 655 | 23.010 | 765 | 0.000 |
| 440 | 68.324 | 550 | 28.114 | 660 | 17.090 | 770 | 0.888 |
| 445 | 85.245 | 555 | 28.688 | 665 | 13.755 | 775 | 0.778 |
| 450 | 76.756 | 560 | 29.933 | 670 | 11.601 | 780 | 0.709 |
| 455 | 59.913 | 565 | 31.713 | 675 | 9.858 | | |

**Sample No. CRT1304251519-007**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25°C – Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-007 | UP | 120.0 | 788 | 94.24 | 0.996 | 3.16 | 3303 | 35.05 |

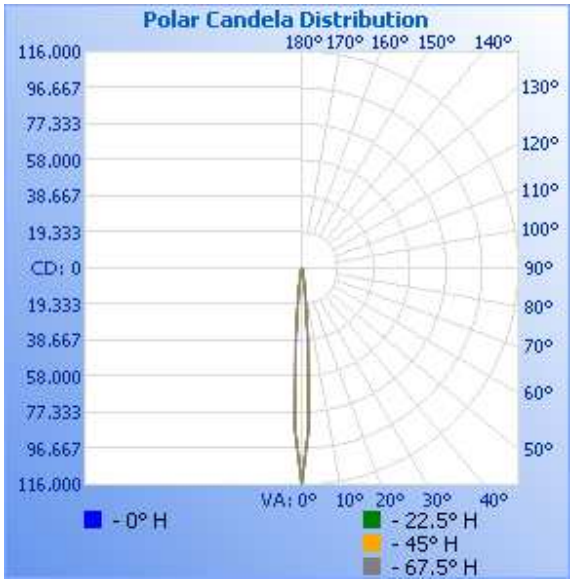
| Intertek Sample No. | Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-007 | 6139 | 64.5 | -60.9 | 0.029 | 0.324 | 0.284 | 0.225 | 0.443 |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1304251519-007 | UP | 120.1 | 784.0 | 93.79 | 0.996 | 3324 | 35.44 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 115486 | 115486 | 115486 | 115486 | 115486 |
| 5 | 35845 | 36062 | 36949 | 37990 | 38882 |
| 10 | 5905 | 6187 | 6423 | 6226 | 5950 |
| 15 | 1033 | 1133 | 1198 | 1080 | 1030 |
| 20 | 465 | 477 | 484 | 476 | 455 |
| 25 | 113 | 117 | 116 | 117 | 112 |
| 30 | 55 | 53 | 54 | 51 | 49 |
| 35 | 33 | 33 | 29 | 30 | 31 |
| 40 | 25 | 20 | 23 | 26 | 18 |
| 45 | 22 | 16 | 19 | 22 | 22 |
| 50 | 14 | 19 | 16 | 17 | 18 |
| 55 | 8 | 9 | 10 | 13 | 11 |
| 60 | 0 | 0 | 0 | 3 | 2 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

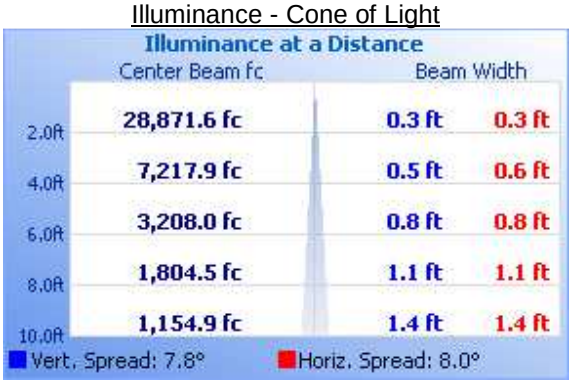


RESULTS OF TESTS (cont'd)

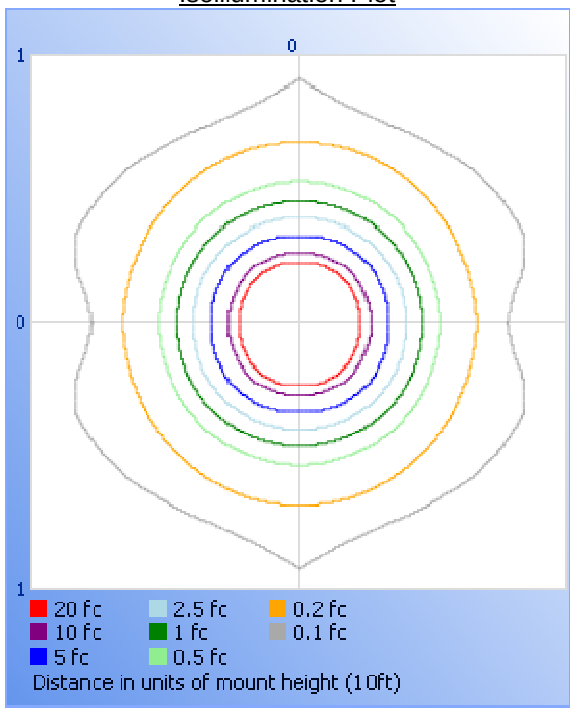
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3279 | 98.7 |
| 0-40 | 3300 | 99.3 |
| 0-60 | 3324 | 100.0 |
| 60-90 | 0.1 | 0.0 |
| 0-90 | 3324 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3324 | 100.0 |

Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

David Ellis
Senior Project Engineer
Lighting Division

Attachment: None

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: June 10, 2013

REPORT NO. 101158242CRT-008

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. D40 LUSTR+
LED MODEL NO. LUXEON REBEL ES

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and LED In-Situ tests.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

Energy Star Version 1.2 (2012): Program Requirements for Luminaires
Energy Star Manufacturer's Guide Version 2.0 (2009): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one sample of model number D40 LUSTR+. The sample was received by Intertek on May 6, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-007.

DATES OF TESTS: June 7, 2013

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D40 LUSTR+ |
| Description: | LED WASH LIGHT |

| Criteria | Result |
|---|---|
| Maximum In-Situ Source Temperature Point (°C) | 74.7 |
| Maximum Power Supply Case Temperature (°C) | 48.3 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Fluke Multimeter | PM2525 | M127 | 10/17/12 | 10/17/13 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

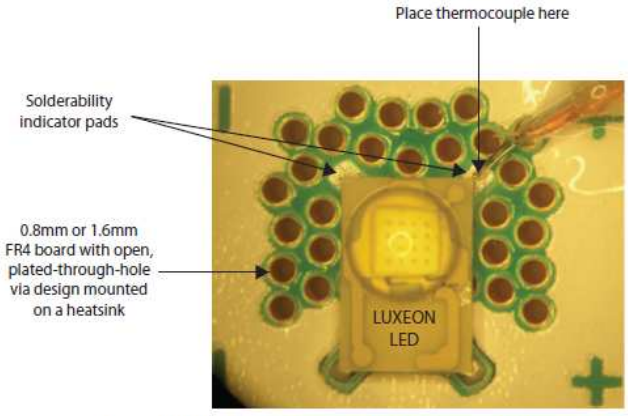
In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Agilent 34970A Data Logger. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TESTS

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: LUXEON REBEL ES




| Color | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|---|
| Neutral-White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |
| Cool-White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |

| Parameter | Cool-White / Neutral-White |
|---|---|
| DC Forward Current (mA) | 1000 mA |
| Peak Pulsed Forward Current (mA) [2] | 1200 mA |
| Average Forward Current (mA) | 1000 mA |
| ESD Sensitivity | < 8000V Human Body Model (HBM)<br>Class 3B JESD22-A114-E |
| LED Junction Temperature[3] | 150°C |
| Operating Case Temperature at 350 mA | -40°C - 135°C |
| Storage Temperature | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 |
| Autoclave Conditions | 121°C at 2 ATM<br>100% Relative Humidity for 96 Hours Maximum |
| Reverse Voltage (Vr) | LUXEON Rebel ES LEDs are not designed to be driven in reverse bias |

RESULTS OF TESTS (cont'd)

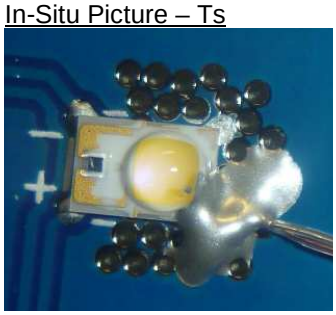
In-Situ Maximum Measured LED Source Temperature - LXML-PM01 GREEN

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5 V
Measured LED Current = 565 mA
Calculated LED Wattage = Vf x Measured LED Current = 2.254W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 127°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-007 | 74.7 | Per diagram above | 127 |

In-Situ Picture – Ts

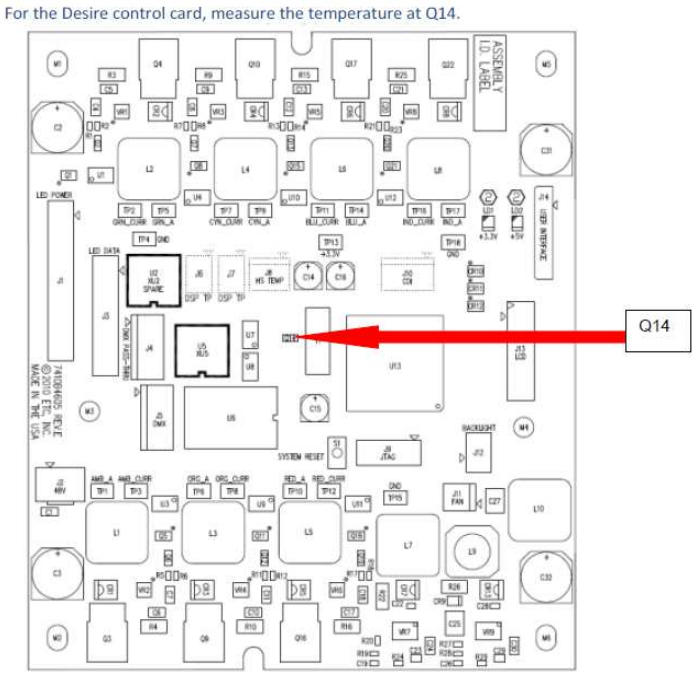


In-Situ Picture – Ts location

RESULTS OF TESTS (cont'd)

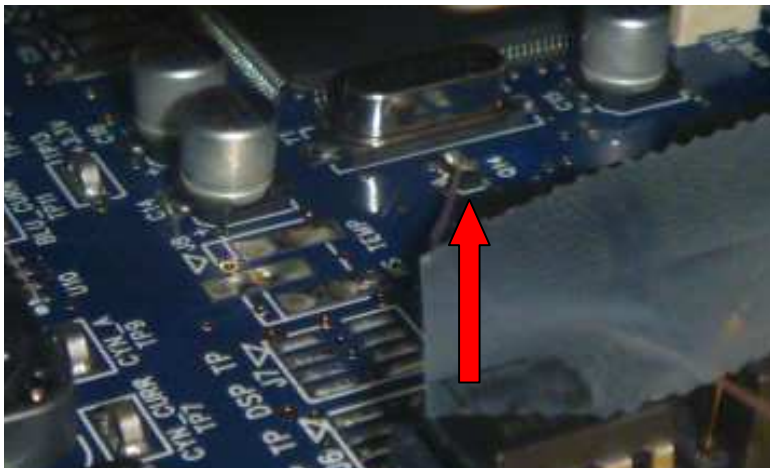
Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:
Driver identified as: Model No. 7410B4605




Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Measured Power Supply Case Temperature (°C) | Location | Maximum Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1304251519-007 | 48.3 | Q14 | 100 |



Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: December 6, 2013

REPORT NO. 101414036CRT-007

TEST OF ONE VIVID-R SL3 LED FIXTURE 11

MODEL NO. VIVID-R SL3 11
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number VIVID-R SL3 11. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-007.

DATES OF TESTS: December 3, 2013 through December 5, 20:

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | VIVID-R SL3 11 |
| Description: | VIVID-R SL3 LED FIXTURE 11 |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 2752 | 3038 |
| Total Power (W) | 97.89 | 97.78 |
| Luminaire Efficacy (LPW) | 28.11 | 31.07 |

| Criteria | Result |
|---|---|
| Power Factor | 0.974 |
| Current ATHD % | 16.84 |
| Correlated Color Temperature (CCT - K) | 3065 |
| Color Rendering Index (CRI - Ra) | 32.1 |
| Color Rendering Index (CRI - R9) | -53.6 |
| DUV | 0.066 |
| Chromaticity Coordinate (x) | 0.362 |
| Chromaticity Coordinate (y) | 0.253 |
| Chromaticity Coordinate (u') | 0.273 |
| Chromaticity Coordinate (v') | 0.428 |
| Maximum In-Situ Source Temperature Point (°C) | 70.4 |
| Maximum In-Situ Driver Case Temperature (°C) | 52.9 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/27/13 | 11/27/14 |
| Sorensen DC Power Supply | DLM150-20E | --- | --- | --- |
| Data Precision | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole-Palmer Hygro Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1360 | 11/27/13 | 11/27/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

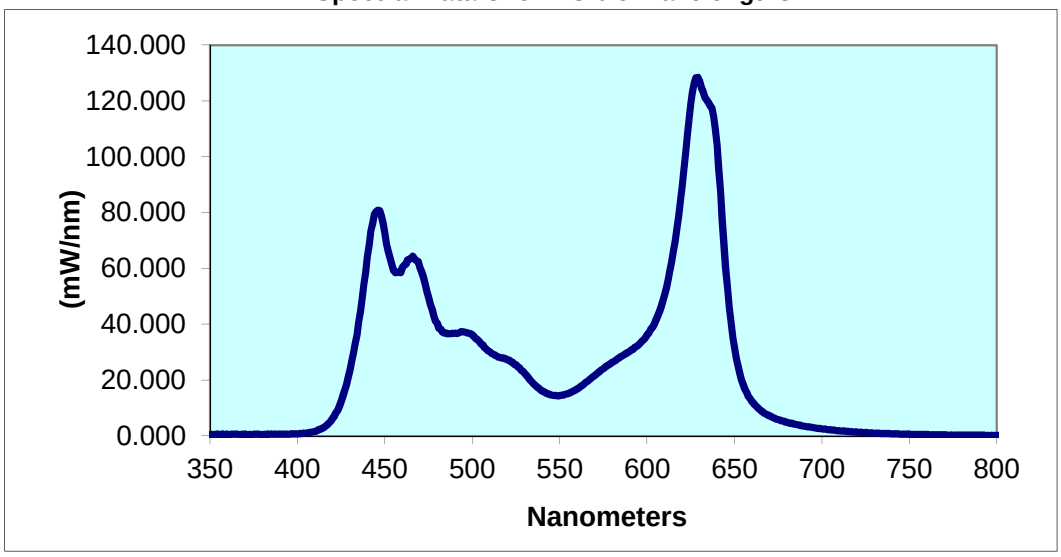
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-007 | UP | 120.0 | 837.3 | 97.89 | 0.974 | 16.84 | 2752 | 28.11 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3065 | 32.1 | -53.6 | 0.066 | 0.362 | 0.253 | 0.273 | 0.428 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.438 | 440 | 64.19 | 530 | 22.37 | 620 | 89.64 | 710 | 1.851 |
| 355 | 0.48 | 445 | 80.37 | 535 | 18.75 | 625 | 119.4 | 715 | 1.605 |
| 360 | 0.617 | 450 | 72.52 | 540 | 16.06 | 630 | 127.1 | 720 | 1.363 |
| 365 | 0.557 | 455 | 59.32 | 545 | 14.68 | 635 | 119.5 | 725 | 1.17 |
| 370 | 0.66 | 460 | 60.42 | 550 | 14.45 | 640 | 104.2 | 730 | 0.988 |
| 375 | 0.561 | 465 | 63.54 | 555 | 15.27 | 645 | 59.94 | 735 | 0.848 |
| 380 | 0.472 | 470 | 60.44 | 560 | 16.89 | 650 | 31.21 | 740 | 0.751 |
| 385 | 0.713 | 475 | 49.45 | 565 | 19.23 | 655 | 18.25 | 745 | 0.644 |
| 390 | 0.655 | 480 | 40.21 | 570 | 21.65 | 660 | 12.29 | 750 | 0.561 |
| 395 | 0.621 | 485 | 36.84 | 575 | 24.14 | 665 | 9.092 | 755 | 0.485 |
| 400 | 0.718 | 490 | 36.81 | 580 | 26.38 | 670 | 7.244 | 760 | 0.443 |
| 405 | 1.042 | 495 | 37.28 | 585 | 28.41 | 675 | 5.843 | 765 | 0.363 |
| 410 | 1.607 | 500 | 36.23 | 590 | 30.25 | 680 | 4.876 | 770 | 0.331 |
| 415 | 3.046 | 505 | 33.02 | 595 | 32.6 | 685 | 4.162 | 775 | 0.286 |
| 420 | 6.164 | 510 | 30.15 | 600 | 36.12 | 690 | 3.51 | 780 | 0.261 |
| 425 | 12.31 | 515 | 28.24 | 605 | 41.5 | 695 | 2.98 | | |
| 430 | 23.61 | 520 | 27.36 | 610 | 50.63 | 700 | 2.584 | | |
| 435 | 40.86 | 525 | 25.43 | 615 | 66.03 | 705 | 2.144 | | |

**Spectral Data Over Visible Wavelengths**



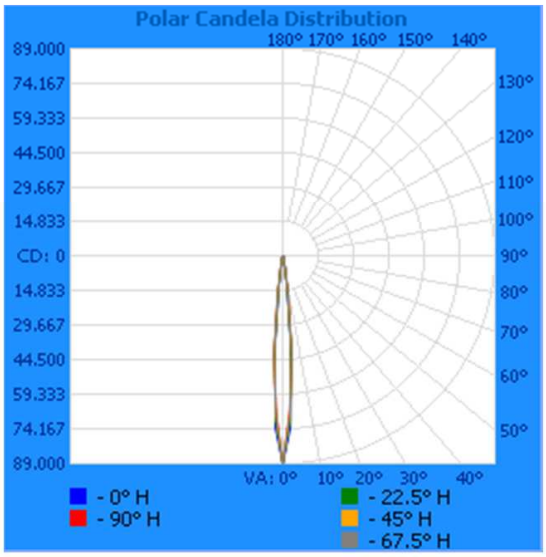
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-007 | UP | 120.0 | 835.9 | 97.78 | 0.975 | 3038 | 31.07 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 88961 | 88961 | 88961 | 88961 | 88961 |
| 5 | 36475 | 38693 | 40393 | 41500 | 43402 |
| 10 | 4093 | 4665 | 5256 | 5372 | 5579 |
| 15 | 652 | 725 | 854 | 807 | 744 |
| 20 | 384 | 383 | 399 | 387 | 348 |
| 25 | 152 | 152 | 158 | 164 | 170 |
| 30 | 70 | 66 | 66 | 63 | 62 |
| 35 | 46 | 40 | 40 | 33 | 34 |
| 40 | 33 | 29 | 26 | 20 | 19 |
| 45 | 28 | 23 | 16 | 14 | 10 |
| 50 | 24 | 20 | 13 | 5 | 5 |
| 55 | 16 | 12 | 5 | 0 | 0 |
| 60 | 7 | 3 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

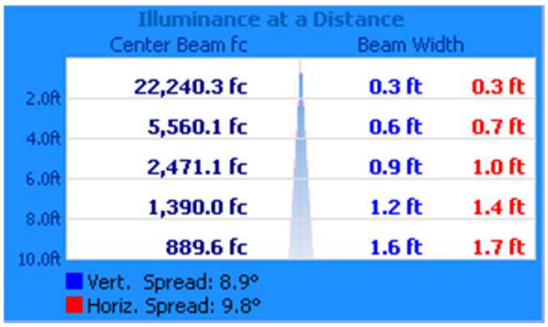


RESULTS OF TEST (cont'd)

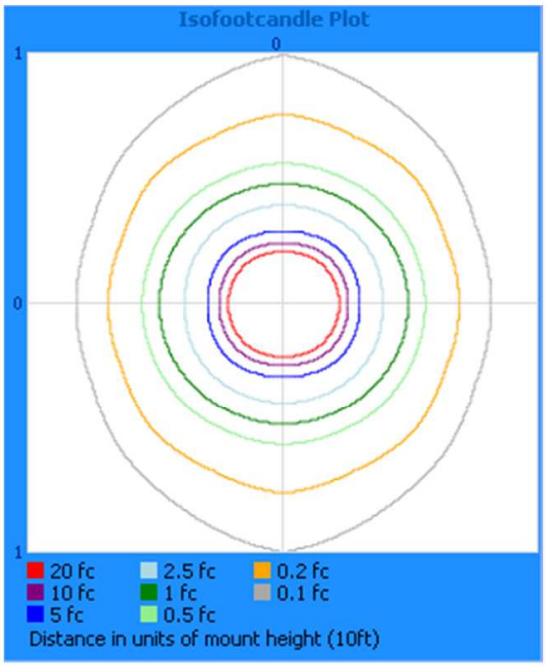
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 2992 | 98.5 |
| 0-40 | 3017 | 99.3 |
| 0-60 | 3038 | 100.0 |
| 60-90 | 0.2 | 0.0 |
| 0-90 | 3038 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3038 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2566 | 84.5 |
| 10-20 | 345.5 | 11.4 |
| 20-30 | 81.2 | 2.7 |
| 30-40 | 25.2 | 0.8 |
| 40-50 | 14.1 | 0.5 |
| 50-60 | 6.0 | 0.2 |
| 60-70 | 0.2 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as: Luxeon Rebel ES (Amber)

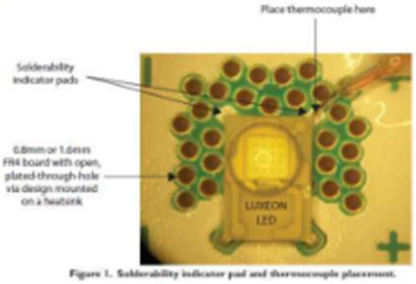



Figure 1. Solderability indicator pad and thermocouple placement.

| Color | Part Number | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|---|---|
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

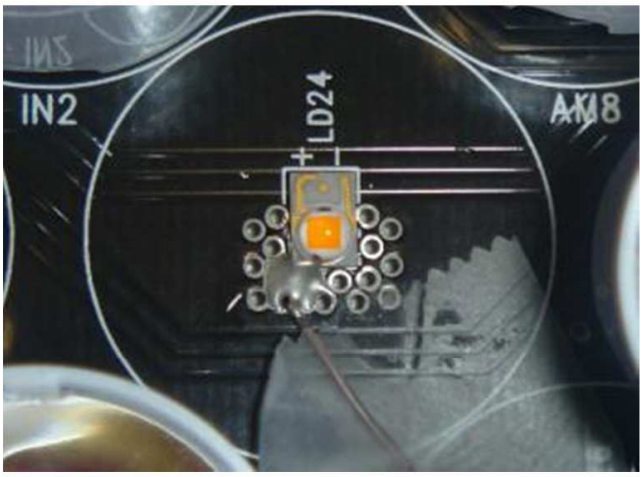
Maximum LED Junction Temperature 135°C

Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 12°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 590mA
Calculated LED Wattage = Vf x Measured LED Current = 2.071W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 110.1°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-007 | 70.4 | Per diagram | 110.1 |

In-Situ Picture – Ts

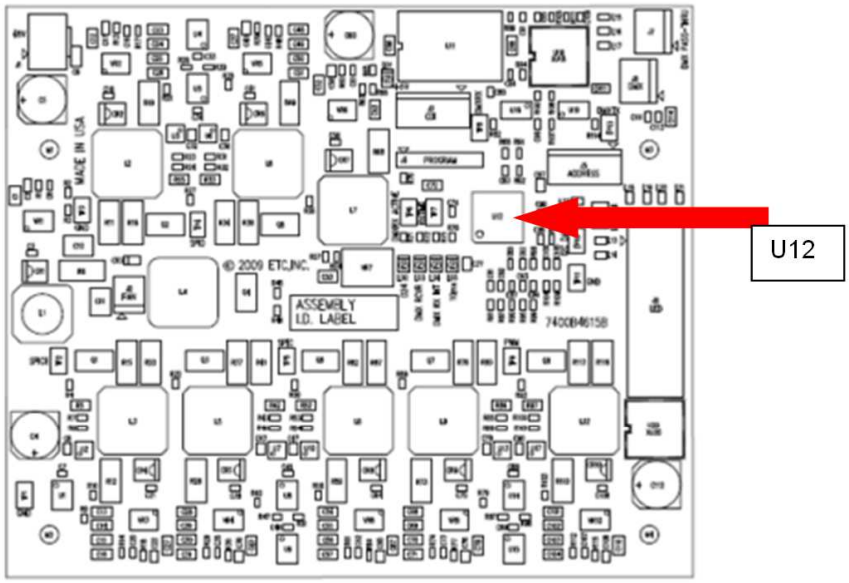



In-Situ Picture – Ts location

RESULTS OF TEST (cont'd)

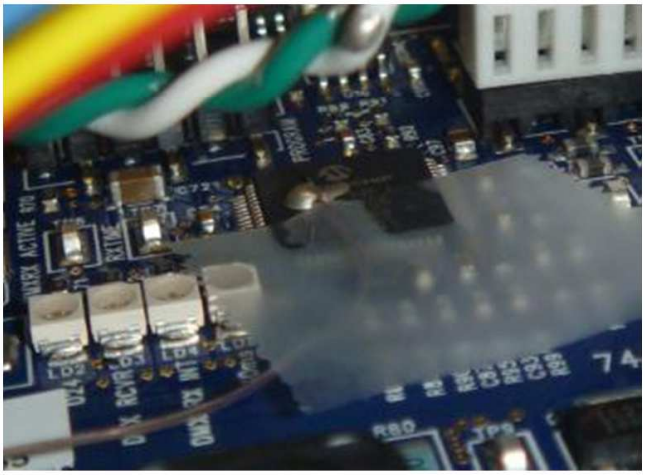
Maximum Measured Power Supply Case Temperature

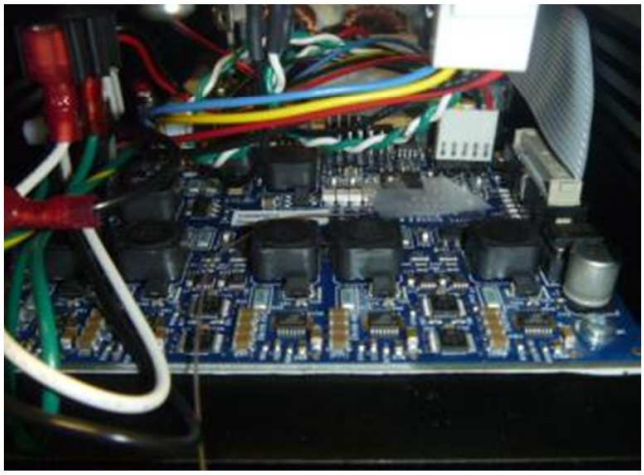
Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-004 | 52.9 | Per diagram | 100 |

Power Supply Picture – Tc location





PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 29, 2013

REPORT NO. 101414036CRT-006

TEST OF ONE D40 STUDIO LED FIXTURE ETL (BLK)

MODEL NO. D40 STUDIO
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 (2008): Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products
ANSI ANSLG C38.377 (2012): Specifications of the Chromaticity of Solid State Lighting Products
Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D40 STUDIO. The sample was received by Intertek on November 7, 2013, in undamaged condition, and one sample was tested as received. The sample designation was CRT1311071604-006.

DATES OF TESTS: November 21, 2013 through November 26, 2013.

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | D40 STUDIO |
| Description: | D40 STUDIO LED FIXTURE ETL (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 3651 | 4073 |
| Total Power (W) | 91.94 | 91.84 |
| Luminaire Efficacy (LPW) | 39.71 | 44.35 |

| Criteria | Result |
|---|---|
| Power Factor | 0.995 |
| Current ATHD (%) | 2.93 |
| Correlated Color Temperature (CCT - K) | 4114 |
| Color Rendering Index (CRI) – Ra | 81.80 |
| Color Rendering Index (CRI) - R9 | 64.70 |
| Duv | 0.010 |
| Chromaticity Coordinate (x) | 0.370 |
| Chromaticity Coordinate (y) | 0.351 |
| Chromaticity Coordinate (u') | 0.229 |
| Chromaticity Coordinate (v') | 0.488 |
| Maximum In-Situ Source Temperature Point (°C) | 74.6 |
| Maximum In-Situ Driver Temperature Point (°C) | 73.9 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Cole Parmer Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

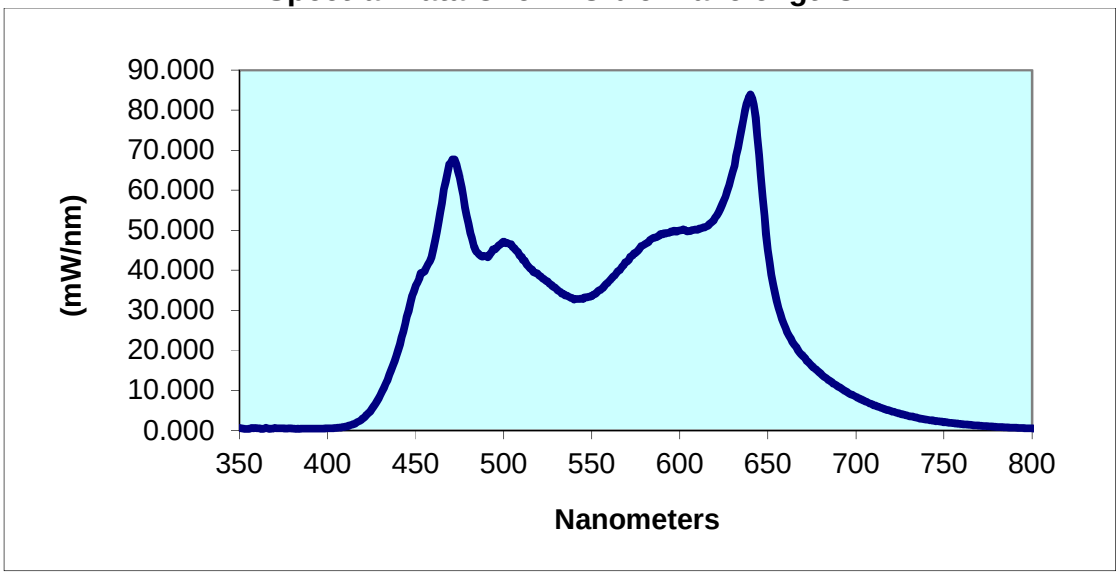
Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the $TMP_{PS}$ or $T_S$ point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.666 | 460 | 45.248 | 570 | 42.212 | 680 | 14.151 |
| 355 | 0.428 | 465 | 56.769 | 575 | 44.349 | 685 | 12.481 |
| 360 | 0.545 | 470 | 66.901 | 580 | 46.520 | 690 | 10.950 |
| 365 | 0.643 | 475 | 63.171 | 585 | 48.141 | 695 | 9.618 |
| 370 | 0.613 | 480 | 51.678 | 590 | 48.951 | 700 | 8.393 |
| 375 | 0.552 | 485 | 44.502 | 595 | 49.672 | 705 | 7.341 |
| 380 | 0.469 | 490 | 43.474 | 600 | 50.011 | 710 | 6.354 |
| 385 | 0.509 | 495 | 45.214 | 605 | 49.841 | 715 | 5.582 |
| 390 | 0.478 | 500 | 47.164 | 610 | 50.162 | 720 | 4.871 |
| 395 | 0.473 | 505 | 45.938 | 615 | 50.959 | 725 | 4.204 |
| 400 | 0.533 | 510 | 43.483 | 620 | 53.196 | 730 | 3.622 |
| 405 | 0.645 | 515 | 40.497 | 625 | 57.438 | 735 | 3.136 |
| 410 | 0.954 | 520 | 38.760 | 630 | 64.645 | 740 | 2.779 |
| 415 | 1.667 | 525 | 37.212 | 635 | 75.317 | 745 | 2.371 |
| 420 | 3.033 | 530 | 35.304 | 640 | 83.891 | 750 | 2.115 |
| 425 | 5.283 | 535 | 33.700 | 645 | 68.790 | 755 | 1.832 |
| 430 | 8.846 | 540 | 32.745 | 650 | 45.204 | 760 | 1.589 |
| 435 | 13.872 | 545 | 32.889 | 655 | 32.299 | 765 | 0.000 |
| 440 | 20.029 | 550 | 33.674 | 660 | 25.567 | 770 | 1.201 |
| 445 | 28.438 | 555 | 35.119 | 665 | 21.445 | 775 | 1.061 |
| 450 | 36.246 | 560 | 37.372 | 670 | 18.571 | 780 | 0.909 |
| 455 | 39.785 | 565 | 39.883 | 675 | 16.106 | | |

**Sample No. 101414036CRT-006**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25°C – Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| 101414036CRT-006 | UP | 120.0 | 769.5 | 91.94 | 0.995 | 2.93 | 3651 | 39.71 |

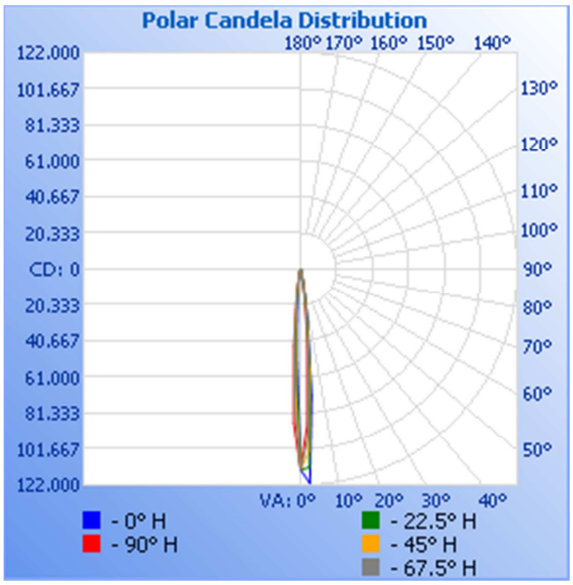
| Intertek Sample No. | Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|---|
| 101414036CRT-006 | 4114 | 81.80 | 64.70 | 0.010 | 0.370 | 0.351 | 0.229 | 0.488 |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| 101414036CRT-006 | UP | 120.1 | 767.7 | 91.84 | 0.997 | 4073 | 44.35 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 113276 | 113276 | 113276 | 113276 | 113276 |
| 5 | 71332 | 64192 | 57167 | 50201 | 43462 |
| 10 | 12758 | 11526 | 10235 | 9102 | 8072 |
| 15 | 2751 | 2537 | 2274 | 2046 | 1832 |
| 20 | 864 | 827 | 779 | 723 | 678 |
| 25 | 238 | 219 | 202 | 194 | 191 |
| 30 | 87 | 83 | 82 | 82 | 81 |
| 35 | 54 | 48 | 49 | 47 | 48 |
| 40 | 28 | 31 | 33 | 34 | 36 |
| 45 | 17 | 22 | 23 | 24 | 32 |
| 50 | 19 | 17 | 21 | 22 | 28 |
| 55 | 7 | 12 | 14 | 14 | 20 |
| 60 | 5 | 5 | 6 | 5 | 3 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

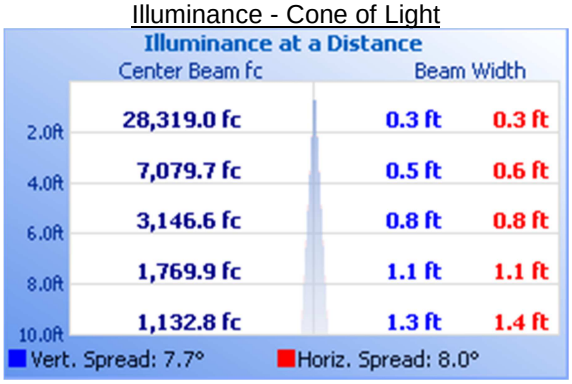


RESULTS OF TESTS (cont'd)

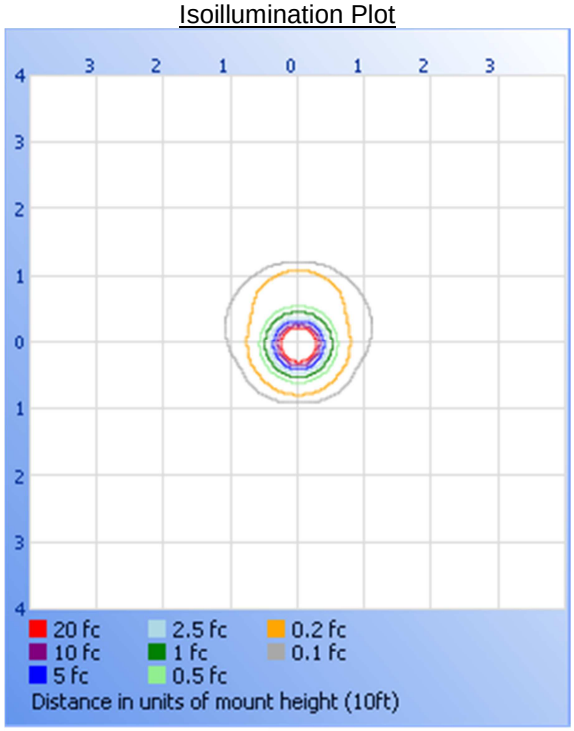
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3995 | 98.1 |
| 0-40 | 4029 | 98.9 |
| 0-60 | 4072 | 100.0 |
| 60-90 | 0.6 | 0.0 |
| 0-90 | 4073 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 4073 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 3174 | 77.9 |
| 10-20 | 708.6 | 17.4 |
| 20-30 | 112.3 | 2.8 |
| 30-40 | 34.0 | 0.8 |
| 40-50 | 28.7 | 0.7 |
| 50-60 | 14.7 | 0.4 |
| 60-70 | 0.6 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TESTS (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: Luxeon Rebel A

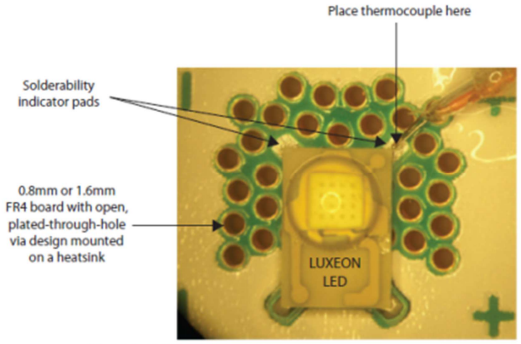


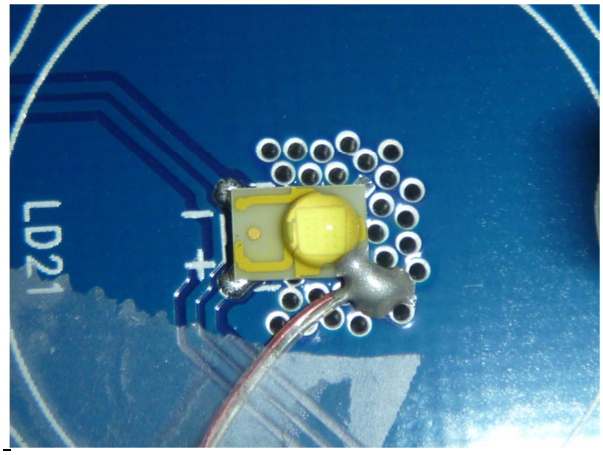

Figure 1. Solderability indicator pad and thermocouple placement.

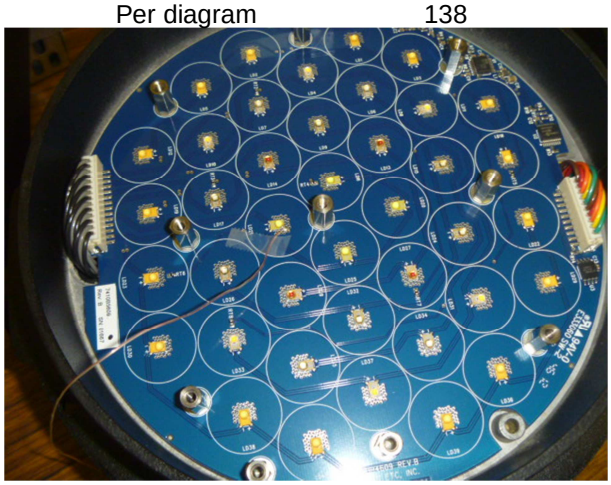
| Nominal ANSI CCT | Forward Voltage $V_f$ [1] (V) Min. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_F / \Delta T_J$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|
| 2700K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 3000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 4000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 5000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |

| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 585mA
Calculated LED Wattage = Vf x Measured LED Current = 2.048W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 138°C

| Intertek Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| 101414036CRT-006 | 74.6 | Per diagram | 138 |



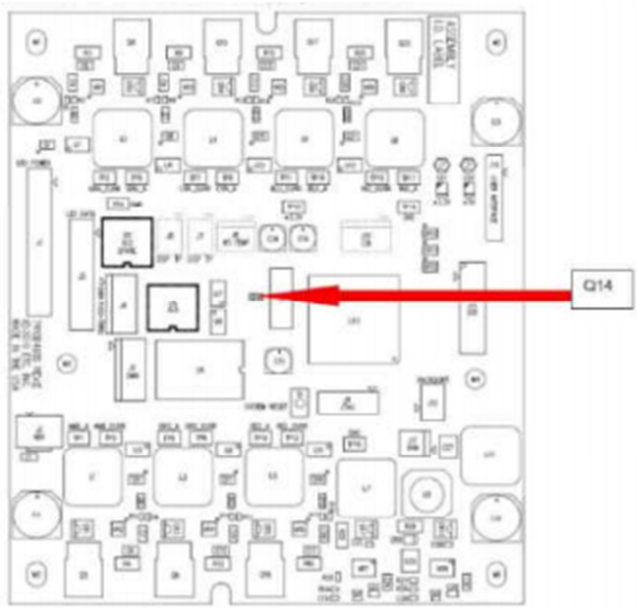



RESULTS OF TESTS (cont'd)

Maximum Measured Power Supply Case Temperature

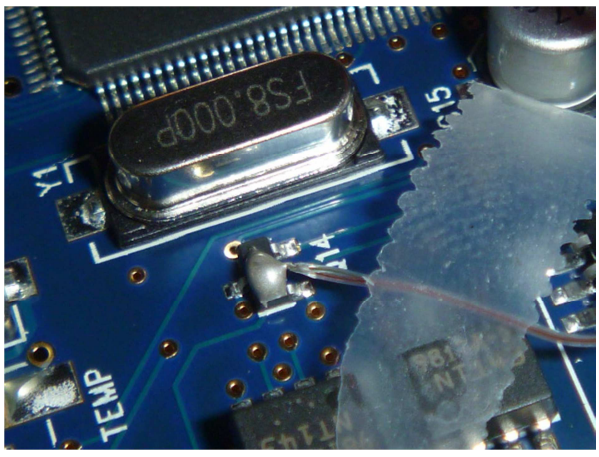
Manufacturer Supplied Documentation:

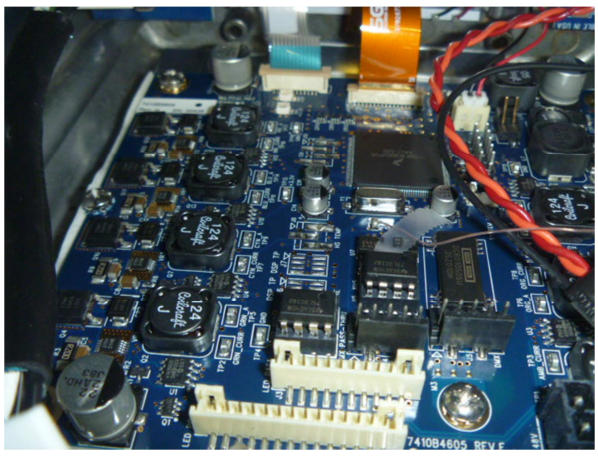
Driver identified as:




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-006 | 73.9 | Per diagram | 100 |

Power Supply Picture – Tc location



Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: May 23, 2013

REPORT NO. 101158242CRT-003

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. D40 STUDIO TUNGSTEN

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79: 2008 Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products

ANSI ANSLG C38.377: 2012 Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D40 STUDIO TUNGSTEN. The sample was received by Intertek on May 6, 2013, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-005.

DATES OF TESTS: May 15, 2013 through May 22, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | D40 STUDIO TUNGSTEN |
| Description: | LED WASH LIGHT |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 4686 | 4804 |
| Total Power (W) | 100.9 | 100.5 |
| Luminaire Efficacy (LPW) | 46.44 | 47.79 |

| Criteria | Result |
|---|---|
| Power Factor | 0.994 |
| Current ATHD (%) | 2.20 |
| Correlated Color Temperature (CCT - K) | 3084 |
| Color Rendering Index (CRI) – Ra | 84.7 |
| Color Rendering Index (CRI) - R9 | 20.6 |
| Duv | 0.001 |
| Chromaticity Coordinate (x) | 0.434 |
| Chromaticity Coordinate (y) | 0.408 |
| Chromaticity Coordinate (u') | 0.247 |
| Chromaticity Coordinate (v') | 0.522 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 05/20/13 | 06/20/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | --- | N1132 | 04/22/13 | 04/22/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Yokogawa Power Analyzer | WT1600 | E462 | 07/06/12 | 07/06/13 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | NA | NA | NA |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Sorensen DC Power Supply | DLM150-20E | N/A | N/A | N/A |
| FLUKE | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

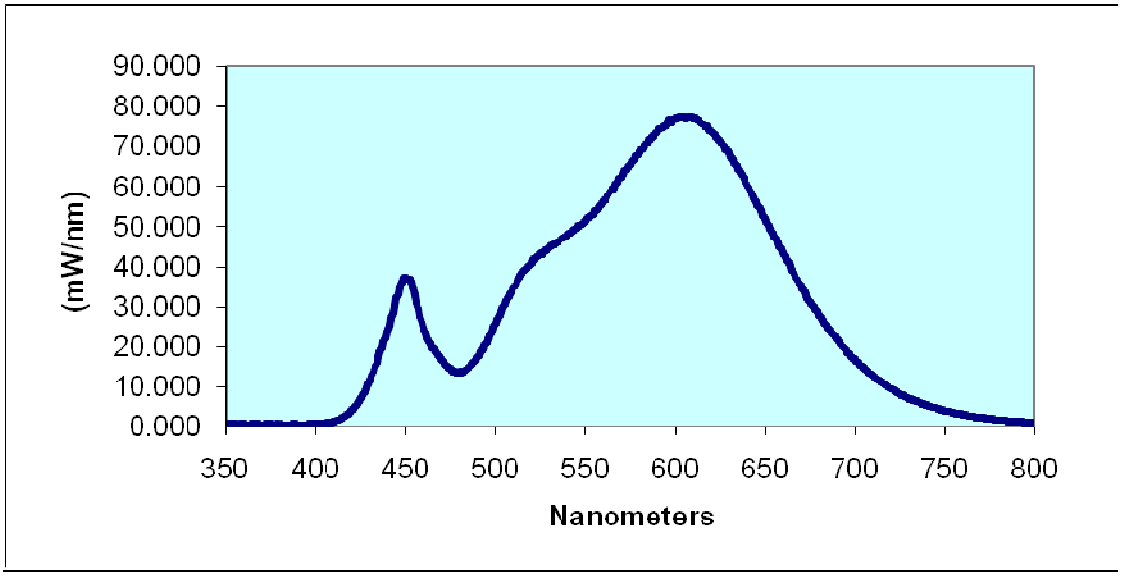
The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.613 | 460 | 24.656 | 570 | 62.168 | 680 | 27.746 |
| 355 | 0.540 | 465 | 19.763 | 575 | 65.265 | 685 | 24.639 |
| 360 | 0.575 | 470 | 16.752 | 580 | 68.504 | 690 | 21.708 |
| 365 | 0.537 | 475 | 14.300 | 585 | 71.499 | 695 | 18.967 |
| 370 | 0.473 | 480 | 13.597 | 590 | 73.537 | 700 | 16.724 |
| 375 | 0.552 | 485 | 14.674 | 595 | 75.372 | 705 | 14.544 |
| 380 | 0.484 | 490 | 17.153 | 600 | 76.998 | 710 | 12.674 |
| 385 | 0.476 | 495 | 20.848 | 605 | 77.249 | 715 | 11.038 |
| 390 | 0.436 | 500 | 25.822 | 610 | 76.932 | 720 | 9.491 |
| 395 | 0.465 | 505 | 30.501 | 615 | 75.764 | 725 | 8.208 |
| 400 | 0.552 | 510 | 34.836 | 620 | 73.899 | 730 | 6.997 |
| 405 | 0.708 | 515 | 38.210 | 625 | 71.450 | 735 | 6.094 |
| 410 | 1.168 | 520 | 40.794 | 630 | 68.095 | 740 | 5.240 |
| 415 | 2.213 | 525 | 43.184 | 635 | 64.178 | 745 | 4.561 |
| 420 | 4.027 | 530 | 44.886 | 640 | 60.295 | 750 | 3.898 |
| 425 | 7.035 | 535 | 46.139 | 645 | 56.188 | 755 | 3.354 |
| 430 | 11.514 | 540 | 47.539 | 650 | 51.960 | 760 | 2.913 |
| 435 | 17.344 | 545 | 49.374 | 655 | 47.432 | 765 | 0.000 |
| 440 | 23.766 | 550 | 51.262 | 660 | 43.176 | 770 | 2.194 |
| 445 | 31.978 | 555 | 53.728 | 665 | 39.061 | 775 | 1.871 |
| 450 | 37.046 | 560 | 56.355 | 670 | 35.250 | 780 | 1.619 |
| 455 | 32.782 | 565 | 59.122 | 675 | 31.294 | | |

**Sample No. CRT1304251519-005**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25°C – Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-005 | UP | 120.0 | 844.6 | 100.9 | 0.994 | 2.20 | 4686 | 46.44 |

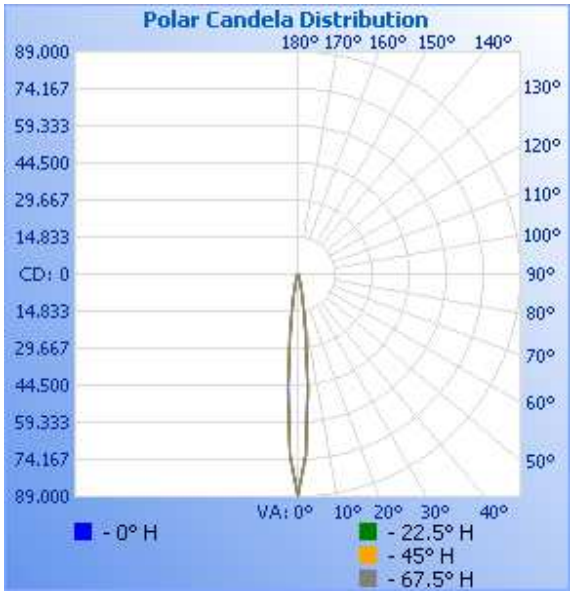
| Intertek Sample No. | Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-005 | 3084 | 84.7 | 20.6 | 0.001 | 0.434 | 0.408 | 0.247 | 0.522 |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1304251519-005 | UP | 120.0 | 841.2 | 100.5 | 0.996 | 4804 | 47.79 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 88709 | 88709 | 88709 | 88709 | 88709 |
| 5 | 44590 | 46050 | 46747 | 46158 | 46566 |
| 10 | 12391 | 12557 | 12791 | 12544 | 13110 |
| 15 | 4078 | 4168 | 3801 | 3156 | 3095 |
| 20 | 1421 | 1452 | 1231 | 1127 | 1161 |
| 25 | 416 | 426 | 413 | 386 | 405 |
| 30 | 152 | 160 | 153 | 133 | 133 |
| 35 | 102 | 95 | 78 | 66 | 76 |
| 40 | 87 | 67 | 64 | 57 | 43 |
| 45 | 68 | 51 | 46 | 40 | 35 |
| 50 | 36 | 39 | 36 | 33 | 30 |
| 55 | 15 | 16 | 21 | 18 | 13 |
| 60 | 2 | 5 | 5 | 6 | 4 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

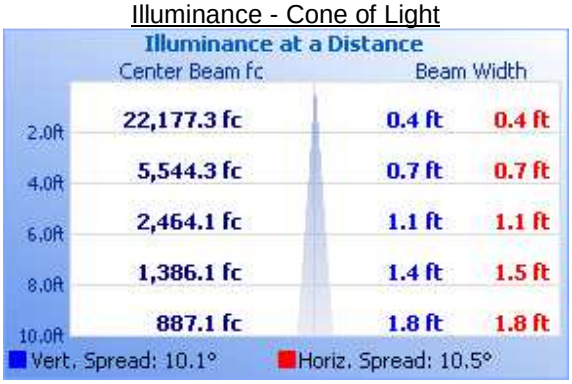


RESULTS OF TESTS (cont'd)

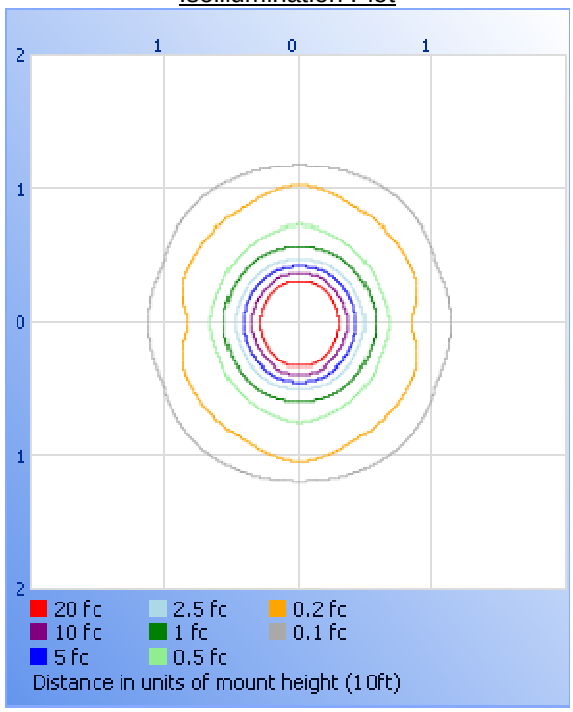
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 4694 | 97.7 |
| 0-40 | 4750 | 98.9 |
| 0-60 | 4803 | 100.0 |
| 60-90 | 0.7 | 0.0 |
| 0-90 | 4804 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 4804 | 100.0 |

Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

David Ellis
Senior Project Engineer
Lighting Division

Attachment: None

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: June 10, 2013

REPORT NO. 101158242CRT-007

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. D40 STUDIO TUNGSTEN
LED MODEL NO. LUXEON REBEL ES

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and LED In-Situ tests.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

Energy Star Version 1.2 (2012): Program Requirements for Luminaires
Energy Star Manufacturer's Guide Version 2.0 (2009): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one sample of model number D40 STUDIO TUNGSTEN. The sample was received by Intertek on May 6, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-005.

DATES OF TESTS: June 7, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | D40 STUDIO TUNGSTEN |
| Description: | LED WASH LIGHT |

| Criteria | Result |
|---|---|
| Maximum In-Situ Source Temperature Point (°C) | 76.9 |
| Maximum Power Supply Case Temperature (°C) | 51.5 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Fluke Multimeter | PM2525 | M127 | 10/17/12 | 10/17/13 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

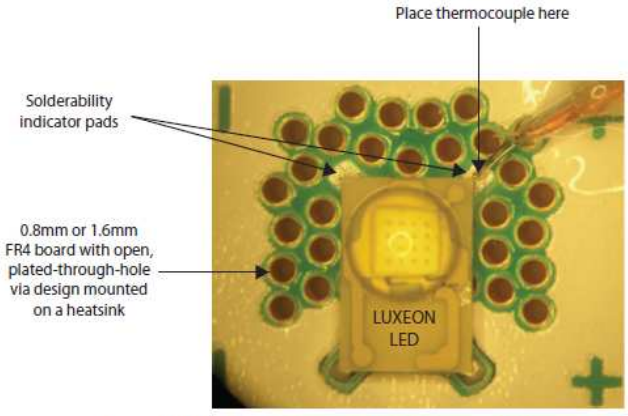
In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Agilent 34970A Data Logger. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TESTS

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: LUXEON REBEL ES




| Color | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|
| Neutral-White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |
| Cool-White | 2.5 | 3.0 | 3.5 | -2.0 to -4.0 | 6 |

| Parameter | Cool-White / Neutral-White |
|---|---|
| DC Forward Current (mA) | 1000 mA |
| Peak Pulsed Forward Current (mA) [2] | 1200 mA |
| Average Forward Current (mA) | 1000 mA |
| ESD Sensitivity | < 8000V Human Body Model (HBM)<br>Class 3B JESD22-A114-E |
| LED Junction Temperature[1] | 150°C |
| Operating Case Temperature at 350 mA | -40°C - 135°C |
| Storage Temperature | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 |
| Autoclave Conditions | 121°C at 2 ATM<br>100% Relative Humidity for 96 Hours Maximum |
| Reverse Voltage (Vr) | LUXEON Rebel ES LEDs are not designed to be driven in reverse bias |

RESULTS OF TESTS (cont'd)

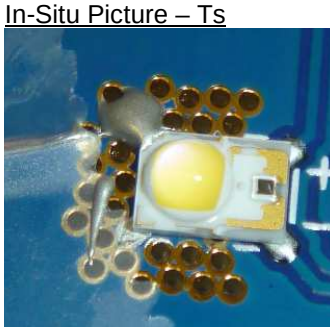
In-Situ Maximum Measured LED Source Temperature - LXML-PM01 GREEN

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5 V
Measured LED Current = 569 mA
Calculated LED Wattage = Vf x Measured LED Current = 2.270W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 127°C

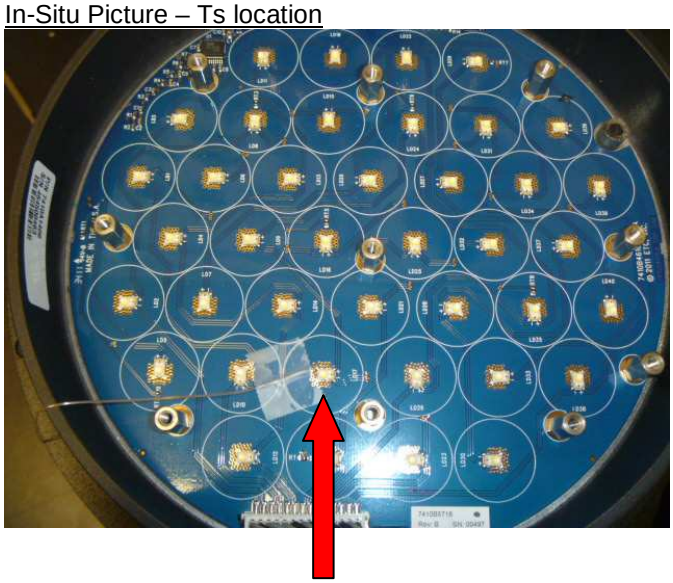
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-005 | 76.9 | Per diagram above | 127 |

In-Situ Picture – Ts



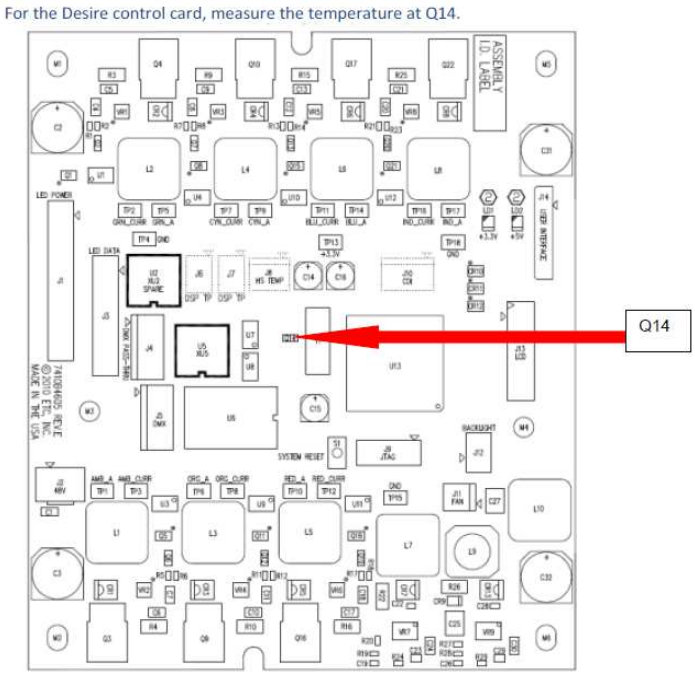
In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

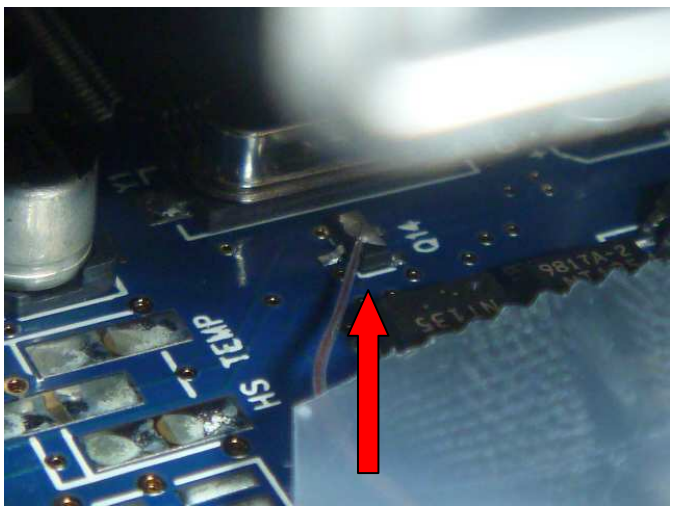
Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:
Driver identified as: Model No. 7410B4605




Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Measured Power Supply Case Temperature (°C) | Location | Maximum Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1304251519-005 | 51.5 | Q14 | 100 |



Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: December 6, 2013

REPORT NO. 101414036CRT-008

TEST OF ONE D40 DAYLIGHT LED FIXTURE ETL (BLK)

MODEL NO. D40 DAYLIGHT
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D40 DAYLIGHT. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-008.

DATES OF TESTS: November 21, 2013 through November 26, 2

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D40 DAYLIGHT |
| Description: | D40 DAYLIGHT LED FIXTURE ETL (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 6008 | 6601 |
| Total Power (W) | 98.55 | 98.38 |
| Luminaire Efficacy (LPW) | 60.96 | 67.1 |

| Criteria | Result |
|---|---|
| Power Factor | 0.995 |
| Current ATHD % | 2.28 |
| Correlated Color Temperature (CCT - K) | 5758 |
| Color Rendering Index (CRI - Ra) | 72.6 |
| Color Rendering Index (CRI - R9) | -5.2 |
| DUV | 0.004 |
| Chromaticity Coordinate (x) | 0.326 |
| Chromaticity Coordinate (y) | 0.345 |
| Chromaticity Coordinate (u') | 0.201 |
| Chromaticity Coordinate (v') | 0.479 |
| Maximum In-Situ Source Temperature Point (°C) | 73.6 |
| Maximum In-Situ Driver Case Temperature (°C) | 71.8 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Fluke Multimeter | 45 | M133 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Cole Parmer Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

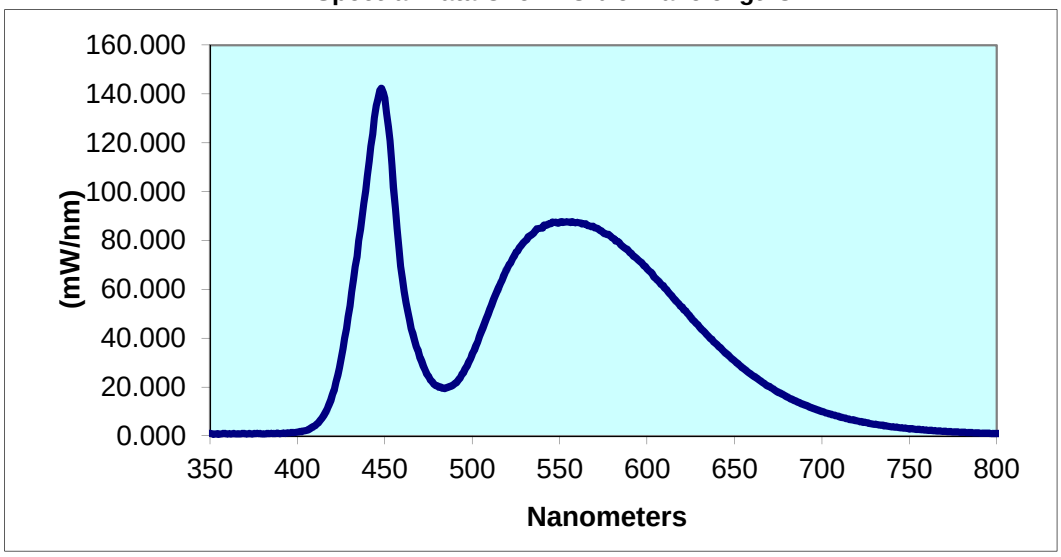
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-008 | UP | 120.0 | 825.2 | 98.55 | 0.995 | 2.28 | 6008 | 60.96 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 5758 | 72.6 | -5.2 | 0.004 | 0.326 | 0.345 | 0.201 | 0.479 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 1.071 | 440 | 106.4 | 530 | 79.74 | 620 | 52.67 | 710 | 7.858 |
| 355 | 0.723 | 445 | 135.3 | 535 | 82.96 | 625 | 48.57 | 715 | 7.007 |
| 360 | 0.855 | 450 | 138 | 540 | 85.1 | 630 | 44.62 | 720 | 6.166 |
| 365 | 0.847 | 455 | 101.8 | 545 | 86.93 | 635 | 40.8 | 725 | 5.476 |
| 370 | 0.961 | 460 | 64.16 | 550 | 87.37 | 640 | 37.36 | 730 | 4.786 |
| 375 | 0.915 | 465 | 43.92 | 555 | 87.51 | 645 | 33.81 | 735 | 4.286 |
| 380 | 0.796 | 470 | 32.31 | 560 | 87.41 | 650 | 30.64 | 740 | 3.74 |
| 385 | 1.079 | 475 | 24.21 | 565 | 86.73 | 655 | 27.65 | 745 | 3.345 |
| 390 | 0.999 | 480 | 20.5 | 570 | 85.16 | 660 | 24.95 | 750 | 2.948 |
| 395 | 1.177 | 485 | 19.73 | 575 | 83.1 | 665 | 22.4 | 755 | 2.616 |
| 400 | 1.544 | 490 | 21.5 | 580 | 81.21 | 670 | 20.15 | 760 | 2.324 |
| 405 | 2.291 | 495 | 25.95 | 585 | 78.28 | 675 | 17.91 | 765 | 2.097 |
| 410 | 4.19 | 500 | 33.58 | 590 | 75.08 | 680 | 16 | 770 | 1.839 |
| 415 | 8.382 | 505 | 42.31 | 595 | 72.03 | 685 | 14.27 | 775 | 1.638 |
| 420 | 16.9 | 510 | 51.93 | 600 | 68.44 | 690 | 12.72 | 780 | 1.474 |
| 425 | 31.53 | 515 | 60.7 | 605 | 64.34 | 695 | 11.25 | | |
| 430 | 53.22 | 520 | 68.52 | 610 | 60.55 | 700 | 10.02 | | |
| 435 | 79.97 | 525 | 75.21 | 615 | 56.48 | 705 | 8.914 | | |

**Spectral Data Over Visible Wavelengths**



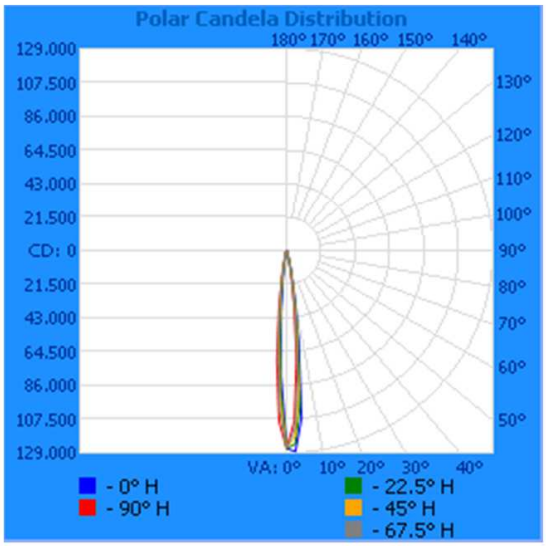
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-008 | UP | 120.0 | 822.9 | 98.38 | 0.996 | 6601 | 67.1 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 126035 | 126035 | 126035 | 126035 | 126035 |
| 5 | 107210 | 97890 | 88563 | 79279 | 70077 |
| 10 | 27377 | 24833 | 22320 | 19637 | 17043 |
| 15 | 6014 | 5277 | 4631 | 3953 | 3478 |
| 20 | 1586 | 1481 | 1348 | 1194 | 1053 |
| 25 | 580 | 526 | 459 | 404 | 344 |
| 30 | 184 | 175 | 164 | 159 | 149 |
| 35 | 119 | 111 | 110 | 108 | 104 |
| 40 | 80 | 88 | 83 | 82 | 88 |
| 45 | 70 | 74 | 70 | 66 | 72 |
| 50 | 74 | 63 | 59 | 55 | 51 |
| 55 | 40 | 38 | 37 | 32 | 29 |
| 60 | 14 | 16 | 12 | 9 | 7 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

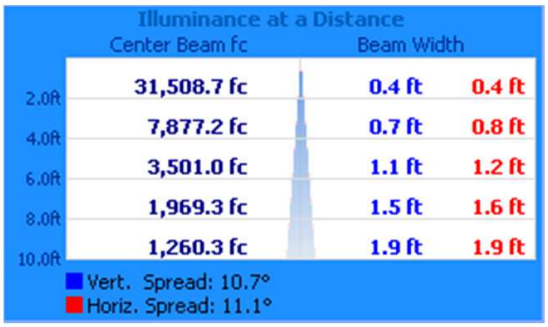


RESULTS OF TEST (cont'd)

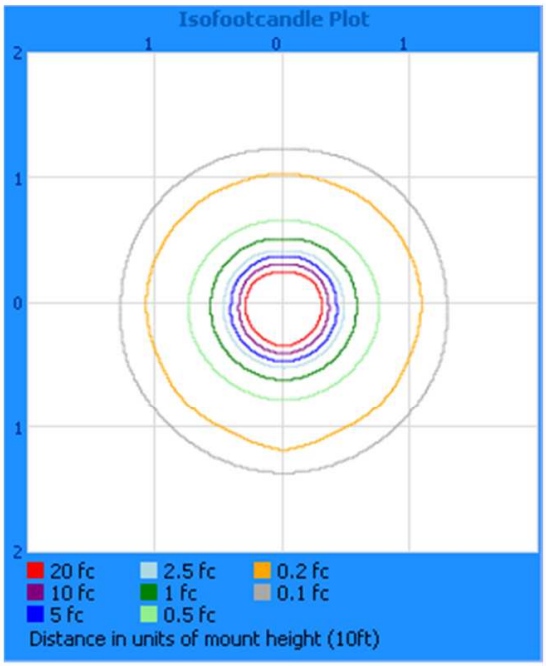
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 6454 | 97.8 |
| 0-40 | 6520 | 98.8 |
| 0-60 | 6600 | 100.0 |
| 60-90 | 1.3 | 0.0 |
| 0-90 | 6601 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 6601 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 4882 | 74.0 |
| 10-20 | 1367 | 20.7 |
| 20-30 | 205.4 | 3.1 |
| 30-40 | 65.1 | 1.0 |
| 40-50 | 51.9 | 0.8 |
| 50-60 | 28.0 | 0.4 |
| 60-70 | 1.3 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as: Luxeon Rebel A

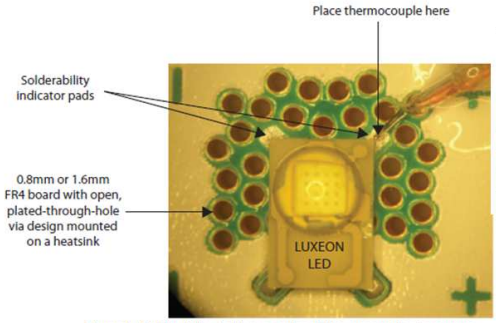


Figure 1. Solderability indicator pad and thermocouple placement.

| Nominal ANSI CCT | Forward Voltage $V_f$ [1] (V) Min. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|
| 2700K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 3000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 4000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 5000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |

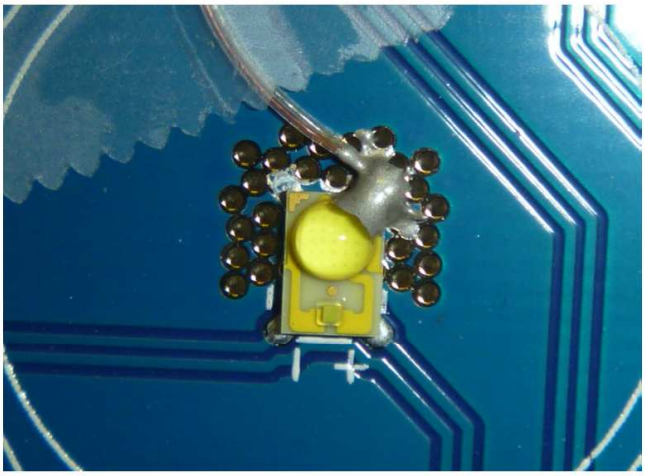
| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 580mA
Calculated LED Wattage = Vf x Measured LED Current = 2.03W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 137.8°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-008 | 73.6 | Per diagram | 137.8 |

In-Situ Picture – Ts

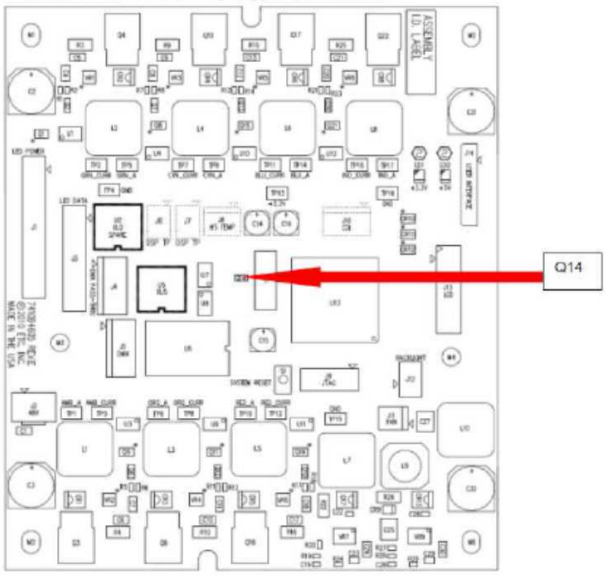


In-Situ Picture – Ts location

RESULTS OF TEST (cont'd)

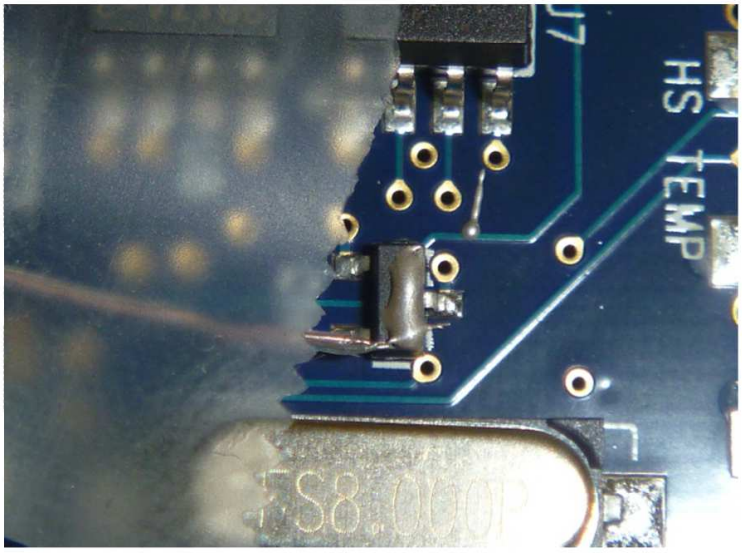
Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-008 | 47.0 | Per diagram | 100 |

Power Supply Picture – Tc location



PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 29, 2013

REPORT NO. 101414036CRT-011

TEST OF ONE D40 VIVID LED FIXTURE ETL (BLK)

MODEL NO. D40 VIVID
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D40 VIVID. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-011.

DATES OF TESTS: November 21, 2013 through November 26, 2013.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D40 VIVID |
| Description: | D40 VIVID LED FIXTURE ETL (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 2899 | 3018 |
| Total Power (W) | 88.85 | 88.96 |
| Luminaire Efficacy (LPW) | 32.63 | 33.93 |

| Criteria | Result |
|---|---|
| Power Factor | 0.995 |
| Current ATHD % | 3.18 |
| Correlated Color Temperature (CCT - K) | 3256 |
| Color Rendering Index (CRI - Ra) | 33.3 |
| Color Rendering Index (CRI - R9) | -69.3 |
| DUV | 0.058 |
| Chromaticity Coordinate (x) | 0.363 |
| Chromaticity Coordinate (y) | 0.265 |
| Chromaticity Coordinate (u') | 0.267 |
| Chromaticity Coordinate (v') | 0.437 |
| Maximum In-Situ Source Temperature Point (°C) | 75.9 |
| Maximum In-Situ Driver Case Temperature (°C) | 61.5 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Leeds & Northup Standard Resistor | Manganin | Y089 | 02/07/13 | 02/07/14 |
| Data Precision Digital Voltmeter | 3600 | V124 | 02/07/13 | 02/07/14 |
| Fluke Multimeter | 45 | M133 | 02/07/13 | 02/07/14 |
| Kikusui DC Power Supply | 35-10L | E160 | --- | --- |
| Sorenson DC Power Supply | DLM150-20E | --- | --- | --- |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| Cole Palmer Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

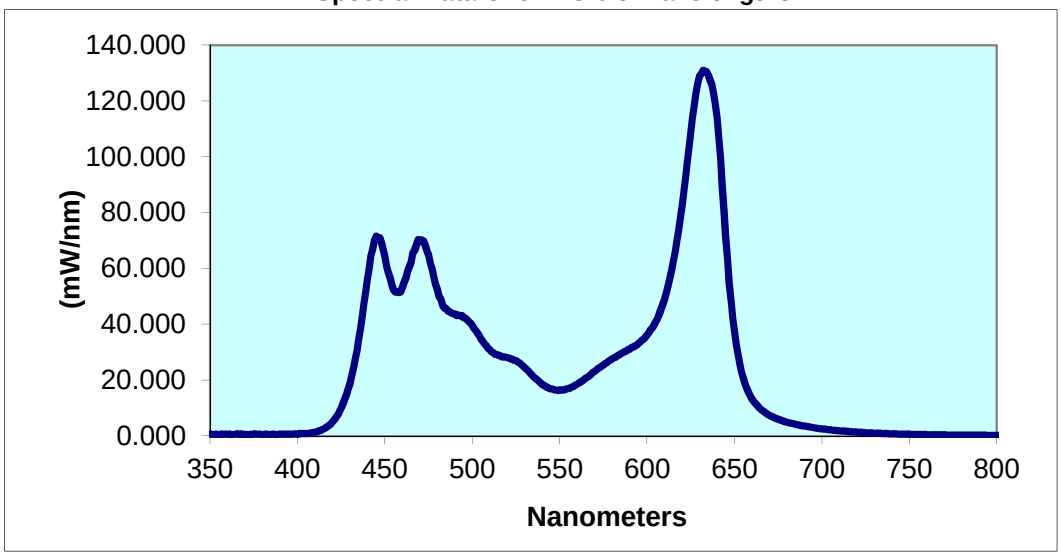
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-011 | UP | 120.0 | 743.7 | 88.85 | 0.995 | 3.18 | 2899 | 32.63 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3256 | 33.3 | -69.3 | 0.058 | 0.363 | 0.265 | 0.267 | 0.437 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.623 | 440 | 55.97 | 530 | 24.44 | 620 | 82.75 | 710 | 1.865 |
| 355 | 0.515 | 445 | 71.53 | 535 | 21.28 | 625 | 109.2 | 715 | 1.599 |
| 360 | 0.647 | 450 | 64.24 | 540 | 18.38 | 630 | 128.9 | 720 | 1.354 |
| 365 | 0.669 | 455 | 52.29 | 545 | 16.86 | 635 | 129.2 | 725 | 1.152 |
| 370 | 0.629 | 460 | 53.39 | 550 | 16.4 | 640 | 113.9 | 730 | 0.981 |
| 375 | 0.648 | 465 | 62.29 | 555 | 16.95 | 645 | 71.32 | 735 | 0.847 |
| 380 | 0.451 | 470 | 69.96 | 560 | 18.6 | 650 | 36.82 | 740 | 0.739 |
| 385 | 0.547 | 475 | 64.65 | 565 | 20.73 | 655 | 20.67 | 745 | 0.625 |
| 390 | 0.563 | 480 | 52.45 | 570 | 23.19 | 660 | 13.25 | 750 | 0.55 |
| 395 | 0.569 | 485 | 45.68 | 575 | 25.44 | 665 | 9.543 | 755 | 0.477 |
| 400 | 0.629 | 490 | 43.68 | 580 | 27.69 | 670 | 7.401 | 760 | 0.425 |
| 405 | 0.874 | 495 | 42.34 | 585 | 29.48 | 675 | 5.936 | 765 | 0.368 |
| 410 | 1.394 | 500 | 39.61 | 590 | 31.08 | 680 | 4.932 | 770 | 0.33 |
| 415 | 2.543 | 505 | 34.72 | 595 | 33.16 | 685 | 4.176 | 775 | 0.286 |
| 420 | 5.055 | 510 | 30.88 | 600 | 36.18 | 690 | 3.517 | 780 | 0.253 |
| 425 | 9.878 | 515 | 28.89 | 605 | 41.06 | 695 | 2.987 | | |
| 430 | 18.79 | 520 | 28.11 | 610 | 49.15 | 700 | 2.539 | | |
| 435 | 34.7 | 525 | 26.94 | 615 | 62.5 | 705 | 2.163 | | |

**Spectral Data Over Visible Wavelengths**



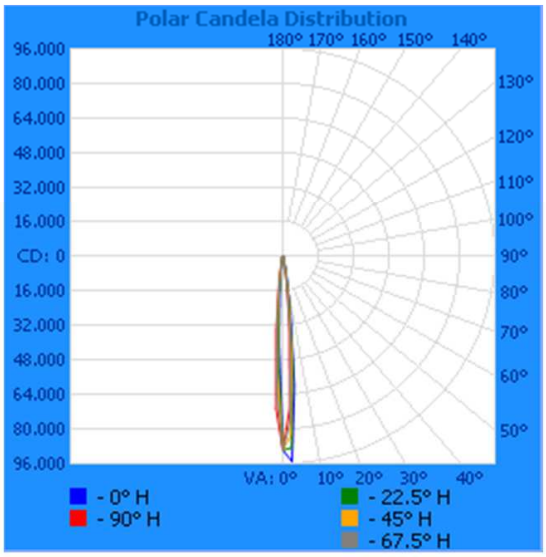
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-011 | UP | 120.0 | 743.7 | 88.96 | 0.996 | 3018 | 33.93 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 89782 | 89782 | 89782 | 89782 | 89782 |
| 5 | 60262 | 52845 | 46814 | 40033 | 34061 |
| 10 | 9870 | 8605 | 7680 | 6610 | 5791 |
| 15 | 1555 | 1414 | 1299 | 1130 | 1012 |
| 20 | 508 | 479 | 471 | 442 | 412 |
| 25 | 166 | 143 | 133 | 121 | 104 |
| 30 | 53 | 48 | 47 | 46 | 44 |
| 35 | 29 | 26 | 25 | 26 | 27 |
| 40 | 14 | 17 | 18 | 18 | 21 |
| 45 | 13 | 16 | 15 | 16 | 20 |
| 50 | 17 | 13 | 13 | 14 | 13 |
| 55 | 8 | 7 | 6 | 5 | 4 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

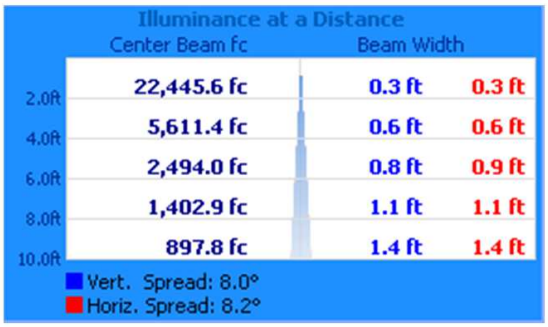


RESULTS OF TEST (cont'd)

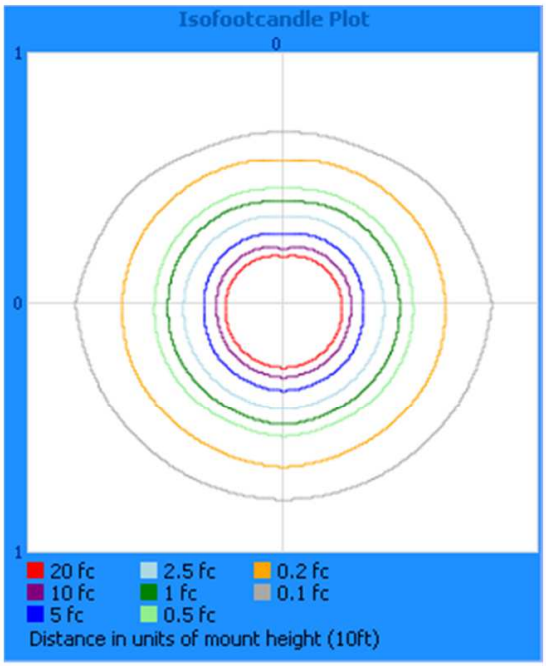
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 2985 | 98.9 |
| 0-40 | 3001 | 99.4 |
| 0-60 | 3018 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 3018 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3018 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2483 | 82.3 |
| 10-20 | 434.3 | 14.4 |
| 20-30 | 67.1 | 2.2 |
| 30-40 | 16.0 | 0.5 |
| 40-50 | 12.2 | 0.4 |
| 50-60 | 4.8 | 0.2 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as:

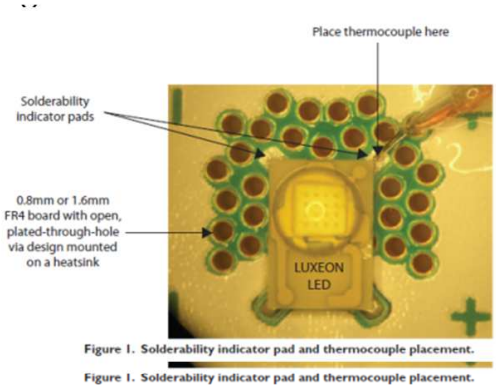


Figure 1. Solderability indicator pad and thermocouple placement.

Figure 1. Solderability indicator pad and thermocouple placement.

| Forward Voltage $V_f$ [1] (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) | Typical Thermal Resistance Junction to Thermal Pad (°C/W) |
|---|---|---|---|---|
| Min. | Typ. | Max. | $\Delta V_f / \Delta T_J$ | $R\theta_{J-C}$ |
| 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |

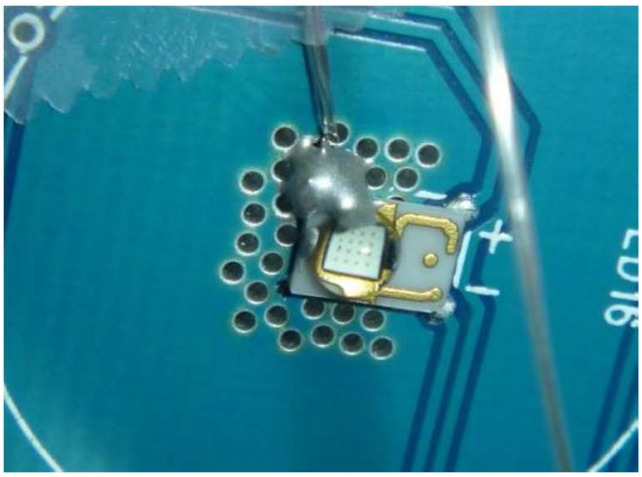
| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 583mA
Calculated LED Wattage = Vf x Measured LED Current = 2.046W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 129.5°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-011 | 75.9 | Per diagram | 129.5 |

In-Situ Picture – Ts



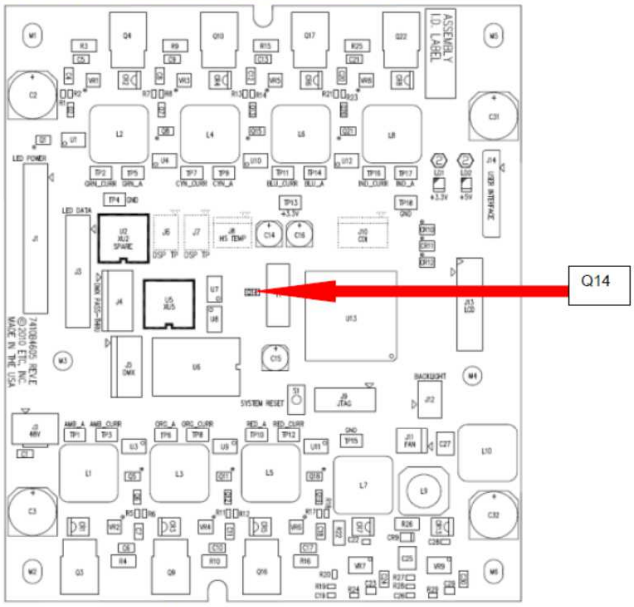
In-Situ Picture – Ts location

RESULTS OF TEST (cont'd)

Maximum Measured Power Supply Case Temperature

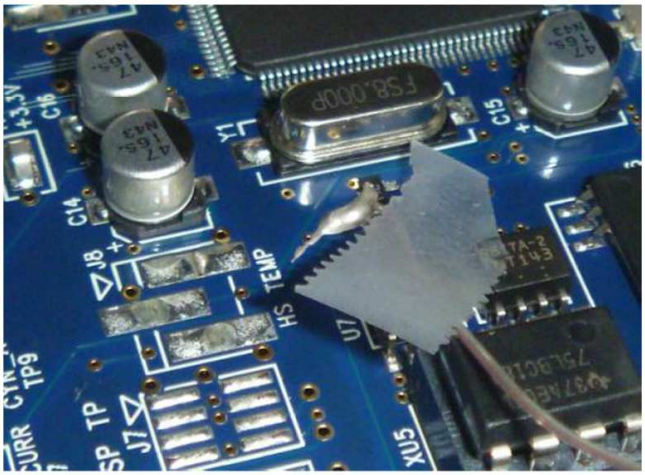
Manufacturer Supplied Documentation

For the Desire control card, measure the temperature at Q14.




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-011 | 61.5 | Per diagram | 100 |

Power Supply Picture – Tc location



PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

# SECTION 5 - Desire D60

Lustr+
Vivid
Studio HD
Tungsten
Daylight

# REPORT

**545 EAST ALGONQUIN ROAD ARLINGTON HEIGHTS, IL 60005**

Project No. G101042610 Date: March 5, 2013

REPORT NO. 101042610CHI-001

TEST OF ONE LED STAGE LUMINAIRE

FIXTURE MODEL NO. DESIRE D60
LED MODEL NO. PHILIPS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500428350.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79: 2008 Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products

ANSI ANSLG C38.377: 2012 Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one prototype sample of model number Desire D60. The sample was received by Intertek on January 16, 2013, in undamaged condition, and one sample was tested as received. The sample designation was 01162013123956

DATES OF TESTS: February 13, 2013 through February 16, 2013

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | Desire D60 |
| Description: | LED Stage and Theatre Luminaire |

| Criteria | Sphere | Goniometer |
|---|---|---|
| Total Lumen Output (Lumens) | 3469 | 3282 |
| Total Power (W) | 88.23 | 89.18 |
| Luminaire Efficacy (LPW) | 39.32 | 36.80 |

| Criteria | Result |
|---|---|
| Power Factor – 120VAC | 0.995 |
| Power Factor – 240VAC | 0.923 |
| Current ATHD – 120VAC | 2.31% |
| Current ATHD – 240VAC | 7.32% |
| Correlated Color Temperature (CCT) | 6579K |
| Color Rendering Index (CRI) - Ra | 66.8 |
| Color Rendering Index (CRI) - R9 | -49.4 |
| Duv | 0.028 |
| Chromaticity Coordinate (x) | 0.319 |
| Chromaticity Coordinate (y) | 0.280 |
| Chromaticity Coordinate (u') | 0.223 |
| Chromaticity Coordinate (v') | 0.440 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Meter | WT210 | 146919 | 12/21/12 | 12/21/13 |
| Omega Thermometer | DPI8-C24 | 146920 | 11/15/12 | 11/15/13 |
| LSI High Speed Mirror Goniometer | 6440T | 146928 | VBU | VBU |
| Newport Hygrometer | iServer | 146961 | 2/23/12 | 2/23/13 |
| Elgar, AC Power Supply | CW1251P | 146918 | VBU | VBU |
| Cole-Parmer Triple Timer | 94440-00 | CHI0041 | 7/19/12 | 7/19/13 |
| Labsphere Spectroradiometer | CDS1100 | CHI0091 | VBU | VBU |
| 3 Meter Sphere | SPR600 | CHI0088 | VBU | VBU |
| Elgar AC Power Supply | CW1251M | 146112 | VBU | VBU |
| Sorenson DC Power Supply | XFR150-8 | 146846 | 4/20/12 | 4/20/13 |
| Newport Temperature and Humidty Recorder | iTHX-SD | 146958 | 2/23/12 | 2/23/13 |
| Yokoga Power Meter | WT1600 | 146769 | 5/18/12 | 5/18/13 |
| Omega Temperature Meter | MDSi8 | 146139 | 7/19/12 | 7/19/13 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Three Meter Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Estimated Total Operating Time

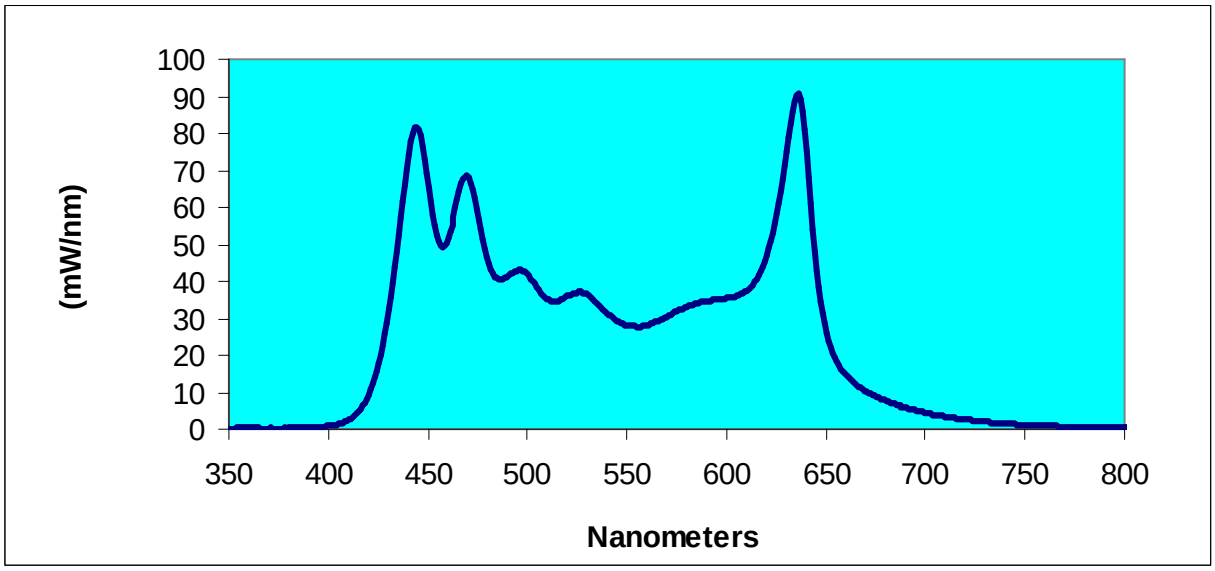
| Model No. | Total Hours |
|---|---|
| Desire D60 | 4 |

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.27 | 460 | 51.79 | 570 | 30.52 | 680 | 7.69 |
| 355 | 0.28 | 465 | 62.19 | 575 | 31.90 | 685 | 6.72 |
| 360 | 0.31 | 470 | 68.56 | 580 | 33.17 | 690 | 5.88 |
| 365 | 0.28 | 475 | 59.63 | 585 | 34.14 | 695 | 5.15 |
| 370 | 0.27 | 480 | 46.55 | 590 | 34.75 | 700 | 4.51 |
| 375 | 0.27 | 485 | 40.97 | 595 | 35.09 | 705 | 3.96 |
| 380 | 0.28 | 490 | 41.33 | 600 | 35.45 | 710 | 3.47 |
| 385 | 0.29 | 495 | 42.90 | 605 | 36.07 | 715 | 3.03 |
| 390 | 0.38 | 500 | 42.15 | 610 | 37.52 | 720 | 2.66 |
| 395 | 0.57 | 505 | 38.51 | 615 | 40.77 | 725 | 2.34 |
| 400 | 0.88 | 510 | 35.18 | 620 | 47.08 | 730 | 2.05 |
| 405 | 1.47 | 515 | 34.66 | 625 | 58.19 | 735 | 1.79 |
| 410 | 2.60 | 520 | 36.00 | 630 | 74.98 | 740 | 1.58 |
| 415 | 4.77 | 525 | 36.94 | 635 | 90.47 | 745 | 1.39 |
| 420 | 9.29 | 530 | 36.50 | 640 | 74.91 | 750 | 1.23 |
| 425 | 18.02 | 535 | 34.26 | 645 | 42.95 | 755 | 1.08 |
| 430 | 32.19 | 540 | 31.52 | 650 | 25.93 | 760 | 0.97 |
| 435 | 52.49 | 545 | 29.34 | 655 | 18.34 | 765 | 0.85 |
| 440 | 74.31 | 550 | 28.16 | 660 | 14.35 | 770 | 0.74 |
| 445 | 81.25 | 555 | 27.82 | 665 | 11.93 | 775 | 0.66 |
| 450 | 65.27 | 560 | 28.18 | 670 | 10.15 | 780 | 0.59 |
| 455 | 50.83 | 565 | 29.16 | 675 | 8.81 | | |

**Sample No. 01162013123956**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25℃ – Integrating Sphere Method

| Intertek Sample No. | Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|---|
| 01162013123956 | 6579 | 66.8 | -49.4 | 0.028 | 0.319 | 0.280 | 0.223 | 0.440 |

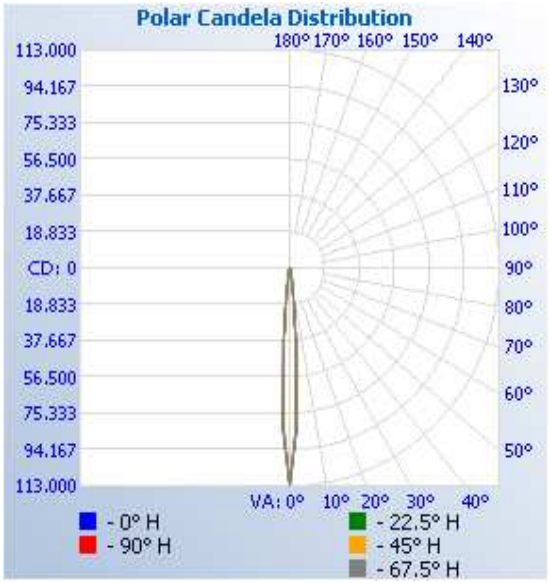
| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| 01162013123956 | Up | 120.0 | 739.0 | 88.23 | 0.995 | 2.31 | 3469 | 39.32 |
| | | 240.0 | | | 0.923 | 7.32 | | |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| 01162013123956 | Up | 120.0 | 747.1 | 89.18 | 0.995 | 3282 | 36.80 |

Intensity (Candlepower) Summary at 25℃ - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 112185 | 112185 | 112185 | 112185 | 112185 |
| 5 | 37445 | 37203 | 37458 | 37022 | 36767 |
| 10 | 5525 | 5771 | 5751 | 5395 | 5182 |
| 15 | 904 | 996 | 1001 | 869 | 839 |
| 20 | 446 | 459 | 456 | 442 | 427 |
| 25 | 130 | 132 | 126 | 123 | 123 |
| 30 | 71 | 68 | 66 | 66 | 66 |
| 35 | 55 | 55 | 51 | 50 | 53 |
| 40 | 50 | 42 | 49 | 46 | 39 |
| 45 | 50 | 42 | 45 | 44 | 39 |
| 50 | 46 | 46 | 39 | 42 | 43 |
| 55 | 39 | 32 | 35 | 37 | 33 |
| 60 | 26 | 22 | 27 | 27 | 22 |
| 65 | 16 | 13 | 14 | 14 | 13 |
| 70 | 7 | 7 | 7 | 7 | 6 |
| 75 | 3 | 3 | 3 | 3 | 3 |
| 80 | 1 | 1 | 1 | 1 | 1 |
| 85 | 1 | 0 | 0 | 1 | 0 |
| 90 | 1 | 0 | 1 | 0 | 0 |

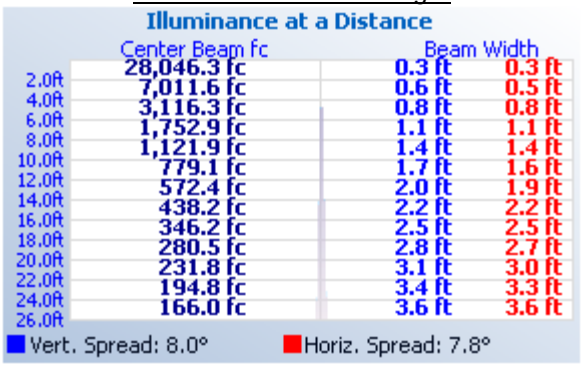


RESULTS OF TESTS (cont'd)

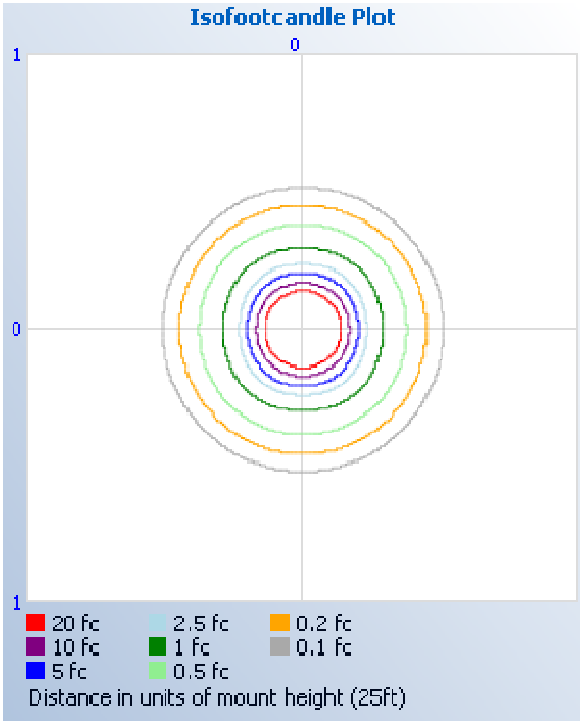
Illumination Plots

Mounting Height: 25 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25℃

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3165 | 96.4 |
| 0-40 | 3198 | 97.5 |
| 0-60 | 3264 | 99.4 |
| 60-90 | 18.4 | 0.6 |
| 0-90 | 3282 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3282 | 100.0 |

Zonal Lumens and Percentages at 25℃

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2679 | 81.6 |
| 10-20 | 406.4 | 12.4 |
| 20-30 | 79.8 | 2.4 |
| 30-40 | 33.3 | 1.0 |
| 40-50 | 34.1 | 1.0 |
| 50-60 | 31.1 | 0.9 |
| 60-70 | 14.4 | 0.4 |
| 70-80 | 3.5 | 0.1 |
| 80-90 | 0.6 | 0.0 |

Pictures (not to scale)

Luminaire mounted on goniophotometer.

LED pattern on luminaire

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Tim Quigley
Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

# REPORT

**545 E. ALGONQUIN ROAD ARLINGTON HEIGHTS, IL 60005**

Project No. G101042629 Date: March 13, 2013

REPORT NO. 101042629CHI-001

IN-SITU TEST OF ONE LED STAGE LIGHT

FIXTURE MODEL NO. DESIRE D60
LED MODEL NO. PHILIPS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

AUTHORIZATION: The testing performed was authorized by signed quote number 500428354.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

Energy Star Version 1.1 (2008): Program Requirements for Solid-State Lighting Luminaires
Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one sample of model number Desire D60 The sample was received by Intertek on January 16, 2013, in undamaged condition, and one sample was tested as received. The sample designation was 01162013123956.

DATE OF TESTS: February 26, 2013 through March 5, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | DESIRE D60 |
| Description: | LED Stage and Theatre Lighting |

| Criteria | Result | |
|---|---|---|
| Maximum In-Situ Source Temperature Point - LEDG REBEL GRN 1812 | 59.0 | Compliant |
| Maximum In -Situ Source Temperature Point - LEDC REBEL CYAN 1812 | 59.8 | Compliant |
| Maximum In -Situ Source Temperature Point - LEDB REBEL BLUE 1812 | 58.4 | Compliant |
| Maximum In -Situ Source Temperature Point - LEDI REBEL INDIGO 1812 | 65.0 | Compliant |
| Maximum In -Situ Source Temperature Point - LEDA REBEL PC-AMBER 1812 | 57.5 | Compliant |
| Maximum In -Situ Source Temperature Point - LEDW REBEL 4300K WHT binTO/TN 1812 | 57.3 | Compliant |
| Maximum In -Situ Source Temperature Point - LEDR REBEL RED LOW Vf 1812 | 54.6 | Compliant |
| Maximum Power Supply Case Temperature | 57.0 | Compliant |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Agilent Data Logger | 34970A | 146764 | 2/21/13 | 2/21/14 |
| Yokogawa Power Meter | WT230 | 146454 | 2/21/13 | 2/21/14 |
| Newport Thermometer/Hygrometer | iTHX-SD | 146959 | 2/21/13 | 2/21/14 |
| Staco Variac | 3PN1510B | 146359 | VBU | VBU |
| General, Wireless Weather Station | DBAR880 | 146558 | 7/23/12 | 7/23/13 |
| Fluke IR Thermometer | 561 HVACPRO | 146578 | VBU | VBU |

TEST METHODS

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Agilent 34970A Data Logger. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

Estimated Total Operating Time

| Model No. | Total Hours |
|---|---|
| Desire D60 | 33 |

RESULTS OF TESTS

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: Philips Luxeon Rebel

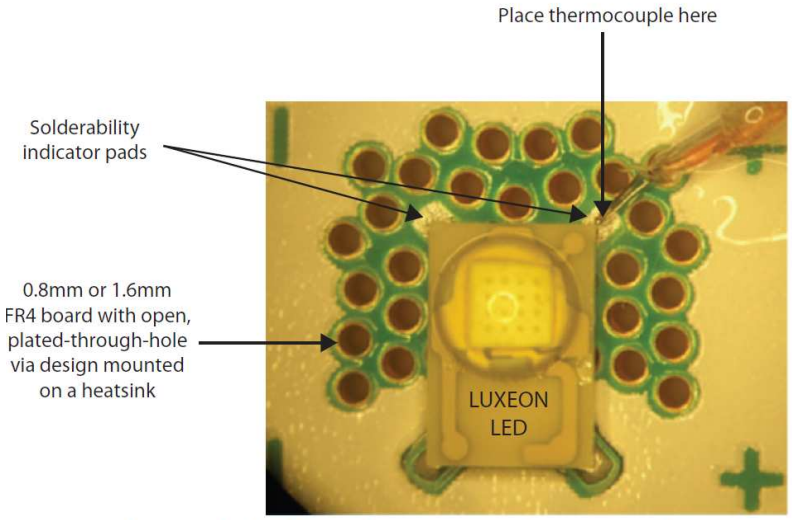



**Figure 1. Solderability indicator pad and thermocouple placement.**

| | | | |
|---|---|---|---|
| CR9180-F | LEDG REBEL GRN 1812 | PHILIPS LIGHTING | LXML-PM01-0090 (C-BIN 4) |
| CR9180-F | LEDG REBEL GRN 1812 | PHILIPS LIGHTING | LXML-PM01-0090 (C-BIN 5) |
| CR9180-F | LEDG REBEL GRN 1812 | PHILIPS LIGHTING | LXML-PM01-0090 (C-BIN 6) |
| CR9180-F | LEDG REBEL GRN 1812 | PHILIPS LIGHTING | LXML-PM01-0090 (C-BIN 3) |
| CR9181-F | LEDC REBEL CYAN 1812 | PHILIPS LIGHTING | LXML-PE01-0070 (C-BIN 1) |
| CR9181-F | LEDC REBEL CYAN 1812 | PHILIPS LIGHTING | LXML-PE01-0070 (C-BIN 2) |
| CR9181-F | LEDC REBEL CYAN 1812 | PHILIPS LIGHTING | LXML-PE01-0070 (C-BIN 3) |
| CR9182-F | LEDB REBEL BLUE 1812 | PHILIPS LIGHTING | LXML-PB01-0030 (C-BIN 3) |
| CR9182-F | LEDB REBEL BLUE 1812 | PHILIPS LIGHTING | LXML-PB01-0030 (C-BIN 4) |
| CR9182-F | LEDB REBEL BLUE 1812 | PHILIPS LIGHTING | LXML-PB01-0030 (C-BIN 5) |
| CR9183-F | LEDI REBEL INDIGO 1812 | PHILIPS LIGHTING | LXML-PR01-0350 (C-BIN 1) |
| CR9183-F | LEDI REBEL INDIGO 1812 | PHILIPS LIGHTING | LXML-PR01-0350 (C-BIN 2) |
| CR9183-F | LEDI REBEL INDIGO 1812 | PHILIPS LIGHTING | LXML-PR01-0350 (C-BIN 3) |
| CR9183-F | LEDI REBEL INDIGO 1812 | PHILIPS LIGHTING | LXML-PR01-0350 (C-BIN 4) |
| CR9183-F | LEDI REBEL INDIGO 1812 | PHILIPS LIGHTING | LXML-PR01-0425 (C-BIN 1) |
| CR9183-F | LEDI REBEL INDIGO 1812 | PHILIPS LIGHTING | LXML-PR01-0425 (C-BIN 2) |
| CR9183-F | LEDI REBEL INDIGO 1812 | PHILIPS LIGHTING | LXML-PR01-0425 (C-BIN 3) |
| CR9183-F | LEDI REBEL INDIGO 1812 | PHILIPS LIGHTING | LXML-PR01-0425 (C-BIN 4) |
| CR9184-F | LEDA REBEL PC-AMBER 1812 | PHILIPS LIGHTING | LXM2-PL01-0000 |
| CR9197-F | LEDW REBEL 4300K WHT binTO/TN 1812 | PHILIPS LIGHTING | LXML-PWN1-0100 (C-BIN TO) |
| CR9197-F | LEDW REBEL 4300K WHT binTO/TN 1812 | PHILIPS LIGHTING | LXML-PWN1-0100 (C-BIN TN) |
| CR9214-F | LEDR REBEL RED LOW Vf 1812 | PHILIPS LIGHTING | LXM2-PD01-0050-2000RL(T/R)(C-BIN4) |

RESULTS OF TESTS (cont'd)

Manufacturer Supplied Documentation:

## Electrical Characteristics at 350 mA for LUXEON Rebel color, Thermal Pad Temperature = 25°C

**Table 3.**

| Color | Part Number | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_J$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|---|
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Royal-Blue | LXML-PR01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red-Orange | LXM2-PH01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

## Absolute Maximum Ratings

**Table 5.**

| Parameter | Green/Cyan/ Blue/Royal Blue | LUXEON Rebel ES Royal Blue/ES Blue | Red/Deep-Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |

Table 11. Applicable for LXML-PXX1 and LXM2-PL01 (PC Amber) emitters.

| | $V_f$ Bins | |
|---|---|---|
| Bin Code | Minimum Forward Voltage (V) | Maximum Forward Voltage (V) |
| A | 2.31 | 2.55 |
| B | 2.55 | 2.79 |
| C | 2.79 | 3.03 |
| D | 3.03 | 3.27 |
| E | 3.27 | 3.51 |

Table 12. Applicable for LXML-PR02-XXXX (ES Royal Blue) and LXML-PB02-XXXX (ES Blue) emitters tested at 700 mA

| Bin Code | Minimum Forward Voltage (V) | Maximum Forward Voltage (V) |
|---|---|---|
| P | 2.50 | 2.75 |
| R | 2.75 | 3.00 |
| S | 3.00 | 3.25 |
| T | 3.25 | 3.50 |

Table 13. Applicable for LXM2-PXX1 (Red and Red Orange) and LXM3-PD01 (Deep Red) emitters tested at 350 mA.

| Bin Code | Minimum Forward Voltage (V) | Maximum Forward Voltage (V) |
|---|---|---|
| V | 1.80 | 2.00 |
| W | 2.00 | 2.20 |
| X | 2.20 | 2.40 |
| Y | 2.40 | 2.60 |
| Z | 2.60 | 2.80 |

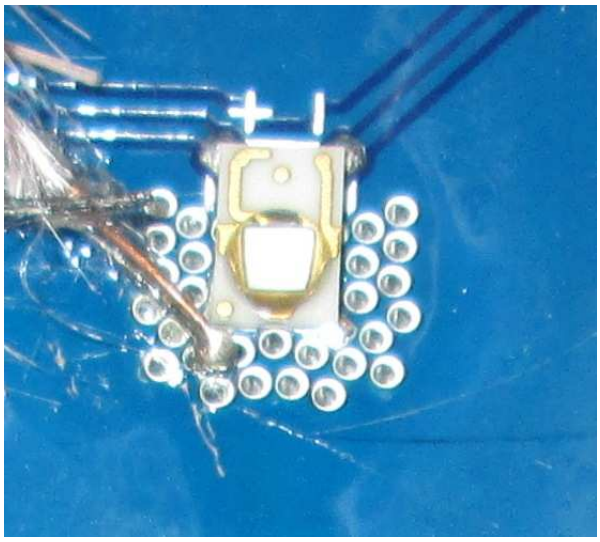
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDG Rebel GRN 1812 LXML-PM01-0090

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10 ℃/W
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 458.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.607
Maximum Source Temperature (Ts) = 134 C°

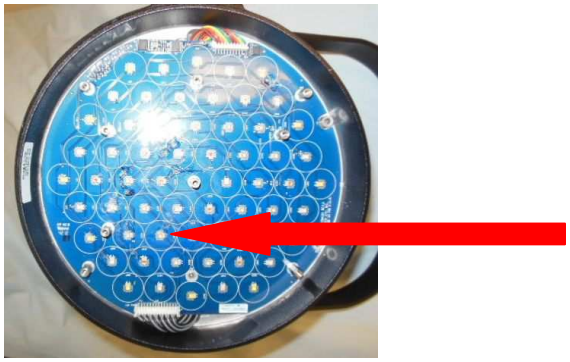
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013123956 | Philips Luxeon LEDG Rebel GRN 1812 LXML-PM01-0090 | 59.0 | Per diagram above | 134 |

In-Situ Picture – Ts point



In-Situ Picture – Array location



RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDC REBEL CYAN 1812 LXML-PE01-0070

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10 ℃/W
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 479.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.680
Maximum Source Temperature (Ts) = 133 C°

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013123956 | Philips Luxeon LEDC REBEL CYAN 1812 LXML-PE01-0070 | 59.8 | Per diagram above | 133 |

In-Situ Picture – Ts point

In-Situ Picture – Array location

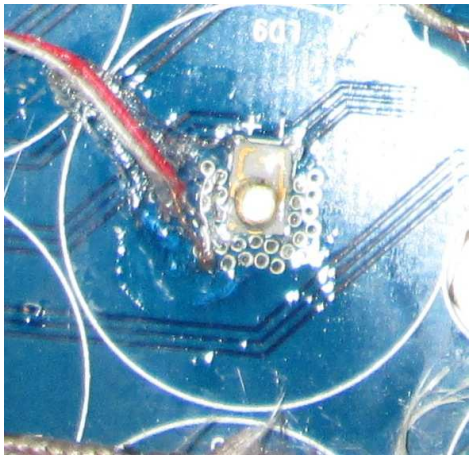
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDB REBEL BLUE 1812 LXML-PB01-0030

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10 ℃/W
Maximum Forward Voltage (Vf) from LED specification = 3.5
Measured LED Current = 480.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.685
Maximum Source Temperature (Ts) = 133 C°

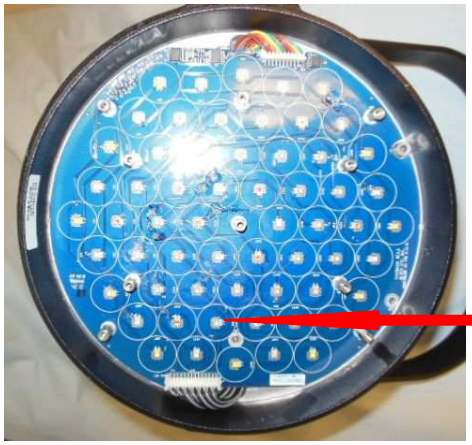
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013123956 | Philips Luxeon LEDB REBEL BLUE 1812 LXML-PB01-0030 | 58.4 | Per diagram above | 133 |

In-Situ Picture – Ts point



In-Situ Picture – Array location



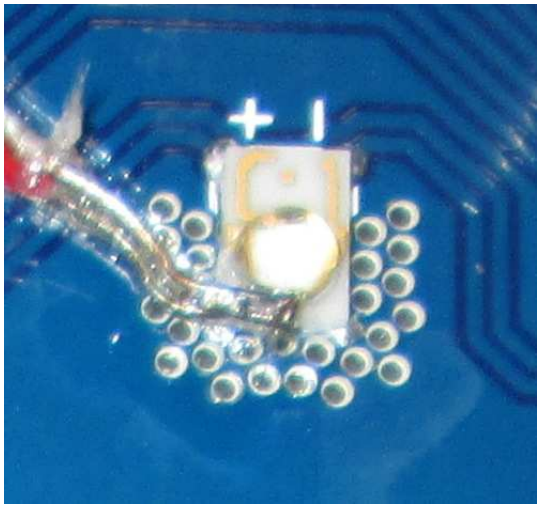
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDI REBEL INDIGO 1812 LXML-PR01-0350

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10 ℃/W
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 485.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.703
Maximum Source Temperature (Ts) = 133 C°

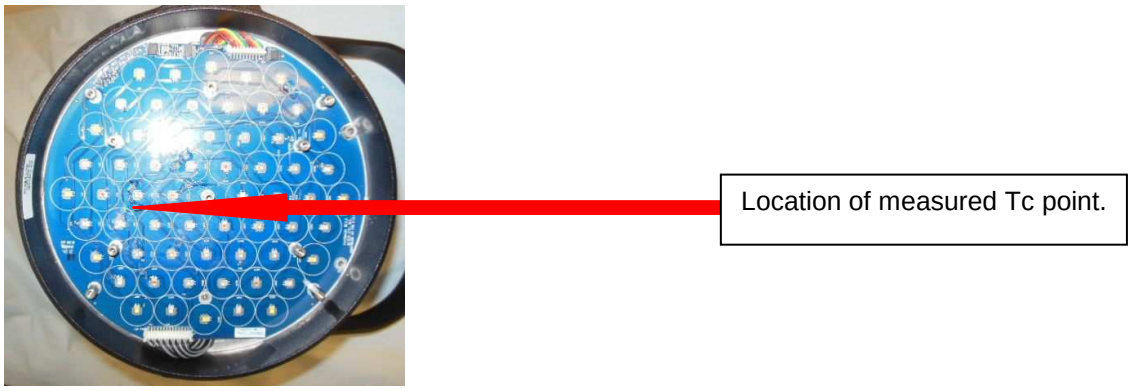
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013123956 | Philips Luxeon LEDI REBEL INDIGO 1812 LXML-PR01-0350 | 65.0 | Per diagram above | 133 |

In-Situ Picture – Ts point



In-Situ Picture – Array location



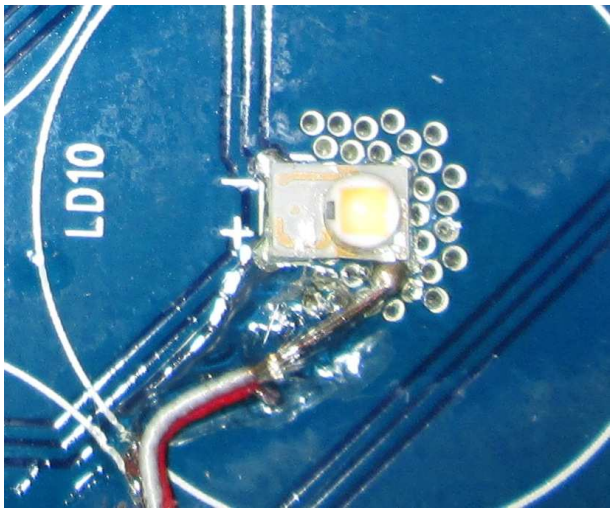
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDA REBEL PC-AMBER 1812 LXM2-PL01-0000

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 130℃
Thermal Resistance Formula from LED specification = 10 ℃/W
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 442.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.550
Maximum Source Temperature (Ts) = 114 C°

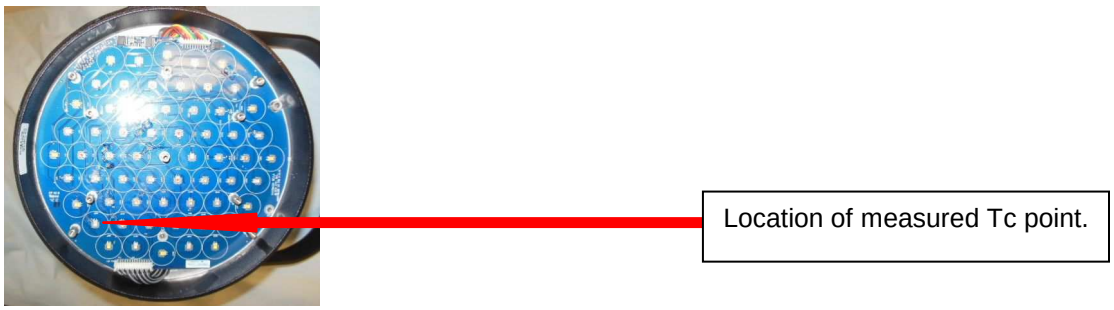
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013123956 | Philips Luxeon LEDA REBEL PC-AMBER 1812 LXM2-PL01-0000 | 57.5 | Per diagram above | 114 |

In-Situ Picture – Ts point




In-Situ Picture – Array location



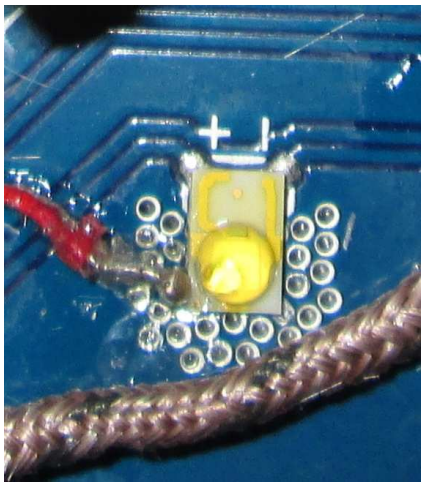
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDW REBEL 4300K WHT binTO/TN 1812 LXML-PWN1-0100

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10 ℃/W
Maximum Forward Voltage (Vf) from LED specification = 3.99
Measured LED Current = 468.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.868
Maximum Source Temperature (Ts) = 131 C°

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013123956 | Philips Luxeon LEDW REBEL 4300K WHT binTO/TN 1812 LXML-PWN1-0100 | 57.3 | Per diagram above | 131 |

In-Situ Picture – Ts point



In-Situ Picture – Array location

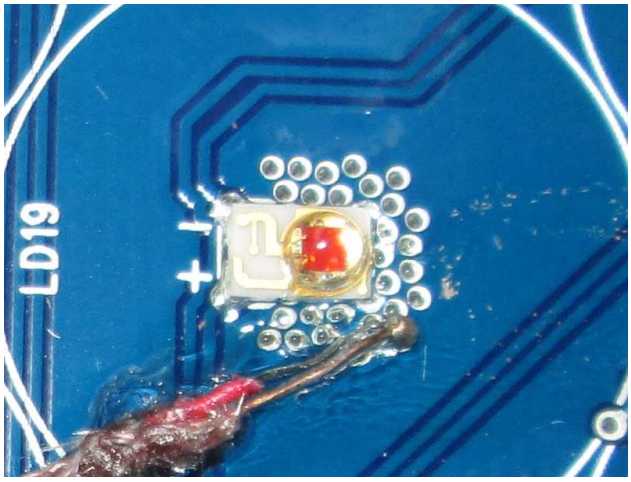
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDR REBEL RED LOW Vf 1812 LXM2-PD01-0050-2000RL(T/R)

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 135℃
Thermal Resistance Formula from LED specification = 8 ℃/W
Maximum Forward Voltage (Vf) from LED specification = 2.8Vdc
Measured LED Current = 401.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.122
Maximum Source Temperature (Ts) = 126 C°

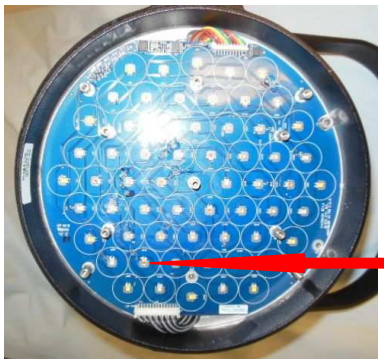
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013123956 | Philips Luxeon LEDR REBEL RED LOW Vf 1812 LXM2-PD01-0050-2000RL(T/R) | 54.6 | Per diagram above | 126 |

In-Situ Picture – Ts point




In-Situ Picture – Array location

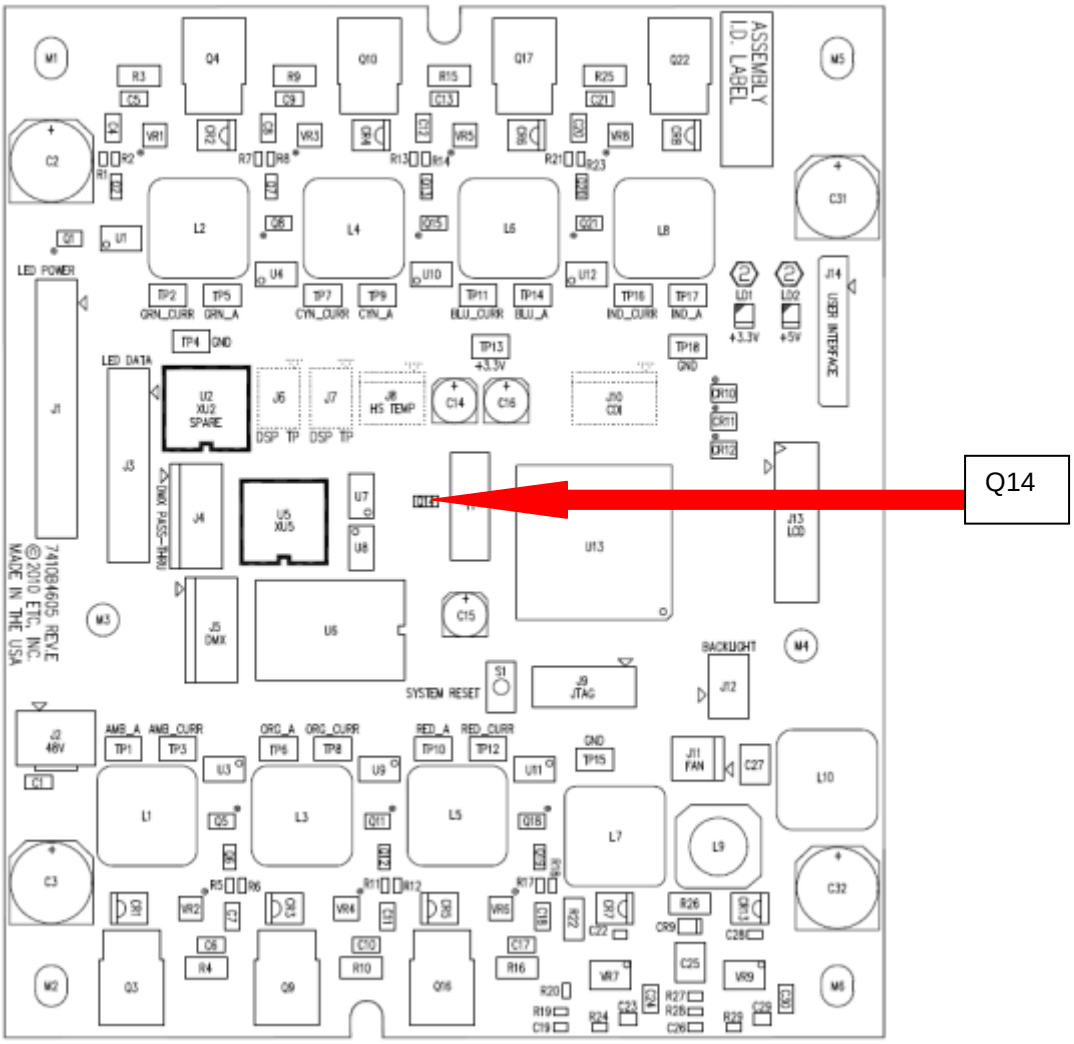


RESULTS OF TESTS (cont'd)

Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:
Driver identified as: Model No. 7410B4605

For the Desire control card, measure the temperature at Q14.

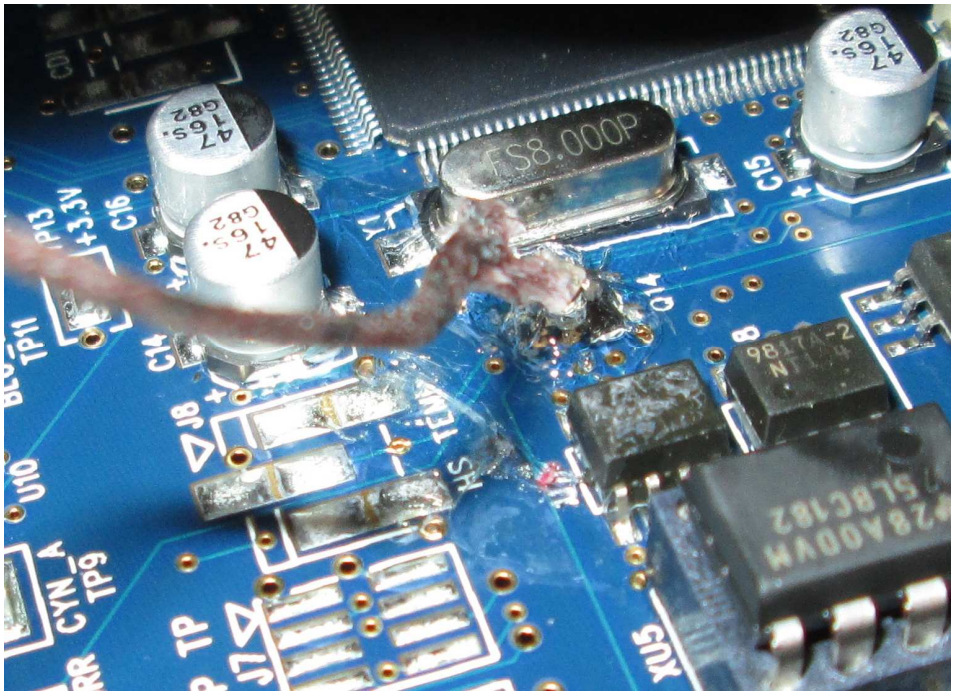



Measured Power Supply Case Temperature

| Sample No. | Model No. | Measured Power Supply Case Temperature (C°) | Location | Maximum Power Supply Case Temperature (C°) |
|---|---|---|---|---|
| 01162013123956 | 7410B4605 | 57.0 | Q14 | 100 |

RESULTS OF TESTS (cont'd)

Power Supply Picture



Pictures (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Tim Quigley
Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 20 , 2013

REPORT NO. 101414036CRT-012

TEST OF ONE D60 VIVID LED FIXTURE ETL (BLK)

MODEL NO. D60 VIVID
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D60 VIVID. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-012.

DATES OF TESTS: November 13, 2013 through November 19, 2

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D60 VIVID |
| Description: | D60 VIVID LED FIXTURE ETL (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 4340 | 3925 |
| Total Power (W) | 127.0 | 126.8 |
| Luminaire Efficacy (LPW) | 34.17 | 30.95 |

| Criteria | Result |
|---|---|
| Power Factor | 0.991 |
| Current ATHD % | 2.45 |
| Correlated Color Temperature (CCT - K) | 3467 |
| Color Rendering Index (CRI - Ra) | 29.1 |
| Color Rendering Index (CRI - R9) | -85.7 |
| DUV | 0.062 |
| Chromaticity Coordinate (x) | 0.355 |
| Chromaticity Coordinate (y) | 0.256 |
| Chromaticity Coordinate (u') | 0.266 |
| Chromaticity Coordinate (v') | 0.429 |
| Maximum In-Situ Source Temperature Point (°C) | 76.4 |
| Maximum In-Situ Driver Case Temperature (°C) | 60.5 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| COLE PARMER-Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Sorensen DC Power Supply | DLM150-20E | --- | --- | --- |
| Data Precision | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 10/21/13 | 11/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole Palmer Thermometer | 03313-85 | T1470 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1355 | 11/26/12 | 11/26/13 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

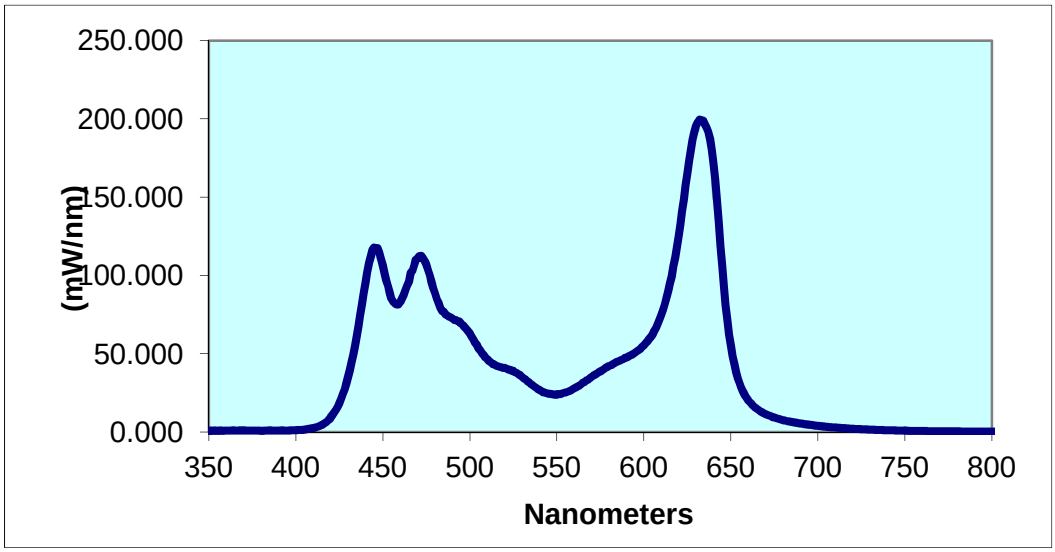
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-012 | UP | 120.0 | 1066 | 127.0 | 0.991 | 2.45 | 4340 | 34.17 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3467 | 29.1 | -85.7 | 0.062 | 0.355 | 0.256 | 0.266 | 0.429 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.689 | 440 | 95.24 | 530 | 35.34 | 620 | 125.5 | 710 | 2.86 |
| 355 | 0.817 | 445 | 117.9 | 535 | 30.7 | 625 | 165.5 | 715 | 2.45 |
| 360 | 0.989 | 450 | 106.3 | 540 | 26.77 | 630 | 195.6 | 720 | 2.114 |
| 365 | 0.848 | 455 | 85.12 | 545 | 24.46 | 635 | 196.8 | 725 | 1.789 |
| 370 | 1.032 | 460 | 83.02 | 550 | 24.05 | 640 | 172.2 | 730 | 1.508 |
| 375 | 0.917 | 465 | 96.06 | 555 | 25.22 | 645 | 105.6 | 735 | 1.338 |
| 380 | 0.727 | 470 | 110.8 | 560 | 27.9 | 650 | 55 | 740 | 1.108 |
| 385 | 1.037 | 475 | 107 | 565 | 31.47 | 655 | 31.08 | 745 | 0.959 |
| 390 | 0.904 | 480 | 88.79 | 570 | 35.18 | 660 | 20.18 | 750 | 0.878 |
| 395 | 0.983 | 485 | 76.75 | 575 | 38.65 | 665 | 14.7 | 755 | 0.733 |
| 400 | 1.096 | 490 | 72.51 | 580 | 42.15 | 670 | 11.41 | 760 | 0.653 |
| 405 | 1.53 | 495 | 69.11 | 585 | 44.99 | 675 | 9.199 | 765 | 0.579 |
| 410 | 2.471 | 500 | 63.06 | 590 | 47.54 | 680 | 7.539 | 770 | 0.476 |
| 415 | 4.503 | 505 | 53.41 | 595 | 50.77 | 685 | 6.418 | 775 | 0.438 |
| 420 | 9.268 | 510 | 46.61 | 600 | 55.3 | 690 | 5.441 | 780 | 0.398 |
| 425 | 18.35 | 515 | 42.46 | 605 | 62.28 | 695 | 4.644 | | |
| 430 | 34.78 | 520 | 40.89 | 610 | 74.64 | 700 | 3.905 | | |
| 435 | 61.55 | 525 | 39.04 | 615 | 94.74 | 705 | 3.346 | | |

**Spectral Data Over Visible Wavelengths**



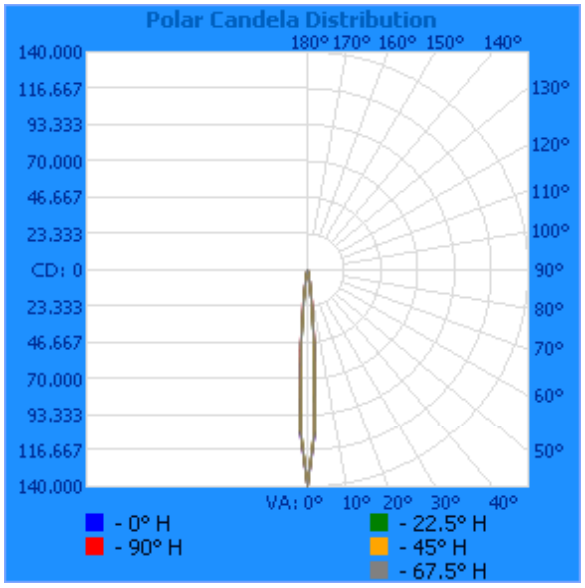
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-012 | UP | 120.1 | 1065 | 126.8 | 0.992 | 3925 | 30.95 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 139057 | 139057 | 139057 | 139057 | 139057 |
| 5 | 47995 | 46665 | 47174 | 47980 | 51173 |
| 10 | 6827 | 6902 | 7162 | 7107 | 7364 |
| 15 | 944 | 1034 | 1083 | 957 | 924 |
| 20 | 341 | 376 | 400 | 376 | 337 |
| 25 | 9 | 17 | 21 | 28 | 24 |
| 30 | 0 | 0 | 0 | 0 | 0 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

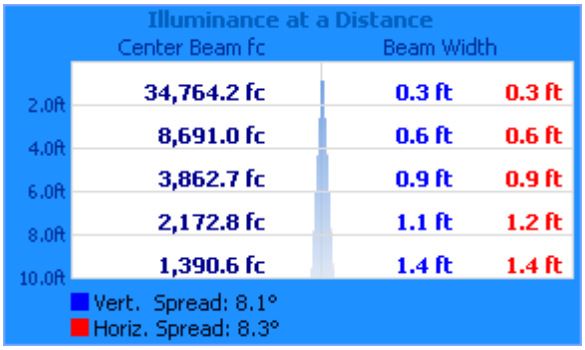


RESULTS OF TEST (cont'd)

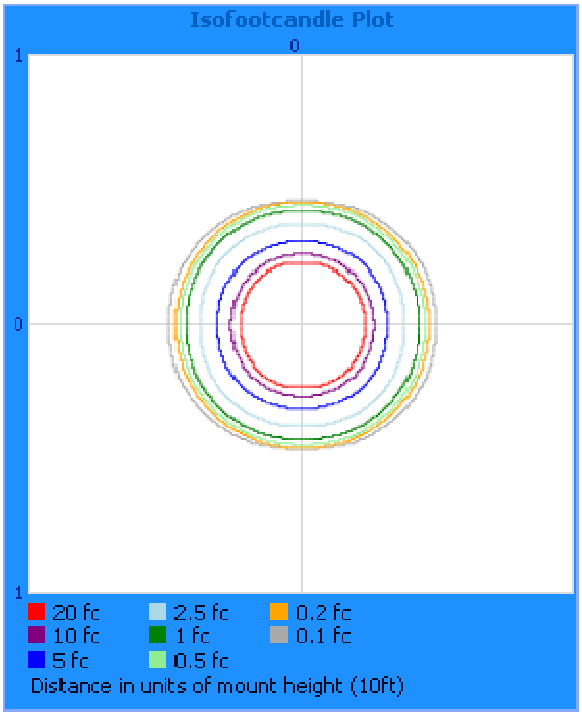
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3925 | 100.0 |
| 0-40 | 3925 | 100.0 |
| 0-60 | 3925 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 3925 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3925 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 3421 | 87.2 |
| 10-20 | 464.5 | 11.8 |
| 20-30 | 39.5 | 1.0 |
| 30-40 | 0.0 | 0.0 |
| 40-50 | 0.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as:

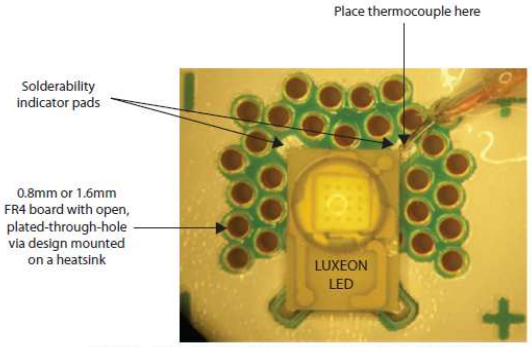


Figure 1. Solderability indicator pad and thermocouple placement.

| Forward Voltage $V_f$ [1] (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) | Typical Thermal Resistance Junction to Thermal Pad (°C/W) |
|---|---|---|---|---|
| Min. | Typ. | Max. | $\Delta V_f / \Delta T_J$ | $R\theta_{J-C}$ |
| 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |

| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

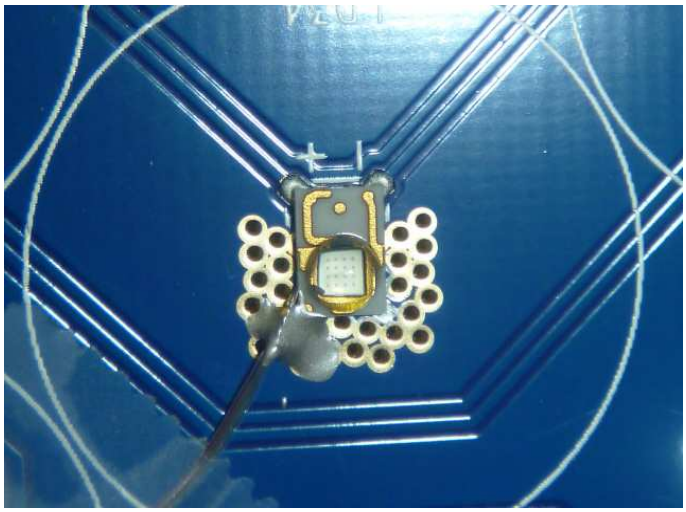
Green LXML-PM01

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 545mA
Calculated LED Wattage = Vf x Measured LED Current = 1.913W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 130.9°C

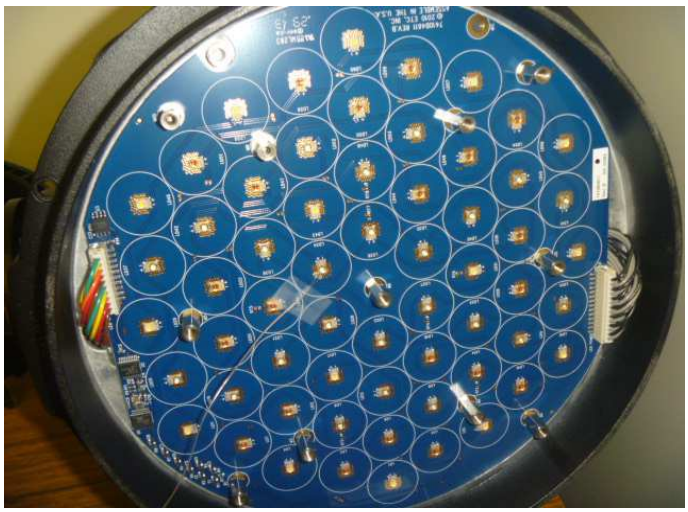
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-012 | 76.4 | Per diagram | 130.9 |

In-Situ Picture – Ts



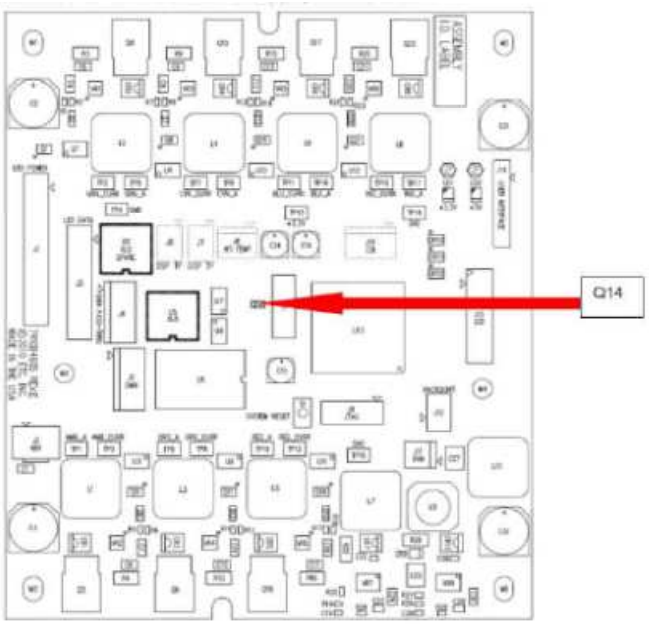
In-Situ Picture – Ts location



RESULTS OF TEST (cont'd)

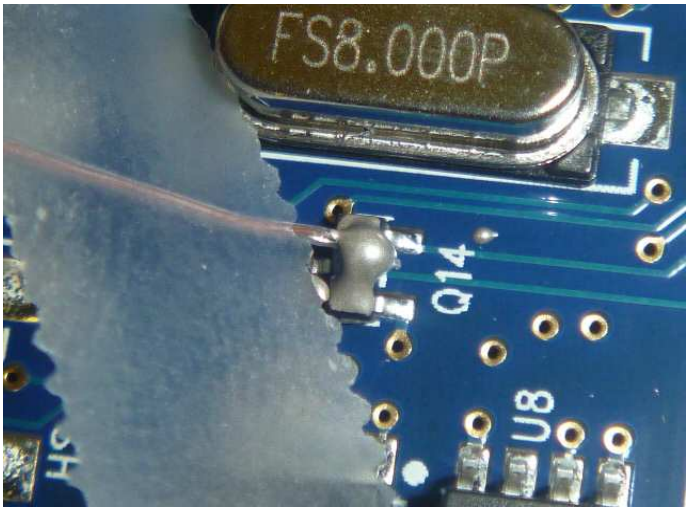
Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-009 | 60.5 | Per diagram | 100 |

Power Supply Picture – Tc location



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 20, 2013

REPORT NO. 101414036CRT-013

TEST OF ONE D60 STUDIO LED FIXTURE ETL (BLK)

MODEL NO. D60 STUDIO
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D60 STUDIO. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-013.

DATES OF TESTS: November 13, 2013 through November 19, 2

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D60 STUDIO |
| Description: | D60 STUDIO LED FIXTURE ETL (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 5744 | 5443 |
| Total Power (W) | 135.3 | 135.1 |
| Luminaire Efficacy (LPW) | 42.45 | 40.29 |

| Criteria | Result |
|---|---|
| Power Factor | 0.990 |
| Current ATHD % | 2.42 |
| Correlated Color Temperature (CCT - K) | 4071 |
| Color Rendering Index (CRI - Ra) | 84.3 |
| Color Rendering Index (CRI - R9) | 79.7 |
| DUV | 0.005 |
| Chromaticity Coordinate (x) | 0.374 |
| Chromaticity Coordinate (y) | 0.364 |
| Chromaticity Coordinate (u') | 0.226 |
| Chromaticity Coordinate (v') | 0.495 |
| Maximum In-Situ Source Temperature Point (°C) | 82.5 |
| Maximum In-Situ Driver Case Temperature (°C) | 65.0 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| COLE PARMER-Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Sorensen DC Power Supply | DLM150-20E | --- | --- | --- |
| Data Precision | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 10/21/13 | 11/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1355 | 11/26/12 | 11/26/13 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

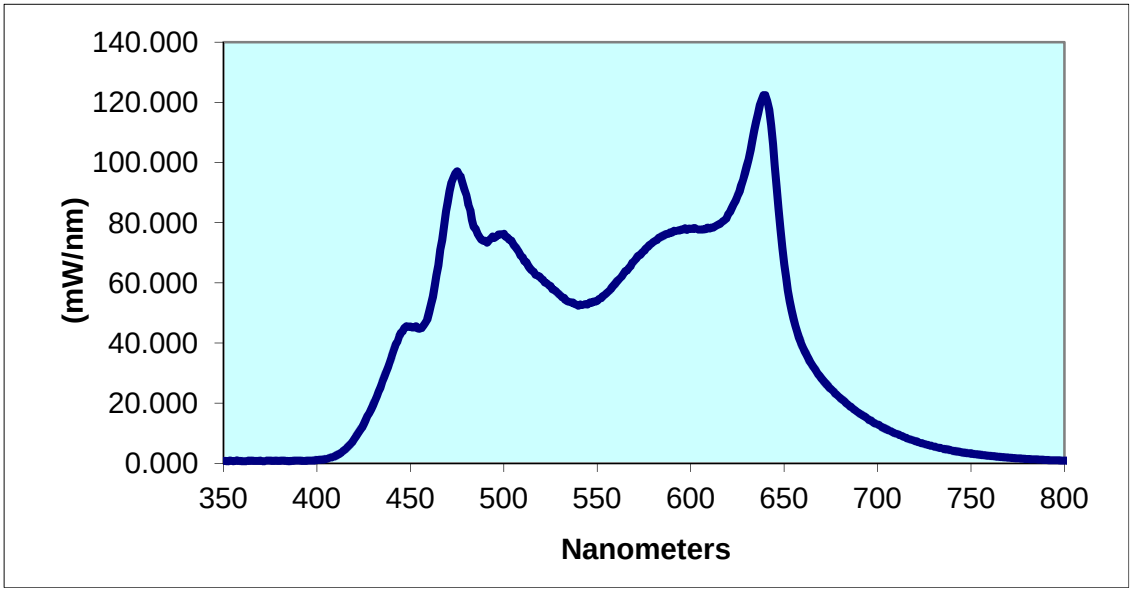
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-013 | UP | 120.0 | 1138 | 135.3 | 0.990 | 2.42 | 5744 | 42.45 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 4071 | 84.3 | 79.7 | 0.005 | 0.374 | 0.364 | 0.226 | 0.495 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.796 | 440 | 35.5 | 530 | 55.91 | 620 | 82.59 | 710 | 9.837 |
| 355 | 0.7 | 445 | 43.36 | 535 | 53.63 | 625 | 88.93 | 715 | 8.589 |
| 360 | 0.707 | 450 | 45.44 | 540 | 52.41 | 630 | 98.87 | 720 | 7.395 |
| 365 | 0.854 | 455 | 44.69 | 545 | 52.84 | 635 | 113.6 | 725 | 6.456 |
| 370 | 0.92 | 460 | 50.12 | 550 | 54.1 | 640 | 122.4 | 730 | 5.605 |
| 375 | 0.903 | 465 | 65.8 | 555 | 56.55 | 645 | 99.13 | 735 | 4.85 |
| 380 | 0.748 | 470 | 86.7 | 560 | 60.11 | 650 | 66.11 | 740 | 4.206 |
| 385 | 0.705 | 475 | 97.08 | 565 | 63.9 | 655 | 48.02 | 745 | 3.641 |
| 390 | 0.855 | 480 | 89.17 | 570 | 67.36 | 660 | 38.55 | 750 | 3.21 |
| 395 | 0.827 | 485 | 77.87 | 575 | 70.57 | 665 | 32.66 | 755 | 2.8 |
| 400 | 1.047 | 490 | 74.03 | 580 | 73.54 | 670 | 28.23 | 760 | 2.454 |
| 405 | 1.447 | 495 | 74.94 | 585 | 75.63 | 675 | 24.71 | 765 | 2.137 |
| 410 | 2.462 | 500 | 76.3 | 590 | 76.64 | 680 | 21.71 | 770 | 1.873 |
| 415 | 4.531 | 505 | 72.96 | 595 | 77.71 | 685 | 19.01 | 775 | 1.613 |
| 420 | 8.037 | 510 | 68.56 | 600 | 77.93 | 690 | 16.76 | 780 | 1.435 |
| 425 | 12.98 | 515 | 64.1 | 605 | 77.7 | 695 | 14.64 | | |
| 430 | 19.27 | 520 | 61.43 | 610 | 78.01 | 700 | 12.89 | | |
| 435 | 27.08 | 525 | 58.87 | 615 | 79.4 | 705 | 11.24 | | |

**Spectral Data Over Visible Wavelengths**



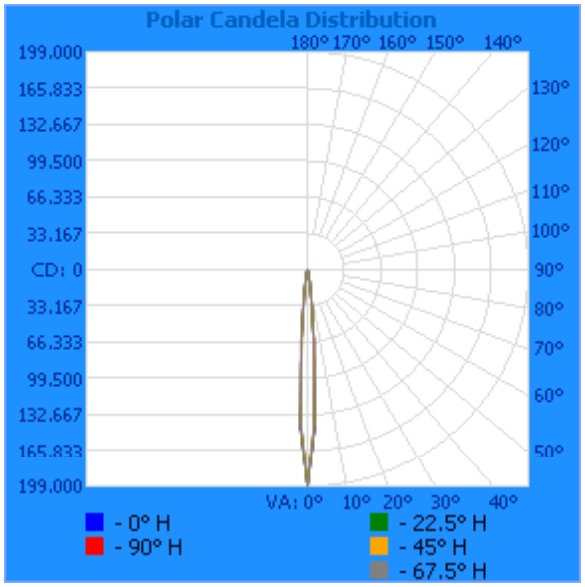
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-013 | UP | 120.0 | 1136 | 135.1 | 0.990 | 5443 | 40.29 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 198042 | 198042 | 198042 | 198042 | 198042 |
| 5 | 61742 | 60888 | 60327 | 61994 | 65648 |
| 10 | 10330 | 10569 | 11042 | 11004 | 10907 |
| 15 | 1646 | 1739 | 1970 | 1866 | 1766 |
| 20 | 680 | 666 | 661 | 632 | 609 |
| 25 | 57 | 66 | 59 | 71 | 65 |
| 30 | 0 | 0 | 0 | 0 | 0 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

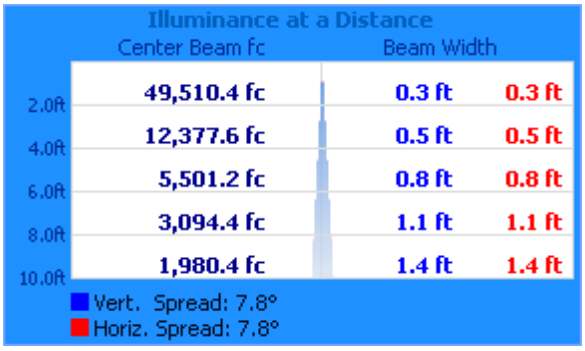


RESULTS OF TEST (cont'd)

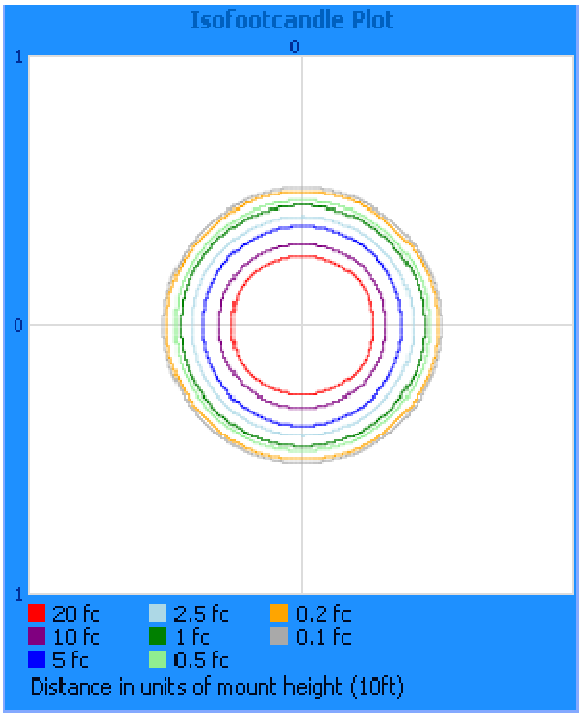
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 5443 | 100.0 |
| 0-40 | 5443 | 100.0 |
| 0-60 | 5443 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 5443 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 5443 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 4597 | 84.5 |
| 10-20 | 779.2 | 14.3 |
| 20-30 | 66.3 | 1.2 |
| 30-40 | 0.0 | 0.0 |
| 40-50 | 0.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as: Luxeon Rebel

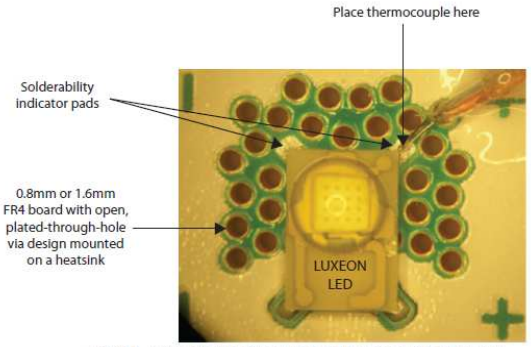


Figure 1. Solderability indicator pad and thermocouple placement.

| Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|
| 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

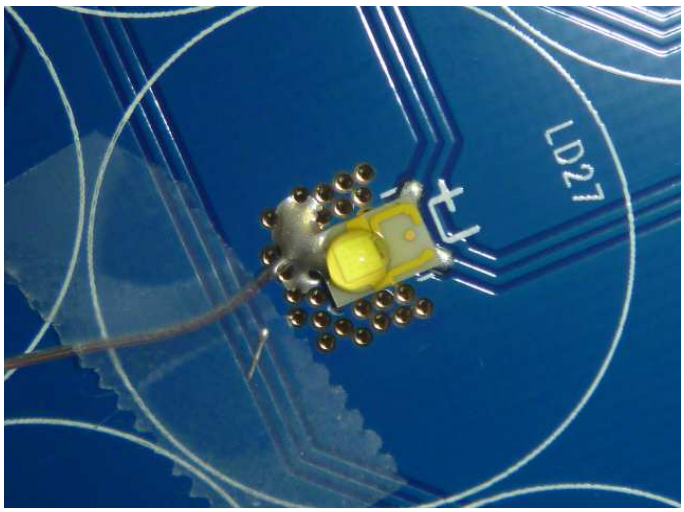
Amber LXML-PL01

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 12°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 588mA
Calculated LED Wattage = Vf x Measured LED Current = 2.064W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 125.2°C

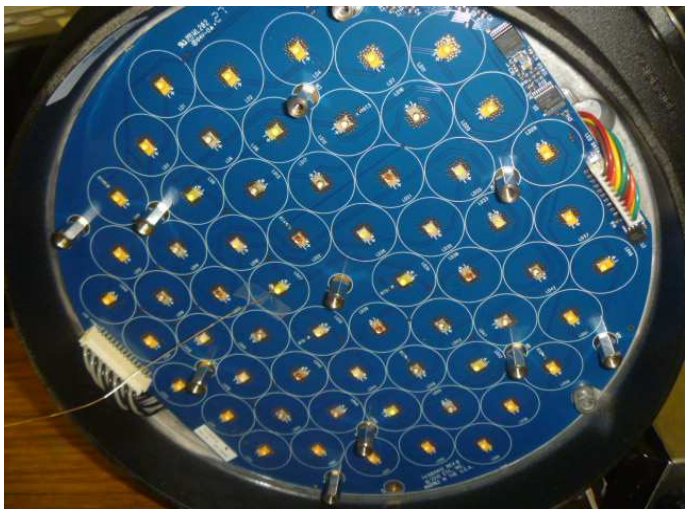
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-013 | 82.5 | Per diagram | 125.2 |

In-Situ Picture – Ts



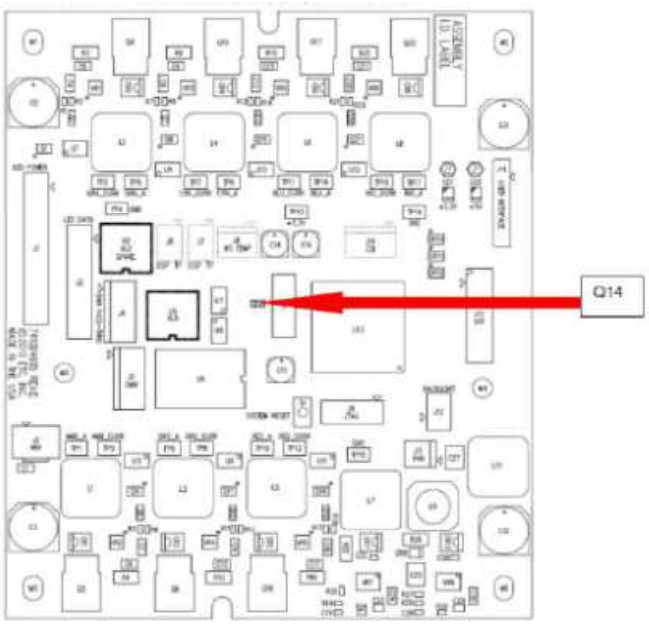

In-Situ Picture – Ts location



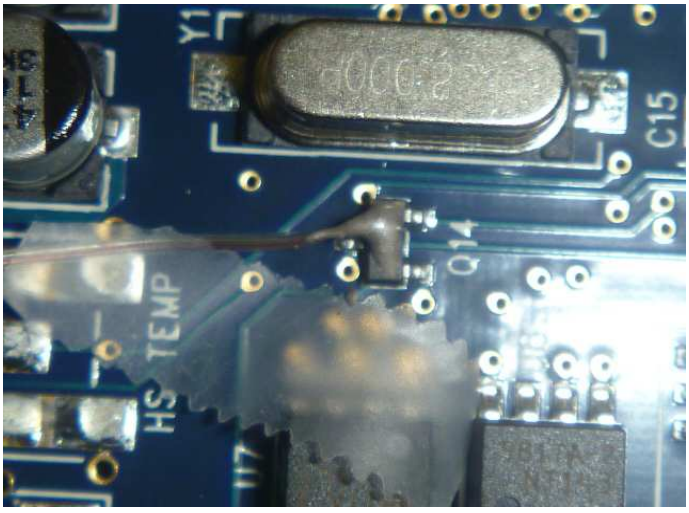
RESULTS OF TEST (cont'd)

Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-009 | 65.0 | Per diagram | 100 |



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: May 23, 2013

REPORT NO. 101158242CRT-001

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. D60 STUDIO TUNGSTEN

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79: 2008 Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products

ANSI ANSLG C38.377: 2012 Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D60 STUDIO TUNGSTEN. The sample was received by Intertek on May 6, 2013, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-001.

DATES OF TESTS: May 15, 2013 through May 22, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | D60 STUDIO TUNGSTEN |
| Description: | LED WASH LIGHT |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 7390 | 6971 |
| Total Power (W) | 137.5 | 136.8 |
| Luminaire Efficacy (LPW) | 53.74 | 50.96 |

| Criteria | Result |
|---|---|
| Power Factor | 0.991 |
| Current ATHD (%) | 2.76 |
| Correlated Color Temperature (CCT - K) | 3049 |
| Color Rendering Index (CRI) – Ra | 85.4 |
| Color Rendering Index (CRI) - R9 | 22.6 |
| Duv | 0.001 |
| Chromaticity Coordinate (x) | 0.435 |
| Chromaticity Coordinate (y) | 0.407 |
| Chromaticity Coordinate (u') | 0.248 |
| Chromaticity Coordinate (v') | 0.522 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 05/20/13 | 06/20/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | --- | N1132 | 04/22/13 | 04/22/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Yokogawa Power Analyzer | WT1600 | E462 | 07/06/12 | 07/06/13 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | NA | NA | NA |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Sorensen DC Power Supply | DLM150-20E | N/A | N/A | N/A |
| FLUKE | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

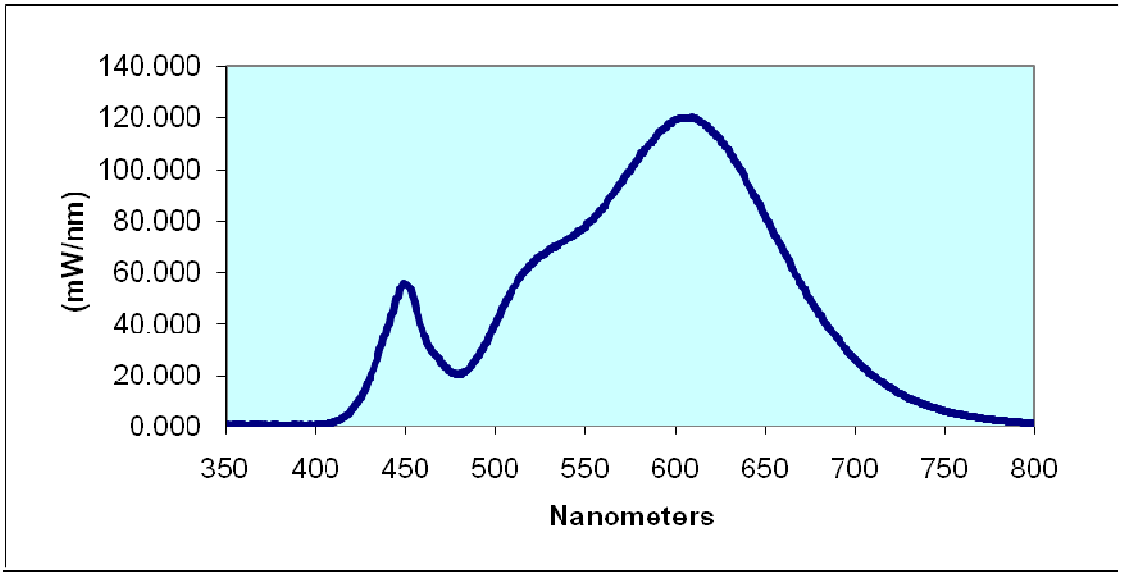
The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.845 | 460 | 35.938 | 570 | 94.720 | 680 | 43.791 |
| 355 | 0.765 | 465 | 29.072 | 575 | 99.549 | 685 | 38.922 |
| 360 | 0.772 | 470 | 24.958 | 580 | 104.844 | 690 | 34.191 |
| 365 | 0.879 | 475 | 21.485 | 585 | 109.463 | 695 | 29.908 |
| 370 | 0.824 | 480 | 20.693 | 590 | 113.118 | 700 | 26.292 |
| 375 | 0.743 | 485 | 22.540 | 595 | 116.234 | 705 | 22.804 |
| 380 | 0.682 | 490 | 26.516 | 600 | 119.102 | 710 | 19.847 |
| 385 | 0.713 | 495 | 32.605 | 605 | 119.687 | 715 | 17.227 |
| 390 | 0.720 | 500 | 40.210 | 610 | 119.666 | 720 | 14.878 |
| 395 | 0.740 | 505 | 47.357 | 615 | 118.149 | 725 | 12.793 |
| 400 | 0.808 | 510 | 54.179 | 620 | 115.532 | 730 | 10.928 |
| 405 | 1.058 | 515 | 59.046 | 625 | 111.870 | 735 | 9.505 |
| 410 | 1.758 | 520 | 63.083 | 630 | 106.881 | 740 | 8.145 |
| 415 | 3.364 | 525 | 66.188 | 635 | 101.029 | 745 | 7.069 |
| 420 | 6.298 | 530 | 68.678 | 640 | 94.931 | 750 | 6.024 |
| 425 | 11.294 | 535 | 70.320 | 645 | 88.539 | 755 | 5.311 |
| 430 | 18.919 | 540 | 72.536 | 650 | 81.936 | 760 | 4.518 |
| 435 | 28.264 | 545 | 74.927 | 655 | 74.853 | 765 | 0.000 |
| 440 | 38.187 | 550 | 77.691 | 660 | 68.310 | 770 | 3.366 |
| 445 | 49.441 | 555 | 81.338 | 665 | 61.720 | 775 | 2.890 |
| 450 | 55.250 | 560 | 85.490 | 670 | 55.594 | 780 | 2.493 |
| 455 | 47.782 | 565 | 89.872 | 675 | 49.406 | | |

**Sample No. CRT1304251519-001**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25°C – Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-001 | UP | 120.0 | 1157 | 137.5 | 0.991 | 2.76 | 7390 | 53.74 |

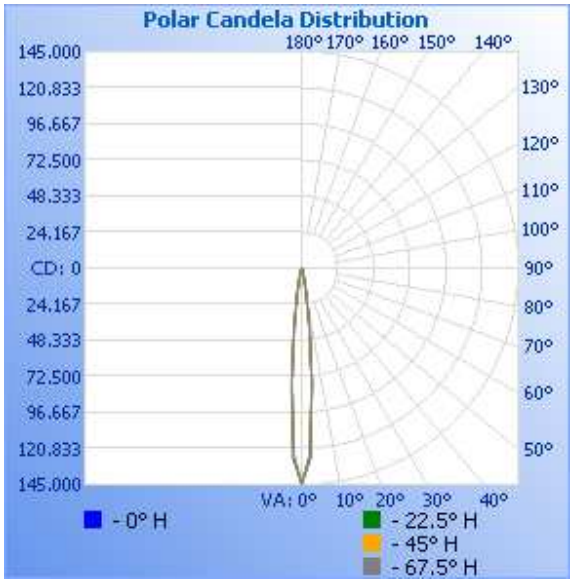
| Intertek Sample No. | Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|---|
| CRT1304251519-001 | 3049 | 85.4 | 22.6 | 0.001 | 0.435 | 0.407 | 0.248 | 0.522 |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1304251519-001 | UP | 119.9 | 1151 | 136.8 | 0.991 | 6971 | 50.96 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 144469 | 144469 | 144469 | 144469 | 144469 |
| 5 | 77131 | 76356 | 77037 | 78701 | 80853 |
| 10 | 17519 | 17648 | 18046 | 18290 | 18453 |
| 15 | 3806 | 3777 | 3677 | 3795 | 3961 |
| 20 | 1186 | 1161 | 1165 | 1198 | 1193 |
| 25 | 294 | 285 | 298 | 298 | 286 |
| 30 | 23 | 19 | 12 | 24 | 26 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

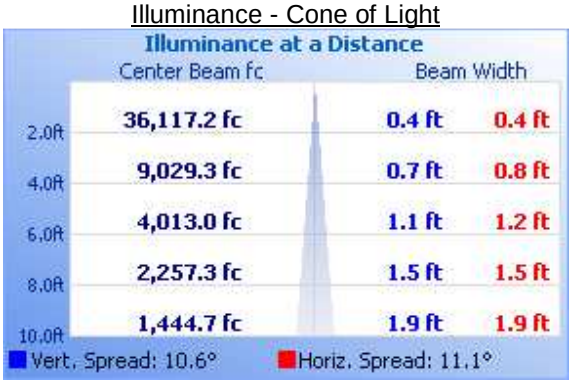


RESULTS OF TESTS (cont'd)

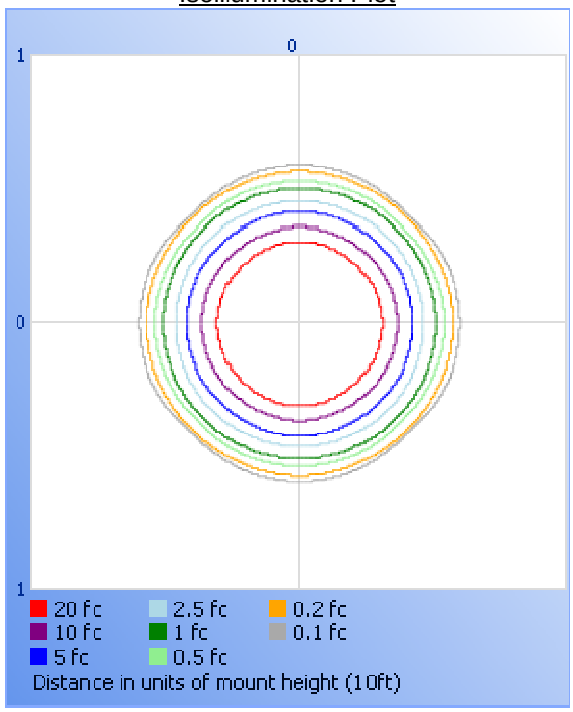
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 6969 | 100.0 |
| 0-40 | 6971 | 100.0 |
| 0-60 | 6971 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 6971 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 6971 | 100.0 |

Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

David Ellis
Senior Project Engineer
Lighting Division

Attachment: None

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: May 30, 2013

REPORT NO. 101158242CRT-006

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. D60 STUDIO TUNGSTEN
LED MODEL NO. LUXEON REBEL LXML-PWN2

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and LED In-Situ tests.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

Energy Star Version 1.2 (2012): Program Requirements for Luminaires
Energy Star Manufacturer's Guide Version 2.0 (2009): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one sample of model number D60 STUDIO TUNGSTEN. The sample was received by Intertek on May 6, in undamaged condition, and one sample was tested as received. The sample designation was CRT1304251519-001.

DATES OF TESTS: May 24, 2013

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D60 STUDIO TUNGSTEN |
| Description: | LED WASH LIGHT |

| Criteria | Result |
|---|---|
| Maximum In-Situ Source Temperature Point – LXML-PM01 GREEN (°C) | 67.0 |
| Maximum In-Situ Source Temperature Point – LXML-PE01 CYAN (°C) | 71.3 |
| Maximum In-Situ Source Temperature Point – LXML-PB01 BLUE (°C) | 62.5 |
| Maximum In-Situ Source Temperature Point – LXML-PR01 ROYAL BLUE (°C) | 61.3 |
| Maximum In-Situ Source Temperature Point – LXM2-PL01 PC AMBER (°C) | 58.6 |
| Maximum In-Situ Source Temperature Point – LXM2-PD01 RED LOW VF (°C) | 65.3 |
| Maximum In-Situ Source Temperature Point – LXML-PWN1 WHITE (°C) | 60.5 |
| Maximum Power Supply Case Temperature | 49.1 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Fluke Multimeter | PM2525 | M127 | 10/17/12 | 10/17/13 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

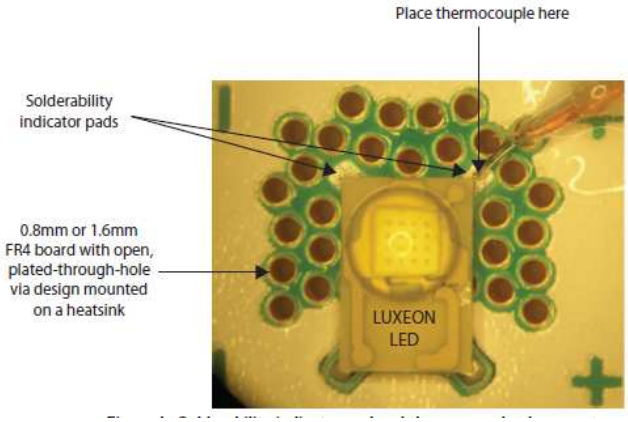
In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Agilent 34970A Data Logger. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TESTS

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: LUXEON REBEL LXML-PWN2




## Electrical Characteristics at 350 mA for LUXEON Rebel color, Thermal Pad Temperature = 25°C

**Table 3.**

| Color | Part Number | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|---|---|
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Royal-Blue | LXML-PR01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red-Orange | LXM2-PH01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

## Absolute Maximum Ratings

**Table 5.**

| Parameter | Green/Cyan/ Blue/Royal Blue | LUXEON Rebel ES Royal Blue/ES Blue | Red/Deep-Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |

RESULTS OF TESTS (cont'd)

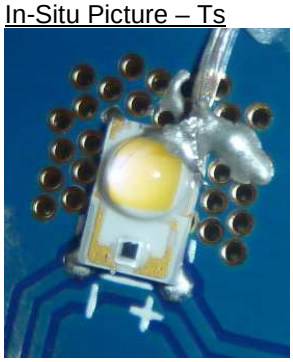
In-Situ Maximum Measured LED Source Temperature - LXML-PM01 GREEN

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5 V
Measured LED Current = 590 mA
Calculated LED Wattage = Vf x Measured LED Current = 2.071W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 129°C

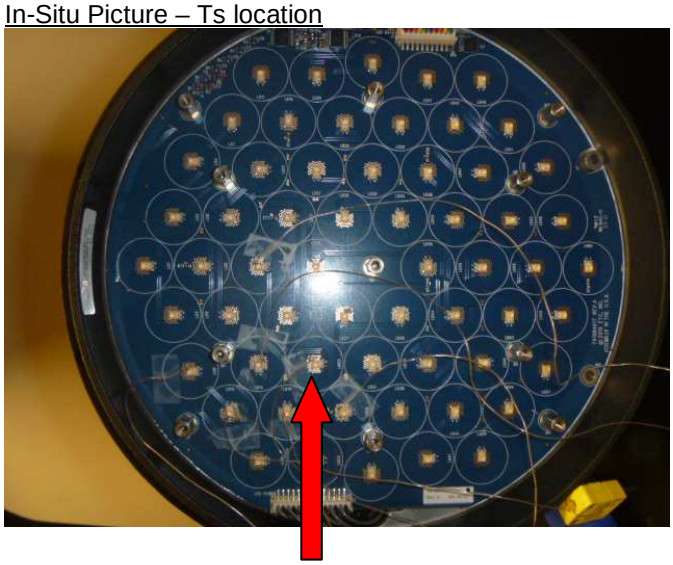
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-001 | 67.0 | Per diagram above | 129 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

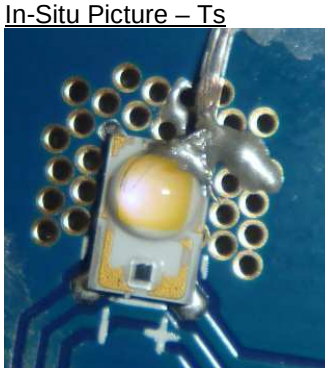
In-Situ Maximum Measured LED Source Temperature - LXML-PE01 CYAN

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5 V
Measured LED Current = 595 mA
Calculated LED Wattage = Vf x Measured LED Current = 2.089W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 129°C

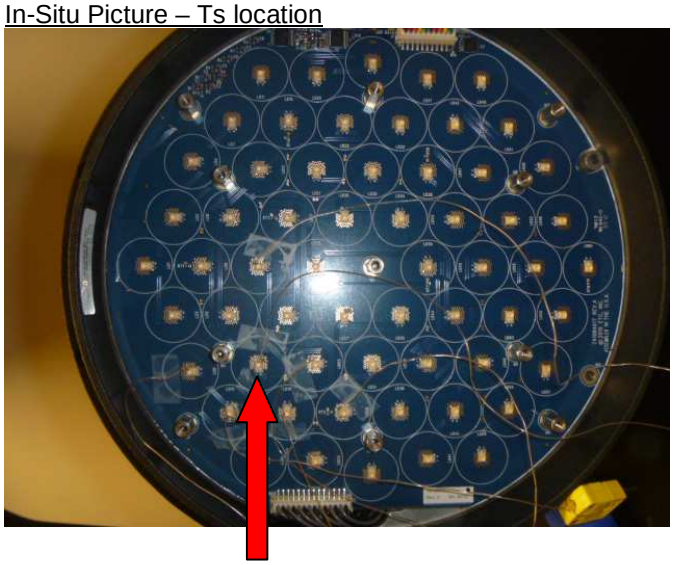
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-001 | 71.3 | Per diagram above | 129 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

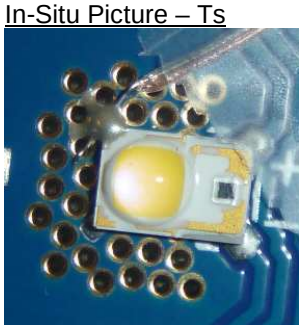
In-Situ Maximum Measured LED Source Temperature - LXML-PB01 BLUE

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 590 mA
Calculated LED Wattage = Vf x Measured LED Current = 2.071W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 129°C

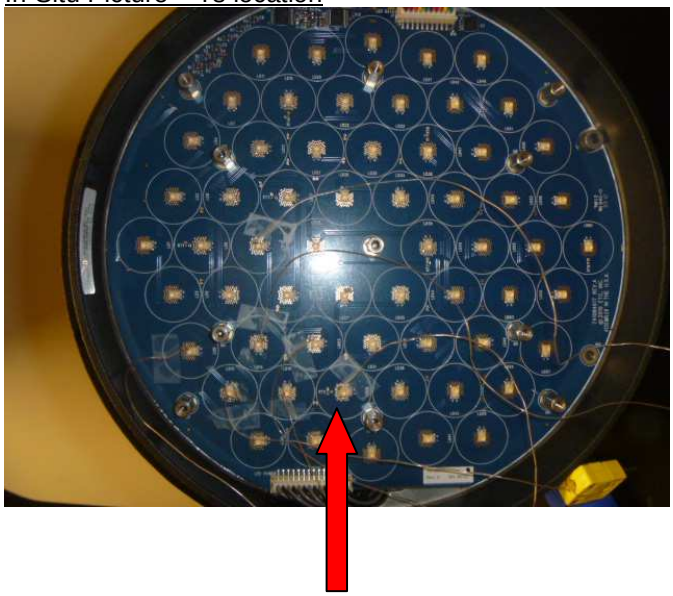
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-001 | 62.5 | Per diagram above | 129 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

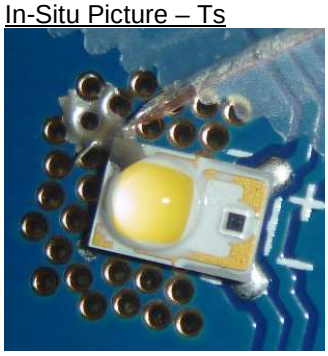
In-Situ Maximum Measured LED Source Temperature - LXML-PR01 ROYAL BLUE

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 590 mA
Calculated LED Wattage = Vf x Measured LED Current = 2.071W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 129°C

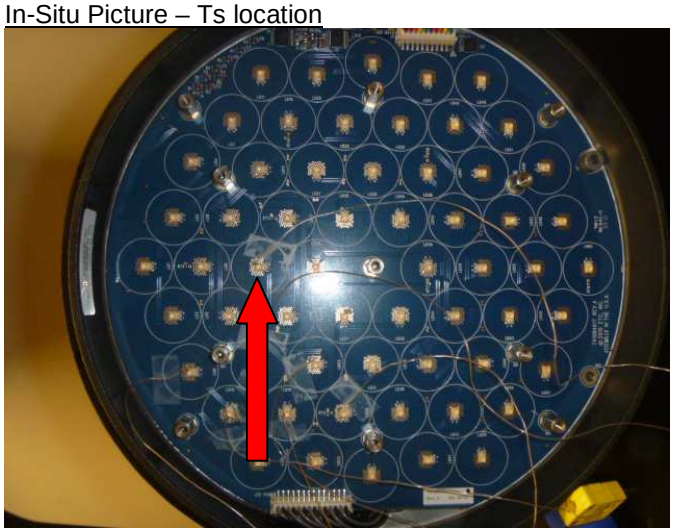
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-001 | 61.3 | Per diagram above | 129 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

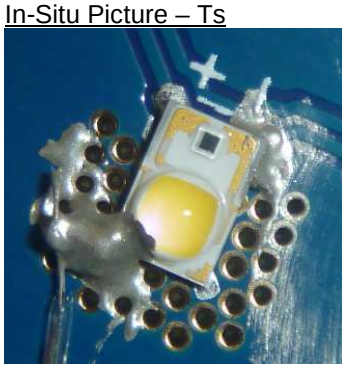
In-Situ Maximum Measured LED Source Temperature - LXM2-PL01 PC AMBER

Maximum Junction Temperature from LED specification (Tj) = 130°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 585 mA
Calculated LED Wattage = Vf x Measured LED Current = 2.053W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 109°C

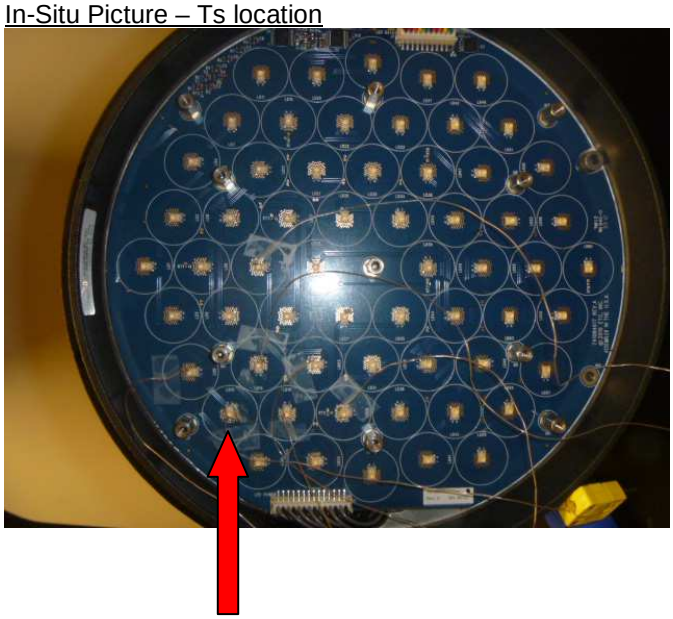
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-001 | 58.6 | Per diagram above | 109 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

In-Situ Maximum Measured LED Source Temperature - LXM2-PD01 RED  LOW VF

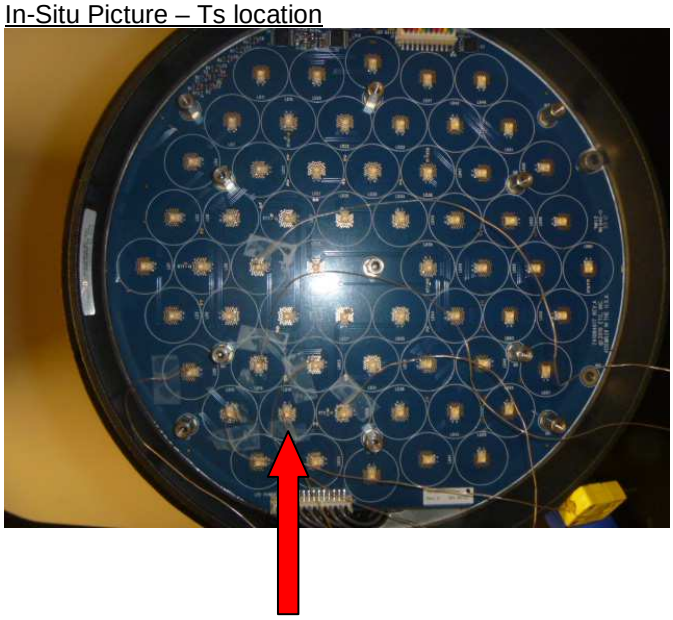
Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 2.8V
Measured LED Current = 588 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.646W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 122°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-001 | 65.3 | Per diagram above | 122 |

In-Situ Picture – Ts

In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

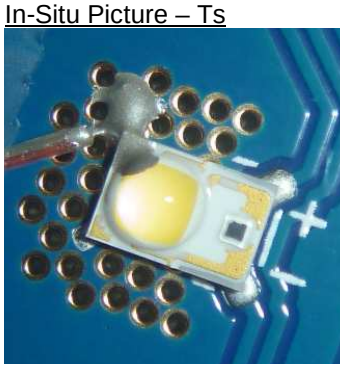
In-Situ Maximum Measured LED Source Temperature - LXML-PWN1 WHITE

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.99V
Measured LED Current = 589 mA
Calculated LED Wattage = Vf x Measured LED Current = 2.350W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 126°C

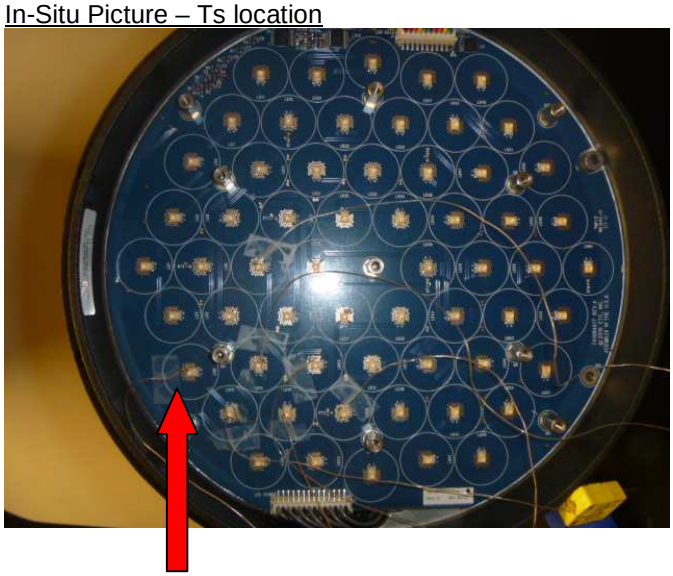
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1304251519-001 | 60.5 | Per diagram above | 126 |

In-Situ Picture – Ts



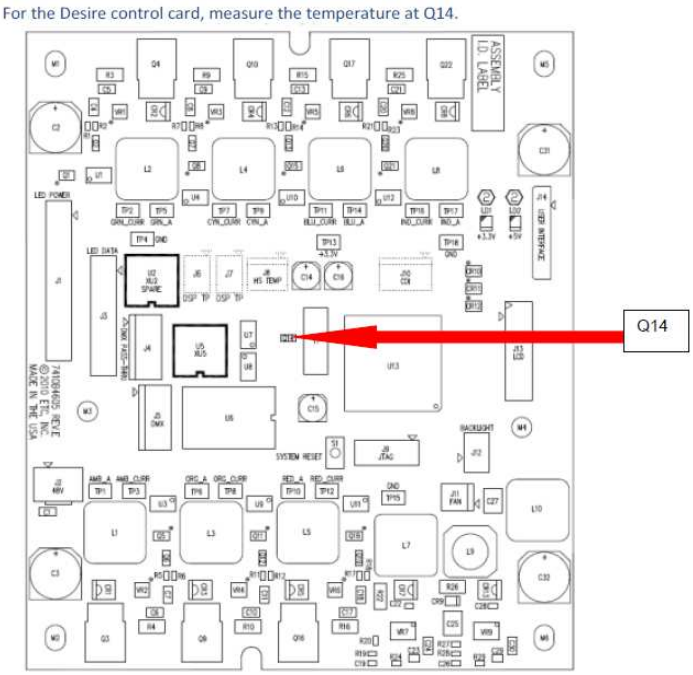
In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

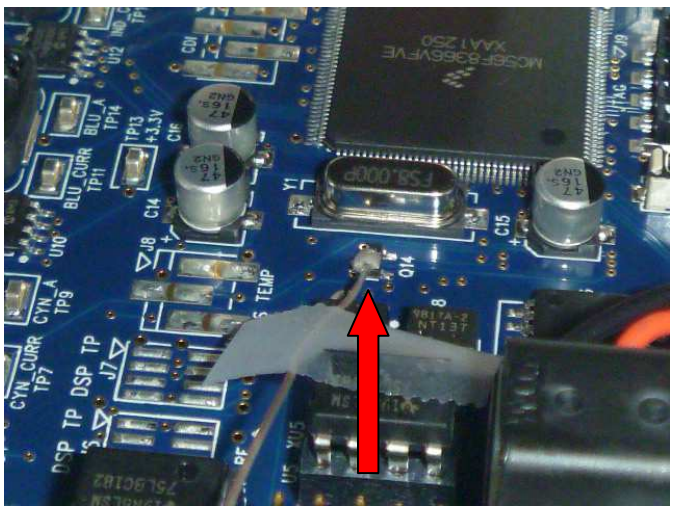
Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:
Driver identified as: Model No. 7410B4605




Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Measured Power Supply Case Temperature (°C) | Location | Maximum Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1304251519-001 | 49.1 | Q14 | 100 |



Picture (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

David Ellis
Senior Project Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: November 20, 2013

REPORT NO. 101414036CRT-009

TEST OF ONE D60 DAYLIGHT LED FIXTURE ETL (BLK)

MODEL NO. D60 DAYLIGHT
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number D60 DAYLIGHT. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-009.

DATES OF TESTS: November 13, 2013 through November 19, 2

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | D60 DAYLIGHT |
| Description: | D60 DAYLIGHT LED FIXTURE ETL (BLK) |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 8943 | 8620 |
| Total Power (W) | 133.1 | 132.6 |
| Luminaire Efficacy (LPW) | 67.19 | 65.01 |

| Criteria | Result |
|---|---|
| Power Factor | 0.991 |
| Current ATHD % | 2.98 |
| Correlated Color Temperature (CCT - K) | 5539 |
| Color Rendering Index (CRI - Ra) | 70.7 |
| Color Rendering Index (CRI - R9) | -3.7 |
| DUV | 0.003 |
| Chromaticity Coordinate (x) | 0.332 |
| Chromaticity Coordinate (y) | 0.347 |
| Chromaticity Coordinate (u') | 0.204 |
| Chromaticity Coordinate (v') | 0.480 |
| Maximum In-Situ Source Temperature Point (°C) | 60.1 |
| Maximum In-Situ Driver Case Temperature (°C) | 47.0 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| COLE PARMER-Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Sorensen DC Power Supply | DLM150-20E | --- | --- | --- |
| Data Precision | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 10/21/13 | 11/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1355 | 11/26/12 | 11/26/13 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

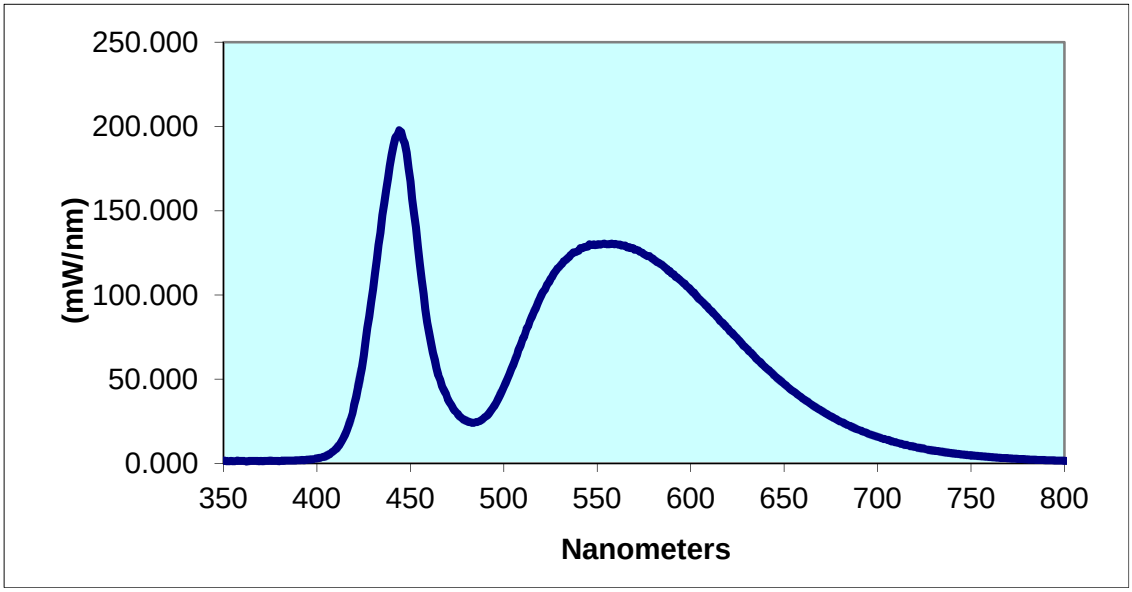
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-009 | UP | 120.0 | 117.4 | 133.1 | 0.991 | 2.98 | 8943 | 67.19 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 5539 | 70.7 | -3.7 | 0.003 | 0.332 | 0.347 | 0.204 | 0.480 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 1.615 | 440 | 183 | 530 | 117.2 | 620 | 79.99 | 710 | 12.33 |
| 355 | 1.171 | 445 | 197 | 535 | 122.4 | 625 | 74.03 | 715 | 10.94 |
| 360 | 1.352 | 450 | 167.4 | 540 | 126 | 630 | 68.15 | 720 | 9.672 |
| 365 | 1.394 | 455 | 118.5 | 545 | 128.8 | 635 | 62.27 | 725 | 8.518 |
| 370 | 1.463 | 460 | 77.57 | 550 | 129.9 | 640 | 57.06 | 730 | 7.53 |
| 375 | 1.498 | 465 | 52.14 | 555 | 130.1 | 645 | 51.91 | 735 | 6.7 |
| 380 | 1.207 | 470 | 38.18 | 560 | 130.3 | 650 | 47.02 | 740 | 5.921 |
| 385 | 1.546 | 475 | 29.7 | 565 | 129.3 | 655 | 42.6 | 745 | 5.256 |
| 390 | 1.774 | 480 | 25.18 | 570 | 126.9 | 660 | 38.42 | 750 | 4.658 |
| 395 | 2.112 | 485 | 24.36 | 575 | 124.1 | 665 | 34.65 | 755 | 4.165 |
| 400 | 2.822 | 490 | 27.46 | 580 | 121.3 | 670 | 31.13 | 760 | 3.721 |
| 405 | 4.39 | 495 | 34.1 | 585 | 117.5 | 675 | 27.78 | 765 | 3.223 |
| 410 | 8.246 | 500 | 45.49 | 590 | 112.5 | 680 | 24.8 | 770 | 2.926 |
| 415 | 16.85 | 505 | 58.41 | 595 | 108.1 | 685 | 22.26 | 775 | 2.597 |
| 420 | 34.8 | 510 | 73.45 | 600 | 103 | 690 | 19.81 | 780 | 2.332 |
| 425 | 63.83 | 515 | 86.92 | 605 | 97.18 | 695 | 17.57 | | |
| 430 | 103.2 | 520 | 99.4 | 610 | 91.43 | 700 | 15.7 | | |
| 435 | 147.4 | 525 | 109.6 | 615 | 85.7 | 705 | 13.91 | | |

**Spectral Data Over Visible Wavelengths**



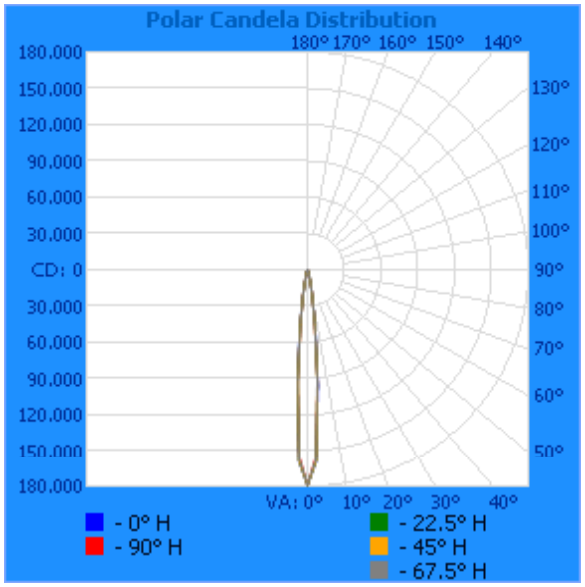
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-009 | UP | 120.0 | 1115 | 132.6 | 0.991 | 8620 | 65.01 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 179133 | 179133 | 179133 | 179133 | 179133 |
| 5 | 95790 | 94049 | 92779 | 91308 | 93244 |
| 10 | 22610 | 22053 | 21704 | 21630 | 21783 |
| 15 | 5441 | 5621 | 5790 | 5417 | 5304 |
| 20 | 1567 | 1592 | 1630 | 1568 | 1524 |
| 25 | 299 | 288 | 280 | 269 | 291 |
| 30 | 51 | 46 | 49 | 45 | 41 |
| 35 | 1 | 9 | 7 | 4 | 23 |
| 40 | 0 | 0 | 8 | 11 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

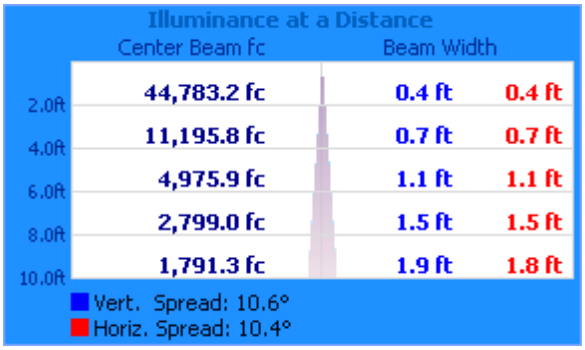


RESULTS OF TEST (cont'd)

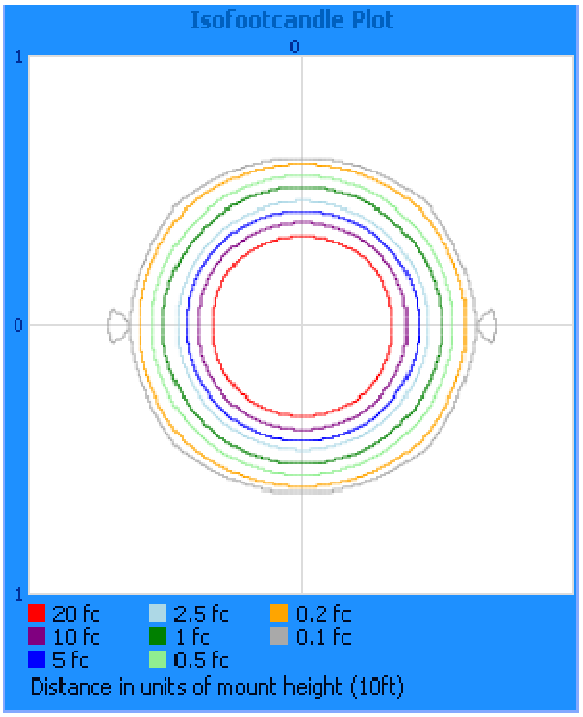
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 8612 | 99.9 |
| 0-40 | 8619 | 100.0 |
| 0-60 | 8620 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 8620 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 8620 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 6493 | 75.3 |
| 10-20 | 1908 | 22.1 |
| 20-30 | 210.7 | 2.4 |
| 30-40 | 7.5 | 0.1 |
| 40-50 | 1.2 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as: Luxeon Rebel

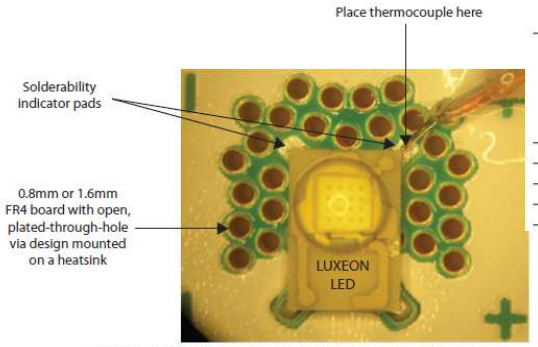


Figure 1. Solderability indicator pad and thermocouple placement.

| Nominal ANSI CCT | Forward Voltage $V_f$ [1] (V) Min. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_F / \Delta T_J$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|
| 2700K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 3000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 4000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 5000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |

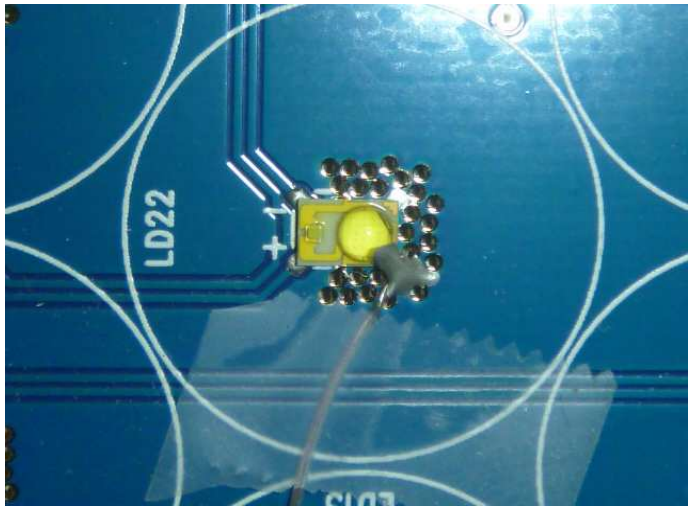
| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 589mA
Calculated LED Wattage = Vf x Measured LED Current = 2.067W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 137.6°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-009 | 60.1 | Per diagram | 137.6 |

In-Situ Picture – Ts

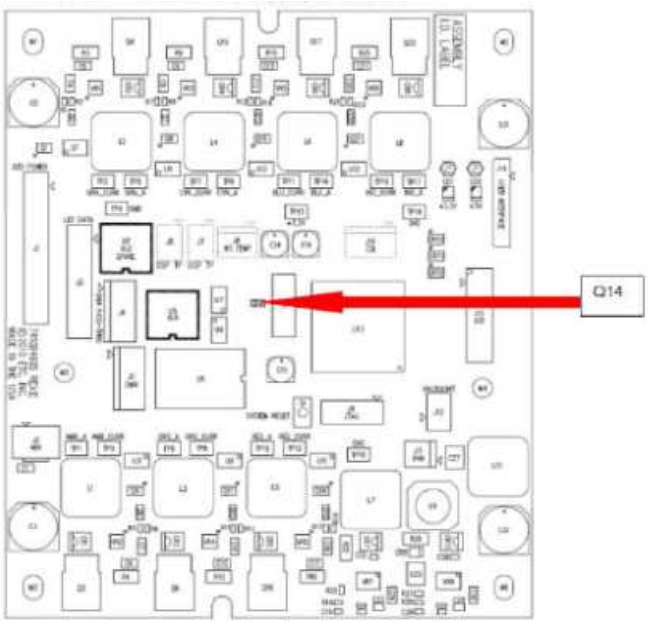



In-Situ Picture – Ts location

RESULTS OF TEST (cont'd)

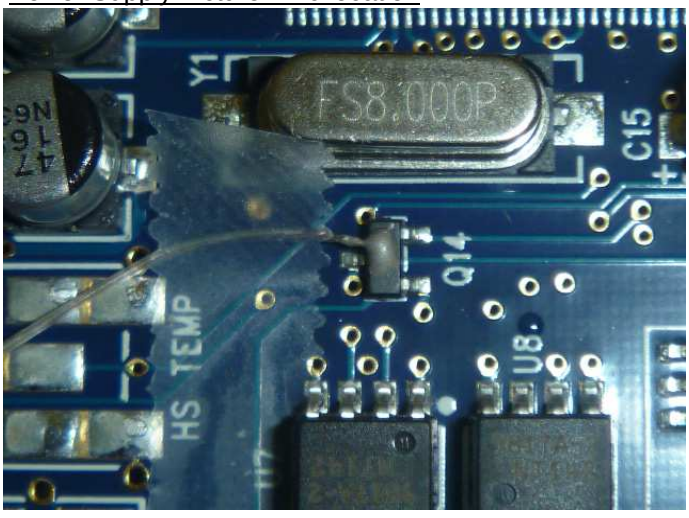
Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-009 | 47.0 | Per diagram | 100 |

Power Supply Picture – Tc location



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

# SECTION 6 - Selador Classic

Lustr 11”
Luster 21”
Vivid-R 11”
Paletta 11”
Pearl 11”

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: December 6, 2013

REPORT NO. 101414036CRT-010

TEST OF ONE LUSTR SL3 LED FIXTURE 11

MODEL NO. LUSTR SL3 11
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number LUSTR SL3 11. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-010.

DATES OF TESTS: December 3, 2013 through December 5, 20:

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | LUSTR SL3 11 |
| Description: | LUSTR SL3 LED FIXTURE 11 |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 2416 | 2643 |
| Total Power (W) | 95.97 | 96.00 |
| Luminaire Efficacy (LPW) | 25.17 | 27.53 |

| Criteria | Result |
|---|---|
| Power Factor | 0.970 |
| Current ATHD % | 19.05 |
| Correlated Color Temperature (CCT - K) | 7370 |
| Color Rendering Index (CRI - Ra) | 47.1 |
| Color Rendering Index (CRI - R9) | -107.3 |
| DUV | 0.025 |
| Chromaticity Coordinate (x) | 0.310 |
| Chromaticity Coordinate (y) | 0.278 |
| Chromaticity Coordinate (u') | 0.217 |
| Chromaticity Coordinate (v') | 0.438 |
| Maximum In-Situ Source Temperature Point (°C) | 72.9 |
| Maximum In-Situ Driver Case Temperature (°C) | 56.5 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/27/13 | 11/27/14 |
| Sorensen DC Power Supply | DLM150-20E | --- | --- | --- |
| Data Precision | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole-Palmer Hygro Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1360 | 11/27/13 | 11/27/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

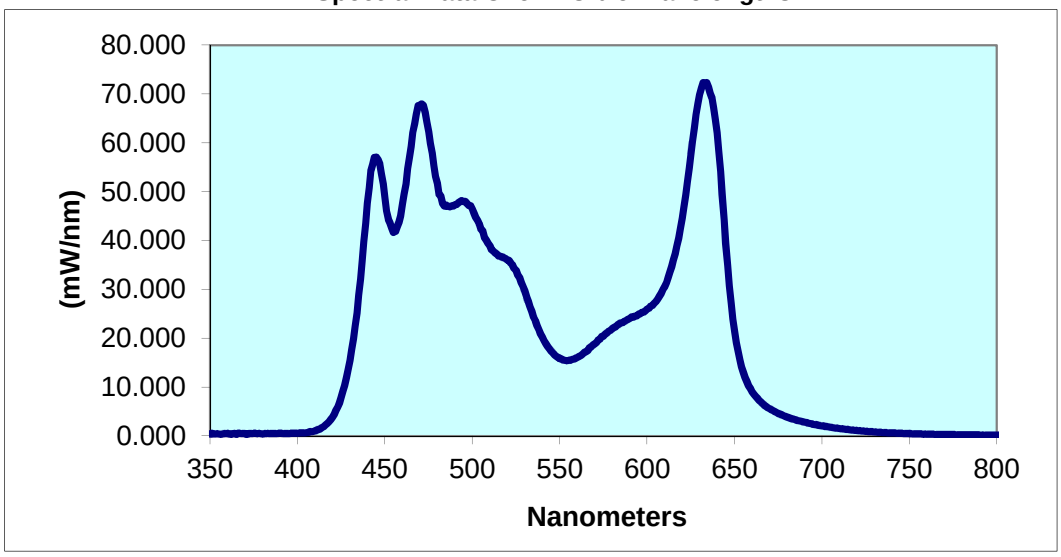
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-010 | UP | 120.0 | 823.6 | 95.97 | 0.970 | 19.05 | 2416 | 25.17 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 7370 | 47.1 | -107.3 | 0.025 | 0.310 | 0.278 | 0.217 | 0.438 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.422 | 440 | 47.65 | 530 | 29.68 | 620 | 44.54 | 710 | 1.505 |
| 355 | 0.391 | 445 | 57.1 | 535 | 24.39 | 625 | 57.64 | 715 | 1.311 |
| 360 | 0.511 | 450 | 48.77 | 540 | 20.2 | 630 | 69.87 | 720 | 1.111 |
| 365 | 0.364 | 455 | 41.66 | 545 | 17.37 | 635 | 71.36 | 725 | 0.938 |
| 370 | 0.452 | 460 | 47.56 | 550 | 15.86 | 640 | 62.04 | 730 | 0.804 |
| 375 | 0.521 | 465 | 59.25 | 555 | 15.46 | 645 | 39.46 | 735 | 0.685 |
| 380 | 0.374 | 470 | 67.7 | 560 | 16.05 | 650 | 21.22 | 740 | 0.59 |
| 385 | 0.441 | 475 | 62.27 | 565 | 17.26 | 655 | 12.99 | 745 | 0.53 |
| 390 | 0.512 | 480 | 51.72 | 570 | 18.91 | 660 | 9.032 | 750 | 0.466 |
| 395 | 0.455 | 485 | 47.26 | 575 | 20.46 | 665 | 6.937 | 755 | 0.407 |
| 400 | 0.527 | 490 | 47.31 | 580 | 21.96 | 670 | 5.599 | 760 | 0.358 |
| 405 | 0.686 | 495 | 47.98 | 585 | 23.12 | 675 | 4.614 | 765 | 0.304 |
| 410 | 1.052 | 500 | 46.47 | 590 | 24.05 | 680 | 3.862 | 770 | 0.281 |
| 415 | 1.937 | 505 | 42.31 | 595 | 24.91 | 685 | 3.299 | 775 | 0.241 |
| 420 | 3.828 | 510 | 38.91 | 600 | 25.96 | 690 | 2.825 | 780 | 0.219 |
| 425 | 7.887 | 515 | 36.86 | 605 | 27.56 | 695 | 2.4 | | |
| 430 | 15.46 | 520 | 36.08 | 610 | 30.57 | 700 | 2.065 | | |
| 435 | 28.83 | 525 | 33.9 | 615 | 35.97 | 705 | 1.765 | | |

**Spectral Data Over Visible Wavelengths**



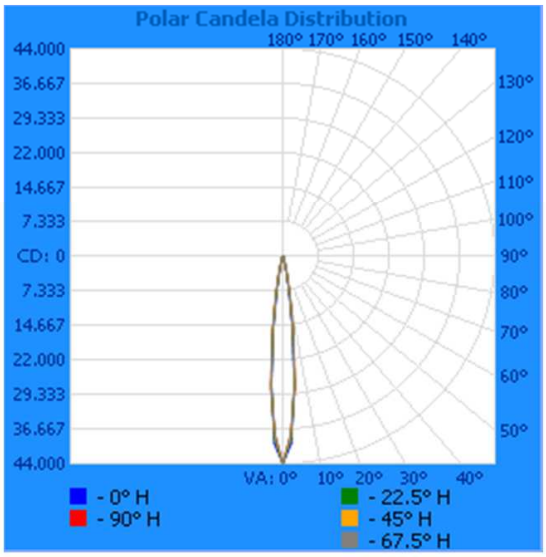
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-010 | UP | 120.0 | 822.5 | 96.00 | 0.972 | 2643 | 27.53 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 43977 | 43977 | 43977 | 43977 | 43977 |
| 5 | 27192 | 27476 | 27765 | 28078 | 28240 |
| 10 | 7202 | 7691 | 8365 | 8451 | 8309 |
| 15 | 1322 | 1488 | 1813 | 1610 | 1467 |
| 20 | 332 | 359 | 416 | 384 | 357 |
| 25 | 114 | 122 | 133 | 136 | 134 |
| 30 | 46 | 45 | 48 | 47 | 44 |
| 35 | 24 | 21 | 21 | 20 | 18 |
| 40 | 13 | 10 | 8 | 5 | 6 |
| 45 | 6 | 3 | 0 | 0 | 0 |
| 50 | 2 | 1 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

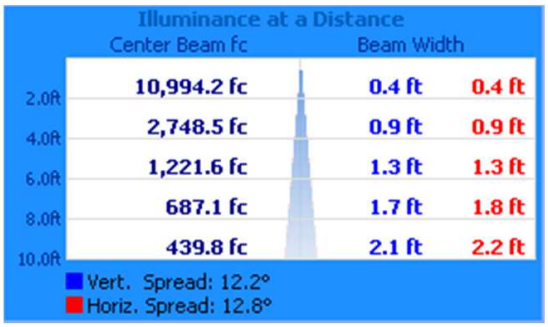


RESULTS OF TEST (cont'd)

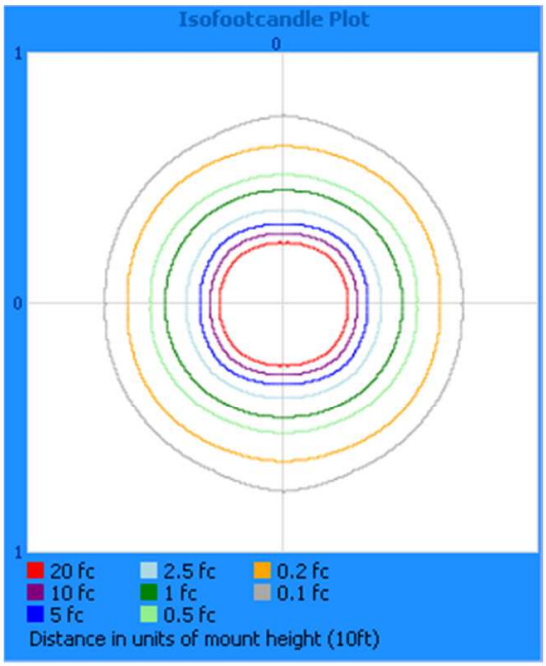
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 2627 | 99.4 |
| 0-40 | 2641 | 99.9 |
| 0-60 | 2643 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 2643 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 2643 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 1951 | 73.8 |
| 10-20 | 607.6 | 23.0 |
| 20-30 | 68.3 | 2.6 |
| 30-40 | 14.1 | 0.5 |
| 40-50 | 2.0 | 0.1 |
| 50-60 | 0.1 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as: Luxeon Rebel ES (Amber)

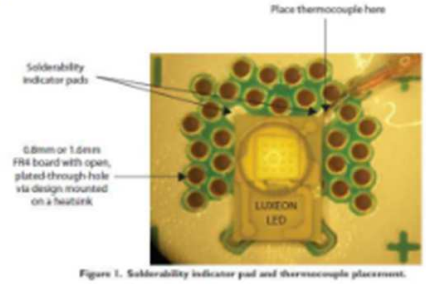



Figure 1. Solderability indicator pad and thermocouple placement.

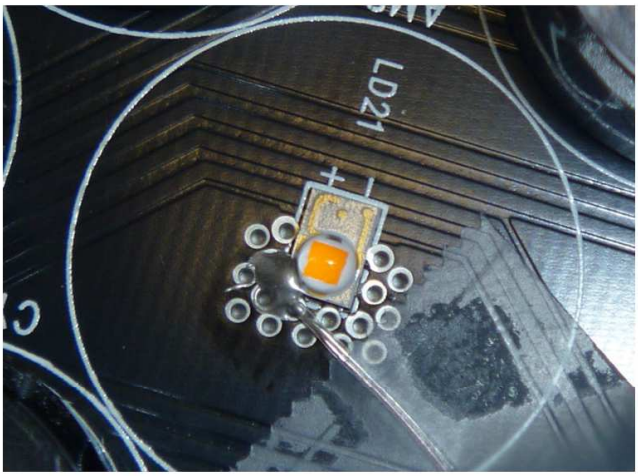
| Color | Part Number | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|---|---|
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

Maximum LED Junction Temperature 135°C
Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 12°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 585mA
Calculated LED Wattage = Vf x Measured LED Current = 2.053W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 110.4°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-010 | 72.9 | Per diagram | 110.4 |

In-Situ Picture – Ts

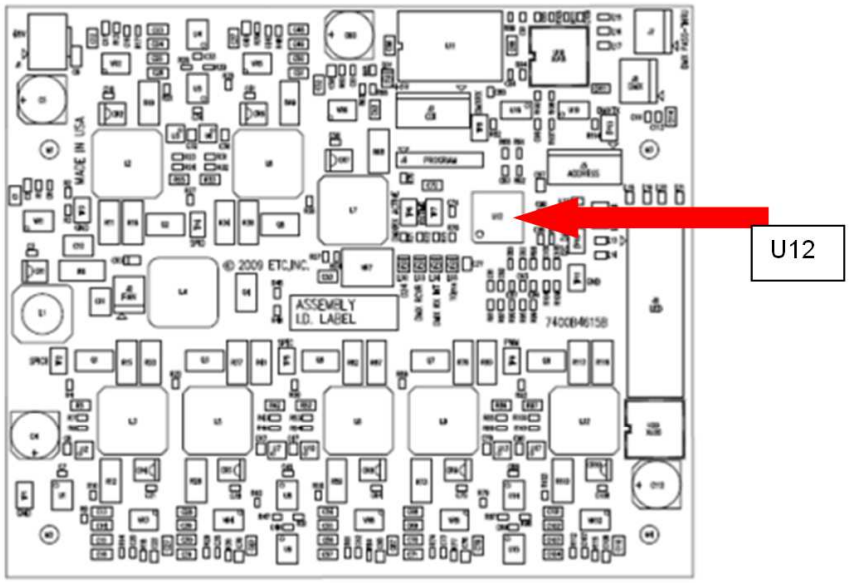


In-Situ Picture – Ts location

RESULTS OF TEST (cont'd)

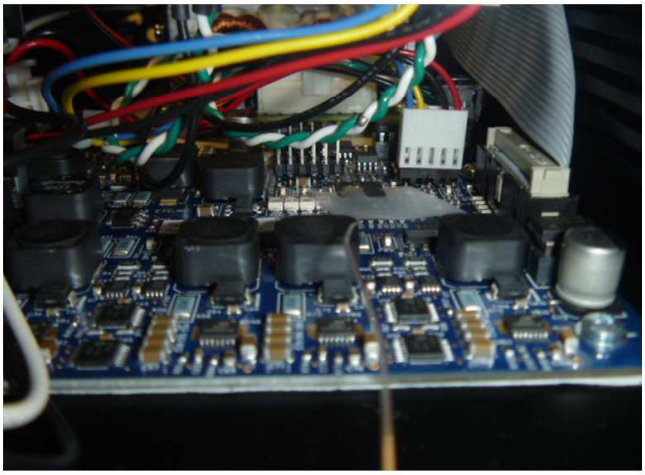
Maximum Measured Power Supply Case Temperature

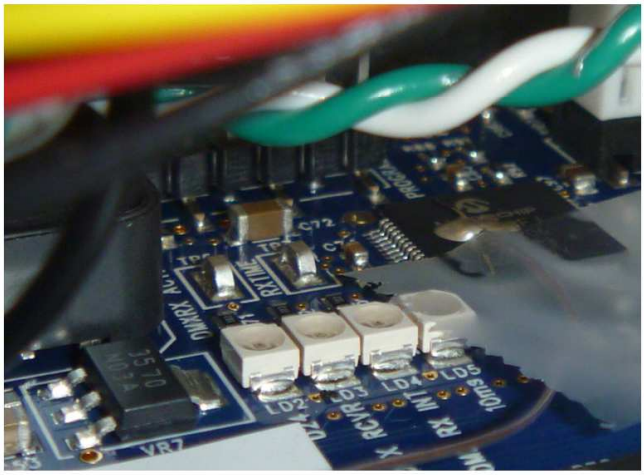
Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-010 | 52.9 | Per diagram | 100 |

Power Supply Picture – Tc location





PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

# REPORT

**545 E. ALGONQUIN ROAD ARLINGTON HEIGHTS, IL 60005**

Project No. G101042657 Date: March 13, 2013

REPORT NO. 101042657CHI-001

IN-SITU TEST OF ONE LED STAGE LIGHT

FIXTURE MODEL NO. LUSTR 21
LED MODEL NO. PHILIPS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

AUTHORIZATION: The testing performed was authorized by signed quote number 500428362.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

Energy Star Version 1.1 (2008): Program Requirements for Solid-State Lighting Luminaires
Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one sample of model number Lustr 21 The sample was received by Intertek on January 16, 2013, in undamaged condition, and one sample was tested as received. The sample designation was 01162013124204.

DATE OF TESTS: February 28, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | Lustr 21 |
| Description: | LED Stage and theatre luminaire |

| Criteria | Result | |
|---|---|---|
| LEDR REBEL RED 1812 LXML-PD01-0040 | 80.3 | Compliant |
| LEDO REBEL ORG 1812 LXML-PH01-0050 | 80.4 | Compliant |
| LEDG REBEL GRN 1812 LXML-PM01-0090 | 79.3 | Compliant |
| LEDC REBEL CYAN 1812 LXML-PE01-0070 | 77.4 | Compliant |
| LEDB REBEL BLUE 1812 LXML-PB01-0030 | 77.6 | Compliant |
| LEDI REBEL INDIGO 1812 LXML-PR01-0350 | 79.0 | Compliant |
| LEDA REBEL PC-AMBER 1812 LXM2-PL01-0000 | 77.0 | Compliant |
| Maximum Power Supply Case Temperature | 63.5 | Compliant |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Agilent Data Logger | 34970A | 146763 | 4/23/12 | 4/23/13 |
| Weather Station | DBAR880 | 146558 | 7/23/12 | 7/23/13 |
| Yokogawa Power Meter | WT210 561 | 146761 | 4/19/12 | 4/19/13 |
| Fluke IR Thermometer | HVACPRO | 146578 | VBU | VBU |
| Staco Variac | 3PN1510B | 146345 | VBU | VBU |

TEST METHODS

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Agilent 34970A Data Logger. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

Estimated Total Operating Time

| Model No. | Total Hours |
|---|---|
| Lustr 21 | 24 |

RESULTS OF TESTS

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: Philips Luxeon Rebel

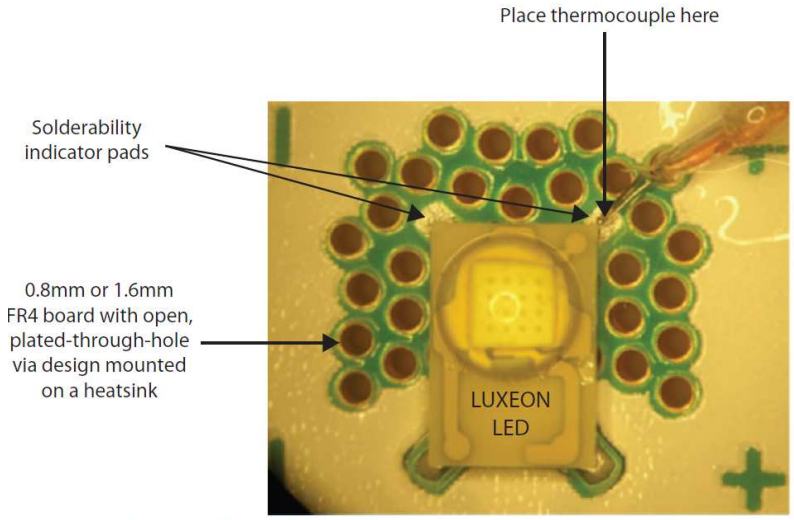



**Figure 1. Solderability indicator pad and thermocouple placement.**

| | | |
|---|---|---|
| CR9177-F | LEDR REBEL RED 1812 | LXML-PD01-0040 (C-BIN 4) |
| CR9177-F | LEDR REBEL RED 1812 | LXML-PD01-0040 (C-BIN 5) |
| CR9178-F | LEDO REBEL ORG 1812 | LXML-PH01-0050 (C-BIN 2) |
| CR9180-F | LEDG REBEL GRN 1812 | LXML-PM01-0090 (C-BIN 4) |
| CR9180-F | LEDG REBEL GRN 1812 | LXML-PM01-0090 (C-BIN 5) |
| CR9180-F | LEDG REBEL GRN 1812 | LXML-PM01-0090 (C-BIN 6) |
| CR9180-F | LEDG REBEL GRN 1812 | LXML-PM01-0090 (C-BIN 3) |
| CR9181-F | LEDC REBEL CYAN 1812 | LXML-PE01-0070 (C-BIN 1) |
| CR9181-F | LEDC REBEL CYAN 1812 | LXML-PE01-0070 (C-BIN 2) |
| CR9181-F | LEDC REBEL CYAN 1812 | LXML-PE01-0070 (C-BIN 3) |
| CR9182-F | LEDB REBEL BLUE 1812 | LXML-PB01-0030 (C-BIN 3) |
| CR9182-F | LEDB REBEL BLUE 1812 | LXML-PB01-0030 (C-BIN 4) |
| CR9182-F | LEDB REBEL BLUE 1812 | LXML-PB01-0030 (C-BIN 5) |
| CR9183-F | LEDI REBEL INDIGO 1812 | LXML-PR01-0350 (C-BIN 1) |
| CR9183-F | LEDI REBEL INDIGO 1812 | LXML-PR01-0350 (C-BIN 2) |
| CR9183-F | LEDI REBEL INDIGO 1812 | LXML-PR01-0350 (C-BIN 3) |
| CR9183-F | LEDI REBEL INDIGO 1812 | LXML-PR01-0350 (C-BIN 4) |
| CR9183-F | LEDI REBEL INDIGO 1812 | LXML-PR01-0425 (C-BIN 1) |
| CR9183-F | LEDI REBEL INDIGO 1812 | LXML-PR01-0425 (C-BIN 2) |
| CR9183-F | LEDI REBEL INDIGO 1812 | LXML-PR01-0425 (C-BIN 3) |
| CR9183-F | LEDI REBEL INDIGO 1812 | LXML-PR01-0425 (C-BIN 4) |
| CR9184-F | LEDA REBEL PC-AMBER 1812 | LXM2-PL01-0000 |

RESULTS OF TESTS (cont'd)

Manufacturer Supplied Documentation:

## Electrical Characteristics at 350 mA for LUXEON Rebel color, Thermal Pad Temperature = 25°C

**Table 3.**

| Color | Part Number | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_J$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|---|
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Royal-Blue | LXML-PR01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red-Orange | LXM2-PH01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

## Absolute Maximum Ratings

**Table 5.**

| Parameter | Green/Cyan/ Blue/Royal Blue | LUXEON Rebel ES Royal Blue/ES Blue | Red/Deep-Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |

Manufacturer Supplied Documentation:

**Table 11. Applicable for LXML-PXX1 and LXM2-PL01 (PC Amber) emitters.**

| | $V_f$ Bins | |
|---|---|---|
| Bin Code | Minimum Forward Voltage (V) | Maximum Forward Voltage (V) |
| A | 2.31 | 2.55 |
| B | 2.55 | 2.79 |
| C | 2.79 | 3.03 |
| D | 3.03 | 3.27 |
| E | 3.27 | 3.51 |

**Table 12. Applicable for LXML-PR02-XXXX (ES Royal Blue) and LXML-PB02-XXXX (ES Blue) emitters tested at 700 mA**

| Bin Code | Minimum Forward Voltage (V) | Maximum Forward Voltage (V) |
|---|---|---|
| P | 2.50 | 2.75 |
| R | 2.75 | 3.00 |
| S | 3.00 | 3.25 |
| T | 3.25 | 3.50 |

**Table 13. Applicable for LXM2-PXX1 (Red and Red Orange) and LXM3-PD01 (Deep Red) emitters tested at 350 mA.**

| Bin Code | Minimum Forward Voltage (V) | Maximum Forward Voltage (V) |
|---|---|---|
| V | 1.80 | 2.00 |
| W | 2.00 | 2.20 |
| X | 2.20 | 2.40 |
| Y | 2.40 | 2.60 |
| Z | 2.60 | 2.80 |

RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDR REBEL RED 1812 LXML-PD01-0040

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 135℃
Thermal Resistance Formula from LED specification = 12 ℃/W
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 557.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.9565
Maximum Source Temperature (Ts) = 112C°

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013124204 | Philips Luxeon LEDR REBEL RED 1812 LXML-PD01-0040 | 80.3 | Per diagram above | 112 |

In-Situ Picture – Ts point

In-Situ Picture – Array location

RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDO REBEL ORG 1812 LXML-PH01-0050

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 12
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 549.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.9256
Maximum Source Temperature (Ts) = 127 C°

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013124204 | Philips Luxeon LEDO REBEL ORG 1812 LXML-PH01-0050 | 80.4 | Per diagram above | 127 |

In-Situ Picture – Ts point

In-Situ Picture – Array location

RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDG REBEL GRN 1812 LXML-PM01-0090

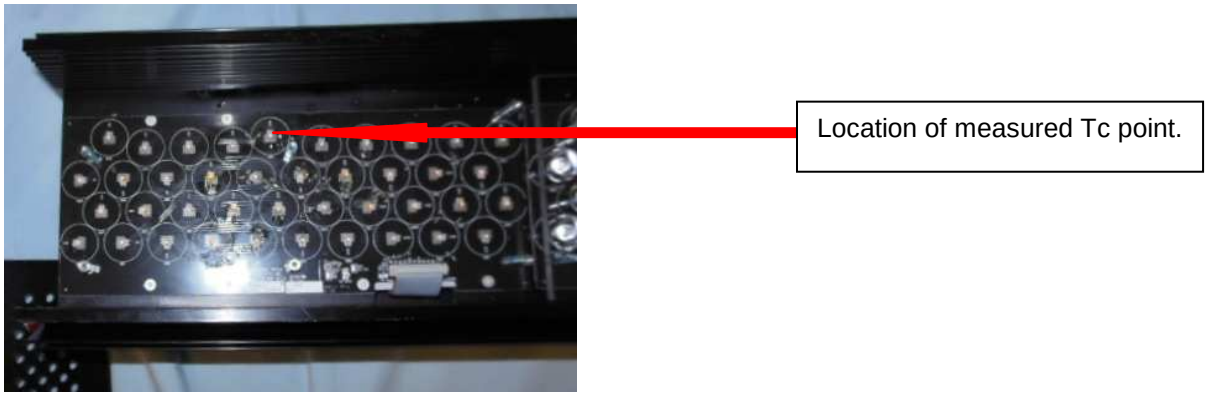
In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 559.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.9614
Maximum Source Temperature (Ts) = 130 C°

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013124204 | Philips Luxeon LEDG Rebel GRN 1812 LXML-PM01-0090 | 79.3 | Per diagram above | 130 |

In-Situ Picture – Ts point

In-Situ Picture – Array location



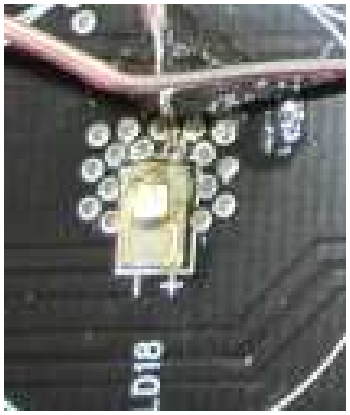
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDC REBEL CYAN 1812 LXML-PE01-0070

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 551.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.9351
Maximum Source Temperature (Ts) = 131C°

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013124204 | Philips Luxeon LEDC REBEL CYAN 1812 LXML-PE01-0070 | 77.4 | Per diagram above | 131 |

In-Situ Picture – Ts point



In-Situ Picture – Array location

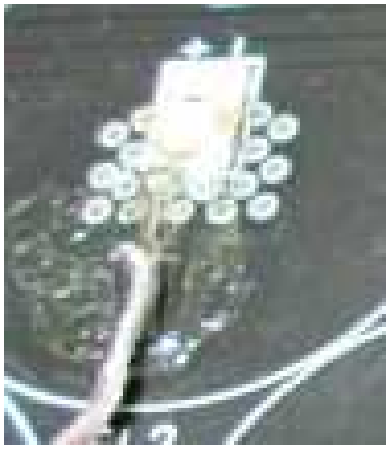
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDB REBEL BLUE 1812 LXML-PB01-0030

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 567.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.9909
Maximum Source Temperature (Ts) = 130 C°

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013124204 | Philips Luxeon LEDB REBEL BLUE 1812 LXML-PB01-0030 | 77.6 | Per diagram above | 130 |

In-Situ Picture – Ts point



In-Situ Picture – Array location

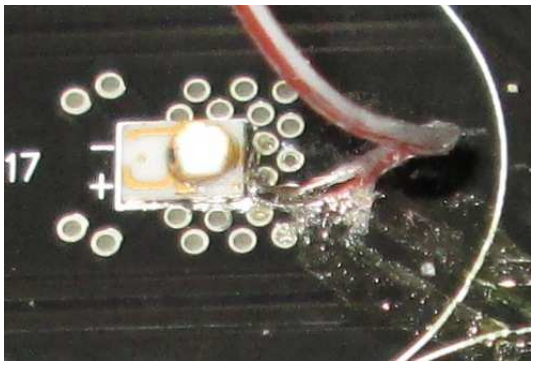
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDI REBEL INDIGO 1812 LXML-PR01-0350

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 10
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 540.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.8958
Maximum Source Temperature (Ts) = 131 C°

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013124204 | Philips Luxeon LEDI REBEL INDIGO 1812 LXML-PR01-0350 | 79.0 | Per diagram above | 131 |

In-Situ Picture – Ts point




In-Situ Picture – Array location

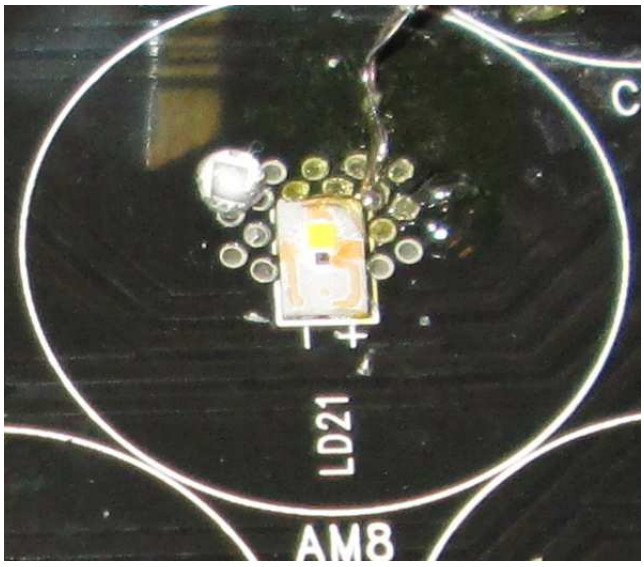
RESULTS OF TESTS (cont'd)
LED Identified as Philips Luxeon LEDA REBEL PC-AMBER 1812 LXM2-PL01-0000

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 130℃
Thermal Resistance Formula from LED specification = 10
Maximum Forward Voltage (Vf) from LED specification = 3.51
Measured LED Current = 552.0 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.9379
Maximum Source Temperature (Ts) = 111 C°

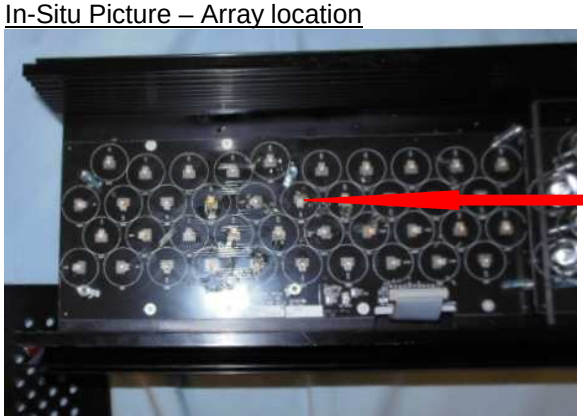
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Model No. | Maximum Measured Source Temperature (C°) | Location | Maximum Rated Source Temperature (C°) |
|---|---|---|---|---|
| 01162013124204 | Philips Luxeon LEDA REBEL PC-AMBER 1812 LXM2-PL01-0000 | 77.0 | Per diagram above | 111 |

In-Situ Picture – Ts point




In-Situ Picture – Array location

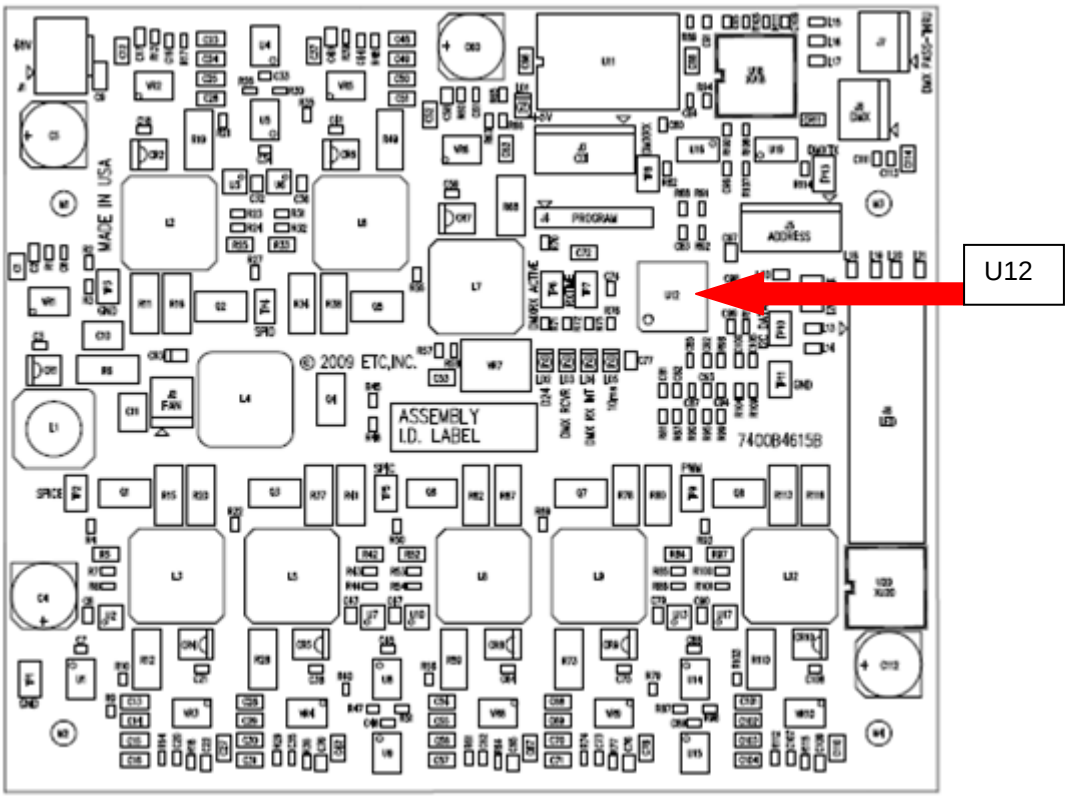


RESULTS OF TESTS (cont'd)

Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:
Driver identified as: 7400B5615, Rev. F

For Selador Classic, measure the temperature at U12.

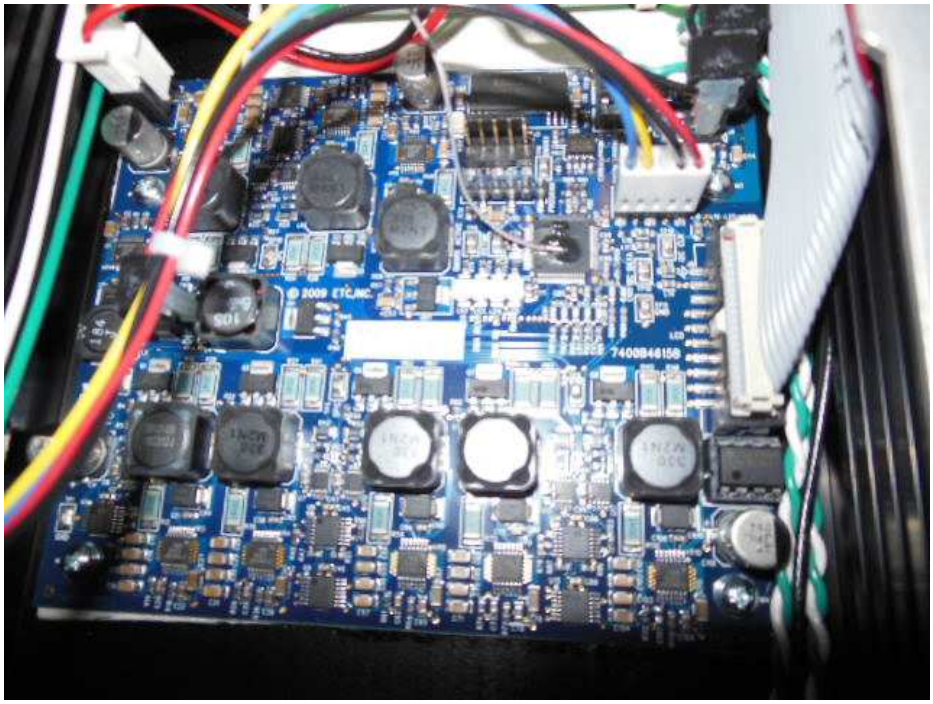



Measured Power Supply Case Temperature

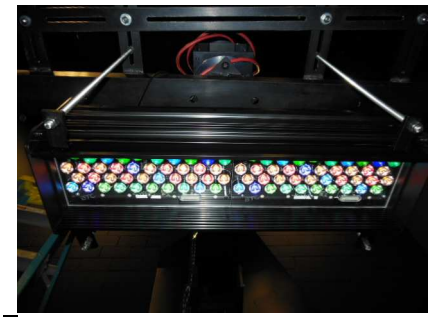
| Sample No. | Model No. | Measured Power Supply Case Temperature (C°) | Location | Maximum Power Supply Case Temperature (C°) |
|---|---|---|---|---|
| 01162013124204 | 7400B5615, Rev. F | 63.5 | U12 | 100 |

RESULTS OF TESTS (cont'd)

Power Supply Picture



Pictures (not to scale)



CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Tim Quigley
Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

# REPORT

**545 EAST ALGONQUIN ROAD ARLINGTON HEIGHTS, IL 60005**

Project No. G101042643 Date: March 4, 2013

REPORT NO. 101042643CHI-001

TEST OF ONE LED STAGE LUMINAIRE

FIXTURE MODEL NO. SELADOR LUSTR LED 21
LED MODEL NO. PHILIPS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500428358.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79: 2008 Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products

ANSI ANSLG C38.377: 2012 Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one prototype sample of model number Selador Lustr LED 21. The sample was received by Intertek on January 16, 2013, in undamaged condition, and one sample was tested as received. The sample designation was 01162013124204.

DATES OF TESTS: February 11, 2013 through February 12, 2013

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | Selador Lustr LED 21 |
| Description: | 21” LED Stage Lighting with two LED modules |

| Criteria | Sphere | Goniometer |
|---|---|---|
| Total Lumen Output (Lumens) | 4638 | 4446 |
| Total Power (W) | 196.20 | 197.1 |
| Luminaire Efficacy (LPW) | 23.64 | 22.56 |

| Criteria | Result |
|---|---|
| Power Factor – 120VAC | 0.964 |
| Power Factor – 240VAC | 0.398 |
| Current ATHD – 120VAC | 23.84% |
| Current ATHD – 240VAC | 43.02% |
| Correlated Color Temperature (CCT) | N/A* |
| Color Rendering Index (CRI) - Ra | 74.5 |
| Color Rendering Index (CRI) - R9 | -17.2 |
| Duv | 0.009 |
| Chromaticity Coordinate (x) | 0.262 |
| Chromaticity Coordinate (y) | 0.283 |
| Chromaticity Coordinate (u’) | 0.179 |
| Chromaticity Coordinate (v’) | 0.434 |

* Note: CCT is a white color parameter and is not accurate for saturated colors.

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Meter | WT210 | 146919 | 12/21/12 | 12/21/13 |
| Omega Thermometer | DPI8-C24 | 146920 | 11/15/12 | 11/15/13 |
| LSI High Speed Mirror Goniometer | 6440T | 146928 | VBU | VBU |
| Newport Hygrometer | iServer | 146961 | 2/23/12 | 2/23/13 |
| Elgar, AC Power Supply | CW1251P | 146918 | VBU | VBU |
| Cole-Parmer Triple Timer | 94440-00 | CHI0041 | 7/19/12 | 7/19/13 |
| Labsphere Spectroradiometer | CDS1100 | CHI0091 | VBU | VBU |
| 3 Meter Sphere | SPR600 | CHI0088 | VBU | VBU |
| Elgar AC Power Supply | CW1251M | 146112 | VBU | VBU |
| Sorenson DC Power Supply | XFR150-8 | 146846 | 4/20/12 | 4/20/13 |
| Newport Temperature and Humidty Recorder | iTHX-SD | 146958 | 2/23/12 | 2/23/13 |
| Yokoga Power Meter | WT1600 | 146769 | 5/18/12 | 5/18/13 |
| Omega Temperature Meter | MDSi8 | 146139 | 7/19/12 | 7/19/13 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Three Meter Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided integral driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Estimated Total Operating Time

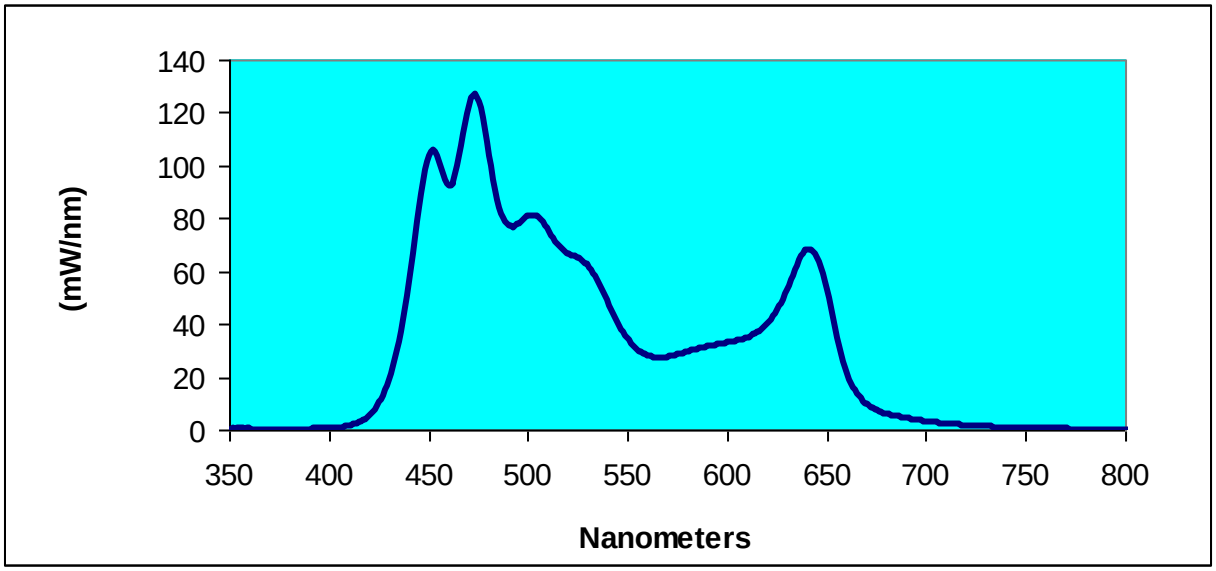
| Model No. | Total Hours |
|---|---|
| Selador Lustr LED 21 | 5 |

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.35 | 460 | 92.21 | 570 | 27.49 | 680 | 5.91 |
| 355 | 0.39 | 465 | 100.04 | 575 | 28.40 | 685 | 5.07 |
| 360 | 0.39 | 470 | 119.52 | 580 | 29.55 | 690 | 4.40 |
| 365 | 0.38 | 475 | 125.99 | 585 | 30.63 | 695 | 3.81 |
| 370 | 0.35 | 480 | 108.92 | 590 | 31.55 | 700 | 3.28 |
| 375 | 0.34 | 485 | 87.29 | 595 | 32.26 | 705 | 2.81 |
| 380 | 0.32 | 490 | 77.58 | 600 | 32.99 | 710 | 2.39 |
| 385 | 0.31 | 495 | 77.67 | 605 | 33.70 | 715 | 2.05 |
| 390 | 0.37 | 500 | 80.64 | 610 | 34.81 | 720 | 1.76 |
| 395 | 0.44 | 505 | 80.77 | 615 | 36.89 | 725 | 1.51 |
| 400 | 0.59 | 510 | 76.08 | 620 | 40.11 | 730 | 1.28 |
| 405 | 0.92 | 515 | 70.30 | 625 | 45.27 | 735 | 1.10 |
| 410 | 1.57 | 520 | 66.90 | 630 | 52.82 | 740 | 0.94 |
| 415 | 2.81 | 525 | 65.15 | 635 | 62.31 | 745 | 0.80 |
| 420 | 5.31 | 530 | 62.47 | 640 | 68.35 | 750 | 0.69 |
| 425 | 10.23 | 535 | 56.73 | 645 | 64.93 | 755 | 0.60 |
| 430 | 18.95 | 540 | 48.91 | 650 | 53.06 | 760 | 0.52 |
| 435 | 33.63 | 545 | 40.93 | 655 | 34.89 | 765 | 0.45 |
| 440 | 55.79 | 550 | 34.63 | 660 | 21.06 | 770 | 0.39 |
| 445 | 84.07 | 555 | 30.45 | 665 | 13.66 | 775 | 0.34 |
| 450 | 103.92 | 560 | 28.14 | 670 | 9.58 | 780 | 0.30 |
| 455 | 101.60 | 565 | 27.27 | 675 | 7.24 | | |

**Sample No. 01162013124204**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25°C – Integrating Sphere Method

| Intertek Sample No. | Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|---|
| 01162013124204 | -- | 74.5 | -17.2 | 0.009 | 0.262 | 0.283 | 0.179 | 0.434 |

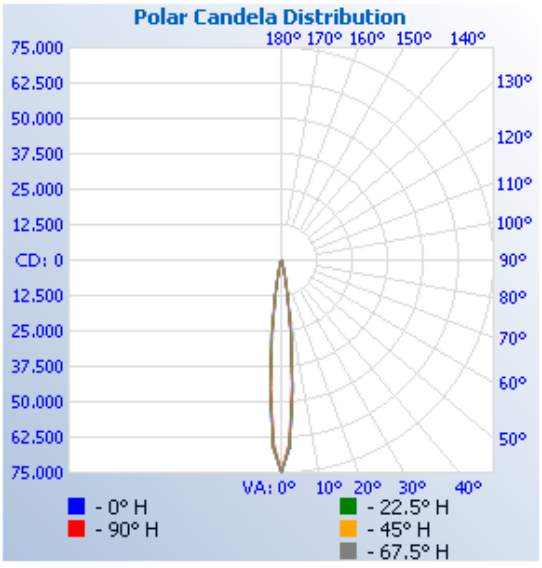
| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| 01162013124204 | Up | 120.0 | 1697 | 196.2 | 0.964 | 23.84 | 4638 | 23.64 |
| | | 240.0 | | | 0.398 | 43.02 | | |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| 01162013124204 | Up | 120.1 | 1695 | 197.1 | 0.969 | 4446 | 22.56 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 74851 | 74851 | 74851 | 74851 | 74851 |
| 5 | 45954 | 45887 | 45250 | 44972 | 43755 |
| 10 | 12975 | 13286 | 13188 | 12560 | 11990 |
| 15 | 2715 | 2864 | 2850 | 2582 | 2381 |
| 20 | 665 | 720 | 733 | 671 | 624 |
| 25 | 258 | 271 | 272 | 262 | 248 |
| 30 | 122 | 125 | 123 | 119 | 111 |
| 35 | 81 | 79 | 75 | 69 | 63 |
| 40 | 62 | 57 | 50 | 44 | 40 |
| 45 | 52 | 45 | 33 | 30 | 27 |
| 50 | 45 | 38 | 25 | 20 | 19 |
| 55 | 40 | 33 | 20 | 15 | 12 |
| 60 | 35 | 28 | 16 | 10 | 5 |
| 65 | 28 | 20 | 11 | 1 | 1 |
| 70 | 20 | 13 | 1 | 1 | 0 |
| 75 | 11 | 5 | 0 | 0 | 0 |
| 80 | 4 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

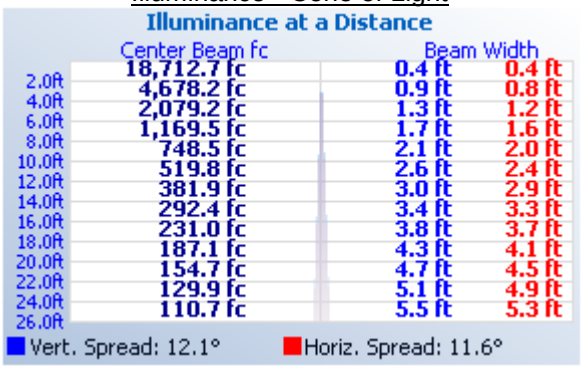


RESULTS OF TESTS (cont'd)

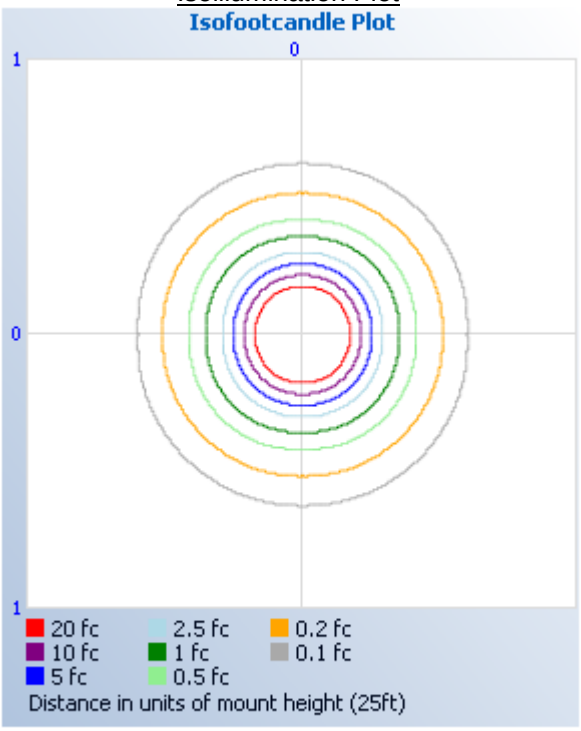
Illumination Plots

Mounting Height: 25 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 4333 | 97.4 |
| 0-40 | 4381 | 98.5 |
| 0-60 | 4431 | 99.7 |
| 60-90 | 15.2 | 0.3 |
| 0-90 | 4446 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 4446 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 3186 | 71.7 |
| 10-20 | 1008 | 22.7 |
| 20-30 | 138.3 | 3.1 |
| 30-40 | 48.1 | 1.1 |
| 40-50 | 29.1 | 0.7 |
| 50-60 | 21.2 | 0.5 |
| 60-70 | 11.5 | 0.3 |
| 70-80 | 3.3 | 0.1 |
| 80-90 | 0.3 | 0.0 |

Pictures (not to scale)

Model Selador Lustr LED 21

Luminaire mounted on goniophotometer

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Tim Quigley
Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: December 6, 2013

REPORT NO. 101414036CRT-007

TEST OF ONE VIVID-R SL3 LED FIXTURE 11

MODEL NO. VIVID-R SL3 11
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number VIVID-R SL3 11. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-007.

DATES OF TESTS: December 3, 2013 through December 5, 20:

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | VIVID-R SL3 11 |
| Description: | VIVID-R SL3 LED FIXTURE 11 |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 2752 | 3038 |
| Total Power (W) | 97.89 | 97.78 |
| Luminaire Efficacy (LPW) | 28.11 | 31.07 |

| Criteria | Result |
|---|---|
| Power Factor | 0.974 |
| Current ATHD % | 16.84 |
| Correlated Color Temperature (CCT - K) | 3065 |
| Color Rendering Index (CRI - Ra) | 32.1 |
| Color Rendering Index (CRI - R9) | -53.6 |
| DUV | 0.066 |
| Chromaticity Coordinate (x) | 0.362 |
| Chromaticity Coordinate (y) | 0.253 |
| Chromaticity Coordinate (u') | 0.273 |
| Chromaticity Coordinate (v') | 0.428 |
| Maximum In-Situ Source Temperature Point (°C) | 70.4 |
| Maximum In-Situ Driver Case Temperature (°C) | 52.9 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/27/13 | 11/27/14 |
| Sorensen DC Power Supply | DLM150-20E | --- | --- | --- |
| Data Precision | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole-Palmer Hygro Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1360 | 11/27/13 | 11/27/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

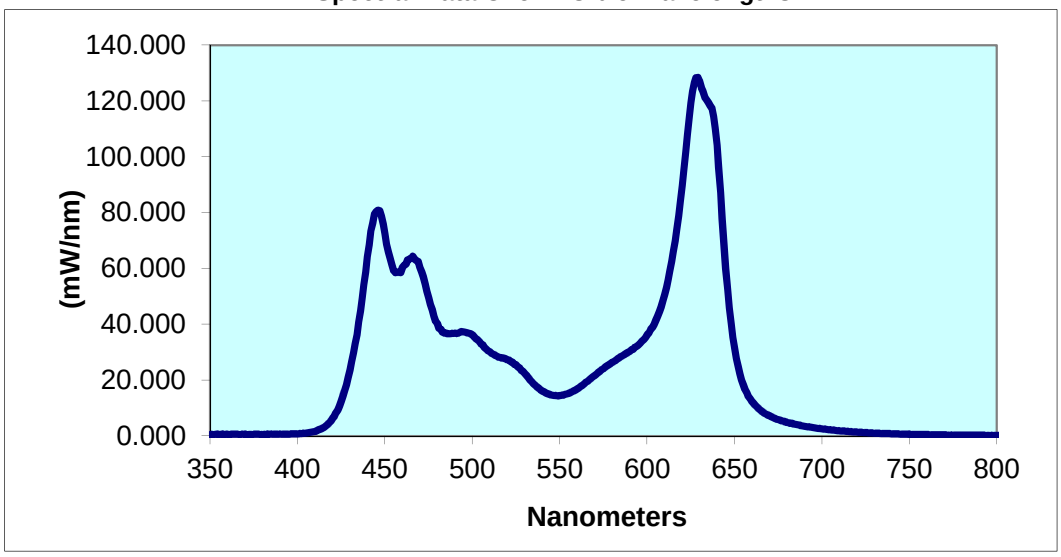
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-007 | UP | 120.0 | 837.3 | 97.89 | 0.974 | 16.84 | 2752 | 28.11 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3065 | 32.1 | -53.6 | 0.066 | 0.362 | 0.253 | 0.273 | 0.428 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.438 | 440 | 64.19 | 530 | 22.37 | 620 | 89.64 | 710 | 1.851 |
| 355 | 0.48 | 445 | 80.37 | 535 | 18.75 | 625 | 119.4 | 715 | 1.605 |
| 360 | 0.617 | 450 | 72.52 | 540 | 16.06 | 630 | 127.1 | 720 | 1.363 |
| 365 | 0.557 | 455 | 59.32 | 545 | 14.68 | 635 | 119.5 | 725 | 1.17 |
| 370 | 0.66 | 460 | 60.42 | 550 | 14.45 | 640 | 104.2 | 730 | 0.988 |
| 375 | 0.561 | 465 | 63.54 | 555 | 15.27 | 645 | 59.94 | 735 | 0.848 |
| 380 | 0.472 | 470 | 60.44 | 560 | 16.89 | 650 | 31.21 | 740 | 0.751 |
| 385 | 0.713 | 475 | 49.45 | 565 | 19.23 | 655 | 18.25 | 745 | 0.644 |
| 390 | 0.655 | 480 | 40.21 | 570 | 21.65 | 660 | 12.29 | 750 | 0.561 |
| 395 | 0.621 | 485 | 36.84 | 575 | 24.14 | 665 | 9.092 | 755 | 0.485 |
| 400 | 0.718 | 490 | 36.81 | 580 | 26.38 | 670 | 7.244 | 760 | 0.443 |
| 405 | 1.042 | 495 | 37.28 | 585 | 28.41 | 675 | 5.843 | 765 | 0.363 |
| 410 | 1.607 | 500 | 36.23 | 590 | 30.25 | 680 | 4.876 | 770 | 0.331 |
| 415 | 3.046 | 505 | 33.02 | 595 | 32.6 | 685 | 4.162 | 775 | 0.286 |
| 420 | 6.164 | 510 | 30.15 | 600 | 36.12 | 690 | 3.51 | 780 | 0.261 |
| 425 | 12.31 | 515 | 28.24 | 605 | 41.5 | 695 | 2.98 | | |
| 430 | 23.61 | 520 | 27.36 | 610 | 50.63 | 700 | 2.584 | | |
| 435 | 40.86 | 525 | 25.43 | 615 | 66.03 | 705 | 2.144 | | |

**Spectral Data Over Visible Wavelengths**



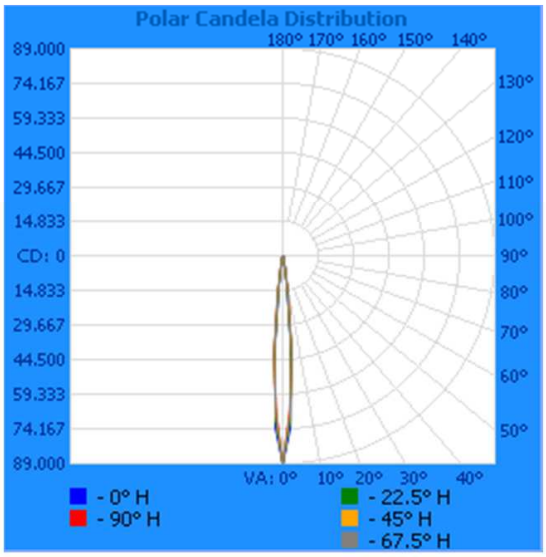
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-007 | UP | 120.0 | 835.9 | 97.78 | 0.975 | 3038 | 31.07 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 88961 | 88961 | 88961 | 88961 | 88961 |
| 5 | 36475 | 38693 | 40393 | 41500 | 43402 |
| 10 | 4093 | 4665 | 5256 | 5372 | 5579 |
| 15 | 652 | 725 | 854 | 807 | 744 |
| 20 | 384 | 383 | 399 | 387 | 348 |
| 25 | 152 | 152 | 158 | 164 | 170 |
| 30 | 70 | 66 | 66 | 63 | 62 |
| 35 | 46 | 40 | 40 | 33 | 34 |
| 40 | 33 | 29 | 26 | 20 | 19 |
| 45 | 28 | 23 | 16 | 14 | 10 |
| 50 | 24 | 20 | 13 | 5 | 5 |
| 55 | 16 | 12 | 5 | 0 | 0 |
| 60 | 7 | 3 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

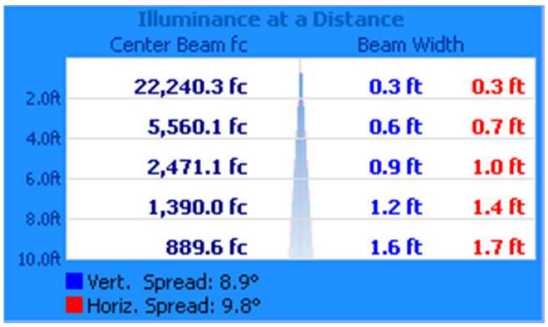


RESULTS OF TEST (cont'd)

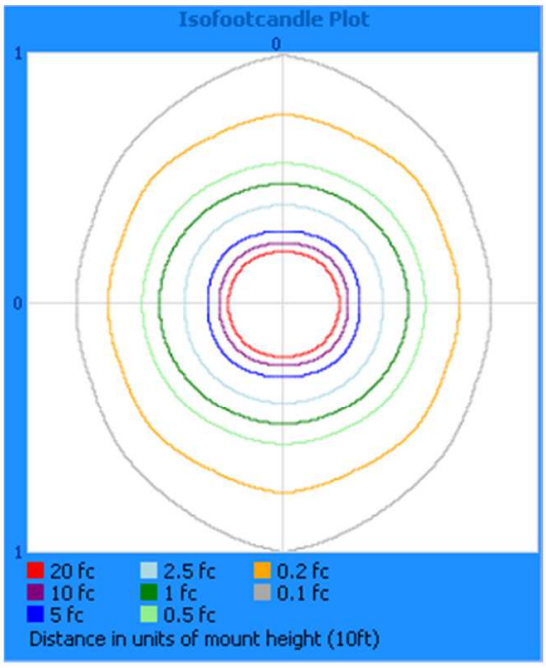
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 2992 | 98.5 |
| 0-40 | 3017 | 99.3 |
| 0-60 | 3038 | 100.0 |
| 60-90 | 0.2 | 0.0 |
| 0-90 | 3038 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3038 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2566 | 84.5 |
| 10-20 | 345.5 | 11.4 |
| 20-30 | 81.2 | 2.7 |
| 30-40 | 25.2 | 0.8 |
| 40-50 | 14.1 | 0.5 |
| 50-60 | 6.0 | 0.2 |
| 60-70 | 0.2 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as: Luxeon Rebel ES (Amber)

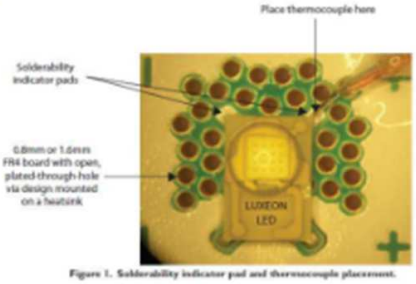



Figure 1. Solderability indicator pad and thermocouple placement.

| Color | Part Number | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|---|---|
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

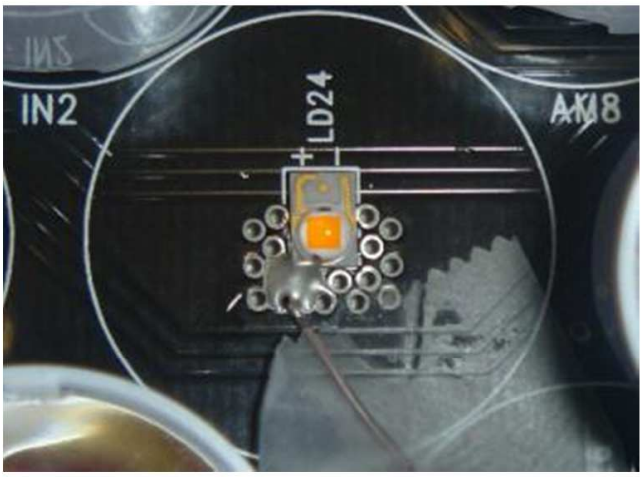
Maximum LED Junction Temperature 135°C

Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 12°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 590mA
Calculated LED Wattage = Vf x Measured LED Current = 2.071W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 110.1°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-007 | 70.4 | Per diagram | 110.1 |

In-Situ Picture – Ts

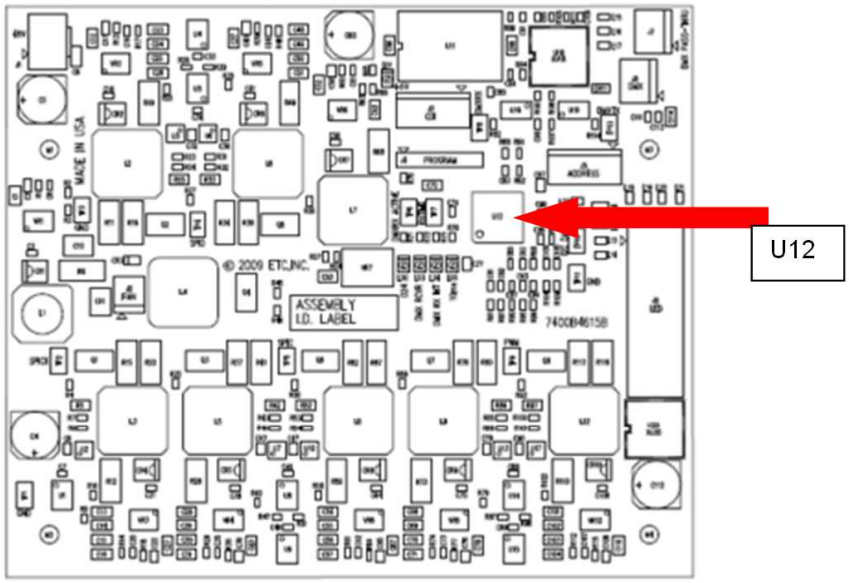



In-Situ Picture – Ts location

RESULTS OF TEST (cont'd)

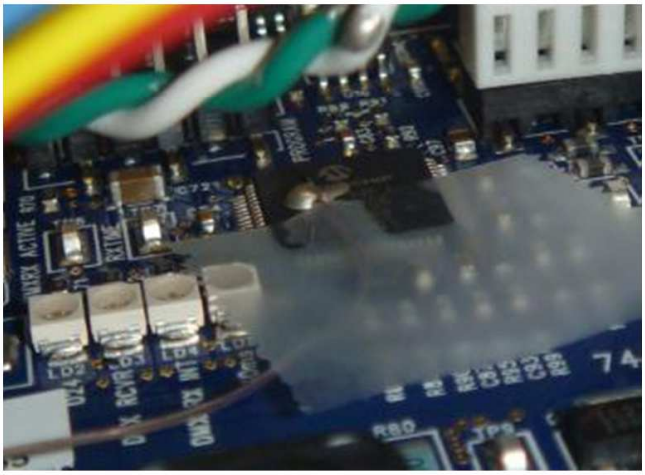
Maximum Measured Power Supply Case Temperature

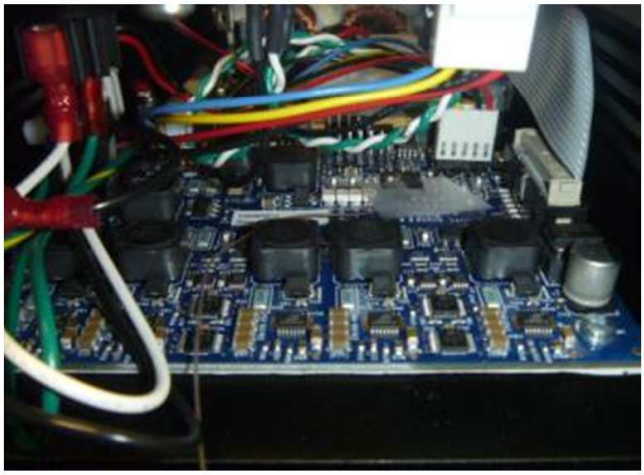
Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-004 | 52.9 | Per diagram | 100 |

Power Supply Picture – Tc location





PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: August 16, 2013

REPORT NO. 101158242CRT-011

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. 11" PALETTA

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79: 2008 Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products

ANSI ANSLG C38.377: 2012 Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number 11" PALETTA. The sample was received by Intertek on August 12, 2013, in undamaged condition, and one sample was tested as received. The sample designation was CRT1308121125-001.

DATES OF TESTS: August 15, 2013

**This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.**

SUMMARY

| | |
|---|---|
| Model No.: | 11” PALETTA |
| Description: | LED WASH LIGHT |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 1123 | 1098 |
| Total Power (W) | 75.30 | 75.15 |
| Luminaire Efficacy (LPW) | 14.91 | 14.61 |

| Criteria | Result |
|---|---|
| Power Factor | 0.974 |
| Current ATHD (%) | 15.51 |
| Duv | 0.064 |
| Chromaticity Coordinate (x) | 0.231 |
| Chromaticity Coordinate (y) | 0.208 |
| Chromaticity Coordinate (u’) | 0.184 |
| Chromaticity Coordinate (v’) | 0.372 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 07/24/13 | 08/24/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Extech Hygro Thermometer | 445703 | T1359 | 11/08/12 | 11/08/13 |
| Fisher Scientific | --- | N1132 | 04/22/13 | 04/22/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temp Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | NA | NA | NA |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Sorensen DC Power Supply | DLM150-20E | N/A | N/A | N/A |
| FLUKE | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

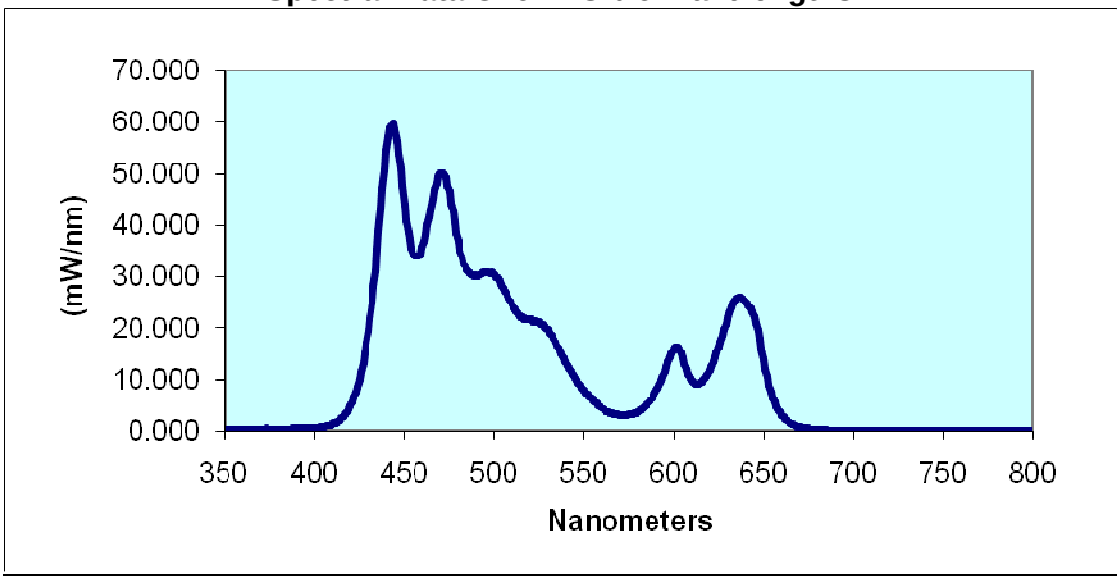
The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

RESULTS OF TESTS

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.208 | 460 | 35.595 | 570 | 2.990 | 680 | 0.231 |
| 355 | 0.272 | 465 | 42.888 | 575 | 3.049 | 685 | 0.150 |
| 360 | 0.344 | 470 | 49.978 | 580 | 3.716 | 690 | 0.103 |
| 365 | 0.298 | 475 | 46.458 | 585 | 5.109 | 695 | 0.082 |
| 370 | 0.375 | 480 | 36.856 | 590 | 7.493 | 700 | 0.067 |
| 375 | 0.350 | 485 | 31.155 | 595 | 11.399 | 705 | 0.055 |
| 380 | 0.336 | 490 | 30.234 | 600 | 15.772 | 710 | 0.049 |
| 385 | 0.362 | 495 | 30.702 | 605 | 14.099 | 715 | 0.046 |
| 390 | 0.380 | 500 | 30.364 | 610 | 9.585 | 720 | 0.040 |
| 395 | 0.445 | 505 | 27.596 | 615 | 9.267 | 725 | 0.040 |
| 400 | 0.528 | 510 | 24.525 | 620 | 11.703 | 730 | 0.037 |
| 405 | 0.724 | 515 | 22.144 | 625 | 16.255 | 735 | 0.030 |
| 410 | 1.190 | 520 | 21.536 | 630 | 21.808 | 740 | 0.031 |
| 415 | 2.363 | 525 | 21.010 | 635 | 25.802 | 745 | 0.023 |
| 420 | 4.909 | 530 | 19.536 | 640 | 25.185 | 750 | 0.027 |
| 425 | 9.879 | 535 | 16.575 | 645 | 22.157 | 755 | 0.028 |
| 430 | 19.291 | 540 | 13.275 | 650 | 13.822 | 760 | 0.028 |
| 435 | 35.270 | 545 | 10.239 | 655 | 6.422 | 765 | 0.000 |
| 440 | 54.536 | 550 | 7.754 | 660 | 3.031 | 770 | 0.028 |
| 445 | 58.491 | 555 | 5.858 | 665 | 1.462 | 775 | 0.028 |
| 450 | 43.860 | 560 | 4.418 | 670 | 0.743 | 780 | 0.024 |
| 455 | 34.177 | 565 | 3.450 | 675 | 0.395 | | |

**Sample No. CRT1308121125-001**
**Spectral Data Over Visible Wavelengths**



RESULTS OF TESTS (cont'd)

Photometric and Electrical Measurements at 25°C – Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|---|
| CRT1308121125-001 | UP | 120.0 | 644.3 | 75.30 | 0.974 | 15.51 | 1123 | 14.91 |

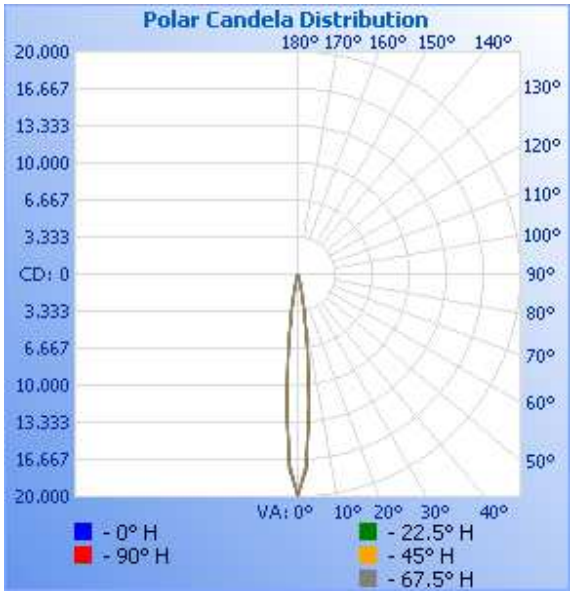
| Intertek Sample No. | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|
| CRT1308121125-001 | 0.064 | 0.231 | 0.208 | 0.184 | 0.372 |

Photometric and Electrical Measurements – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage (Vac) | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1308121125-001 | UP | 120.0 | 641.5 | 75.15 | 0.976 | 1098 | 14.61 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 19920 | 19920 | 19920 | 19920 | 19920 |
| 5 | 11793 | 11684 | 11694 | 11652 | 11631 |
| 10 | 3162 | 3227 | 3323 | 3212 | 3095 |
| 15 | 615 | 664 | 712 | 630 | 596 |
| 20 | 162 | 170 | 181 | 168 | 159 |
| 25 | 47 | 49 | 51 | 48 | 47 |
| 30 | 10 | 10 | 8 | 7 | 6 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

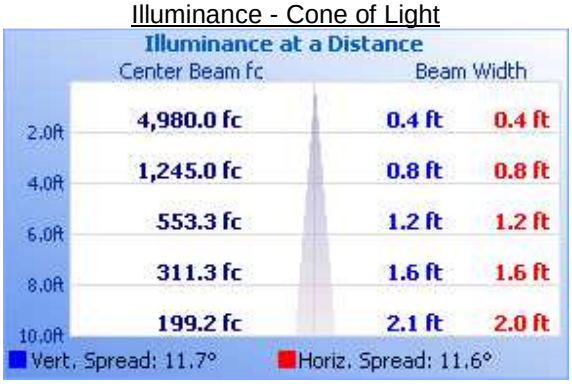


RESULTS OF TESTS (cont'd)

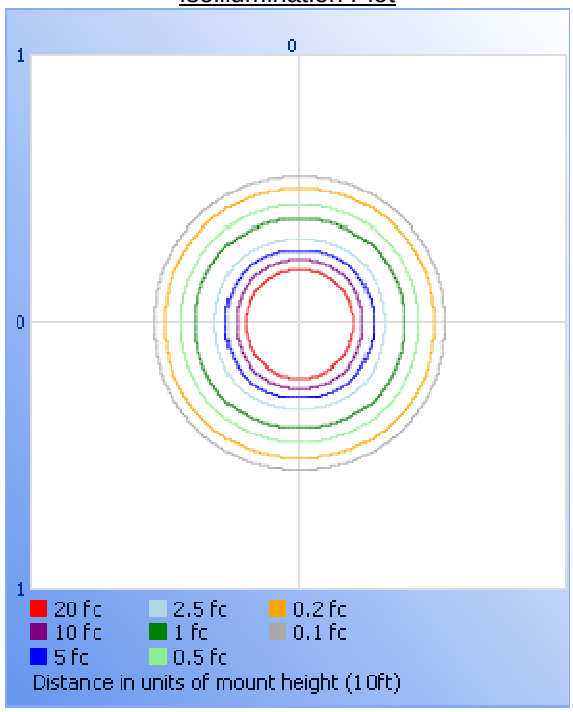
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light



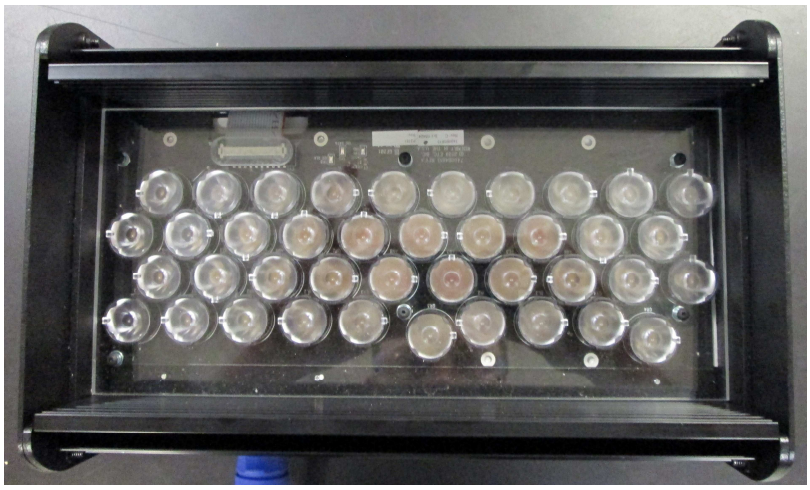

Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 1097 | 99.9 |
| 0-40 | 1098 | 100.0 |
| 0-60 | 1098 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 1098 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 1098 | 100.0 |

Picture (not to scale)



CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

# REPORT

**3933 US ROUTE 11 CORTLAND, NEW YORK 13045**

Project No. G101158242 Date: August 20, 2013

REPORT NO. 101158242CRT-012

TEST OF ONE LED WASH LIGHT

FIXTURE MODEL NO. 11" PALETTA
LED MODEL NO. LUXEON REBEL LXML-PWN2

RENDERED TO

ELECTRONIC THEATER CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and LED In-Situ tests.

LABORATORY NOTE: The laboratory that conducted the testing detailed in this report has been Qualified, Verified, and Recognized for LM-79 Testing for ENERGY STAR for SSL by US DOE's CALiPER program.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500447562.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

Energy Star Version 1.2 (2012): Program Requirements for Luminaires
Energy Star Manufacturer's Guide Version 2.0 (2009): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one sample of model number 11" PALETTA. The sample was received by Intertek on August 12, in undamaged condition, and one sample was tested as received. The sample designation was CRT1308121125-001.

DATES OF TESTS: August 16, 2013 through August 20, 2013

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | 11” PALETTA |
| Description: | LED WASH LIGHT |

| Criteria | Result |
|---|---|
| Maximum In-Situ Source Temperature Point – LXML-PM01 GREEN (°C) | 71.0 |
| Maximum In-Situ Source Temperature Point – LXML-PE01 CYAN (°C) | 68.3 |
| Maximum In-Situ Source Temperature Point – LXML-PB01 BLUE (°C) | 73.7 |
| Maximum In-Situ Source Temperature Point – LXML-PR01 ROYAL BLUE (°C) | 70.4 |
| Maximum In-Situ Source Temperature Point – LXM2-PL01 PC AMBER (°C) | 71.8 |
| Maximum In-Situ Source Temperature Point – LXML-PD01 RED(°C) | 71.8 |
| Maximum In-Situ Source Temperature Point – LXML-PH01 ORANGE (°C) | 70.3 |
| Maximum Power Supply Case Temperature | 70.6 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Calibration Date | Calibration Due Date |
|---|---|---|---|---|
| Extech Hygro-Thermometer | 445703 | T1366 | 11/08/12 | 11/08/13 |
| Fluke Multimeter | PM2525 | M127 | 10/17/12 | 10/17/13 |
| Fluke Temp Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

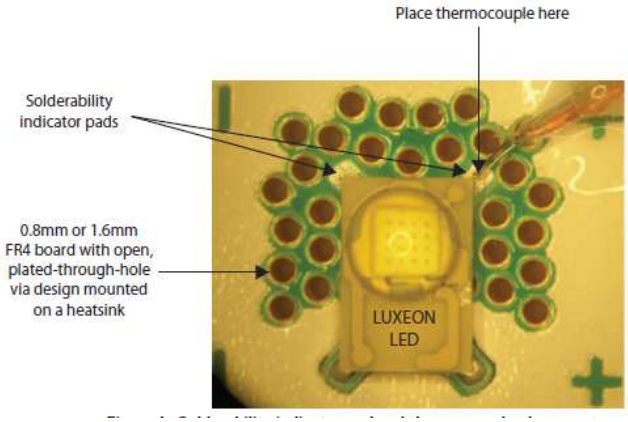
In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke Temperature Meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TESTS

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED identified as: LUXEON REBEL LXML-PWN2




## Electrical Characteristics at 350 mA for LUXEON Rebel color, Thermal Pad Temperature = 25°C

**Table 3.**

| Color | Part Number | Forward Voltage $V_f$ [1] (V) Min. | Typ. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|---|---|
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Royal-Blue | LXML-PR01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| Red-Orange | LXM2-PH01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |

## Absolute Maximum Ratings

**Table 5.**

| Parameter | Green/Cyan/ Blue/Royal Blue | LUXEON Rebel ES Royal Blue/ES Blue | Red/Deep-Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |

RESULTS OF TESTS (cont'd)

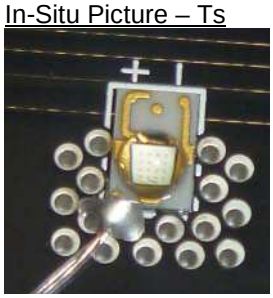
In-Situ Maximum Measured LED Source Temperature - LXML-PM01 GREEN

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 561 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.969W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 130°C

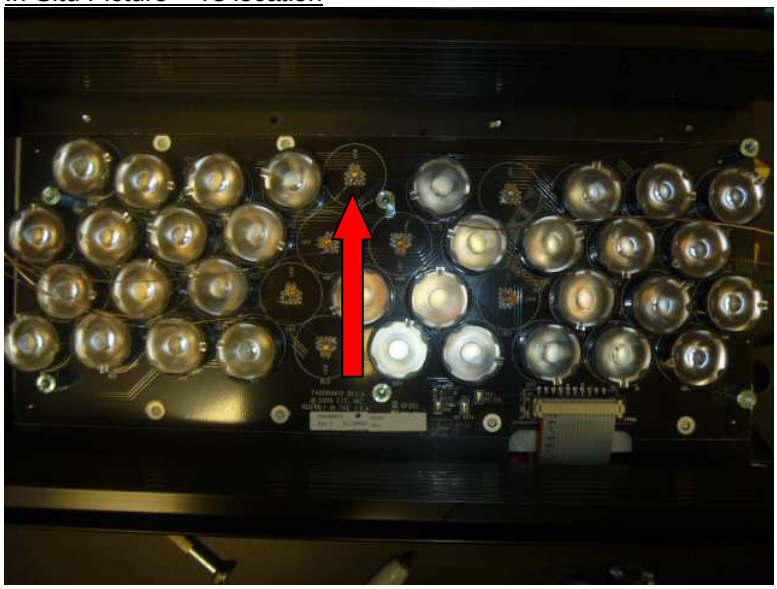
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1308121125-001 | 71.0 | Per diagram above | 130 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

In-Situ Maximum Measured LED Source Temperature - LXML-PE01 CYAN

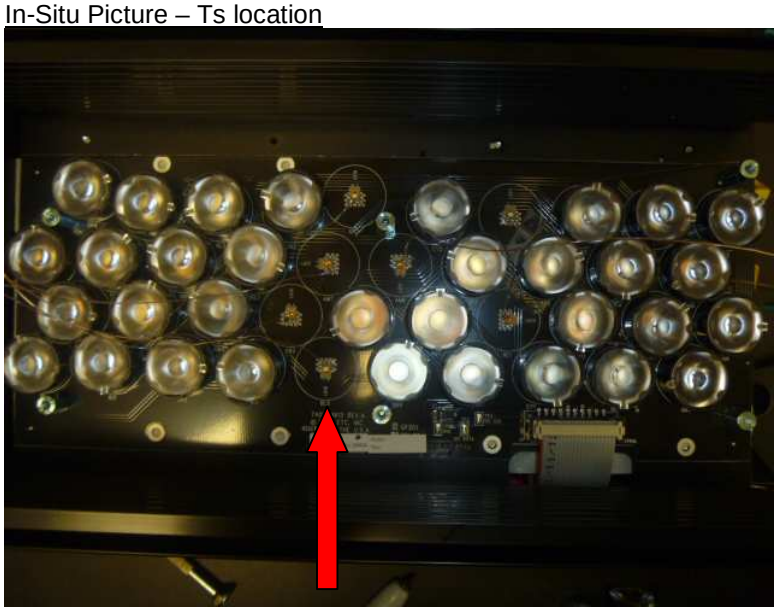
Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 548 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.924W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 131°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1308121125-001 | 68.3 | Per diagram above | 131 |

In-Situ Picture – Ts

In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

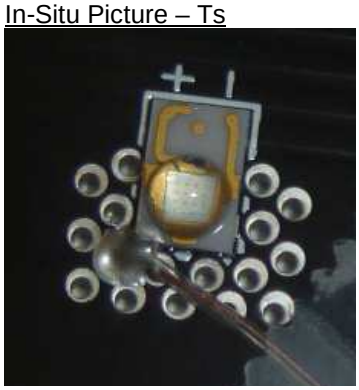
In-Situ Maximum Measured LED Source Temperature - LXML-PB01 BLUE

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 566 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.987W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 130°C

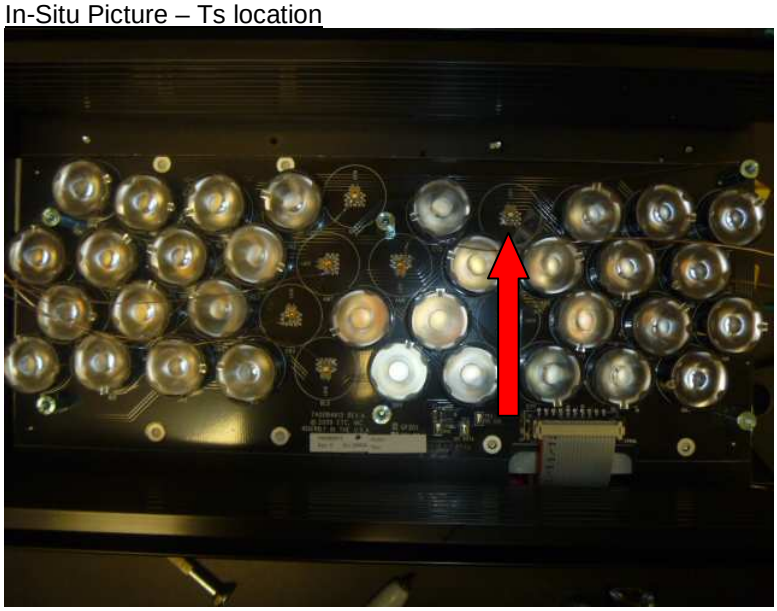
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1308121125-001 | 73.7 | Per diagram above | 130 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

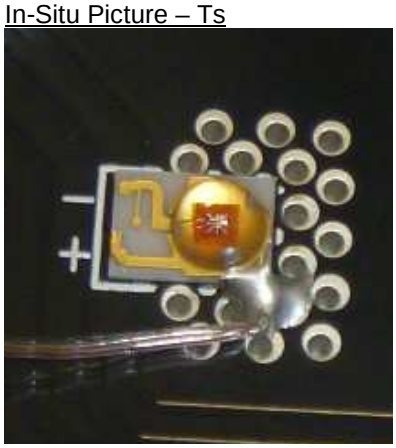
In-Situ Maximum Measured LED Source Temperature - LXML-PR01 ROYAL BLUE

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 10°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 540 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.895W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 131°C

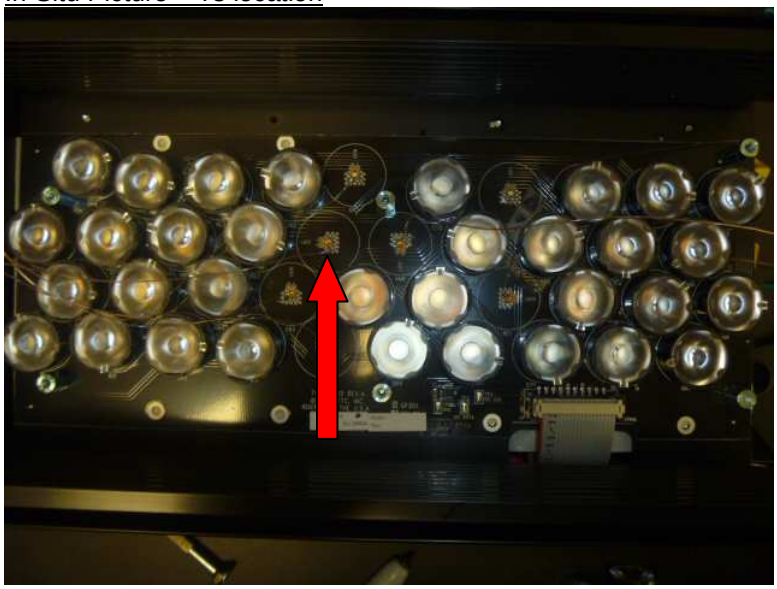
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1308121125-001 | 70.4 | Per diagram above | 131 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

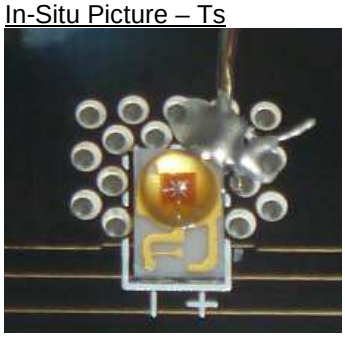
In-Situ Maximum Measured LED Source Temperature - LXM2-PL01 PC AMBER

Maximum Junction Temperature from LED specification (Tj) = 130°C
Thermal Resistance Formula from LED specification = 12°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51 V
Measured LED Current = 554 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.945W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 107°C

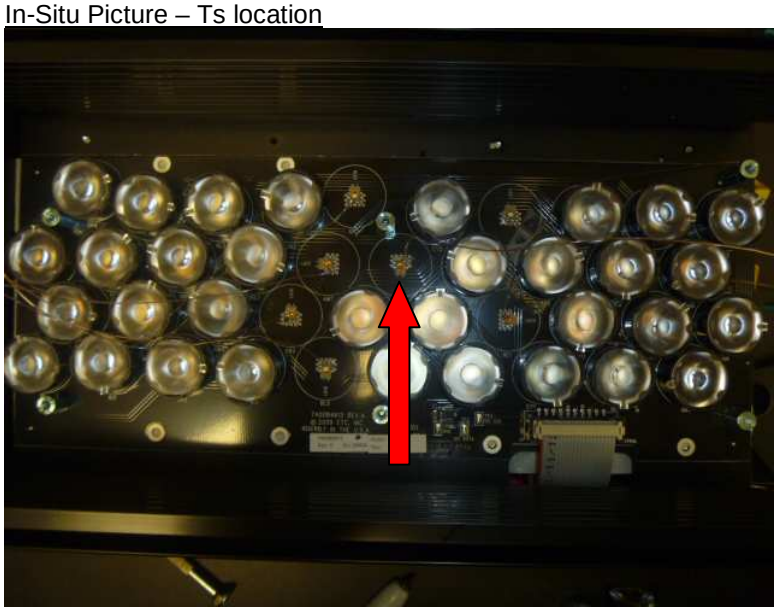
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1308121125-001 | 71.8 | Per diagram above | 107 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

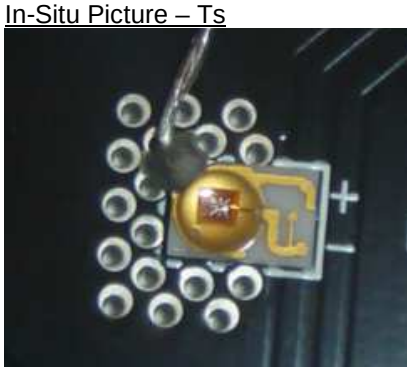
In-Situ Maximum Measured LED Source Temperature - LXM2-PD01 RED

Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 12°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 555 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.948W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 112°C

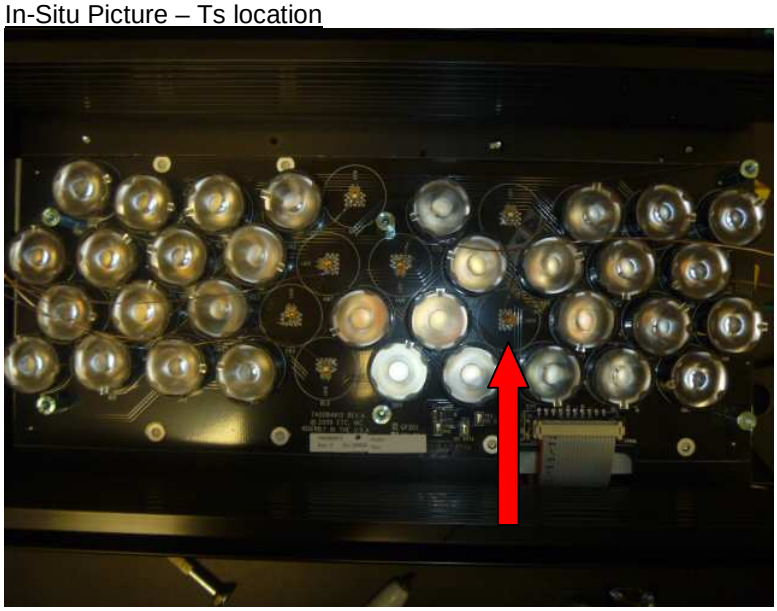
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1308121125-001 | 71.8 | Per diagram above | 112 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



RESULTS OF TESTS (cont'd)

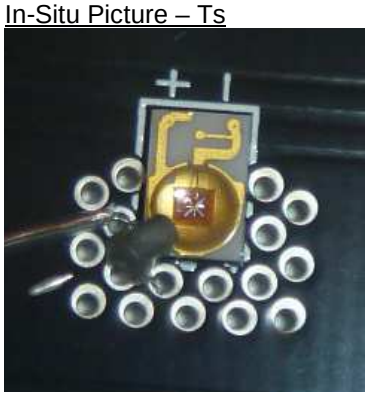
In-Situ Maximum Measured LED Source Temperature - LXML-PH01 ORANGE

Maximum Junction Temperature from LED specification (Tj) = 135°C
Thermal Resistance Formula from LED specification = 12°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.51V
Measured LED Current = 549 mA
Calculated LED Wattage = Vf x Measured LED Current = 1.927W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 112°C

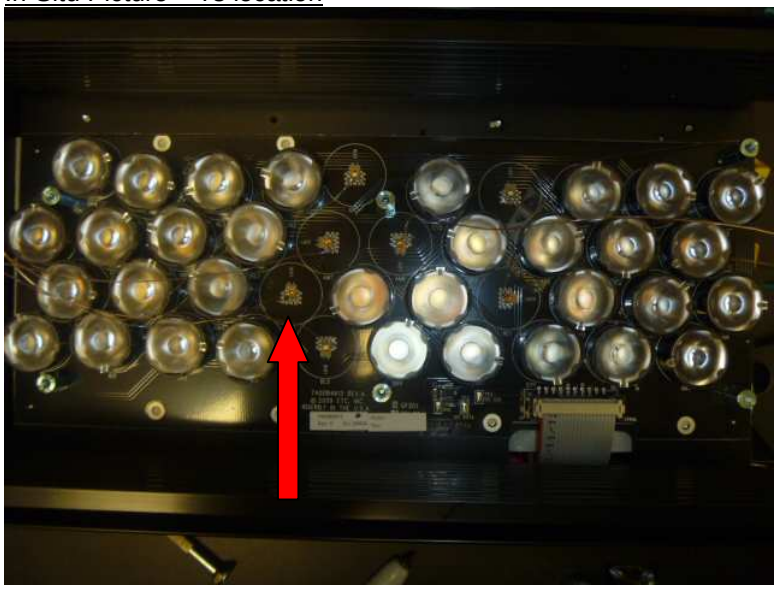
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature(°C) |
|---|---|---|---|
| CRT1308121125-001 | 70.3 | Per diagram above | 112 |

In-Situ Picture – Ts



In-Situ Picture – Ts location



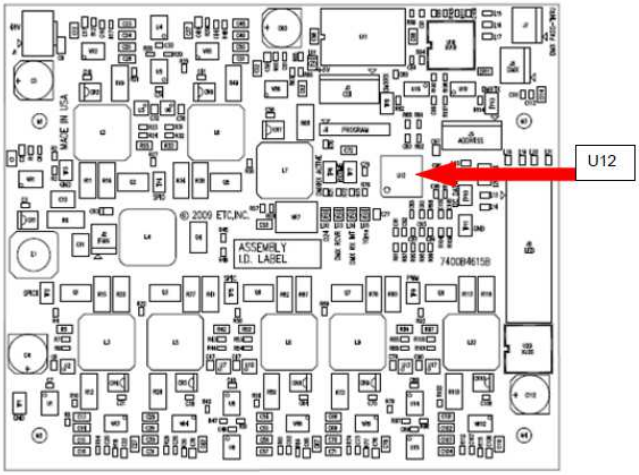
RESULTS OF TESTS (cont'd)

Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:
Driver identified as: Model No. 7400B5615

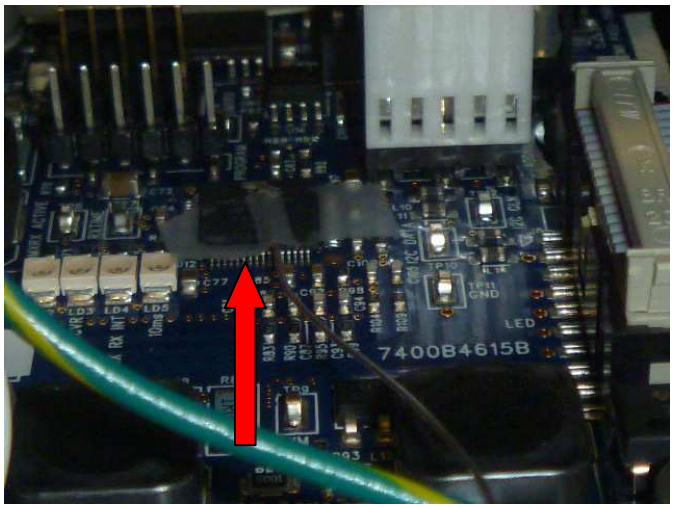
Manufacturer Supplied Documentation:
Driver identified as: 7400B5615, Rev. F

For Selador Classic, measure the temperature at U12.

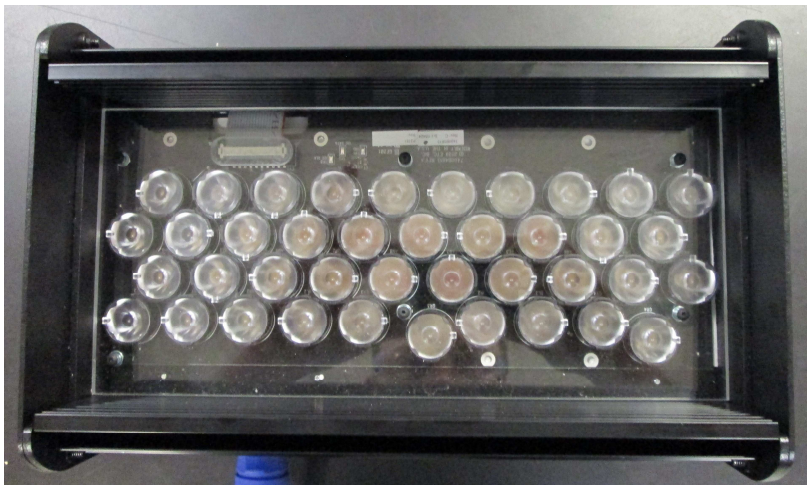



Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Measured Power Supply Case Temperature (°C) | Location | Maximum Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1308121125-001 | 70.6 | U12 | 100 |



Picture (not to scale)



CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Kenda Branch
Engineer
Lighting Division

Report Reviewed By:

Denis Niggli
Project Engineer
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101414036 Date: December 6, 2013

REPORT NO. 101414036CRT-003

TEST OF ONE PEARL SL3 FIXTURE 11

MODEL NO. PEARL SL3 11
LED MODEL NO. LUMILEDS LUXEON REBEL

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500491028.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number PEARL SL3 11. The sample was received by Intertek on November 7, 2013, in undamaged condition and one sample was tested as received. The sample designation was CRT1311071604-003.

DATES OF TESTS: December 3, 2013 through December 5, 20:

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | PEARL SL3 11 |
| Description: | PEARL SL3 FIXTURE 11 |

| Criteria | Result | |
|---|---|---|
| | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 4363 | 4942 |
| Total Power (W) | 98.76 | 98.74 |
| Luminaire Efficacy (LPW) | 44.18 | 50.05 |

| Criteria | Result |
|---|---|
| Power Factor | 0.975 |
| Current ATHD % | 17.56 |
| Correlated Color Temperature (CCT - K) | 4338 |
| Color Rendering Index (CRI - Ra) | 80.6 |
| Color Rendering Index (CRI - R9) | 14.0 |
| DUV | 0.002 |
| Chromaticity Coordinate (x) | 0.366 |
| Chromaticity Coordinate (y) | 0.364 |
| Chromaticity Coordinate (u') | 0.221 |
| Chromaticity Coordinate (v') | 0.493 |
| Maximum In-Situ Source Temperature Point (°C) | 70.7 |
| Maximum In-Situ Driver Case Temperature (°C) | 57.3 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Analyzer | WT1600 | E474 | 03/15/13 | 03/15/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/15/13 | 03/15/14 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/27/13 | 11/27/14 |
| Sorensen DC Power Supply | DLM150-20E | --- | --- | --- |
| Data Precision | 45 | M133 | 02/07/13 | 02/07/14 |
| Data Precision | 3600 | V124 | 02/07/13 | 02/07/14 |
| Leeds & Northrup Co. Current Shunt | 37637 | 1089 | 02/07/13 | 02/07/14 |
| LSI High Speed Mirror Goniometer | 6440 | --- | 11/21/13 | 12/21/13 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/13 | 04/17/14 |
| Cole-Palmer Hygro Thermometer | 03313-85 | T1469 | 05/31/13 | 05/31/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/13/13 | 08/13/14 |
| M-D Building Products | Smart Tool | L112 | 02/13/13 | 02/13/14 |
| Extech Hygro-Thermometer | 445703 | T1360 | 11/27/13 | 11/27/14 |
| Fluke Multimeter | 87 V | D590 | 03/28/13 | 03/28/14 |
| Fluke Temperature Meter | 53 II | D588 | 03/15/13 | 03/15/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

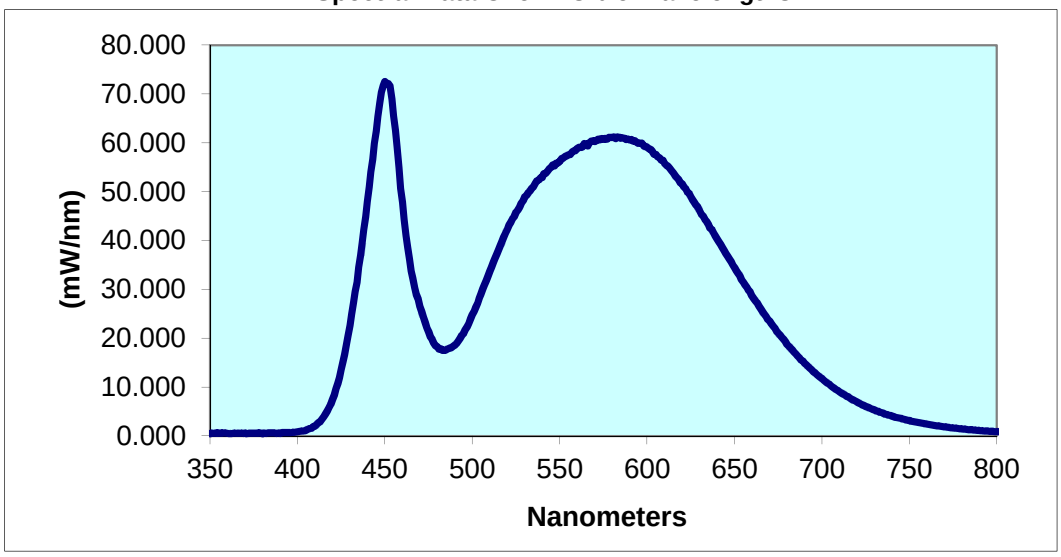
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1311071604-003 | UP | 120.0 | 842.8 | 98.76 | 0.975 | 17.56 | 4363 | 44.18 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 4338 | 80.6 | 14.0 | 0.002 | 0.366 | 0.364 | 0.221 | 0.493 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.46 | 440 | 47.96 | 530 | 48.98 | 620 | 51.58 | 710 | 9.077 |
| 355 | 0.522 | 445 | 62.4 | 535 | 51.01 | 625 | 48.99 | 715 | 7.98 |
| 360 | 0.52 | 450 | 72.57 | 540 | 52.98 | 630 | 46.11 | 720 | 6.978 |
| 365 | 0.547 | 455 | 65.55 | 545 | 54.96 | 635 | 43.24 | 725 | 6.069 |
| 370 | 0.59 | 460 | 47.83 | 550 | 56.2 | 640 | 40.34 | 730 | 5.304 |
| 375 | 0.534 | 465 | 33.77 | 555 | 57.55 | 645 | 37.37 | 735 | 4.667 |
| 380 | 0.459 | 470 | 26.55 | 560 | 58.86 | 650 | 34.39 | 740 | 4.067 |
| 385 | 0.622 | 475 | 21.25 | 565 | 59.69 | 655 | 31.47 | 745 | 3.583 |
| 390 | 0.566 | 480 | 18.24 | 570 | 60.29 | 660 | 28.67 | 750 | 3.142 |
| 395 | 0.68 | 485 | 17.65 | 575 | 60.75 | 665 | 25.96 | 755 | 2.754 |
| 400 | 0.855 | 490 | 18.64 | 580 | 61.08 | 670 | 23.52 | 760 | 2.412 |
| 405 | 1.172 | 495 | 20.98 | 585 | 60.97 | 675 | 21.03 | 765 | 2.115 |
| 410 | 2.026 | 500 | 24.81 | 590 | 60.62 | 680 | 18.72 | 770 | 1.837 |
| 415 | 3.926 | 505 | 29 | 595 | 59.88 | 685 | 16.77 | 775 | 1.637 |
| 420 | 7.44 | 510 | 33.75 | 600 | 59.1 | 690 | 14.91 | 780 | 1.449 |
| 425 | 13.54 | 515 | 38.04 | 605 | 57.64 | 695 | 13.17 | | |
| 430 | 22.65 | 520 | 42.2 | 610 | 55.76 | 700 | 11.72 | | |
| 435 | 34.56 | 525 | 45.83 | 615 | 53.91 | 705 | 10.32 | | |

**Spectral Data Over Visible Wavelengths**



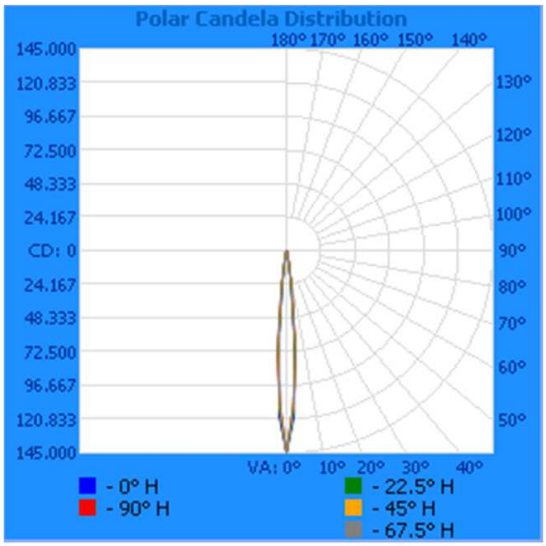
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1311071604-003 | UP | 120.0 | 842.2 | 98.74 | 0.977 | 4942 | 50.05 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 144598 | 144598 | 144598 | 144598 | 144598 |
| 5 | 60806 | 64097 | 68345 | 72110 | 74556 |
| 10 | 7221 | 7992 | 8836 | 9096 | 9425 |
| 15 | 1036 | 1183 | 1475 | 1389 | 1240 |
| 20 | 473 | 445 | 503 | 456 | 379 |
| 25 | 76 | 86 | 98 | 101 | 105 |
| 30 | 0 | 0 | 0 | 0 | 0 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

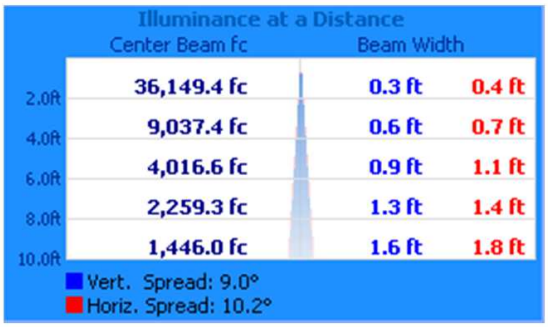


RESULTS OF TEST (cont'd)

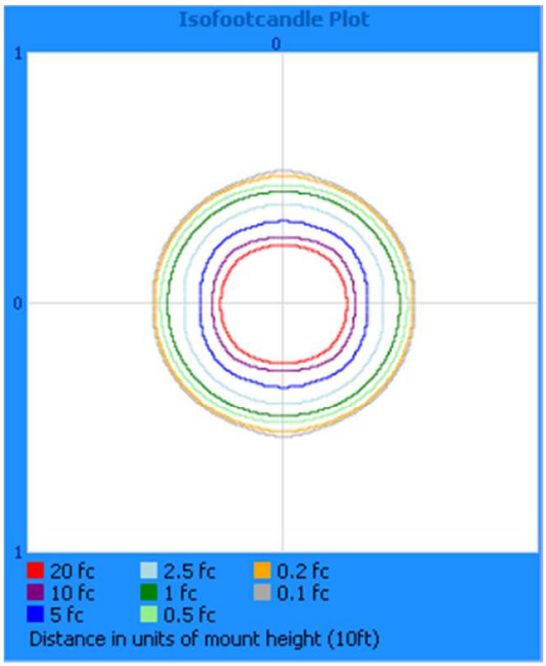
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 4942 | 100.0 |
| 0-40 | 4942 | 100.0 |
| 0-60 | 4942 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 4942 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 4942 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 4304 | 87.1 |
| 10-20 | 575.8 | 11.7 |
| 20-30 | 62.1 | 1.3 |
| 30-40 | 0.0 | 0.0 |
| 40-50 | 0.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED model identified as:

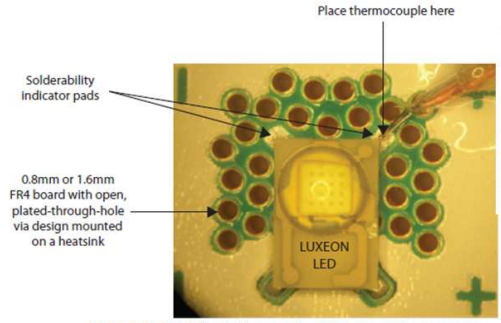


Figure 1. Solderability indicator pad and thermocouple placement.

| Nominal ANSI CCT | Forward Voltage $V_f$ [1] (V) Min. | Max. | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{j-c}$ |
|---|---|---|---|---|
| 2700K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 3000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 4000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |
| 5000K | 2.5 | 3.5 | -1.8 to -2.4 | 6 |

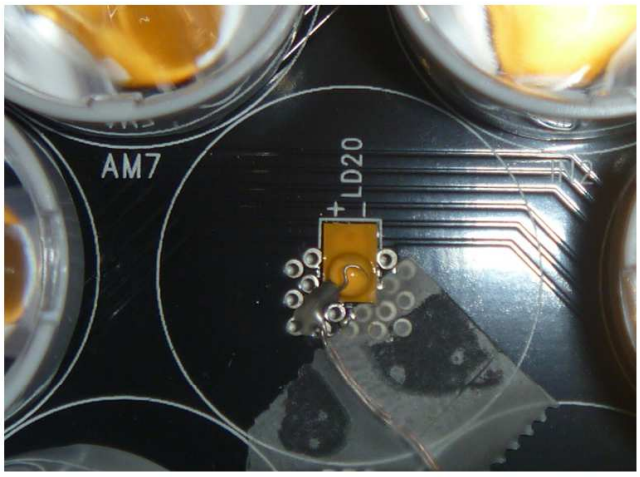
| | |
|---|---|
| LED Junction Temperature [1] | 150°C |

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 6°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.5V
Measured LED Current = 579mA
Calculated LED Wattage = Vf x Measured LED Current = 2.026W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 137.8°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1311071604-003 | 70.7 | Per diagram | 137.8 |

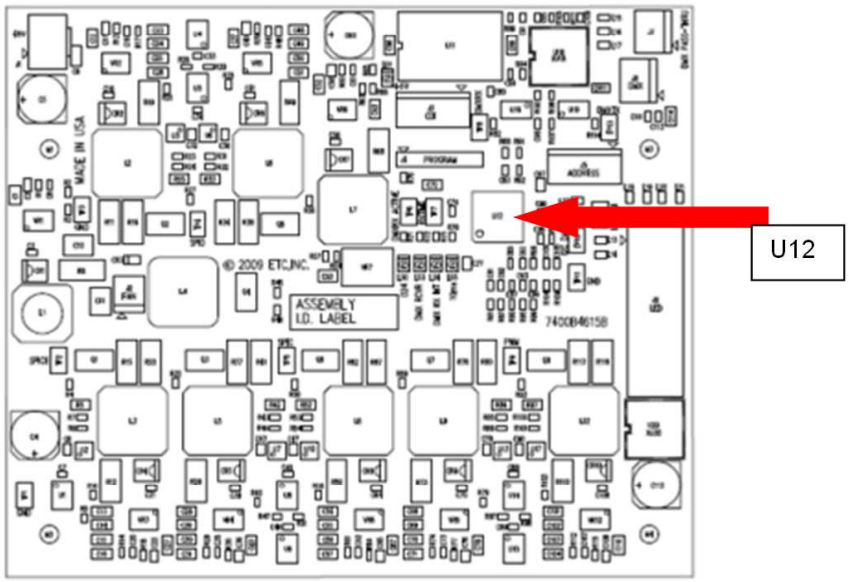
In-Situ Picture – Ts




In-Situ Picture – Ts location

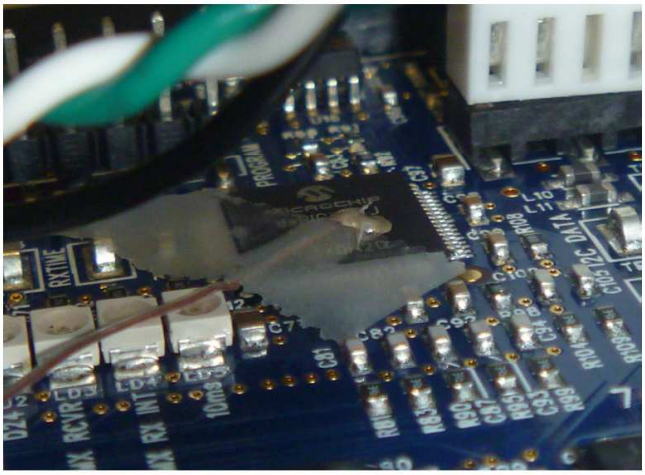
Maximum Measured Power Supply Case Temperature

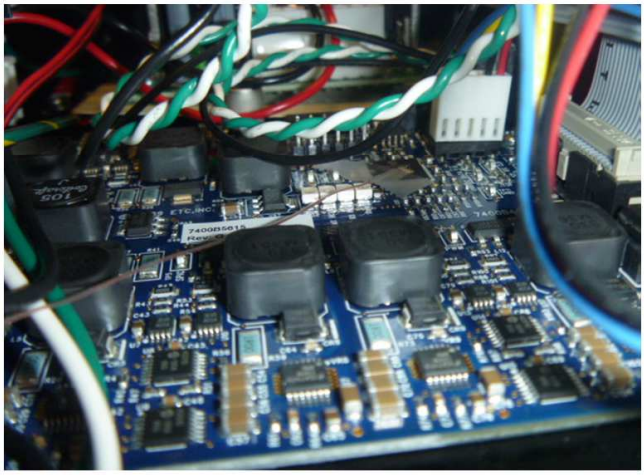
Manufacturer Supplied Documentation




| Sample No. | Maximum Measured Case Temperature (°C) | Location | Maximum Rated Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1311071604-003 | 52.9 | Per diagram | 100 |

Power Supply Picture – Tc location





PICTURE (not to scale`

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

# SECTION 7 - ColorSource Family

PAR
PAR (Deep Blue)
Spot
Spot (Deep Blue)
Linear 1

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G101701272 Date: September 17, 2014

REPORT NO. 101701272CRT-001

TEST OF ONE THEATRICAL LED LUMINAIRE

MODEL NO. CSPAR (COLORSOURCE PAR)
LED MODEL NO. LUXEON Z (LXZ1-PM01 / LXZ1-PD02 / LXZ1-PX01 / LXZ1-PB01)
DRIVER MODEL NO. SLPOWER LB115S48K

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC.
3031 PLEASANT VIEW ROAD MIDDLETON WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500532605.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

Energy Star Version 1.2 (2012): Program Requirements for Luminaires (Light Fixtures)

Energy Star Manufacturer's Guide Version 2.1 (2010): Guide for Qualifying Solid State Lighting Luminaires

DESCRIPTION OF SAMPLE: The client submitted one prototype sample of model number CSPAR (ColorSource PAR). The sample was received by Intertek on August 26, 2014, in undamaged condition and one sample was tested as received. The sample designation was CRT1408261152-001.

DATES OF TESTS: September 12, 2014 through September 17, 2014.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | CSPAR (ColorSource PAR) |
| Description: | Theatrical Led Luminaire |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 3050 | 3215 |
| Total Power (W) | 86.78 | 105.4 |
| Luminaire Efficacy (LPW) | 35.15 | 30.50 |

| Criteria | Result |
|---|---|
| Power Factor | 0.995 |
| Current ATHD % | 4.29 |
| Correlated Color Temperature (CCT - K) | 3901 |
| Color Rendering Index (CRI - Ra) | 64.8 |
| Color Rendering Index (CRI - R9) | -69.0 |
| DUV | 0.004 |
| Chromaticity Coordinate (x) | 0.382 |
| Chromaticity Coordinate (y) | 0.370 |
| Chromaticity Coordinate (u') | 0.229 |
| Chromaticity Coordinate (v') | 0.499 |
| Maximum In-Situ Source Temperature Point (°C) - Blue LED | 58.3 |
| Maximum Power Supply Case Temperature (°C) | 58.8 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date | Date Used |
|---|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 08/25/14 | 09/25/14 | 09/15/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU | 09/15/14 |
| Yokogawa Power Analyzer | WT210 | E464 | 04/17/14 | 04/17/15 | 09/15/14 |
| ExTech Hygro Thermometer | 445703 | T1357 | 11/25/13 | 11/25/14 | 09/15/14 |
| Fisher Scientific | 14-649-9 | N1405 | 08/25/14 | 08/25/15 | 09/15/14 |
| M-D Building Products | Smart Tool | L112 | 03/14/14 | 03/15/15 | 09/15/14 |
| Extech Hygro-Thermometer | 445703 | E474 | 12/30/13 | 12/30/14 | 09/17/14 |
| Fluke Multimeter | 87 | N307 | 03/20/14 | 03/20/15 | 09/17/14 |
| Fluke Temp Meter | 53 II | T1318 | 03/21/14 | 03/21/15 | 09/17/14 |
| Yokogawa Power Analyzer | WT1600 | E474 | 03/07/14 | 03/07/15 | 09/12/14 |
| LABSPHERE 3M | W/ CDS 1100 | N307 | VBU | VBU | 09/12/14 |
| Fluke Temperature Meter | 53 II | T1318 | 03/21/14 | 03/21/15 | 09/12/14 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU | 09/12/14 |
| Extech Hygro-Thermometer | 445703 | T1366 | 11/27/13 | 11/27/14 | 09/12/14 |
| SORENSEN POWER SUPPLY | XFR 150-8 | --- | VBU | VBU | 09/12/14 |
| NIST Spectral Flux Standard Source | RF1024 | --- | 09/18/10 | 100 hrs of use | 09/12/14 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

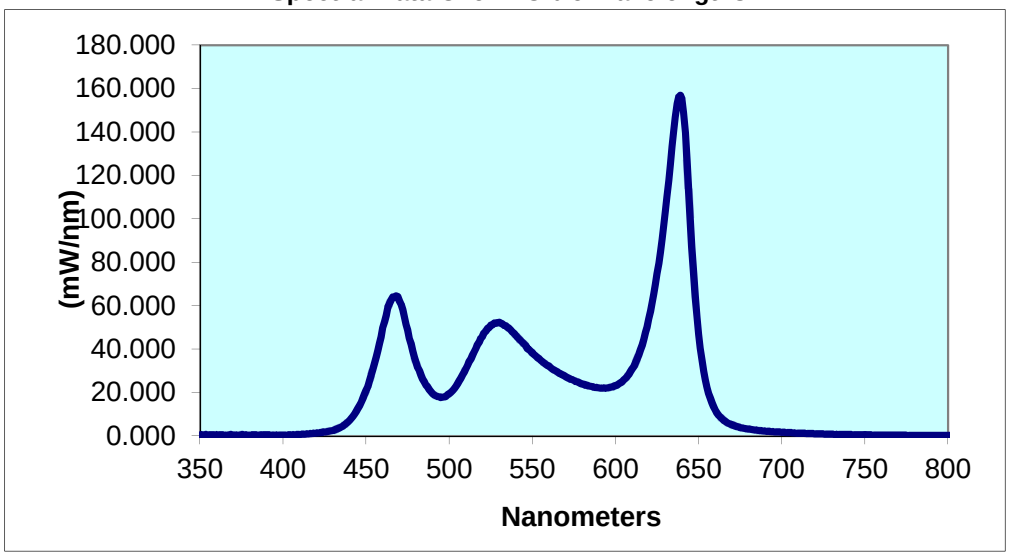
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1408261152-001 | UP | 120.0 | 727.0 | 86.78 | 0.995 | 4.29 | 3050 | 35.15 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3901 | 64.8 | -69.0 | 0.004 | 0.382 | 0.370 | 0.229 | 0.499 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.402 | 440 | 7.224 | 530 | 52.32 | 620 | 53.78 | 710 | 1.258 |
| 355 | 0.417 | 445 | 12.54 | 535 | 49.98 | 625 | 74.80 | 715 | 1.093 |
| 360 | 0.541 | 450 | 21.03 | 540 | 46.33 | 630 | 103.4 | 720 | 0.955 |
| 365 | 0.302 | 455 | 33.25 | 545 | 42.32 | 635 | 140.2 | 725 | 0.837 |
| 370 | 0.465 | 460 | 49.90 | 550 | 38.24 | 640 | 155.2 | 730 | 0.730 |
| 375 | 0.538 | 465 | 62.40 | 555 | 34.88 | 645 | 100.2 | 735 | 0.648 |
| 380 | 0.269 | 470 | 62.24 | 560 | 32.13 | 650 | 46.94 | 740 | 0.573 |
| 385 | 0.278 | 475 | 48.86 | 565 | 29.52 | 655 | 22.46 | 745 | 0.515 |
| 390 | 0.460 | 480 | 33.91 | 570 | 27.32 | 660 | 11.92 | 750 | 0.455 |
| 395 | 0.309 | 485 | 24.87 | 575 | 25.48 | 665 | 7.313 | 755 | 0.374 |
| 400 | 0.373 | 490 | 19.56 | 580 | 23.96 | 670 | 5.169 | 760 | 0.376 |
| 405 | 0.528 | 495 | 17.66 | 585 | 22.90 | 675 | 3.924 | 765 | 0.311 |
| 410 | 0.682 | 500 | 19.44 | 590 | 22.08 | 680 | 3.118 | 770 | 0.269 |
| 415 | 0.915 | 505 | 24.02 | 595 | 22.16 | 685 | 2.630 | 775 | 0.242 |
| 420 | 1.337 | 510 | 31.00 | 600 | 23.19 | 690 | 2.207 | 780 | 0.240 |
| 425 | 1.820 | 515 | 38.75 | 605 | 25.77 | 695 | 1.906 | | |
| 430 | 2.568 | 520 | 45.93 | 610 | 30.79 | 700 | 1.655 | | |
| 435 | 4.185 | 525 | 51.07 | 615 | 39.60 | 705 | 1.432 | | |

**Spectral Data Over Visible Wavelengths**



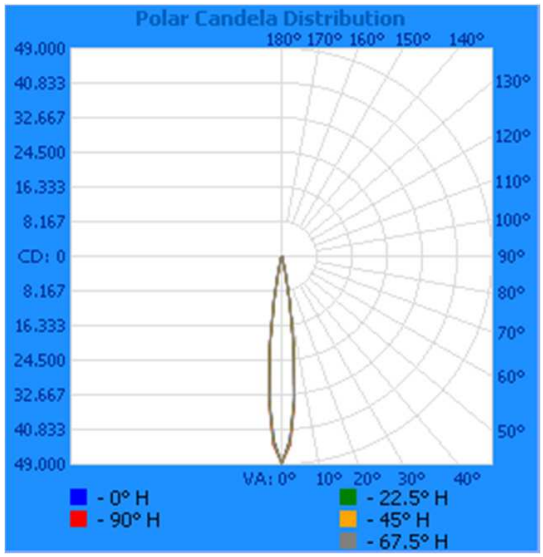
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1408261152-001 | UP | 120.0 | 880.6 | 105.4 | 0.997 | 3215 | 30.50 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 48786 | 48786 | 48786 | 48786 | 48786 |
| 5 | 33454 | 33636 | 33834 | 34066 | 34360 |
| 10 | 10643 | 10336 | 9980 | 9761 | 9725 |
| 15 | 1501 | 1375 | 1364 | 1254 | 1216 |
| 20 | 480 | 466 | 465 | 458 | 449 |
| 25 | 213 | 209 | 211 | 203 | 198 |
| 30 | 130 | 128 | 130 | 129 | 129 |
| 35 | 59 | 57 | 56 | 52 | 53 |
| 40 | 30 | 28 | 27 | 28 | 28 |
| 45 | 16 | 22 | 24 | 20 | 15 |
| 50 | 14 | 7 | 12 | 6 | 14 |
| 55 | 8 | 3 | 4 | 4 | 10 |
| 60 | 4 | 0 | 0 | 0 | 3 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 1 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

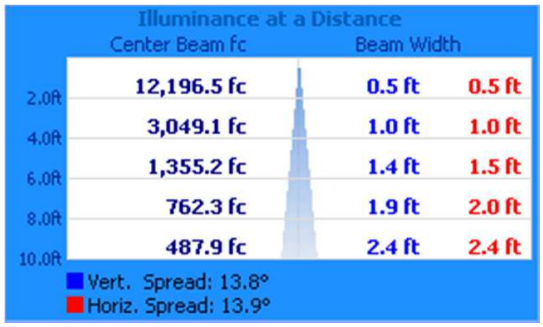


RESULTS OF TEST (cont'd)

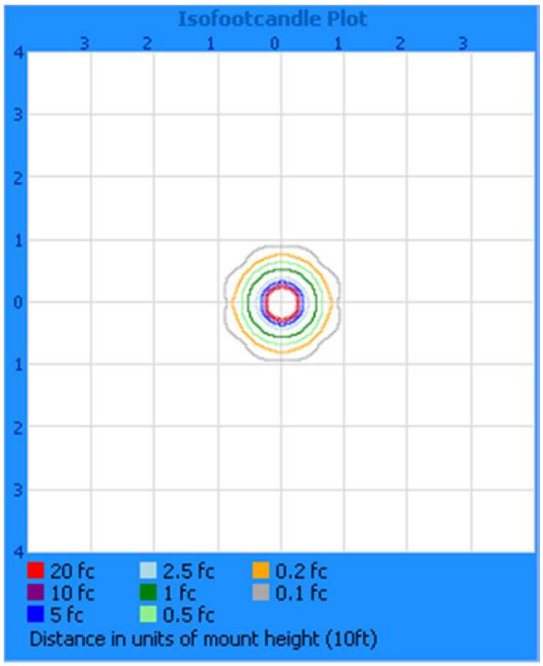
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3154 | 98.1 |
| 0-40 | 3195 | 99.4 |
| 0-60 | 3215 | 100.0 |
| 60-90 | 0.2 | 0.0 |
| 0-90 | 3215 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3215 | 100.0 |

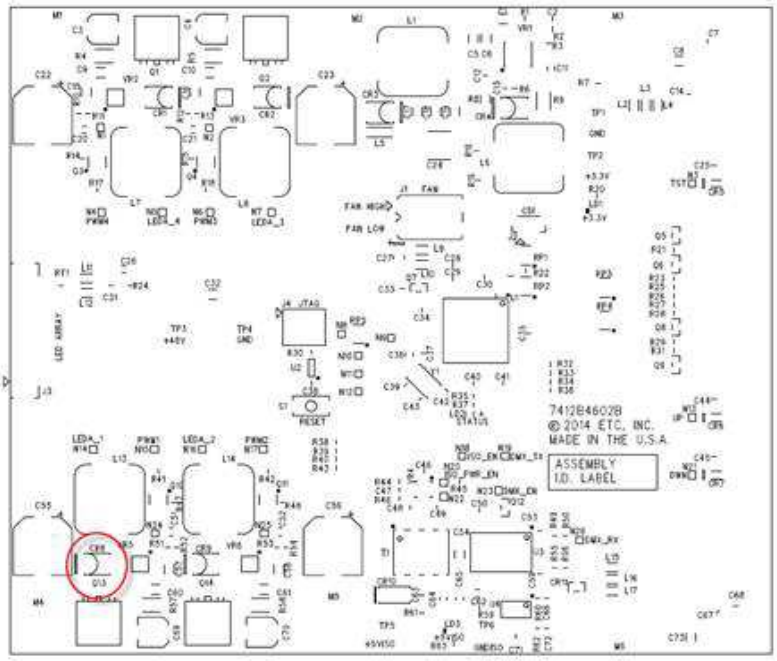
Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2403 | 74.7 |
| 10-20 | 645.2 | 20.1 |
| 20-30 | 106.6 | 3.3 |
| 30-40 | 40.4 | 1.3 |
| 40-50 | 15.3 | 0.5 |
| 50-60 | 4.6 | 0.1 |
| 60-70 | 0.2 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

Maximum Measured Power Supply Case Temperature

Manufacturer Supplied Documentation:

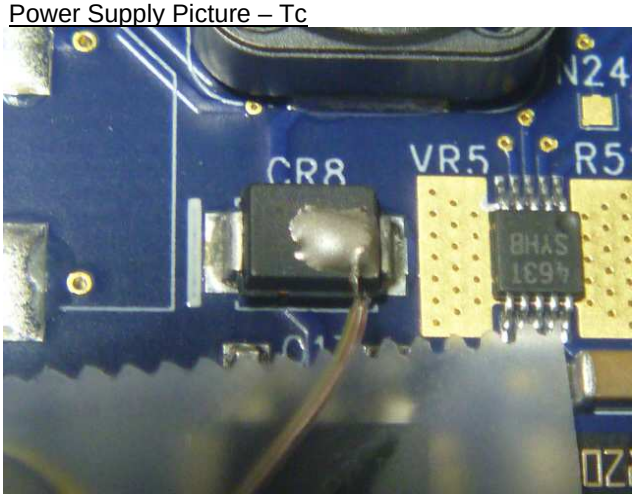


Max Temp: 100°C

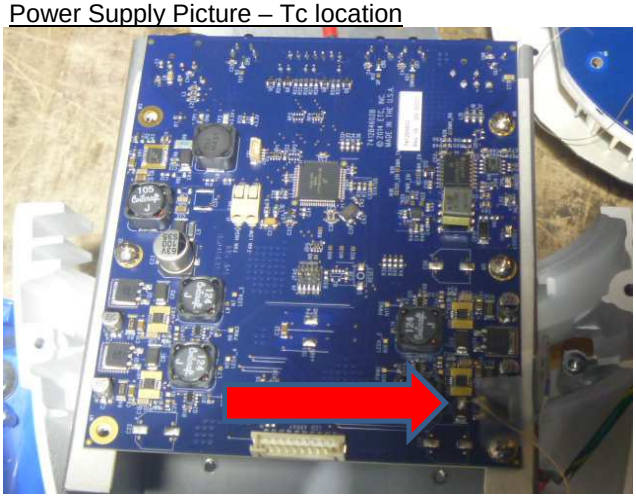
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Measured Power Supply Case Temperature (°C) | Location | Maximum Power Supply Case Temperature (°C) |
|---|---|---|---|
| CRT1408261152-001 | 58.8 | Per specs above | 100 |

Power Supply Picture – Tc




Power Supply Picture – Tc location



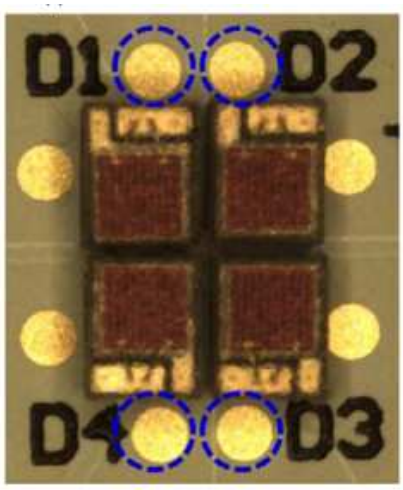
RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

LED Idientified as Luxeon Z (Blue - LXZ1-PB01)

| Parameter | Green/Cyan/ Blue/Royal Blue |
|---|---|
| DC Forward Current (mA) | 1000 |
| Peak Pulsed Forward Current (mA) | 1000 |
| ESD Sensitivity | |
| LED Junction Temperature [1] | 150°C |
| Operating Case Temperature at 500 mA | -40°C - 135°C |
| Storage Temperature | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 |
| Autoclave Conditions | 121°C at 2 ATM |




| Color | Part Number | Forward Voltage $V_f$ [1] (V) @ 500 mA | | | Typical Forward Voltage @ 700 mA $V_f$ | Typical Temperature Coefficient of Forward Voltage (mV/°C) $\Delta V_F / \Delta T_j$ | Typical Thermal Resistance Junction to Case (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|---|---|
| | | Minimum | Typical | Maximum | | | |
| Green | LXZ1-PM01 | 2.50 | 3.05 | 3.50 | 3.15 | -2 to -4 | 5 |
| Cyan | LXZ1-PE01 | 2.50 | 2.95 | 3.50 | 3.05 | -2 to -4 | 5 |
| Blue | LXZ1-PB01 | 2.50 | 3.15 | 3.50 | 3.25 | -2 to -4 | 5 |
| Royal Blue | LXZ1-PR01 | 2.50 | 2.90 | 3.50 | 2.95 | -2 to -4 | 5 |
| Red | LXZ1-PD01 | 1.75 | 2.20 | 2.75 | 2.40 | -2 to -4 | 8 |
| Hot Tested Red | LXZ1-PD02 | 1.75 | 2.15 | 2.75 | 2.35 | -2 to -4 | 8 |
| Deep Red | LXZ1-PA01 | 1.75 | 2.25 | 2.75 | 2.40 | -2 to -4 | 8 |
| Red-Orange | LXZ1-PH01 | 1.75 | 2.20 | 2.75 | 2.40 | -2 to -4 | 8 |
| Red-Orange | LXZ1-PH02 | 1.75 | 2.15 | 2.75 | 2.35 | -2 to -4 | 8 |
| Amber | LXZ1-PL01 | 1.75 | 2.15 | 2.75 | 2.35 | -2 to -4 | 8 |
| Hot Amber | LXZ1-PL03 | 1.75 | 2.10 | 2.75 | 2.30 | -2 to -4 | 8 |
| Lime | LXZ1-PX01 | 2.50 | 2.85 | 3.50 | 2.80 | -2 to -4 | 5 |

RESULTS OF TEST (cont'd)

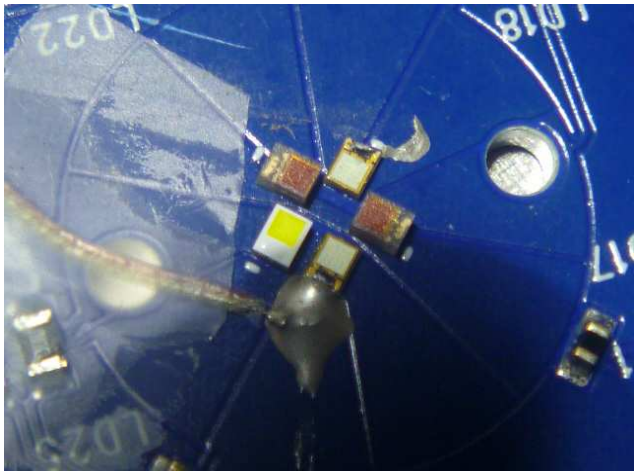
In-Situ Maximum Measured LED Source Temperature

Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 5°C/W
Maximum Forward Voltage (Vf) from LED specification = 3.25V
Measured LED Current = 570mA
Calculated LED Wattage = Vf x Measured LED Current = 1.852W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 140.7°C

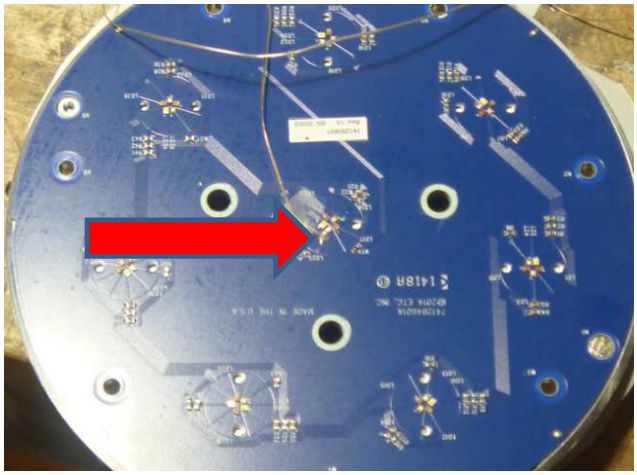
Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| CRT1408261152-001 | 58.3 | Per specs above | 140.7 |

In-Situ Picture – Ts



In-Situ Picture – Ts locatior



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Vladimir Kozak
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Manager
Lighting Division

Attachment: None

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G102501090 Date: April 15, 2016

REPORT NO. 102501090CRT-003

TEST OF ONE LED LUMINAIRE

MODEL NO. COLORSOURCE PAR DEEP BLUE
LED MODEL NO. LUMILEDS LUXEON Z
DRIVER MODEL NO. LB115S

RENDERED TO:

ELECTRONIC THEATRE CONTROLS INC
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

TESTS: Electrical and Photometric tests as required to the IESNA test standard and Insitu Temperature Measurement Tests.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION The testing performed was authorized by signed quote number Qu-00675915.

STANDARDS USED:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting
ANSI NEMA ANSLG C78.377: 2015: Specifications of the Chromaticity of Solid State Lighting Products
UL 1598-2009: Standard for Safety - Luminaires.

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number ColorSource Par Deep Blue. The sample was received by Intertek on March 22, 2016 in undamaged condition and one sample was tested as received. The sample designation was CRT1603221320-001.

DATE OF TESTS: March 30, 2016 through April 13, 2016.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY:

| MODEL NO. ColorSource Par Deep Blue |
|---|
| DESCRIPTION: LED LUMINAIRE |

| | Results | |
|---|---|---|
| Criteria | Integrating Sphere | Goniophotometer |
| Light Output (Lumens) | 3069 | 2965 |
| Total Power (W) | 105.7 | 105.6 |
| Lumen Efficacy (Lm/W) | 29.00 | 28.10 |
| Power Factor ( ) | 0.996 | 0.997 |

| Criteria | Results |
|---|---|
| Current ATHD (%) | 4.02 |
| Correlated Color Temp. (CCT-K) | 21854 |
| Color Rendering Index (CRI - Ra) | 29.3 |
| CRI - R9 | -271.9 |
| DUV ( ) | 0.052 |
| Chromaticity Coordinate (x) | 0.293 |
| Chromaticity Coordinate (y) | 0.218 |
| Chromaticity Coordinate (u') | 0.233 |
| Chromaticity Coordinate (v') | 0.390 |
| Maximum LED Source In-Situ Temperature (°C) | 84.1 |
| Maximum LED Source In-Situ Temperature (°C) | 80.2 |
| Maximum LED Source In-Situ Temperature (°C) | 83.9 |
| Maximum LED Source In-Situ Temperature (°C) | 81.9 |
| Maximum Driver Source In-Situ Temperature (°C) | 39.1 |

Note: See page 8 for LED and measured insitu temperature matching.

EQUIPMENT LIST

| Equipment Used | Model No. | Control No. | Last Cal. | Cal. Due |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 04/04/16 | 05/04/16 |
| Elgar AC Power Supply | CW1251 | --- | VBU | VBU |
| Sorenson DC Power Supply | XG 150-10 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/20/15 | 04/20/16 |
| ExTech Hygro Thermometer | 445703 | T1359 | 02/04/16 | 02/04/17 |
| NIST Luminous Intensity Standard Source | NBS10322 | N1427 | 12/12/14 | 12/12/16 |
| NIST Luminous Intensity Standard Source | NBS10215 | N1432 | 12/12/14 | 12/12/16 |
| NIST Luminous Intensity Standard Source | 960629-3 | N1428 | 12/12/14 | 12/12/16 |
| NIST Luminous Flux Standard Source | NBS10428 | N1424 | 12/17/14 | 12/17/16 |
| Elgar AC Power Supply | CW1251 | --- | VBU | VBU |
| Sorenson DC Power Supply | XFR 150-8 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | ZE475 | 06/09/15 | 06/09/16 |
| Fisher Scientific Stopwatch | 14-649-9 | N1316 | 01/15/16 | 01/15/17 |
| Fluke Thermometer | 52 | T801 | 11/18/15 | 11/18/16 |
| 3M Integrating Sphere Spectrometer System | CDS 1100 | --- | 03/02/16 | 04/02/16 |
| Pearson Current Monitor | 411 | A203 | 06/26/15 | 06/15/16 |
| Secondary Spectral Intensity Standard Source | BS5186 | RF5186 | 01/27/16 | 01/27/17 |
| Secondary Luminous Flux Standard Source | BS3616 | -- | 01/27/16 | 01/27/17 |
| Secondary Luminous Flux Standard Source | BS4116 | -- | 01/27/16 | 01/27/17 |
| Extech Hygro-Thermometer | 445703 | T1366 | 02/04/16 | 02/04/17 |
| Fluke Multimeter | 87 V | D590 | 04/22/15 | 04/22/16 |
| Fluke Temperature Meter | 53 II | D588 | 04/22/15 | 04/22/16 |

TEST METHODS:

Seasoning in Sample Orientation – LED Products
No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method
A Labsphere Model CDS 1100 CCD Array Spectroradiometer and two meter or ten foot sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Stabilization procedures to LM-79 were followed. Electrical measurements including voltage, current, and power were measured using a power analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical measurements – Distribution Method
A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for the SSL sample.

Ambient temperature was measured equal to the height of the sample mounted on the goniometer equipment. The SSL sample was operated on the client provided driver at rated input volts in its designated orientation. The SSL sample was allowed to stabilize for at least thirty minutes before measurements were made. Stabilization procedures to LM-79 were followed. Electrical measurements including voltage, current, and power were measured using a power analyzer.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature
Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598, UL 153, or UL 1993 as applicable.

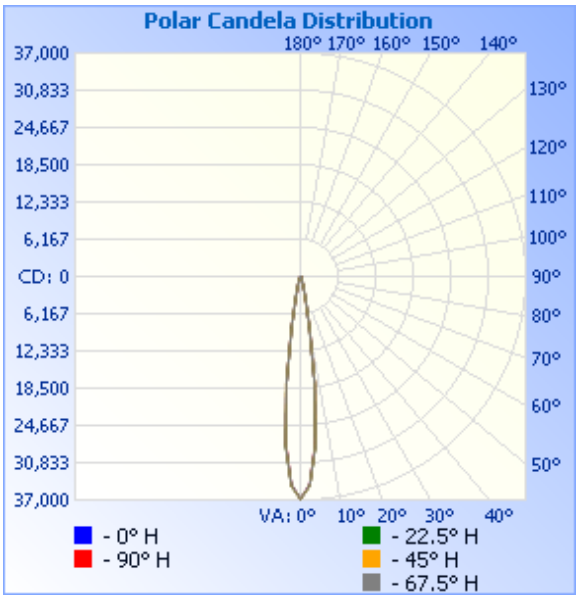
RESULTS:

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Control No. | Base Orientation | Input Voltage (VAC) | Input Current (mA) | Input Power (W) | Input Power Factor ( ) | Light Output (Lumens) | Lumen Efficacy (lm/W) |
|---|---|---|---|---|---|---|---|
| CRT1603221320-001 | Base Up | 120.08 | 882.40 | 105.6 | 0.997 | 2965 | 28.10 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 36817 | 36817 | 36817 | 36817 | 36817 |
| 5 | 27948 | 28085 | 28076 | 28132 | 27759 |
| 10 | 8390 | 8391 | 8246 | 8276 | 8470 |
| 15 | 2035 | 2058 | 2076 | 2066 | 2084 |
| 20 | 796 | 778 | 783 | 783 | 786 |
| 25 | 264 | 259 | 255 | 259 | 263 |
| 30 | 134 | 131 | 129 | 127 | 130 |
| 35 | 81 | 78 | 75 | 76 | 76 |
| 40 | 59 | 58 | 55 | 54 | 59 |
| 45 | 47 | 46 | 49 | 50 | 48 |
| 50 | 46 | 43 | 47 | 43 | 51 |
| 55 | 45 | 37 | 43 | 38 | 45 |
| 60 | 40 | 23 | 26 | 26 | 35 |
| 65 | 27 | 8 | 10 | 8 | 20 |
| 70 | 15 | 0 | 0 | 0 | 7 |
| 75 | 3 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

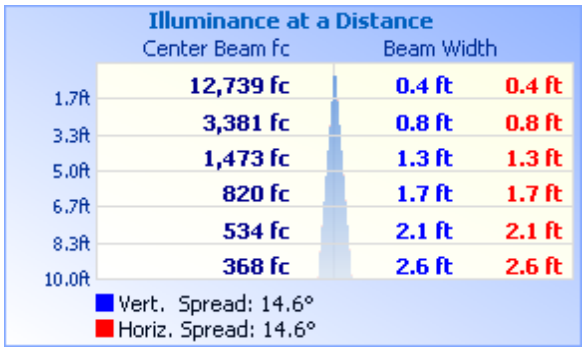


RESULTS:

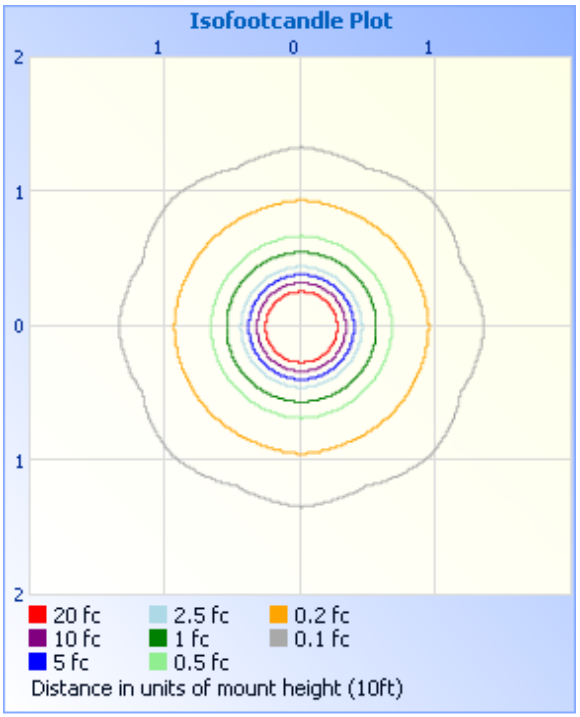
Illumination Plots

Mounting Height: 10ft

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 2827 | 95.3 |
| 0-40 | 2879 | 97.1 |
| 0-60 | 2952 | 99.5 |
| 60-90 | 13.7 | 0.5 |
| 0-90 | 2965 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 2965 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 1957 | 66.0 |
| 10-20 | 724.6 | 24.4 |
| 20-30 | 145.3 | 4.9 |
| 30-40 | 51.2 | 1.7 |
| 40-50 | 37.7 | 1.3 |
| 50-60 | 35.3 | 1.2 |
| 60-70 | 13.0 | 0.4 |
| 70-80 | 0.8 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS:

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Sphere Method

| Intertek Control No. | Base Orientation | Input Voltage (VAC) | Input Current (mA) | Input Power (W) | Input Power Factor ( ) | Current ATHD (%) |
|---|---|---|---|---|---|---|
| CRT1603221320-001 | Base Up | 120.01 | 884.3 | 105.69 | 0.996 | 4.02 |

| Light Output (Lumens) | Lumen Efficacy (lm/W) | Correlated Color Temperature - CCT (K) | CRI -Ra | CRI -R9 | DUV ( ) |
|---|---|---|---|---|---|
| 3069 | 29.00 | 21854 | 29.3 | -271.9 | 0.052 |

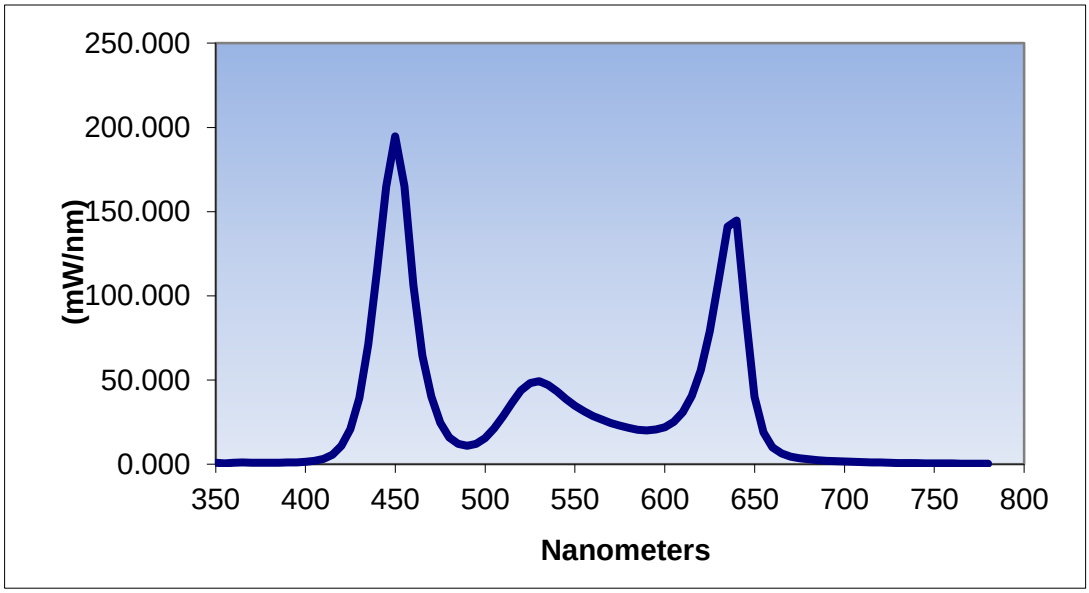
| CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|
| 0.293 | 0.218 | 0.233 | 0.390 |

RESULTS:

**Spectral Distribution Over Visible Wavelengths**

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.735 | 460 | 106.351 | 570 | 24.339 | 680 | 2.825 |
| 355 | 0.490 | 465 | 64.479 | 575 | 22.787 | 685 | 2.365 |
| 360 | 0.838 | 470 | 40.223 | 580 | 21.461 | 690 | 2.042 |
| 365 | 0.982 | 475 | 24.479 | 585 | 20.447 | 695 | 1.757 |
| 370 | 0.768 | 480 | 15.755 | 590 | 20.093 | 700 | 1.536 |
| 375 | 0.871 | 485 | 11.967 | 595 | 20.485 | 705 | 1.362 |
| 380 | 0.848 | 490 | 10.859 | 600 | 21.891 | 710 | 1.167 |
| 385 | 0.885 | 495 | 11.972 | 605 | 25.161 | 715 | 1.024 |
| 390 | 1.007 | 500 | 15.365 | 610 | 30.927 | 720 | 0.909 |
| 395 | 0.974 | 505 | 21.100 | 615 | 40.492 | 725 | 0.781 |
| 400 | 1.351 | 510 | 28.283 | 620 | 55.702 | 730 | 0.714 |
| 405 | 1.906 | 515 | 36.383 | 625 | 78.747 | 735 | 0.609 |
| 410 | 3.040 | 520 | 43.800 | 630 | 109.412 | 740 | 0.552 |
| 415 | 5.629 | 525 | 48.220 | 635 | 141.065 | 745 | 0.467 |
| 420 | 10.801 | 530 | 49.289 | 640 | 144.754 | 750 | 0.437 |
| 425 | 20.742 | 535 | 47.081 | 645 | 89.695 | 755 | 0.389 |
| 430 | 39.187 | 540 | 43.268 | 650 | 39.958 | 760 | 0.349 |
| 435 | 71.154 | 545 | 38.718 | 655 | 18.774 | 765 | 0.330 |
| 440 | 115.930 | 550 | 34.680 | 660 | 9.988 | 770 | 0.282 |
| 445 | 164.807 | 555 | 31.502 | 665 | 6.282 | 775 | 0.241 |
| 450 | 194.609 | 560 | 28.617 | 670 | 4.479 | 780 | 0.235 |
| 455 | 164.999 | 565 | 26.379 | 675 | 3.459 | | |

**Spectral Data Over Visible Wavelengths**



RESULTS:

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj)- LXZ1-PM01 = 150°C
Maximum Junction Temperature from LED specification (Tj)- LXZ1-PX01 = 135°C
Maximum Junction Temperature from LED specification (Tj)- LXZ1-PD02 = 135°C
Maximum Junction Temperature from LED specification (Tj)- LXZ1-PR01 = 150°C

Thermal Resistance Formula from LED specification- LXZ1-PM01 = 5°C/W
Thermal Resistance Formula from LED specification- LXZ1-PX01 = 5°C/W
Thermal Resistance Formula from LED specification- LXZ1-PD02 = 8°C/W
Thermal Resistance Formula from LED specification- LXZ1-PR01 = 5°C/W

Maximum Forward Voltage (Vf) from LED specification- LXZ1-PM01 = 3.5V
Maximum Forward Voltage (Vf) from LED specification- LXZ1-PX01 = 3.5V
Maximum Forward Voltage (Vf) from LED specification- LXZ1-PD02 = 2.75V
Maximum Forward Voltage (Vf) from LED specification- LXZ1-PR01 = 3.5V

Measured LED Current- LXZ1-PM01 = 955mA
Measured LED Current- LXZ1-PX01 = 665mA
Measured LED Current- LXZ1-PD02 = 670mA
Measured LED Current- LXZ1-PR01 = 958mA

Calculated LED Wattage- LXZ1-PM01 = Vf x Measured LED Current = 3.343W
Calculated LED Wattage- LXZ1-PX01 = Vf x Measured LED Current = 2.328W
Calculated LED Wattage- LXZ1-PD02 = Vf x Measured LED Current = 1.843W
Calculated LED Wattage- LXZ1-PR01 = Vf x Measured LED Current = 3.353W

Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance)- LXZ1-PM01= 133.3°C
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance)- LXZ1-PX01 = 123.4°C
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance)- LXZ1-PD02= 120.3°C
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance)- LXZ1-PR01 = 133.2°C

RESULTS:

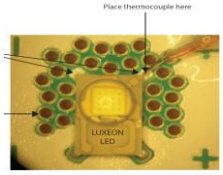
LED Source Temperature Manufacturer's Spec

Table 3. Electrical and thermal characteristics for LUXEON Z Colors at 500mA at junction temperature.

| PART NUMBER | FORWARD VOLTAGE (V) [1] | | | TYPICAL TEMPERATURE COEFFICIENT OF FORWARD VOLTAGE [2] (mV/°C) | TYPICAL THERMAL RESISTANCE — JUNCTION TO SOLDER PAD (°C/W) |
|---|---|---|---|---|---|
| | MINIMUM | TYPICAL | MAXIMUM | | |
| LXZ1-PA01 | 1.75 | 2.20 | 2.75 | -2 to -4 | 8 |
| LXZ1-PD01 | 1.75 | 2.20 | 2.75 | -2 to -4 | 8 |
| LXZ1-PD02 | 1.75 | 2.20 | 2.75 | -2 to -4 | 8 |
| LXZ1-PH01 | 1.75 | 2.20 | 2.75 | -2 to -4 | 8 |
| LXZ1-PH02 | 1.75 | 2.20 | 2.75 | -2 to -4 | 8 |
| LXZ1-PL01 | 1.75 | 2.15 | 2.75 | -2 to -4 | 8 |
| LXZ1-PL02 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |
| LXZ1-PL03 | 1.75 | 2.25 | 2.75 | -2 to -4 | 8 |
| LXZ1-PX01 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |
| LXZ1-PM01 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |
| LXZ1-PE01 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |
| LXZ1-PB01 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |
| LXZ1-PR01 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |

| PARAMETER | GREEN/CYAN/ BLUE/ROYAL BLUE | COLD-TESTED RED/DEEP-RED/ RED-ORANGE/AMBER |
|---|---|---|
| DC Forward Current [1,2] | 1000mA | 700mA |
| Peak Pulsed Forward Current [1,3] | 1000mA | 700mA |
| LED Junction Temperature [1] (DC & Pulse) | 150°C | 135°C |
| ESD Sensitivity (ANSI/ESDA/JEDEC JS-001-2012) | Class 3B | Class 3B |
| Operating Case Temperature [1] | -40°C to 135°C | -40°C to 120°C |
| LED Storage Temperature | -40°C to 135°C | -40°C to 135°C |
| Soldering Temperature | JEDEC 020c 260°C | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 | 3 |
| Autoclave Conditions | -121°C at 2 ATM | 100% Relative Humidity for 96 hours Maximum |
| Reverse Voltage ($V_{reverse}$) | LUXEON LEDs are not designed to be driven in reverse bias | |

| PARAMETER | HOT-TESTED RED/ RED-ORANGE/AMBER [4] | PC AMBER/LIME |
|---|---|---|
| DC Forward Current [1,2] | 1000/700mA | 700mA |
| Peak Pulsed Forward Current [1,3] | 1000/700mA | 700mA |
| LED Junction Temperature [1] (DC & Pulse) | 120°C/135°C | 135°C |
| ESD Sensitivity (ANSI/ESDA/JEDEC JS-001-2012) | Class 3B | Class 3B |
| Operating Case Temperature [1] | -40°C to 120°C | -40°C to 110°C |
| LED Storage Temperature | -40°C to 120°C | -40°C to 135°C |
| Soldering Temperature | JEDEC 020c 260°C | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 | 3 |
| Autoclave Conditions | 100% Relative Humidity for 96 hours Maximum | |
| Reverse Voltage ($V_{reverse}$) | LUXEON LEDs are not designed to be driven in reverse bias | |



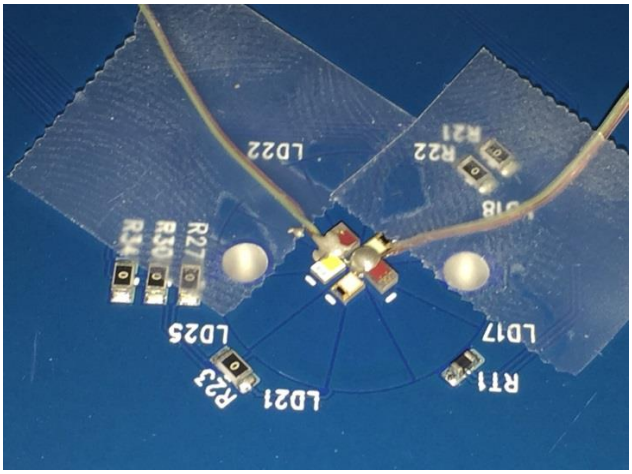
RESULTS:

Maximum Measured Manufacturer Designated Source Temperature

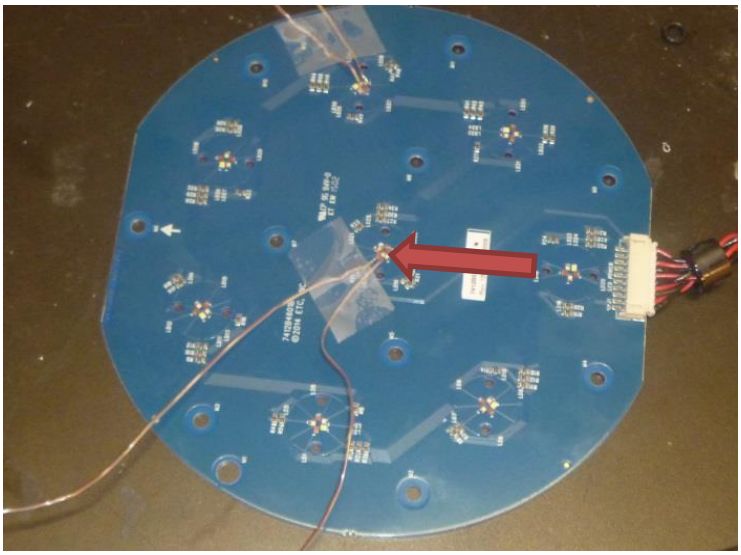
Location of Thermocouple: Per Diagram

| Intertek Sample No. | Maximum Measured Source Temperature | LED Measured | Maximum Rated Source Temperature |
|---|---|---|---|
| CRT1603221320-001 | 84.1 | LXZ1-PM01 | 133.3 |
| | 80.2 | LXZ1-PX01 | 123.4 |
| | 83.9 | LXZ1-PD02 | 120.3 |
| | 81.9 | LXZ1-PR01 | 133.2 |

In-Situ Picture – Ts
Lime and Red



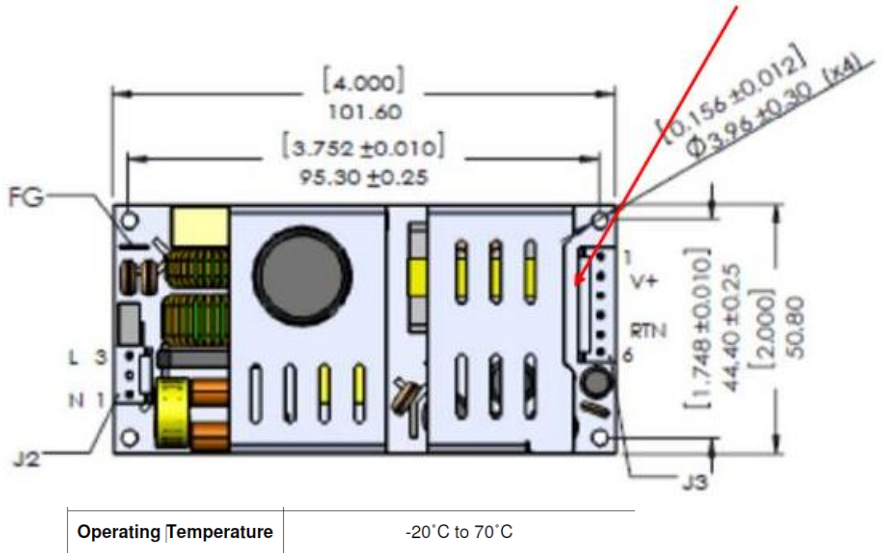

In-Situ Picture – Ts location
Green and Blue



RESULTS:

In-Situ Maximum Measured Driver Source Temperature

Driver Source Manufacturer's Spec
Driver Identified As: LB115S

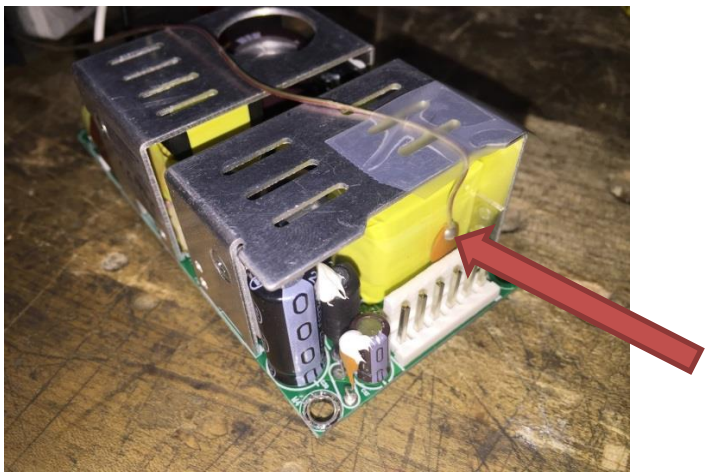



| | |
|---|---|
| Operating Temperature | -20˚C to 70˚C |

cap is located below between the heatsink and the 6 pin connector location

Maximum Measured Manufacturer Designated Driver Source Temperature

| Intertek Sample No. | Maximum Measured Driver Temperature °C | Location of Thermocouple | Maximum Rated Driver Temperature °C |
|---|---|---|---|
| CRT1603221320-001 | 39.1 | Per Diagram | 70.0 |



PRODUCT PICTURE:

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Ryan Siddon

Ryan Siddon
Engineer
Lighting Division

Attachments: IES File - CRT1603221320-001

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G102049957 Date: April 22, 2014

REPORT NO. 102049957CRT-002

TEST OF ONE CORD-CONNECTED LED STAGE AND STUDIO LUMINAIRE

MODEL NO. COLORSOURCE SPOT
DRIVER MODEL NO. SL POWER ELECTRONICS CORP. LB240S56K
LED MODEL NO. LUMILEDS LUXEON REBEL LXML-PM01-0090
LED MODEL NO. LUMILEDS LUXEON REBEL LXML-PX02-0000
LED MODEL NO. LUMILEDS LUXEON REBEL LXM5- PD01-0050/60/80
LED MODEL NO. LUMILEDS LUXEON REBEL LXML-PB02-0060/70/80

RENDERED TO

ELECTRONIC THEATRE CONTROLS, INC.
3031 PLEASANT VIEW ROAD
MIDDLETON, WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number 500585945.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

UL 1598-2009: Underwriters Laboratories Inc. Standard for Safety - Luminaires.

DESCRIPTION OF SAMPLE: The client submitted one production sample of model ColorSource SPOT with lens tube. The sample was received by Intertek on April 1, 2015, in undamaged condition and one sample was tested as received. The sample designations were CRT1504011219-001 (sample) and CRT1504011219-002 (lens tube).

DATES OF TESTS: April 20, 2015 through April 22, 2015.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | ColorSource SPOT |
| Description: | Cord-Connected LED Stage and Studio Luminaire |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 5936 | 6000 |
| Total Power (W) | 169.6 | 169.8 |
| Lumen Efficacy (LPW) | 35.00 | 35.34 |

| Criteria | Result |
|---|---|
| Power Factor at 120Vac | 0.992 |
| Power Factor at 240Vac | 0.938 |
| Current ATHD % at 120Vac | 6.44 |
| Current ATHD % at 240Vac | 13.23 |
| Correlated Color Temperature (CCT - K) | 5237 |
| Color Rendering Index (CRI - Ra) | 60.0 |
| Color Rendering Index (CRI - R9) | -95.0 |
| DUV | 0.013 |
| Chromaticity Coordinate (x) | 0.337 |
| Chromaticity Coordinate (y) | 0.321 |
| Chromaticity Coordinate (u') | 0.218 |
| Chromaticity Coordinate (v') | 0.468 |
| Maximum In-Situ Source Temperature Point (°C) | 72.9 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date |
|---|---|---|---|---|
| Yokogawa Power Analyzer | WT1600 | E474 | 03/06/15 | 03/06/16 |
| 3M Integrating Sphere | W/ CDS 1100 | N307 | VBU | VBU |
| Fluke Temperature Meter | 53 II | T1318 | 03/18/15 | 03/18/16 |
| Elgar Power Supply | CW1251 | --- | --- | --- |
| Extech Hygro-Thermometer | 445703 | T1366 | 12/10/14 | 12/10/15 |
| SORENSEN POWER SUPPLY | XFR 150-8 | --- | VBU | VBU |
| NIST Spectral Flux Standard Source | RF1482 | N1422 | 12/17/14 | 100 hrs of use |
| LSI High Speed Mirror Goniometer | 6440 | --- | 04/08/15 | 05/08/15 |
| Elgar Power Supply | CW1251 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/20/15 | 04/20/16 |
| ExTech Hygro Thermometer | 445703 | T1357 | 12/10/14 | 12/10/15 |
| Fisher Scientific Stopwatch | 14-649-9 | N1405 | 08/25/14 | 08/25/15 |
| M-D Building Products Digital Level | Smart Tool | L112 | 03/25/15 | 03/25/16 |
| Extech Hygro-Thermometer | 445703 | T1366 | 12/10/14 | 12/10/15 |
| Fluke Multimeter | 87 | D589 | 03/17/15 | 03/17/16 |
| Fluke Temperature Meter | 53 II | N1324 | 03/18/15 | 03/18/16 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

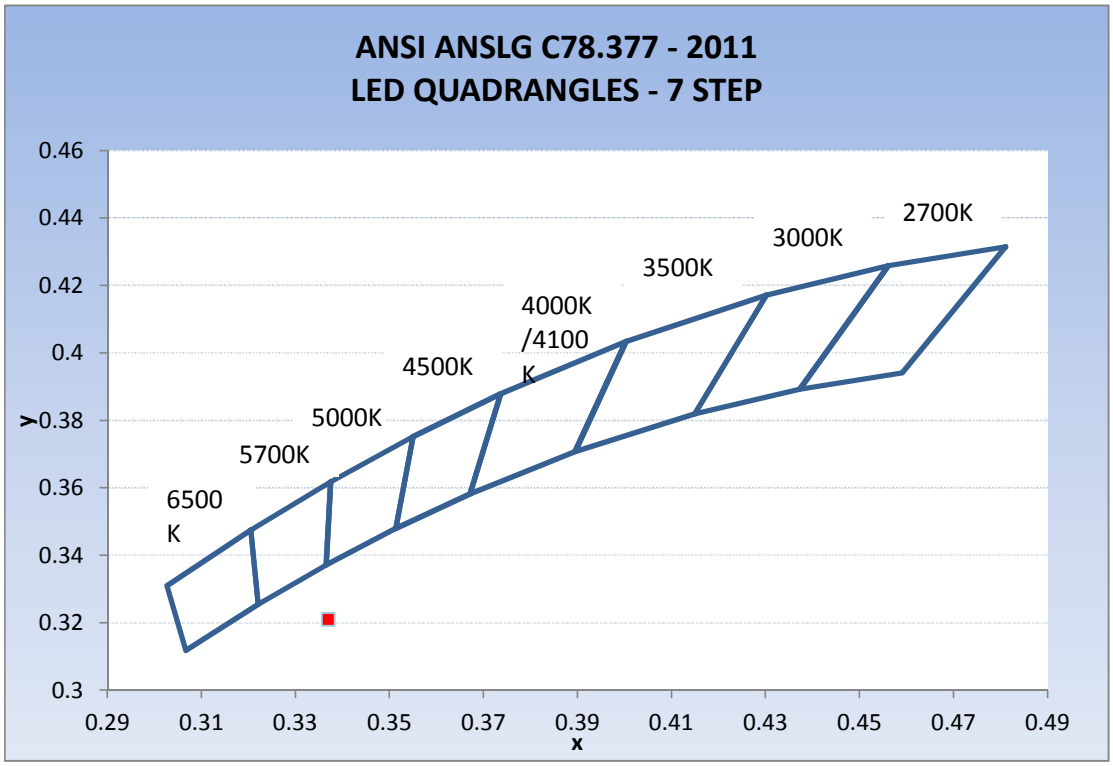
RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| CRT1504011219-001,-00 | UP | 120.0 | 1424 | 169.6 | 0.992 | 6.44 | 5936 | 35 |
| | | 240.0 | 734.5 | 165.4 | 0.938 | 13.23 | | |

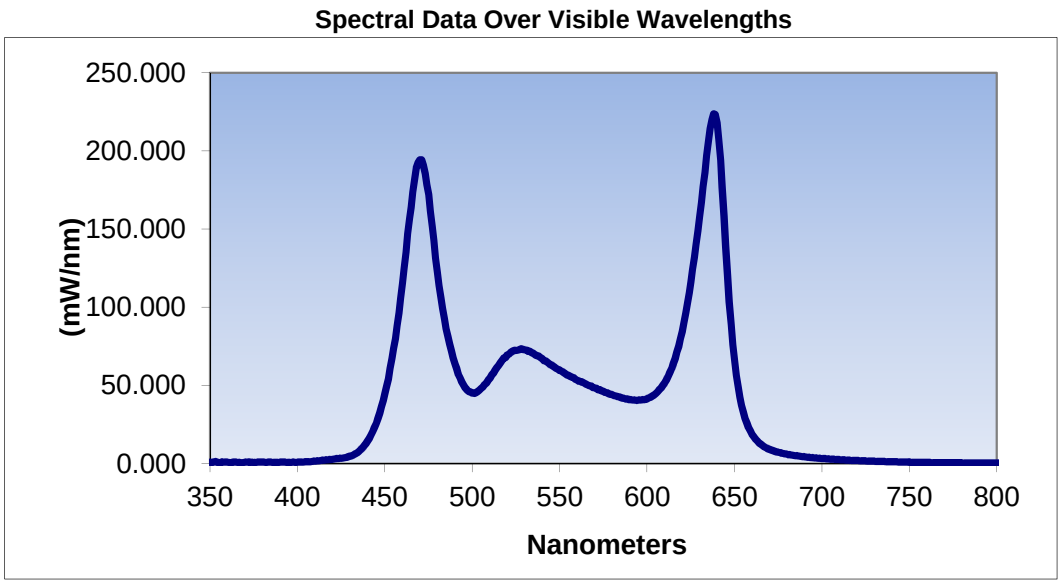
| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 5237 | 60.0 | -95.0 | 0.013 | 0.337 | 0.321 | 0.218 | 0.468 |

**ANSI C78.377 SSL Chromaticity (2011 Version)**



Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.919 | 440 | 14.226 | 530 | 72.820 | 620 | 84.555 | 710 | 2.492 |
| 355 | 0.581 | 445 | 25.772 | 535 | 70.552 | 625 | 116.194 | 715 | 2.199 |
| 360 | 0.938 | 450 | 44.970 | 540 | 67.184 | 630 | 157.681 | 720 | 1.920 |
| 365 | 0.974 | 455 | 73.555 | 545 | 63.179 | 635 | 206.537 | 725 | 1.665 |
| 370 | 0.841 | 460 | 115.606 | 550 | 59.767 | 640 | 217.777 | 730 | 1.469 |
| 375 | 0.770 | 465 | 163.926 | 555 | 56.491 | 645 | 139.080 | 735 | 1.292 |
| 380 | 0.850 | 470 | 194.349 | 560 | 53.310 | 650 | 65.340 | 740 | 1.098 |
| 385 | 0.886 | 475 | 172.240 | 565 | 50.851 | 655 | 32.916 | 745 | 0.988 |
| 390 | 0.899 | 480 | 122.136 | 570 | 48.389 | 660 | 18.783 | 750 | 0.869 |
| 395 | 0.790 | 485 | 86.279 | 575 | 46.274 | 665 | 12.421 | 755 | 0.769 |
| 400 | 0.857 | 490 | 63.774 | 580 | 44.207 | 670 | 9.117 | 760 | 0.687 |
| 405 | 1.148 | 495 | 50.153 | 585 | 42.288 | 675 | 7.244 | 765 | 0.609 |
| 410 | 1.561 | 500 | 45.233 | 590 | 40.874 | 680 | 6.040 | 770 | 0.521 |
| 415 | 2.184 | 505 | 48.100 | 595 | 40.720 | 685 | 5.024 | 775 | 0.453 |
| 420 | 2.831 | 510 | 54.560 | 600 | 41.581 | 690 | 4.358 | 780 | 0.442 |
| 425 | 3.421 | 515 | 62.886 | 605 | 45.069 | 695 | 3.784 | | |
| 430 | 4.761 | 520 | 69.486 | 610 | 51.669 | 700 | 3.235 | | |
| 435 | 7.792 | 525 | 72.513 | 615 | 63.913 | 705 | 2.878 | | |



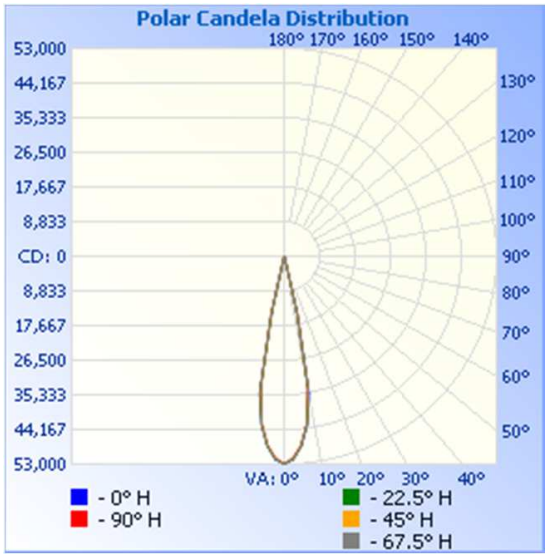
RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (Lumens Per Watt) |
|---|---|---|---|---|---|---|---|
| CRT1504011219-001,-00 | UP | 120.0 | 1425 | 169.8 | 0.993 | 6000 | 35.34 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 52885 | 52885 | 52885 | 52885 | 52885 |
| 5 | 48565 | 48455 | 48290 | 48418 | 48217 |
| 10 | 35491 | 34577 | 34623 | 34508 | 33579 |
| 15 | 442 | 471 | 570 | 777 | 1279 |
| 20 | 113 | 110 | 109 | 107 | 111 |
| 25 | 53 | 50 | 49 | 48 | 50 |
| 30 | 22 | 22 | 19 | 20 | 20 |
| 35 | 2 | 2 | 2 | 3 | 3 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |
| 95 | 0 | 0 | 0 | 0 | 0 |
| 100 | 0 | 0 | 0 | 0 | 0 |
| 105 | 0 | 0 | 0 | 0 | 0 |
| 110 | 0 | 0 | 0 | 0 | 0 |
| 115 | 0 | 0 | 0 | 0 | 0 |
| 120 | 0 | 0 | 0 | 0 | 0 |
| 125 | 0 | 0 | 0 | 0 | 0 |
| 130 | 0 | 0 | 0 | 0 | 0 |
| 135 | 0 | 0 | 0 | 0 | 0 |
| 140 | 0 | 0 | 0 | 0 | 0 |
| 145 | 0 | 0 | 0 | 0 | 0 |
| 150 | 0 | 0 | 0 | 0 | 0 |
| 155 | 0 | 0 | 0 | 0 | 0 |
| 160 | 0 | 0 | 0 | 0 | 0 |
| 165 | 0 | 0 | 0 | 0 | 0 |
| 170 | 0 | 0 | 0 | 0 | 0 |
| 175 | 0 | 0 | 0 | 0 | 0 |
| 180 | 0 | 0 | 0 | 0 | 0 |

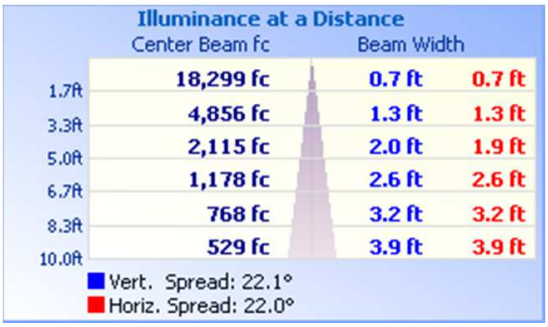


RESULTS OF TEST (cont'd)

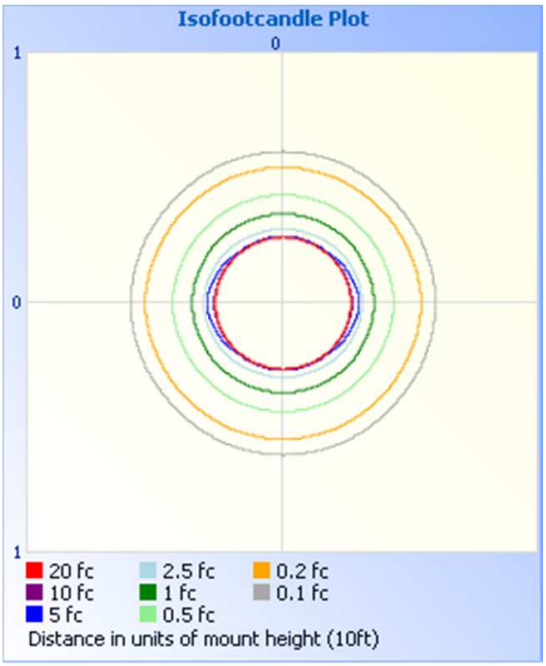
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 5997 | 99.9 |
| 0-40 | 6000 | 100.0 |
| 0-60 | 6000 | 100.0 |
| 60-90 | 0 | 0.0 |
| 70-100 | 0 | 0.0 |
| 90-120 | 0 | 0.0 |
| 0-90 | 5999.7 | 100.0 |
| 90-180 | 0 | 0.0 |
| 0-180 | 0 | 0.0 |

Zonal Lumens and Percentages at 25°C

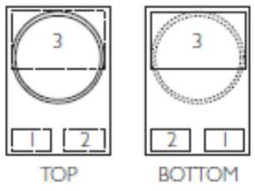
| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 4169 | 69.5 |
| 10-20 | 1803 | 30.1 |
| 20-30 | 24.2 | 0.4 |
| 30-40 | 3.0 | 0.1 |
| 40-50 | 0 | 0.0 |
| 50-60 | 0 | 0.0 |
| 60-70 | 0 | 0.0 |
| 70-80 | 0 | 0.0 |
| 80-90 | 0 | 0.0 |
| 90-100 | 0 | 0.0 |
| 100-110 | 0 | 0.0 |
| 110-120 | 0 | 0.0 |
| 120-130 | 0 | 0.0 |
| 130-140 | 0 | 0.0 |
| 140-150 | 0 | 0.0 |
| 150-160 | 0 | 0.0 |
| 160-170 | 0 | 0.0 |
| 170-180 | 0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperatur

Manufacturer Supplied Documentation:

LED model identified as:




| PAD | FUNCTION |
|---|---|
| 1 | CATHODE |
| 2 | ANODE |
| 3 | THERMAL |

Figure 3. Solder pad layout.

## Electrical Characteristics

### Electrical Characteristics for LUXEON Rebel and LUXEON Rebel ES Colors at Test Current and Temperature [1]

Table 4.

| Color | Part Number | Forward Voltage $V_f$ (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_F / \Delta T_J$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{J-C}$ |
|---|---|---|---|---|---|---|
| | | Minimum | Typical | Maximum | | |
| Lime | LXML-PX02 | 2.60 | 2.75 | 3.00 | -2.0 to -4.0 | 6 |
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB02 | 2.50 | 2.95 | 3.50 | - 2.0 to - 4.0 | 6 |
| | LXML-PB01 | 2.55 | 2.95 | 3.51 | - 2.0 to - 4.0 | 10 |
| Red | LXML-PD01 | 2.31 | 2.90 | 3.51 | - 2.0 to - 4.0 | 12 |
| | LXM2-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| | LXM5-PD01 | 1.80 | 2.10 | 2.40 | - 2.0 to - 4.0 | 6.5 |

Notes for Table 4:
1. LUXEON Rebel ES Lime is tested at 350mA/85°C. LUXEON Rebel ES Royal Blue and ES Blue are tested at 700mA/25°C. All other LUXEON Rebel color emitters are tested at 350mA/25°C.
2. Measured between $T_J$ = 25°C and $T_J$ = 110°C at test current.
3. Lumileds maintains a tolerance of ±0.06V on forward voltage measurements.

## Absolute Maximum Ratings

Table 5.

| Parameter | Green/Cyan/Blue/ Royal Blue | ES Royal Blue/ES Blue/Lime | Red/Deep Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |
| Operating Case Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 120°C | -40°C - 110°C |
| Storage Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 | 3 | 3 | 3 |
| Autoclave Conditions | 121°C at 2 ATM 100% Relative Humidity for 96 Hours Maximum | | | |
| Reverse Voltage (Vr) | LUXEON Rebel Color Portfolio LEDs are not designed to be driven in reverse bias. | | | |

Notes for Table 5:
1. Proper current derating must be observed to maintain junction temperature below the maximum.
2. Pulsed operation of Rebel ES Colors with a peak drive current of 1200 mA is acceptable if the pulse on-time is ≤ 5 ms per cycle and the duty cycle is ≤ 50%.

Note for Table 7:
1. All temperatures refer to the application Printed Circuit Board (PCB), measured on the surface adjacent to the package body.

| LED Color Designation | Green | Lime | Red | Blue |
|---|---|---|---|---|
| Maximum Junction Temperature from LED specification (Tj), °C | 150 | 150 | 135 | 150 |
| Thermal Resistance Formula from LED specification, °C/W | 10 | 6 | 6.5 | 6 |
| Maximum Forward Voltage (Vf) from LED specification, V | 3.51 | 3 | 2.4 | 3.51 |
| Measured LED Current, mA | 598 | 974 | 256 | 975 |
| Calculated LED Wattage = Vf x Measured LED Current W | 2.1 | 2.9 | 0.6 | 3.4 |
| Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance), °C | 129 | 132 | 131 | 129.5 |

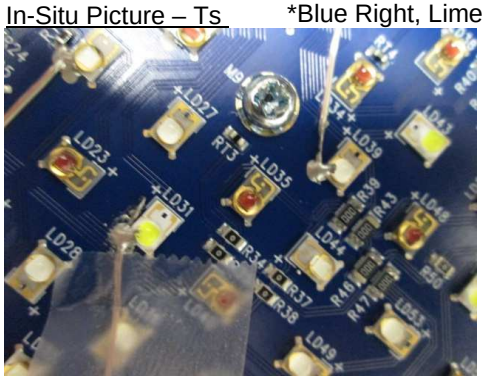
Maximum Measured Manufacturer Designated Source Temperature - two highest temperatures

| Sample No. | LED Designation | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|---|
| CRT15040112 | Lime | 70.0 | Per Table 7 note | 132.5 |
| 19-001, -002 | Red | 72.9 | Per Table 7 note | 131.0 |

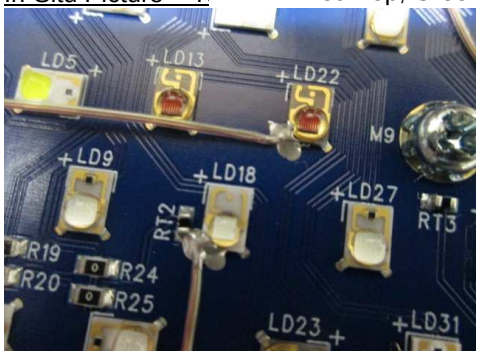
RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

In-Situ Picture – Ts *Blue Right, Lime Left

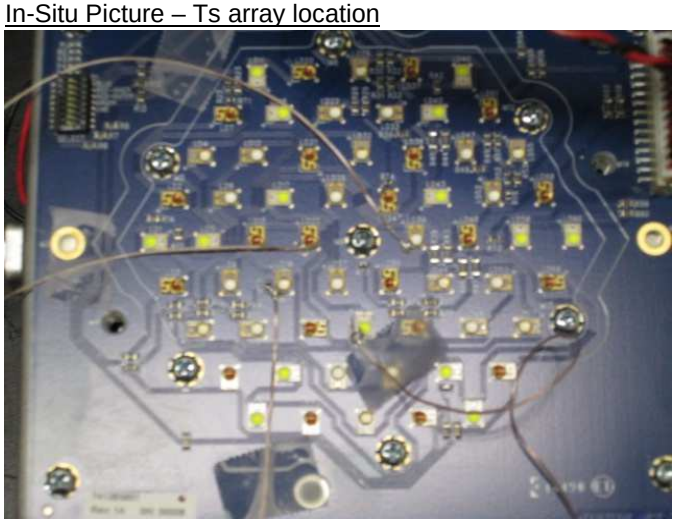



In-Situ Picture – T: *Red Top, Green Bottom




Product Picture (not to scale)

In-Situ Picture – Ts array location



CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Brittnie Stull
Project Engineer
Lighting Division

Report Reviewed By:

Jacki Swiernik
Staff Engineer
Lighting Division

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G102501090 Date: April 15, 2016

REPORT NO. 102501090CRT-002

TEST OF ONE LED LUMINAIRE

MODEL NO. COLORSOURCE SPOT DEEP BLUE
LED MODEL NO. LUMILEDS REBEL
DRIVER MODEL NO. LB240

RENDERED TO:

ELECTRONIC THEATRE CONTROLS INC
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

TESTS: Electrical and Photometric tests as required to the IESNA test standard and Insitu Temperature Measurement Tests.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION The testing performed was authorized by signed quote number Qu-00675915.

STANDARDS USED:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting
ANSI NEMA ANSLG C78.377: 2015: Specifications of the Chromaticity of Solid State Lighting Products
UL 1598-2009: Standard for Safety - Luminaires.

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number ColorSource Spot Deep Blue. The sample was received by Intertek on March 18, 2016 in undamaged condition and one sample was tested as received. The sample designation was CRT1603181057-001.

DATE OF TESTS: March 30, 2016 through April 13, 2016.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY:

| MODEL NO. ColorSource Spot Deep Blue |
|---|
| DESCRIPTION: LED LUMINAIRE |

| | Results | |
|---|---|---|
| Criteria | Integrating Sphere | Goniophotometer |
| Light Output (Lumens) | 5854 | 5972 |
| Total Power (W) | 171.4 | 172.0 |
| Lumen Efficacy (Lm/W) | 34.20 | 34.70 |
| Power Factor ( ) | 0.993 | 0.995 |

| Criteria | Results |
|---|---|
| Current ATHD (%) | 4.60 |
| Correlated Color Temp. (CCT-K) | 22000 |
| Color Rendering Index (CRI - Ra) | 37.2 |
| CRI - R9 | -233.3 |
| DUV ( ) | 0.053 |
| Chromaticity Coordinate (x) | 0.292 |
| Chromaticity Coordinate (y) | 0.216 |
| Chromaticity Coordinate (u') | 0.233 |
| Chromaticity Coordinate (v') | 0.388 |
| Maximum LED Source In-Situ Temperature (°C) | 79.3 |
| Maximum LED Source In-Situ Temperature (°C) | 78.5 |
| Maximum LED Source In-Situ Temperature (°C) | 74.3 |
| Maximum LED Source In-Situ Temperature (°C) | 76.6 |
| Maximum Driver Source In-Situ Temperature (°C) | 54.6 |

Note: See page 8 for LED and measured insitu temperature matching.

EQUIPMENT LIST

| Equipment Used | Model No. | Control No. | Last Cal. | Cal. Due |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 04/04/16 | 05/04/16 |
| Elgar AC Power Supply | CW1251 | --- | VBU | VBU |
| Sorenson DC Power Supply | XG 150-10 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 04/20/15 | 04/20/16 |
| ExTech Hygro Thermometer | 445703 | T1359 | 02/04/16 | 02/04/17 |
| NIST Luminous Intensity Standard Source | NBS10322 | N1427 | 12/12/14 | 12/12/16 |
| NIST Luminous Intensity Standard Source | NBS10215 | N1432 | 12/12/14 | 12/12/16 |
| NIST Luminous Intensity Standard Source | 960629-3 | N1428 | 12/12/14 | 12/12/16 |
| NIST Luminous Flux Standard Source | NBS10428 | N1424 | 12/17/14 | 12/17/16 |
| Elgar AC Power Supply | CW1251 | --- | VBU | VBU |
| Sorenson DC Power Supply | XFR 150-8 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | ZE475 | 06/09/15 | 06/09/16 |
| Fisher Scientific Stopwatch | 14-649-9 | N1316 | 01/15/16 | 01/15/17 |
| Fluke Thermometer | 52 | T801 | 11/18/15 | 11/18/16 |
| 3M Integrating Sphere Spectrometer System | CDS 1100 | --- | 03/02/16 | 04/02/16 |
| Pearson Current Monitor | 411 | A203 | 06/26/15 | 06/15/16 |
| Secondary Spectral Intensity Standard Source | BS5186 | RF5186 | 01/27/16 | 01/27/17 |
| Secondary Luminous Flux Standard Source | BS3616 | -- | 01/27/16 | 01/27/17 |
| Secondary Luminous Flux Standard Source | BS4116 | -- | 01/27/16 | 01/27/17 |
| Extech Hygro-Thermometer | 445703 | T1366 | 02/04/16 | 02/04/17 |
| Fluke Multimeter | 87 V | D590 | 04/22/15 | 04/22/16 |
| Fluke Temperature Meter | 53 II | D588 | 04/22/15 | 04/22/16 |

TEST METHODS:

Seasoning in Sample Orientation – LED Products
No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method
A Labsphere Model CDS 1100 CCD Array Spectroradiometer and two meter or ten foot sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Stabilization procedures to LM-79 were followed. Electrical measurements including voltage, current, and power were measured using a power analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical measurements – Distribution Method
A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for the SSL sample.

Ambient temperature was measured equal to the height of the sample mounted on the goniometer equipment. The SSL sample was operated on the client provided driver at rated input volts in its designated orientation. The SSL sample was allowed to stabilize for at least thirty minutes before measurements were made. Stabilization procedures to LM-79 were followed. Electrical measurements including voltage, current, and power were measured using a power analyzer.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature
Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598, UL 153, or UL 1993 as applicable.

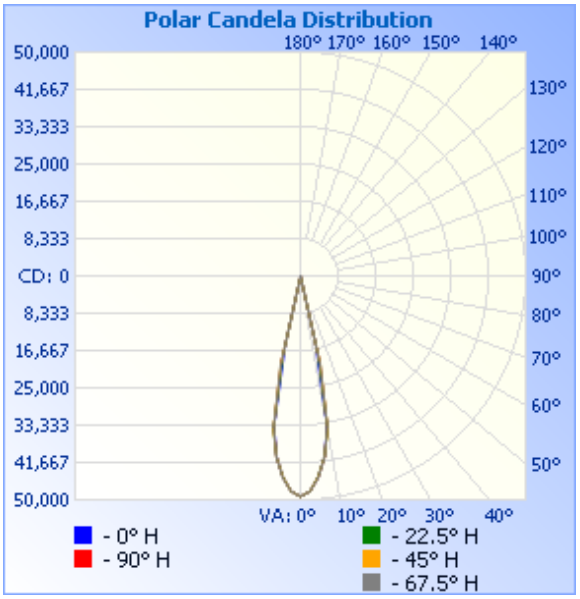
RESULTS:

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Control No. | Base Orientation | Input Voltage (VAC) | Input Current (mA) | Input Power (W) | Input Power Factor ( ) | Light Output (Lumens) | Lumen Efficacy (lm/W) |
|---|---|---|---|---|---|---|---|
| CRT1603181057-001 | Base Up | 120.04 | 1440.5 | 172.0 | 0.995 | 5972 | 34.70 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 49207 | 49207 | 49207 | 49207 | 49207 |
| 5 | 45080 | 44978 | 45080 | 45248 | 44996 |
| 10 | 34075 | 34579 | 34650 | 34283 | 34322 |
| 15 | 98 | 90 | 93 | 93 | 98 |
| 20 | 28 | 23 | 25 | 24 | 26 |
| 25 | 0 | 0 | 0 | 0 | 0 |
| 30 | 0 | 0 | 0 | 0 | 0 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

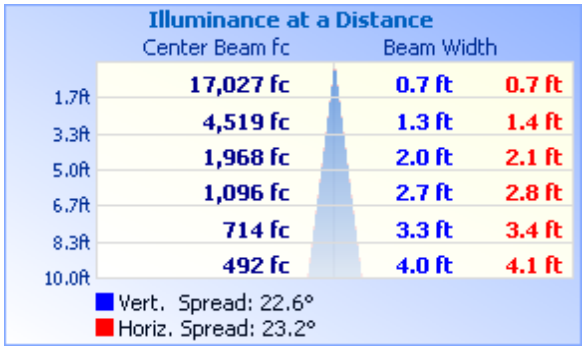


RESULTS:

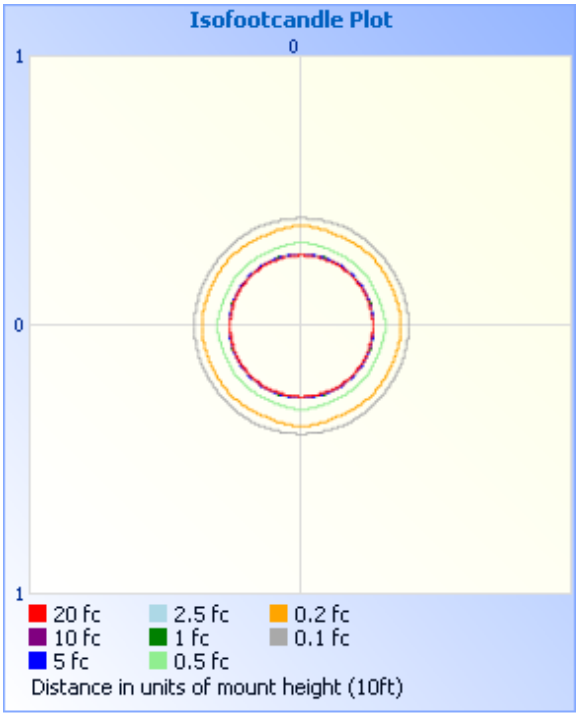
Illumination Plots

Mounting Height: 10ft

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 5972 | 100.0 |
| 0-40 | 5972 | 100.0 |
| 0-60 | 5972 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 5972 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 5972 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 3952 | 66.2 |
| 10-20 | 2018 | 33.8 |
| 20-30 | 1.9 | 0.0 |
| 30-40 | 0.0 | 0.0 |
| 40-50 | 0.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS:

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Sphere Method

| Intertek Control No. | Base Orientation | Input Voltage (VAC) | Input Current (mA) | Input Power (W) | Input Power Factor ( ) | Current ATHD (%) |
|---|---|---|---|---|---|---|
| CRT1603181057-001 | Base Up | 120.02 | 1437.6 | 171.4 | 0.993 | 4.60 |

| Light Output (Lumens) | Lumen Efficacy (lm/W) | Correlated Color Temperature - CCT (K) | CRI -Ra | CRI -R9 | DUV ( ) |
|---|---|---|---|---|---|
| 5854 | 34.20 | 22000 | 37.2 | -233.3 | 0.053 |

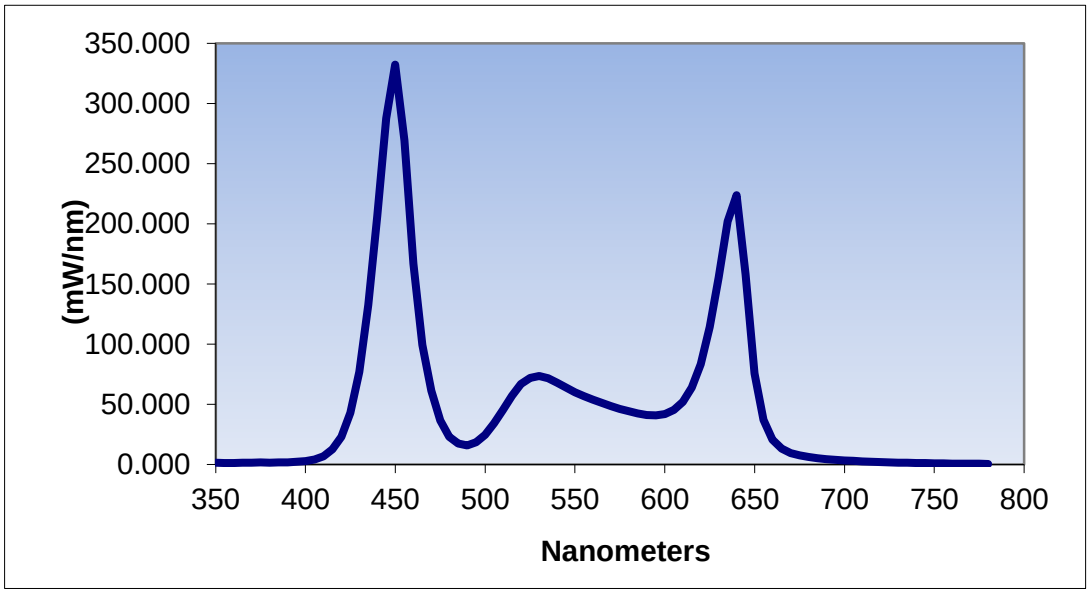
| CIE 31’ Chromaticity Coordinate (x) | CIE 31’ Chromaticity Coordinate (y) | CIE 76’ Chromaticity Coordinate (u') | CIE 76’ Chromaticity Coordinate (v') |
|---|---|---|---|
| 0.292 | 0.216 | 0.233 | 0.388 |

RESULTS:

**Spectral Distribution Over Visible Wavelengths**

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 1.303 | 460 | 166.849 | 570 | 48.527 | 680 | 6.078 |
| 355 | 1.055 | 465 | 99.430 | 575 | 46.220 | 685 | 5.092 |
| 360 | 1.175 | 470 | 61.393 | 580 | 44.213 | 690 | 4.406 |
| 365 | 1.486 | 475 | 36.439 | 585 | 42.295 | 695 | 3.801 |
| 370 | 1.391 | 480 | 23.051 | 590 | 41.000 | 700 | 3.286 |
| 375 | 1.596 | 485 | 17.333 | 595 | 40.743 | 705 | 2.896 |
| 380 | 1.355 | 490 | 15.754 | 600 | 41.825 | 710 | 2.544 |
| 385 | 1.764 | 495 | 18.432 | 605 | 45.366 | 715 | 2.223 |
| 390 | 1.800 | 500 | 24.576 | 610 | 52.083 | 720 | 1.902 |
| 395 | 2.096 | 505 | 34.053 | 615 | 64.023 | 725 | 1.706 |
| 400 | 2.803 | 510 | 45.321 | 620 | 83.335 | 730 | 1.519 |
| 405 | 4.182 | 515 | 56.911 | 625 | 114.111 | 735 | 1.313 |
| 410 | 6.814 | 520 | 66.942 | 630 | 155.437 | 740 | 1.130 |
| 415 | 12.536 | 525 | 71.957 | 635 | 201.994 | 745 | 1.005 |
| 420 | 23.033 | 530 | 73.587 | 640 | 223.842 | 750 | 0.881 |
| 425 | 42.794 | 535 | 71.733 | 645 | 158.180 | 755 | 0.809 |
| 430 | 77.031 | 540 | 67.905 | 650 | 75.738 | 760 | 0.695 |
| 435 | 132.681 | 545 | 63.962 | 655 | 36.981 | 765 | 0.623 |
| 440 | 206.326 | 550 | 59.999 | 660 | 20.499 | 770 | 0.547 |
| 445 | 287.680 | 555 | 56.877 | 665 | 13.125 | 775 | 0.492 |
| 450 | 332.366 | 560 | 53.808 | 670 | 9.440 | 780 | 0.430 |
| 455 | 269.484 | 565 | 51.183 | 675 | 7.394 | | |

**Spectral Data Over Visible Wavelengths**



RESULTS:

In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj)- LXML-PM01-0090 = 150°C
Maximum Junction Temperature from LED specification (Tj)- LXML-PX02-0000 = 150°C
Maximum Junction Temperature from LED specification (Tj)- LXM5-PD01-0050/60/80 = 135°C
Maximum Junction Temperature from LED specification (Tj)- LXML-PR01-0350/0425/0500 = 150°C

Thermal Resistance Formula from LED specification- LXML-PM01-0090 = 10°C/W
Thermal Resistance Formula from LED specification- LXML-PX02-0000 = 6°C/W
Thermal Resistance Formula from LED specification- LXM5-PD01-0050/60/80 = 6.5°C/W
Thermal Resistance Formula from LED specification- LXML-PR01-0350/0425/0500 = 10°C/W

Maximum Forward Voltage (Vf) from LED specification- LXML-PM01-0090 = 3.51V
Maximum Forward Voltage (Vf) from LED specification- LXML-PX02-0000 = 3V
Maximum Forward Voltage (Vf) from LED specification- LXM5-PD01-0050/60/80 = 2.6V
Maximum Forward Voltage (Vf) from LED specification- LXML-PR01-0350/0425/0500 = 3.51V

Measured LED Current- LXML-PM01-0090 = 999mA
Measured LED Current- LXML-PX02-0000 = 986mA
Measured LED Current- LXM5-PD01-0050/60/80 = 667mA
Measured LED Current- LXML-PR01-0350/0425/0500 = 1003mA

Calculated LED Wattage = Vf x Measured LED Current- LXML-PM01-0090 = 3.506W
Calculated LED Wattage = Vf x Measured LED Current- LXML-PX02-0000 = 2.958W
Calculated LED Wattage = Vf x Measured LED Current- LXM5-PD01-0050/60/80 = 1.734W
Calculated LED Wattage = Vf x Measured LED Current- LXML-PR01-0350/0425/0500 = 3.521W

Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance)- LXML-PM01-0090 = 114.9°C
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance)- LXML-PX02-0000 = 132.3°C
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance)- LXM5- PD01-0050/60/80 = 123.7°C
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance)- LXML-PR01-0350/0425/0500 = 114.8°C

RESULTS:

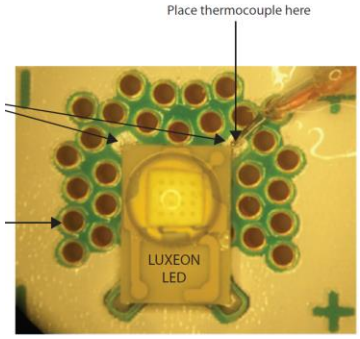
LED Source Temperature Manufacturer's Spec
LED Identified As: LUMILEDS REBEL

Table 4.

| Color | Part Number | Forward Voltage $V_f$ (V) | | | Typical Temperature Coefficient of Forward Voltage [2] (mV/°C) $\Delta V_f / \Delta T_j$ | Typical Thermal Resistance Junction to Thermal Pad (°C/W) $R\theta_{JC}$ |
|---|---|---|---|---|---|---|
| | | Minimum | Typical | Maximum | | |
| Lime | LXML-PX02 | 2.60 | 2.75 | 3.00 | -2.0 to -4.0 | 6 |
| Green | LXML-PM01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Cyan | LXML-PE01 | 2.55 | 2.90 | 3.51 | - 2.0 to - 4.0 | 10 |
| Blue | LXML-PB02<br>LXML-PB01 | 2.50<br>2.55 | 2.95<br>2.95 | 3.50<br>3.51 | - 2.0 to - 4.0<br>- 2.0 to - 4.0 | 6<br>10 |
| Royal Blue | LXML-PR02<br>LXML-PR01 | 2.50<br>2.55 | 2.90<br>2.95 | 3.50<br>3.51 | - 2.0 to - 4.0<br>- 2.0 to - 4.0 | 6<br>10 |
| Red | LXML-PD01<br>LXM2-PD01<br>LXM5-PD01 | 2.31<br>1.80<br>1.80 | 2.90<br>2.10<br>2.10 | 3.51<br>2.80<br>2.60 | - 2.0 to - 4.0<br>- 2.0 to - 4.0<br>- 2.0 to - 4.0 | 12<br>8<br>6.5 |
| Deep Red | LXM3-PD01 | 1.80 | 2.10 | 2.80 | - 2.0 to - 4.0 | 8 |
| Red-Orange | LXML-PH01<br>LXM2-PH01<br>LXM5-PH01 | 2.31<br>1.80<br>1.80 | 2.90<br>2.10<br>2.10 | 3.51<br>2.80<br>2.60 | - 2.0 to - 4.0<br>- 2.0 to - 4.0<br>- 2.0 to - 4.0 | 12<br>8<br>6.5 |
| PC Amber | LXM2-PL01 | 2.55 | 3.05 | 3.51 | - 2.0 to - 4.0 | 10 |
| Amber | LXML-PL01<br>LXM5-PL01 | 2.31<br>1.80 | 2.90<br>2.10 | 3.51<br>2.60 | - 2.0 to - 4.0<br>- 2.0 to - 4.0 | 12<br>6.5 |

Table 5.

| Parameter | Green/Cyan/Blue/ Royal Blue | ES Royal Blue/ES Blue/Lime | Red/Deep Red Red-Orange/Amber | PC Amber |
|---|---|---|---|---|
| DC Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| Peak Pulsed Forward Current (mA) | 1000 | 1200 | 700 | 700 |
| Average Forward Current (mA) | 1000 | 1000 | 700 | 700 |
| ESD Sensitivity | < 8000V Human Body Model (HBM) Class 3A JESD22-A114-B | | | |
| LED Junction Temperature [1] | 150°C | 150°C | 135°C | 130°C |
| Operating Case Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 120°C | -40°C - 110°C |
| Storage Temperature | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C | -40°C - 135°C |
| Soldering Temperature | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C | JEDEC 020c 260°C |
| Allowable Reflow Cycles | 3 | 3 | 3 | 3 |
| Autoclave Conditions | 121°C at 2 ATM<br>100% Relative Humidity for 96 Hours Maximum | | | |
| Reverse Voltage (Vr) | LUXEON Rebel Color Portfolio LEDs are not designed to be driven in reverse bias. | | | |



RESULTS:

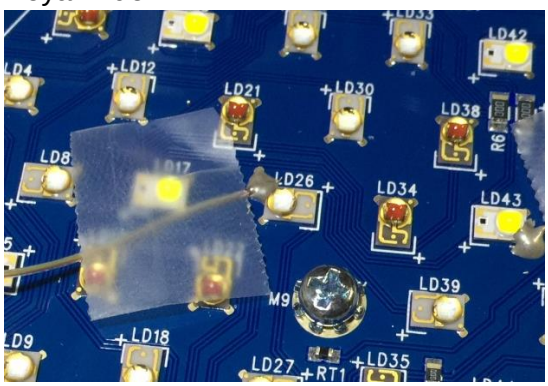
Maximum Measured Manufacturer Designated Source Temperature

Location of Thermocouple: Per Diagram

| Intertek Sample No. | Maximum Measured Source Temperature | LED Measured | Maximum Rated Source Temperature |
|---|---|---|---|
| CRT1603181057-001 | 79.3 | LXML-PM01-0090 | 114.9 |
| | 78.5 | LXML-PX02-0000 | 132.3 |
| | 74.3 | LXM5- PD01-0050/60/80 | 123.7 |
| | 76.6 | LXML-PR01-0350/0425/0500 | 114.8 |

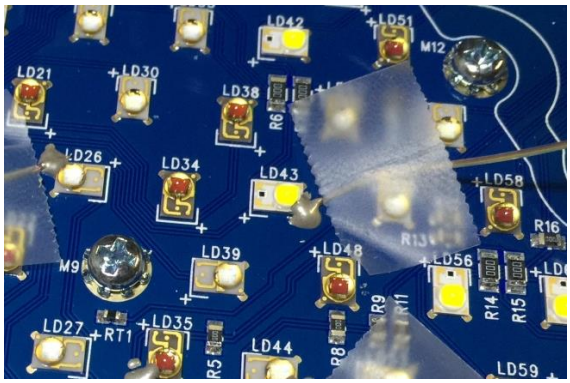
In-Situ Picture – Ts

Royal Blue

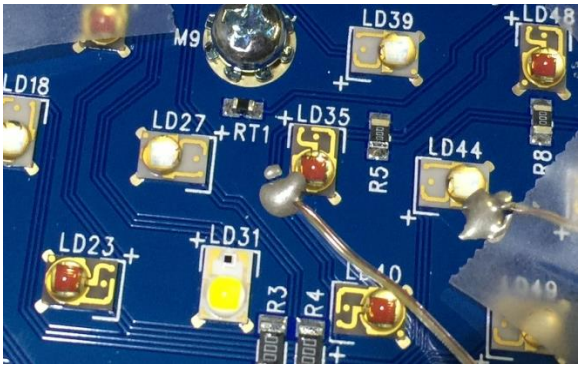



In-Situ Picture – Ts location

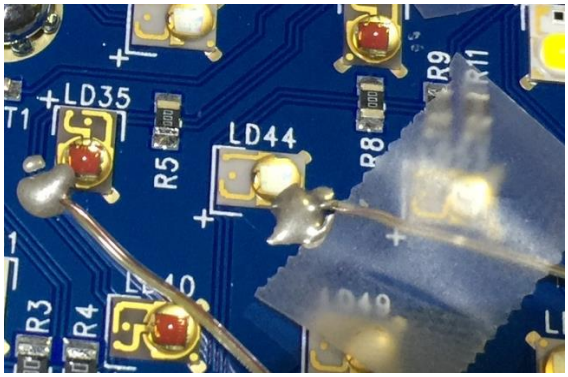
Lime

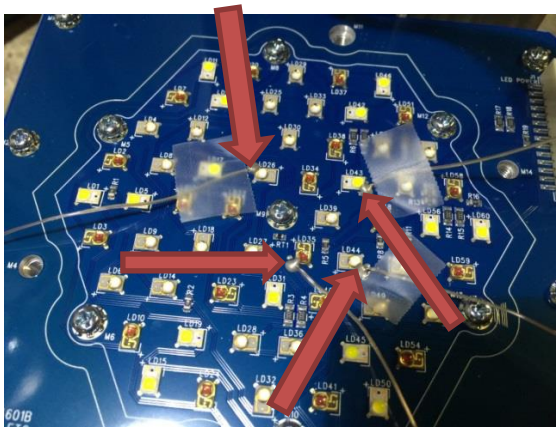



Red




Green

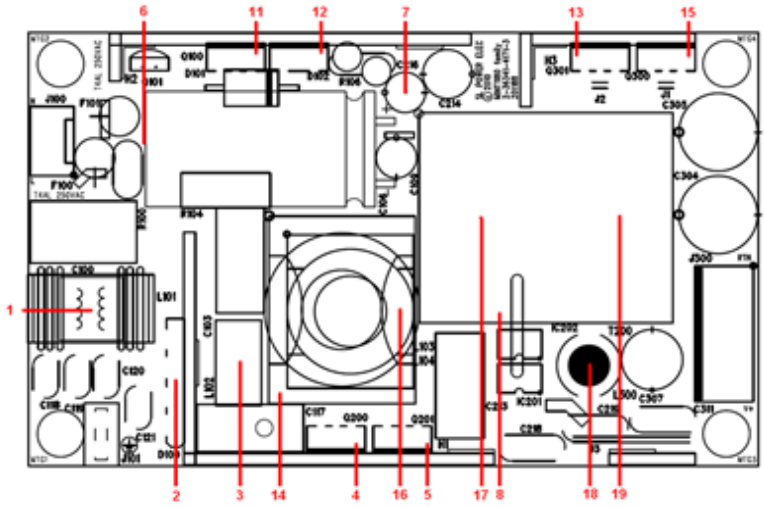


RESULTS:

In-Situ Maximum Measured Driver Source Temperature

Driver Source Manufacturer's Spec
Driver Identified As: LB240

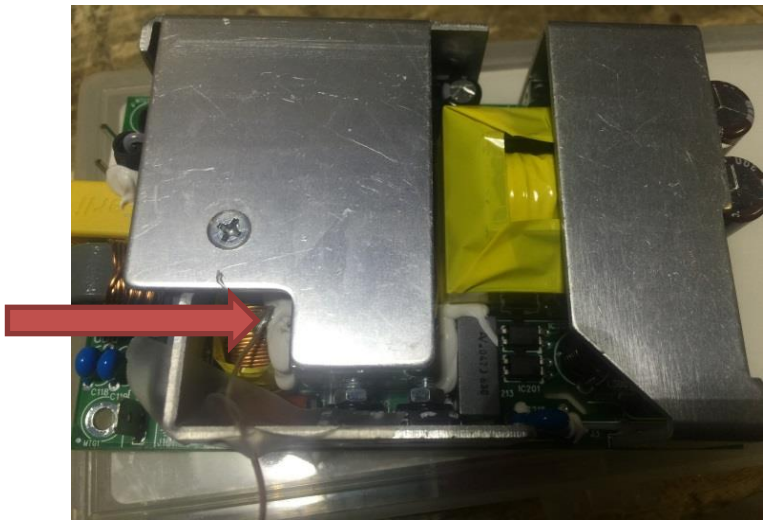



The hottest point in our testing was L102 which is position #3.

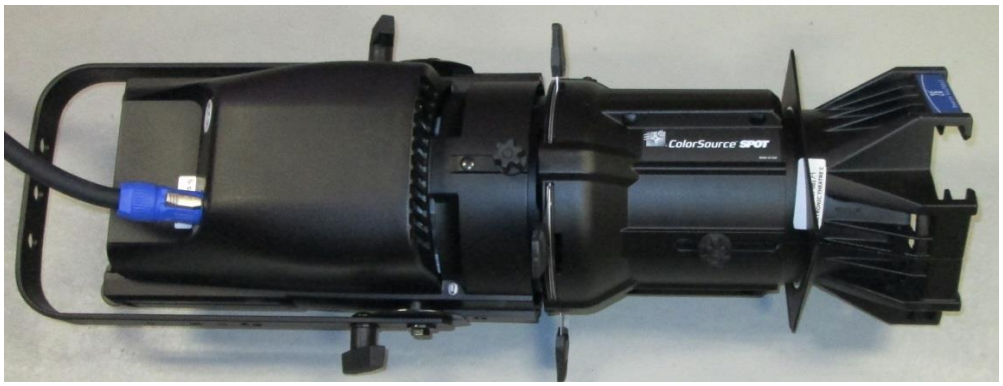
| | |
|---|---|
| **Operating Temperature** | -10℃ to +70℃( See Below Chart)<br>Start Up at -40℃ |

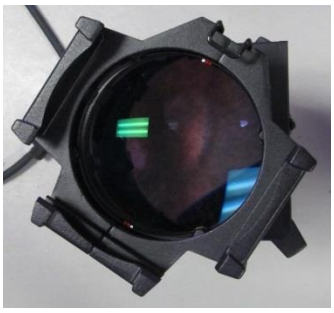
Maximum Measured Manufacturer Designated Driver Source Temperature

| Intertek Sample No. | Maximum Measured Driver Temperature °C | Location of Thermocouple | Maximum Rated Driver Temperature °C |
|---|---|---|---|
| CRT1603181057-001 | 54.6 | Per Diagram | 70.0 |



PRODUCT PICTURE:



CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Ryan Siddon

Ryan Siddon
Engineer
Lighting Division

Attachments: IES File - CRT1603181057-001

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G102489751 Date: July 22, 2016

REPORT NO. 102489751CRT-001

TEST OF ONE THEATRICAL LED LUMINAIRE

MODEL NO. COLORSOURCE LINEAR 1
LED MODEL NO. LUMILEDS LXZ1-PM01 - GREEN - QTY - 8
LED MODEL NO. LUMILEDS LXZ1-PD02 - RED - QTY - 16
LED MODEL NO. LUMILEDS LXZ1-PX01 - LIME - QTY - 8
LED MODEL NO. LUMILEDS LXZ1-PB01 - BLUE - QTY - 8

DRIVER MODEL NO. ELECTRONIC THEATRE CONTROLS COLORSOURCE LINEAR 1

RENDERED TO:

ELECTRONIC THEATRE CONTROLS INC.
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

TESTS: Electrical and Photometric tests as required to the IESNA test standard and Insitu Temperature Measurement Tests.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION The testing performed was authorized by signed quote number Qu-00675928.

STANDARDS USED:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting
ANSI NEMA ANSLG C78.377: 2011: Specifications of the Chromaticity of Solid State Lighting Products
UL 1598-2009: Standard for Safety - Luminaires.

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number COLORSOURCE LINEAR 1. The sample was received by Intertek on July 15, 2016 in undamaged condition and one sample was tested as received. The sample designation was CRT1607151108-001.

DATE OF TESTS: July 18, 2016 through July 20, 2016.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY:

| MODEL NO. COLORSOURCE LINEAR 1 |
|---|
| DESCRIPTION: THEATRICAL LED LUMINAIRE |

| | Results | |
|---|---|---|
| Criteria | Integrating Sphere | Goniophotometer |
| Light Output (Lumens) | 3314 | 3269 |
| Total Power (W) | 107.9 | 108.0 |
| Lumen Efficacy (Lm/W) | 30.71 | 30.26 |
| Power Factor ( ) | 0.996 | 0.997 |

| Criteria | Results |
|---|---|
| Current ATHD (%) | 4.51 |
| Correlated Color Temp. (CCT-K) | 4687 |
| Color Rendering Index (CRI - Ra) | 49.5 |
| CRI - R9 | -132.2 |
| DUV ( ) | 0.018 |
| Chromaticity Coordinate (x) | 0.349 |
| Chromaticity Coordinate (y) | 0.321 |
| Chromaticity Coordinate (u') | 0.227 |
| Chromaticity Coordinate (v') | 0.469 |
| Maximum LED Source In-Situ Temperature (℃) | 68.7 |
| Maximum LED Source In-Situ Temperature (℃) | 58.2 |
| Maximum LED Source In-Situ Temperature (℃) | 58.0 |
| Maximum LED Source In-Situ Temperature (℃) | 71.3 |
| Maximum Driver Source In-Situ Temperature (℃) | 41.5 |

Note: See page 9 for LED and measured insitu temperature matching.

EQUIPMENT LIST

| Equipment Used | Model No. | Control No. | Last Cal. | Cal. Due |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 07/06/16 | 08/06/16 |
| Elgar AC Power Supply | CW1251 | --- | VBU | VBU |
| Sorenson DC Power Supply | XG 150-10 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 05/02/16 | 05/02/17 |
| Omega Thermometer | DPi8-C24 | M263 | 05/02/16 | 05/02/17 |
| NIST Luminous Intensity Standard Source | NBS10322 | N1427 | 12/12/14 | 12/12/16 |
| NIST Luminous Intensity Standard Source | NBS10215 | N1432 | 12/12/14 | 12/12/16 |
| NIST Luminous Intensity Standard Source | 960629-3 | N1428 | 12/12/14 | 12/12/16 |
| NIST Luminous Flux Standard Source | NBS10428 | N1424 | 12/17/14 | 12/17/16 |
| Elgar AC Power Supply | CW1251 | --- | VBU | VBU |
| Sorenson DC Power Supply | XFR 150-8 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E473 | 05/10/16 | 05/10/17 |
| Fluke Thermometer | 53 II | N1324 | 04/07/16 | 04/07/17 |
| Fisher Scientific Stopwatch | 14-649-9 | N1316 | 01/15/16 | 01/15/17 |
| 3M Integrating Sphere Spectrometer System | CDS 1100 | --- | 07/05/16 | 08/05/16 |
| Pearson Current Monitor | 411 | A183 | 05/09/16 | 05/02/17 |
| Secondary Spectral Intensity Standard Source | BS5186 | RF5186 | 01/27/16 | 01/27/17 |
| Secondary Luminous Flux Standard Source | BS3616 | -- | 01/27/16 | 01/27/17 |
| Secondary Luminous Flux Standard Source | BS4116 | -- | 01/27/16 | 01/27/17 |
| Secondary Luminous Flux Standard Source | 6836 | -- | 01/27/16 | 01/27/17 |
| Extech Hygro-Thermometer | 445703 | T1366 | 02/04/16 | 02/04/17 |
| Fluke Multimeter | 87 V | M226 | 11/19/15 | 11/19/16 |
| Fluke Temperature Meter | 53 II | N1324 | 04/07/16 | 04/07/17 |

TEST METHODS:

Seasoning in Sample Orientation – LED Products
No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method
A Labsphere Model CDS 1100 CCD Array Spectroradiometer and two meter or ten foot sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Stabilization procedures to LM-79 were followed. Electrical measurements including voltage, current, and power were measured using a power analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical measurements – Distribution Method
A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for the SSL sample.

Ambient temperature was measured equal to the height of the sample mounted on the goniometer equipment. The SSL sample was operated on the client provided driver at rated input volts in its designated orientation. The SSL sample was allowed to stabilize for at least thirty minutes before measurements were made. Stabilization procedures to LM-79 were followed. Electrical measurements including voltage, current, and power were measured using a power analyzer.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature
Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598, UL 153, or UL 1993 as applicable.

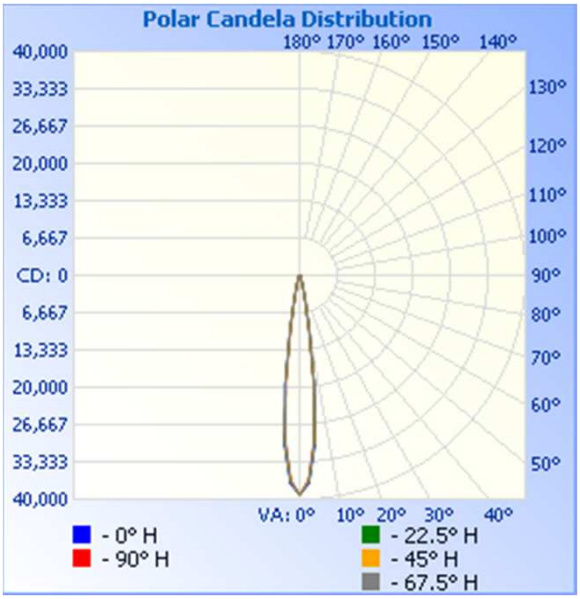
RESULTS:

Photometric and Electrical Measurements at Ambient Temperature (25℃ +/- 1℃) – Distribution Method

| Intertek Control No. | Base Orientation | Input Voltage (VAC) | Input Current (mA) | Input Power (W) | Input Power Factor ( ) | Light Output (Lumens) | Lumen Efficacy (lm/W) |
|---|---|---|---|---|---|---|---|
| CRT1607151108-001 | Base Up | 120.01 | 903.4 | 108.0 | 0.997 | 3269 | 30.26 |

Intensity (Candlepower) Summary at 25℃ - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 39218 | 39218 | 39218 | 39218 | 39218 |
| 5 | 30484 | 30164 | 29818 | 29675 | 29622 |
| 10 | 9786 | 9511 | 9412 | 9347 | 9169 |
| 15 | 2465 | 2436 | 2349 | 2316 | 2260 |
| 20 | 914 | 899 | 876 | 860 | 886 |
| 25 | 310 | 303 | 300 | 300 | 301 |
| 30 | 146 | 150 | 153 | 155 | 156 |
| 35 | 86 | 95 | 94 | 98 | 101 |
| 40 | 67 | 66 | 73 | 74 | 81 |
| 45 | 55 | 56 | 61 | 68 | 72 |
| 50 | 58 | 56 | 64 | 64 | 80 |
| 55 | 44 | 44 | 51 | 49 | 87 |
| 60 | 25 | 26 | 25 | 34 | 27 |
| 65 | 7 | 9 | 6 | 0 | 9 |
| 70 | 0 | 0 | 0 | 0 | 3 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

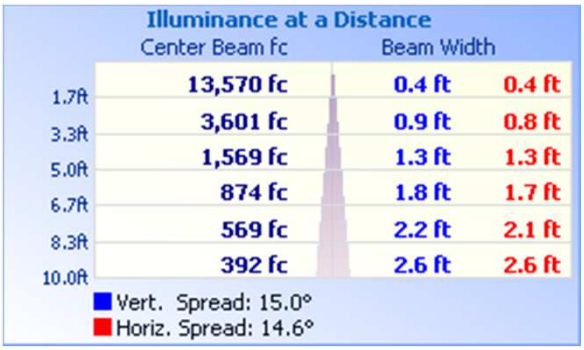


RESULTS:

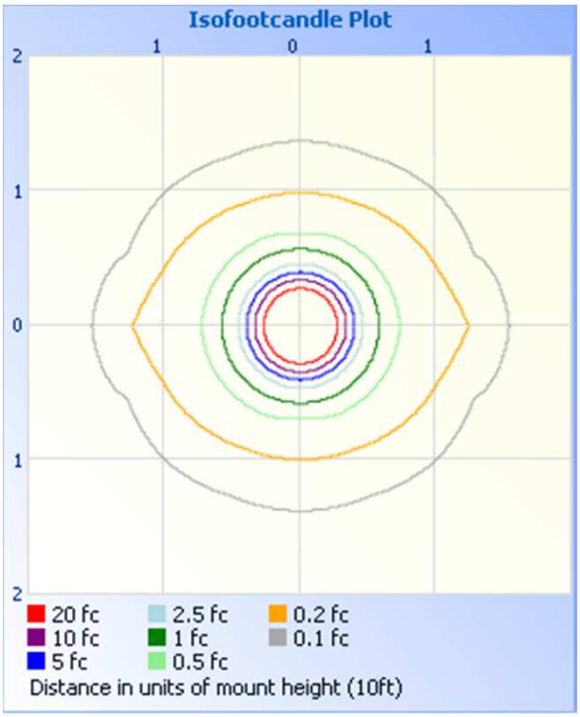
Illumination Plots

Mounting Height: 10ft

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25℃

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 3104.3 | 95.0 |
| 0-40 | 3166.4 | 96.9 |
| 0-60 | 3259.7 | 99.7 |
| 60-90 | 9.1 | 0.3 |
| 0-90 | 3268.8 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 3268.8 | 100.0 |

Zonal Lumens and Percentages at 25℃

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 2106.4 | 64.4 |
| 10-20 | 830.6 | 25.4 |
| 20-30 | 167.3 | 5.1 |
| 30-40 | 62.1 | 1.9 |
| 40-50 | 49.6 | 1.5 |
| 50-60 | 43.7 | 1.3 |
| 60-70 | 9.0 | 0.3 |
| 70-80 | 0.1 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS:

Photometric and Electrical Measurements at Ambient Temperature (25℃ +/- 1℃) – Sphere Method

| Intertek Control No. | Base Orientation | Input Voltage (VAC) | Input Current (mA) | Input Power (W) | Input Power Factor ( ) | Current ATHD (%) |
|---|---|---|---|---|---|---|
| CRT1607151108-001 | Base Up | 120.04 | 903.0 | 107.91 | 0.996 | 4.51 |

| Light Output (Lumens) | Lumen Efficacy (lm/W) | Correlated Color Temperature - CCT (K) | CRI -Ra | CRI -R9 | DUV ( ) |
|---|---|---|---|---|---|
| 3314.0 | 30.7 | 4687 | 49.5 | -132.2 | 0.018 |

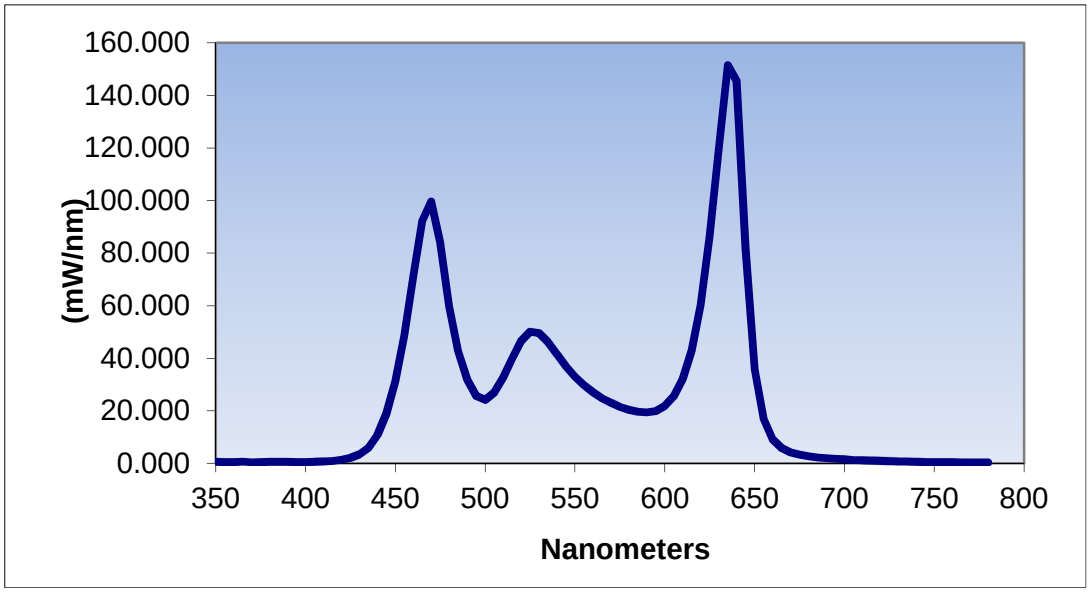
| CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|
| 0.349 | 0.321 | 0.227 | 0.469 |

RESULTS:

**Spectral Distribution Over Visible Wavelengths**

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.477 | 460 | 70.982 | 570 | 23.069 | 680 | 2.694 |
| 355 | 0.431 | 465 | 92.151 | 575 | 21.499 | 685 | 2.275 |
| 360 | 0.416 | 470 | 99.653 | 580 | 20.381 | 690 | 1.953 |
| 365 | 0.635 | 475 | 84.178 | 585 | 19.655 | 695 | 1.687 |
| 370 | 0.416 | 480 | 59.782 | 590 | 19.366 | 700 | 1.455 |
| 375 | 0.444 | 485 | 42.714 | 595 | 19.927 | 705 | 1.298 |
| 380 | 0.451 | 490 | 31.809 | 600 | 21.817 | 710 | 1.094 |
| 385 | 0.458 | 495 | 25.621 | 605 | 25.491 | 715 | 0.975 |
| 390 | 0.491 | 500 | 24.133 | 610 | 32.029 | 720 | 0.833 |
| 395 | 0.442 | 505 | 26.948 | 615 | 42.905 | 725 | 0.772 |
| 400 | 0.406 | 510 | 32.605 | 620 | 60.191 | 730 | 0.669 |
| 405 | 0.456 | 515 | 39.802 | 625 | 86.215 | 735 | 0.591 |
| 410 | 0.647 | 520 | 46.525 | 630 | 119.862 | 740 | 0.522 |
| 415 | 0.919 | 525 | 50.028 | 635 | 151.521 | 745 | 0.449 |
| 420 | 1.377 | 530 | 49.535 | 640 | 145.375 | 750 | 0.403 |
| 425 | 2.071 | 535 | 46.187 | 645 | 81.795 | 755 | 0.351 |
| 430 | 3.485 | 540 | 41.568 | 650 | 35.872 | 760 | 0.332 |
| 435 | 5.865 | 545 | 36.919 | 655 | 16.964 | 765 | 0.286 |
| 440 | 10.694 | 550 | 32.976 | 660 | 9.129 | 770 | 0.274 |
| 445 | 18.843 | 555 | 29.828 | 665 | 5.899 | 775 | 0.226 |
| 450 | 31.240 | 560 | 27.027 | 670 | 4.189 | 780 | 0.217 |
| 455 | 48.618 | 565 | 24.760 | 675 | 3.280 | | |

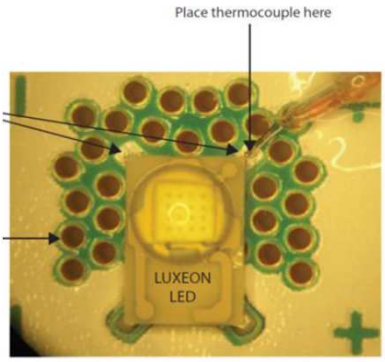
**Spectral Data Over Visible Wavelengths**



RESULTS:

| PART NUMBER | FORWARD VOLTAGE (V) [1] | | | TYPICAL TEMPERATURE COEFFICIENT OF FORWARD VOLTAGE [2] (mV/°C) | TYPICAL THERMAL RESISTANCE — JUNCTION TO SOLDER PAD (°C/W) |
|---|---|---|---|---|---|
| | MINIMUM | TYPICAL | MAXIMUM | | |
| LXZ1-PM01 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |
| LXZ1-PD02 | 1.75 | 2.20 | 2.75 | -2 to -4 | 8 |
| LXZ1-PX01 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |
| LXZ1-PB01 | 2.50 | 2.85 | 3.50 | -2 to -4 | 5 |

| PARAMETER | GREEN/CYAN/ BLUE/ROYAL BLUE | COLD-TESTED RED/DEEP-RED/ RED-ORANGE/AMBER |
|---|---|---|
| LED Junction Temperature [1] (DC & Pulse) | 150°C | 135°C |




| | | |
|---|---|---|
| Measured LED Current: | Lumileds LXZ1-PM01 - Green | 975.0 |
| Measured LED Current: | Lumileds LXZ1-PD02 - Red | 675.0 |
| Measured LED Current: | Lumileds LXZ1-PX01 - Lime | 678.0 |
| Measured LED Current: | Lumileds LXZ1-PB01 - Blue | 974.0 |

| | | |
|---|---|---|
| Calculated LED Wattage = Vf x Measured LED Current: | LXZ1-PM01 | 3.413 |
| Calculated LED Wattage = Vf x Measured LED Current: | LXZ1-PD02 | 1.856 |
| Calculated LED Wattage = Vf x Measured LED Current: | LXZ1-PX01 | 2.373 |
| Calculated LED Wattage = Vf x Measured LED Current: | LXZ1-PB01 | 3.409 |

| | | |
|---|---|---|
| Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) | LXZ1-PM01 | 132.9 |
| Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) | LXZ1-PD02 | 120.2 |
| Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) | LXZ1-PX01 | 138.1 |
| Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) | LXZ1-PB01 | 133.0 |

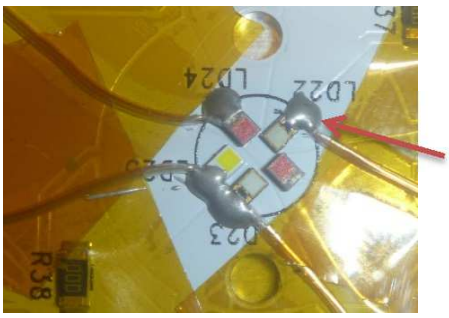
RESULTS:

Maximum Measured Manufacturer Designated Source Temperature

| Intertek Sample No. | Maximum Measured Source Temperature | Location of Thermocouple | Maximum Rated Source Temperature |
|---|---|---|---|
| CRT1607151108-001 | 68.7 | Per Diagram | 132.9 |
| | 58.2 | | 120.2 |
| | 58.0 | | 138.1 |
| | 71.3 | | 133.0 |

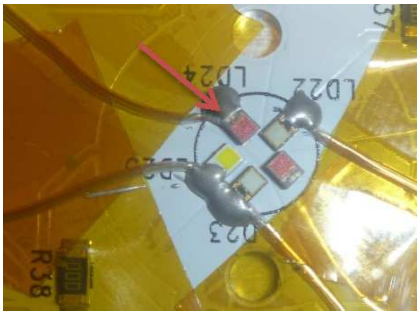
In-Situ Picture – Ts

Lumileds LXZ1-PM01 - Greer



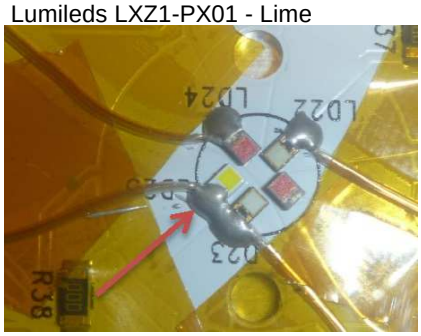
In-Situ Picture – Ts

Lumileds LXZ1-PD02 - Red



In-Situ Picture – Ts

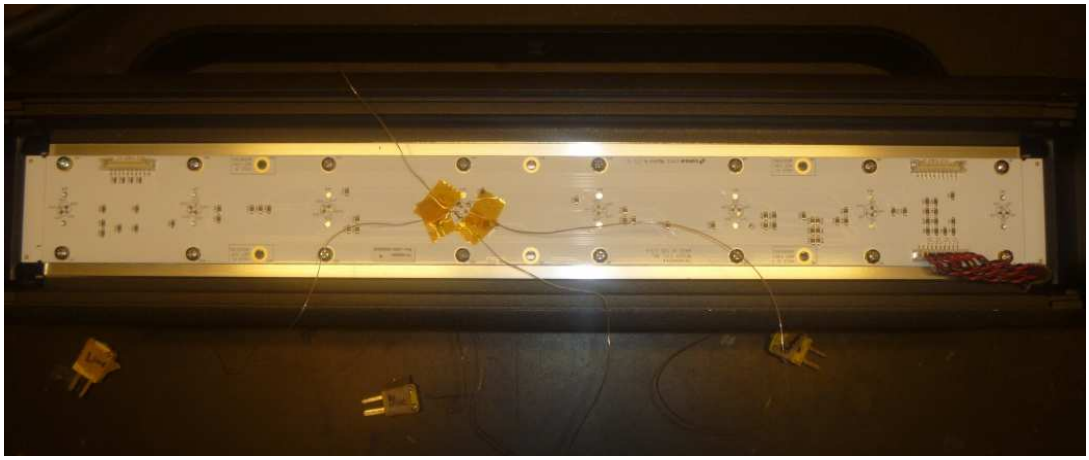
Lumileds LXZ1-PX01 - Lime



In-Situ Picture – Ts

Lumileds LXZ1-PB01 - Blue

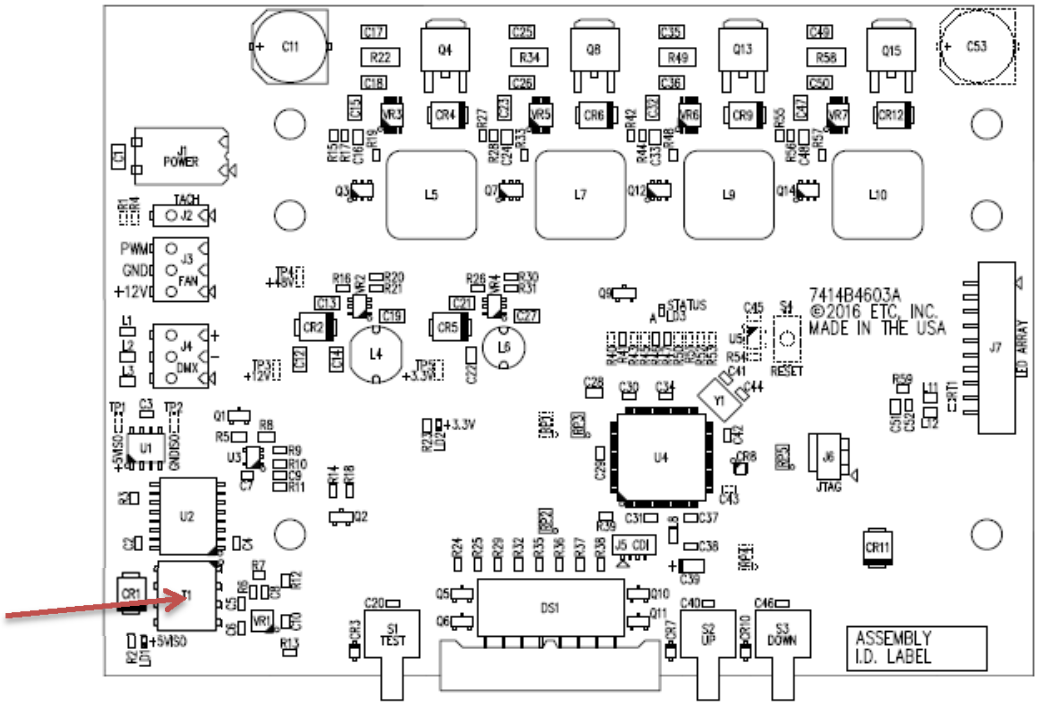
In-Situ Picture – Ts location



RESULTS:

In-Situ Maximum Measured Driver Source Temperature

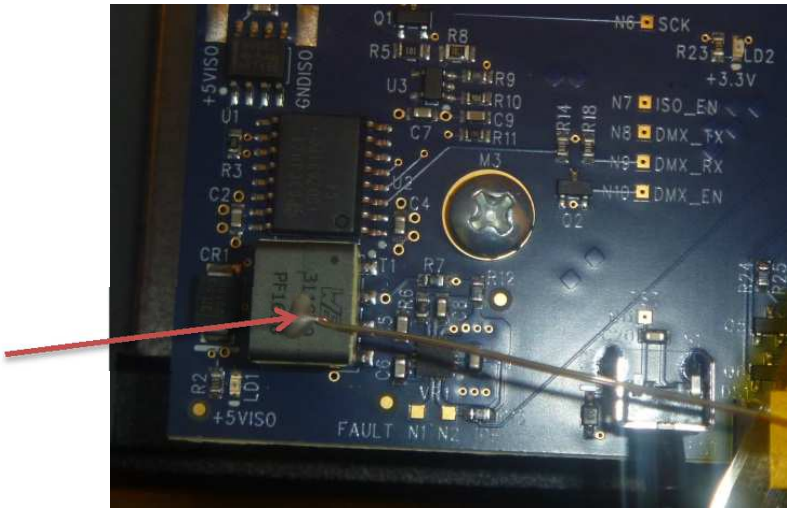
Driver Source Manufacturer's Spec
Driver Identified As: ELECTRONIC THEATRE CONTROLS COLORSOURCE LINEAR 1




Maximum Measured Manufacturer Designated Driver Source Temperature

| Intertek Sample No. | Maximum Measured Driver Temperature ℃ | Location of Thermocouple | Maximum Rated Driver Temperature ℃ |
|---|---|---|---|
| CRT1607151108-001 | 41.5 | T1 | 70.0 |

In-Situ Picture – Td



PRODUCT PICTURE:

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation.

In Charge Of Tests:

Melanie Brittain

Melanie Brittain
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Supervisor
Lighting Division

Attachments: IES File - CRT1607151108-001

# SECTION 8 - Source 4WRD

80 CRI
90 CRI

FOR THE SCOPE OF
ACCREDITATION UNDER NVLAP LAB
CODE 100402-0.

# REPORT

**3933 US ROUTE 11, CORTLAND, NEW YORK 13045**

Project No. G102442057 Date: February 26, 2016

REPORT NO. 102442057CRT-002

TEST OF ONE LED RETROFIT

MODEL NO. S4WRD AND S4WRDFB
LED MODEL NO. CREE HX-C, CUSTOM TO ETC, CREE USING 1350 CHIP
DRIVER MODEL NO. ERP HX1, CUSTOM TO ETC

RENDERED TO:

ELECTRONIC THEATRE CONTROLS INC
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

TESTS: Electrical and Photometric tests as required to the IESNA test standard and Insitu Temperature Measurement Tests.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION The testing performed was authorized by signed quote number Qu-00668268.

STANDARDS USED:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting
ANSI NEMA ANSLG C78.377: 2015: Specifications of the Chromaticity of Solid State Lighting Products
UL 1598-2009: Standard for Safety - Luminaires.

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number S4WRD AND S4WRDFB. The sample was received by Intertek on February 5, 2016 in undamaged condition and one sample was tested as received. The sample designation was CRT1602050826-001.

DATE OF TESTS: February 12, 2016 through February 18, 2016.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY:

| MODEL NO. S4WRD AND S4WRDFB |
|---|
| DESCRIPTION: LED RETROFIT |

| | Results | |
|---|---|---|
| Criteria | Integrating Sphere | Goniophotometer |
| Light Output (Lumens) | 5406.3 | 5551.2 |
| Total Power (W) | 156.80 | 155.39 |
| Lumen Efficacy (Lm/W) | 34.5 | 35.7 |
| Power Factor ( ) | 0.991 | 0.992 |

| Criteria | Results |
|---|---|
| Current ATHD (%) | 10.97 |
| Correlated Color Temp. (CCT-K) | 2968 |
| Color Rendering Index (CRI - Ra) | 78.8 |
| CRI - R9 | -4.3 |
| DUV ( ) | 0.001 |
| Chromaticity Coordinate (x) | 0.440 |
| Chromaticity Coordinate (y) | 0.407 |
| Chromaticity Coordinate (u') | 0.251 |
| Chromaticity Coordinate (v') | 0.523 |
| Maximum LED Source In-Situ Temperature (℃) | 62.9 |
| Maximum Driver Source In-Situ Temperature (℃) | 64.1 |

EQUIPMENT LIST

| Equipment Used | Model No. | Control No. | Last Cal. | Cal. Due |
|---|---|---|---|---|
| LSI High Speed Mirror Goniometer | 6440 | --- | 2/2/2016 | 3/2/2016 |
| Elgar AC Power Supply | CW1251 | --- | VBU | VBU |
| Sorenson DC Power Supply | XG 150-10 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT210 | E464 | 4/20/2015 | 4/20/2016 |
| ExTech Hygro Thermometer | 445703 | T1550 | 1/8/2016 | 1/8/2017 |
| M-D Building Products Digital Level | Smart Tool | L112 | 3/25/2015 | 3/25/2016 |
| NIST Luminous Intensity Standard Source | NBS10322 | N1427 | 12/12/2014 | 12/12/2016 |
| NIST Luminous Intensity Standard Source | NBS10215 | N1432 | 12/12/2014 | 12/12/2016 |
| NIST Luminous Intensity Standard Source | 960629-3 | N1428 | 12/12/2014 | 12/12/2016 |
| Secondary Luminous Flux Standard Source | NBS10428 | N1424 | 12/17/2014 | 12/17/2016 |
| Elgar AC Power Supply | CW1251 | --- | VBU | VBU |
| Sorenson DC Power Supply | XFR 150-8 | --- | VBU | VBU |
| Yokogawa Power Analyzer | WT1600 | E474 | 3/6/2015 | 3/6/2016 |
| Fisher Scientific Stopwatch | 14-649-9 | N1316 | 1/15/2016 | 1/15/2017 |
| Fluke Thermometer | 53II | T1318 | 3/18/2015 | 3/18/2016 |
| 3M Integrating Sphere Spectrometer System | CDS 1100 | --- | 2/1/2016 | 3/1/2016 |
| Pearson Current Monitor | 411 | A138 | 3/6/2015 | 3/6/2016 |
| NIST Spectral Intensity Standard Source | RF1482 | N1422 | 12/17/2014 | 100 Hrs of use |
| NIST Luminous Flux Standard Source | 150-14 | N1423 | 12/17/2014 | 12/17/2016 |
| NIST Luminous Flux Standard Source | NBS10428 | N1424 | 12/17/2014 | 12/17/2016 |
| NIST Luminous Flux Standard Source | BS8830 | N1425 | 12/17/2014 | 12/17/2016 |
| Extech Hygro Thermometer | 445715 | T1550 | 1/8/2016 | 1/8/2017 |
| Fluke Multimeter | 87 V | D589 | 3/17/2015 | 3/17/2016 |
| Fluke Temperature Meter | 53 II | N1324 | 3/18/2015 | 3/18/2016 |

TEST METHODS:

Seasoning in Sample Orientation – LED Products
No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method
A Labsphere Model CDS 1100 CCD Array Spectroradiometer and two meter or ten foot sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Stabilization procedures to LM-79 were followed. Electrical measurements including voltage, current, and power were measured using a power analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical measurements – Distribution Method
A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for the SSL sample.

Ambient temperature was measured equal to the height of the sample mounted on the goniometer equipment. The SSL sample was operated on the client provided driver at rated input volts in its designated orientation. The SSL sample was allowed to stabilize for at least thirty minutes before measurements were made. Stabilization procedures to LM-79 were followed. Electrical measurements including voltage, current, and power were measured using a power analyzer.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature
Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598, UL 153, or UL 1993 as applicable.

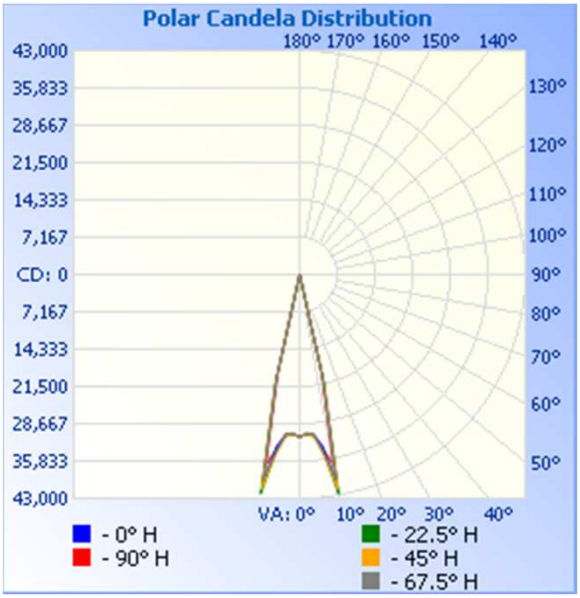
RESULTS:

Photometric and Electrical Measurements at Ambient Temperature (25℃ +/- 1℃) – Distribution Method

| Intertek Control No. | Base Orientation | Input Voltage (VAC) | Input Current (mA) | Input Power (W) | Input Power Factor ( ) | Light Output (Lumens) | Lumen Efficacy (lm/W) |
|---|---|---|---|---|---|---|---|
| CRT1602050826-001 | Base Up | 120.02 | 1305.8 | 155.39 | 0.992 | 5551.2 | 35.7 |

Intensity (Candlepower) Summary at 25℃ - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 31166 | 31166 | 31166 | 31166 | 31166 |
| 5 | 30810 | 31206 | 31029 | 30743 | 30914 |
| 10 | 40480 | 42651 | 41613 | 39357 | 37073 |
| 15 | 128 | 124 | 116 | 89 | 94 |
| 20 | 16 | 14 | 19 | 2 | 2 |
| 25 | 0 | 0 | 0 | 0 | 0 |
| 30 | 0 | 0 | 0 | 0 | 0 |
| 35 | 0 | 0 | 0 | 0 | 0 |
| 40 | 0 | 0 | 0 | 0 | 0 |
| 45 | 0 | 0 | 0 | 0 | 0 |
| 50 | 0 | 0 | 0 | 0 | 0 |
| 55 | 0 | 0 | 0 | 0 | 0 |
| 60 | 0 | 0 | 0 | 0 | 0 |
| 65 | 0 | 0 | 0 | 0 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

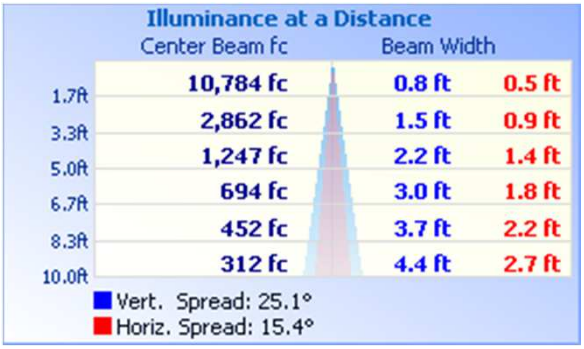


RESULTS:

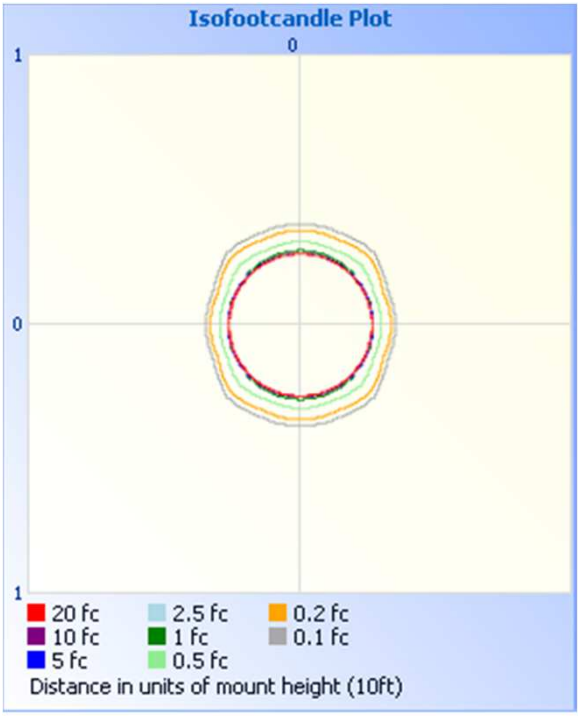
Illumination Plots

Mounting Height: 10ft

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25℃

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 5551.2 | 100.0 |
| 0-40 | 5551.2 | 100.0 |
| 0-60 | 5551.2 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 5551.2 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 5551.2 | 100.0 |

Zonal Lumens and Percentages at 25℃

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 3282.6 | 59.1 |
| 10-20 | 2268.1 | 40.9 |
| 20-30 | 0.5 | 0.0 |
| 30-40 | 0.0 | 0.0 |
| 40-50 | 0.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS:

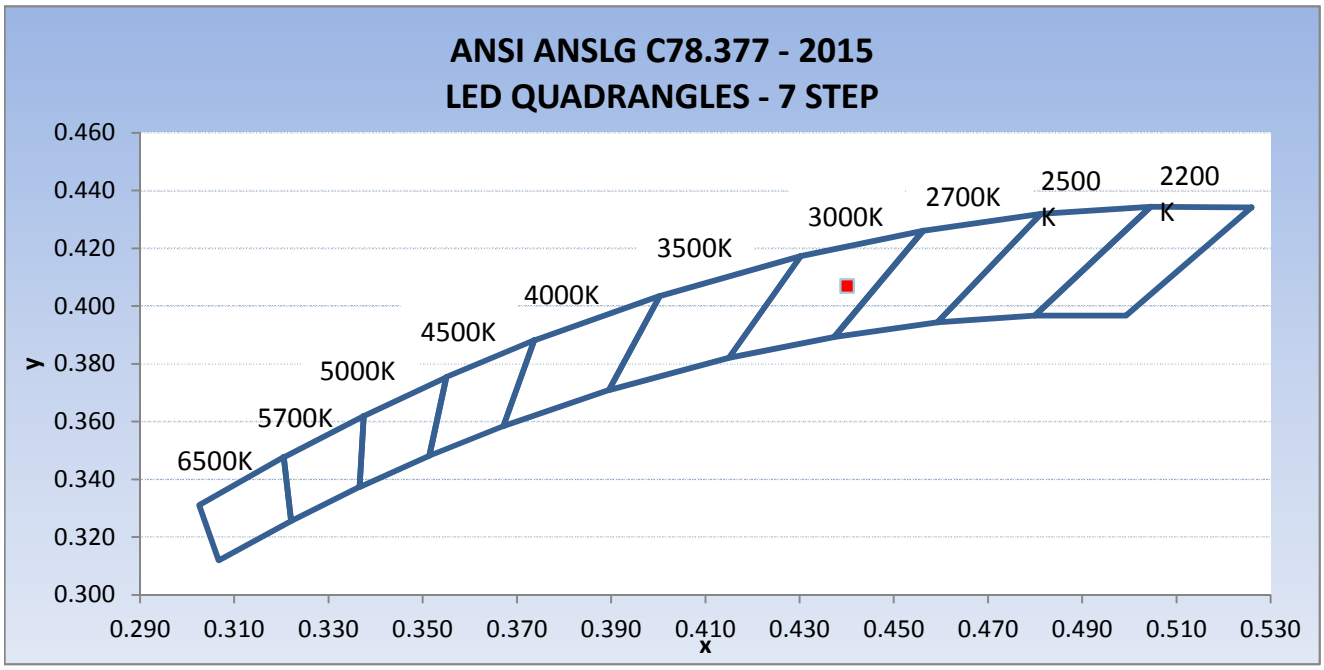
Photometric and Electrical Measurements at Ambient Temperature (25℃ +/- 1℃) – Sphere Method

| Intertek Control No. | Base Orientation | Input Voltage (VAC) | Input Current (mA) | Input Power (W) | Input Power Factor ( ) | Current ATHD (%) |
|---|---|---|---|---|---|---|
| CRT1602050826-001 | Base Up | 120.03 | 1318.1 | 156.80 | 0.991 | 10.97 |

| Light Output (Lumens) | Lumen Efficacy (lm/W) | Correlated Color Temperature - CCT (K) | CRI -Ra | CRI -R9 | DUV ( ) |
|---|---|---|---|---|---|
| 5406.3 | 34.5 | 2968 | 78.8 | -4.3 | 0.001 |

| CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|
| 0.440 | 0.407 | 0.251 | 0.523 |

**ANSI C78.377 SSL Chromaticity (2015 Version)**

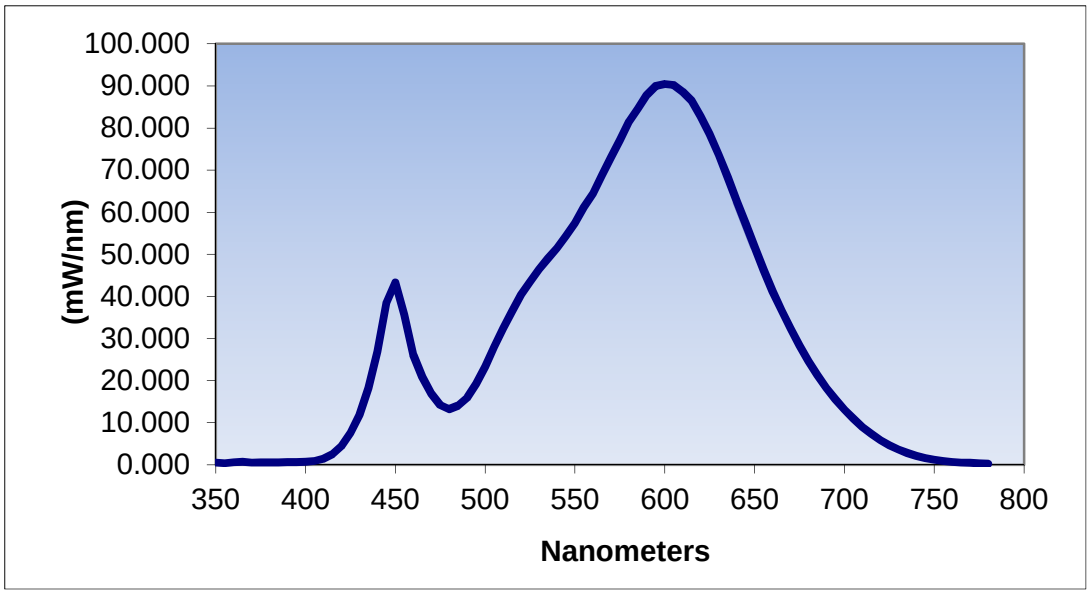


RESULTS:

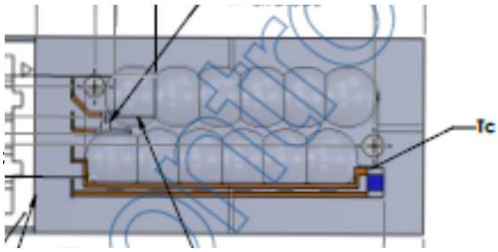
**Spectral Distribution Over Visible Wavelengths**

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|
| 350 | 0.543 | 460 | 26.123 | 570 | 73.046 | 680 | 24.649 |
| 355 | 0.366 | 465 | 20.862 | 575 | 77.155 | 685 | 21.350 |
| 360 | 0.580 | 470 | 16.965 | 580 | 81.450 | 690 | 18.267 |
| 365 | 0.687 | 475 | 14.179 | 585 | 84.604 | 695 | 15.588 |
| 370 | 0.521 | 480 | 13.241 | 590 | 87.881 | 700 | 13.068 |
| 375 | 0.510 | 485 | 14.063 | 595 | 89.994 | 705 | 10.931 |
| 380 | 0.501 | 490 | 15.941 | 600 | 90.447 | 710 | 8.991 |
| 385 | 0.517 | 495 | 19.220 | 605 | 90.214 | 715 | 7.338 |
| 390 | 0.620 | 500 | 23.348 | 610 | 88.669 | 720 | 5.824 |
| 395 | 0.595 | 505 | 28.029 | 615 | 86.488 | 725 | 4.641 |
| 400 | 0.684 | 510 | 32.351 | 620 | 82.804 | 730 | 3.611 |
| 405 | 0.912 | 515 | 36.477 | 625 | 78.564 | 735 | 2.793 |
| 410 | 1.437 | 520 | 40.412 | 630 | 73.573 | 740 | 2.117 |
| 415 | 2.511 | 525 | 43.463 | 635 | 68.279 | 745 | 1.586 |
| 420 | 4.463 | 530 | 46.452 | 640 | 62.617 | 750 | 1.185 |
| 425 | 7.594 | 535 | 49.048 | 645 | 57.222 | 755 | 0.908 |
| 430 | 11.845 | 540 | 51.460 | 650 | 51.700 | 760 | 0.691 |
| 435 | 18.211 | 545 | 54.333 | 655 | 46.251 | 765 | 0.537 |
| 440 | 26.871 | 550 | 57.384 | 660 | 41.210 | 770 | 0.407 |
| 445 | 38.477 | 555 | 61.175 | 665 | 36.677 | 775 | 0.293 |
| 450 | 43.282 | 560 | 64.441 | 670 | 32.361 | 780 | 0.226 |
| 455 | 35.635 | 565 | 68.781 | 675 | 28.343 | | |

**Spectral Data Over Visible Wavelengths**



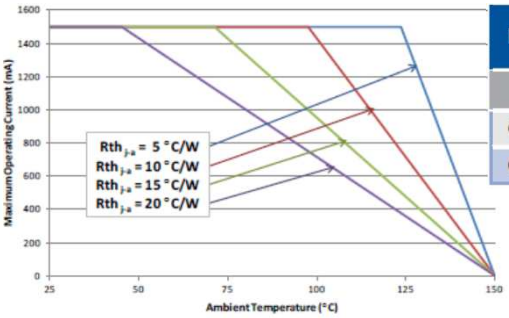
RESULTS:




In-Situ Maximum Measured LED Source Temperature
Maximum Junction Temperature from LED specification (Tj) = 150℃
Thermal Resistance Formula from LED specification = 20℃/W
Maximum Forward Voltage (Vf) from LED specification = 3.4V
Measured LED Current = 487.5mA
Calculated LED Wattage = Vf x Measured LED Current = 1.658W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 116.9℃

LED Source Temperature Manufacturer's Spec
LED Identified As: CREE 1350

| Maximum Ratings at $T_A$ = 25°C Note 1 | CxxxEZ1350-Sxxx00-x |
|---|---|
| DC Forward Current | 1500 mA |
| Peak Forward Current (1/10 duty cycle @ 1 kHz) | 2000 mA |
| LED Junction Temperature | 150°C |
| Reverse Voltage | 5 V |
| Operating Temperature Range | -40°C to +100°C |
| LED Chip Storage Temperature | -40°C to +120°C |
| Recommended Die Sheet Storage Conditions | ≤30°C / ≤85% RH |

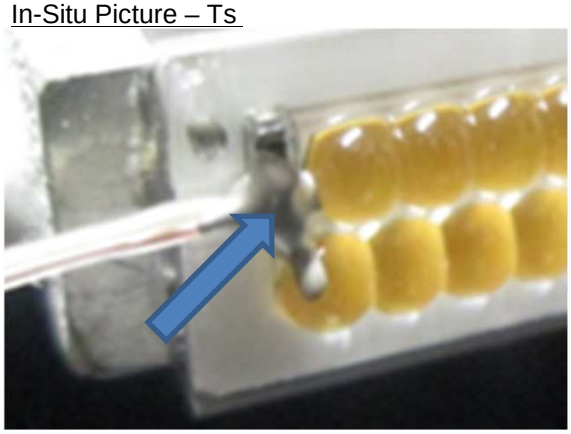



| Part Number | Forward Voltage ($V_f$, V) | | |
|---|---|---|---|
| | Min. | Typ. | Max. |
| C450EZ1350-S*xxxx*00-*x* | 2.6 | 3.0 | 3.4 |
| C460EZ1350-S*xxxx*00-*x* | 2.6 | 3.0 | 3.4 |

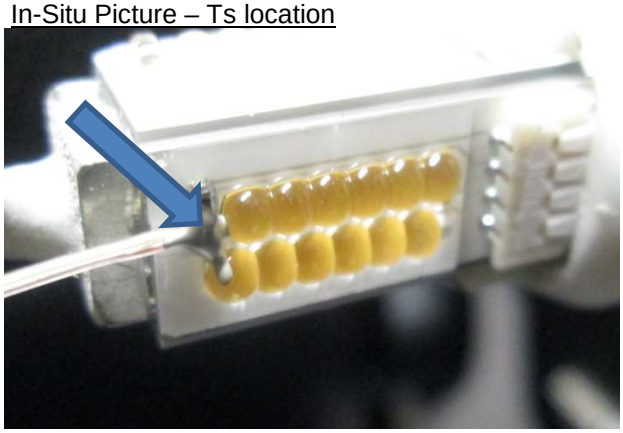
Maximum Measured Manufacturer Designated Source Temperature

| Intertek Sample No. | Maximum Measured Source Temperature | Location of Thermocouple | Maximum Rated Source Temperature |
|---|---|---|---|
| CRT1602050826-001 | 62.9 | Per Diagram | 116.9 |

In-Situ Picture – Ts



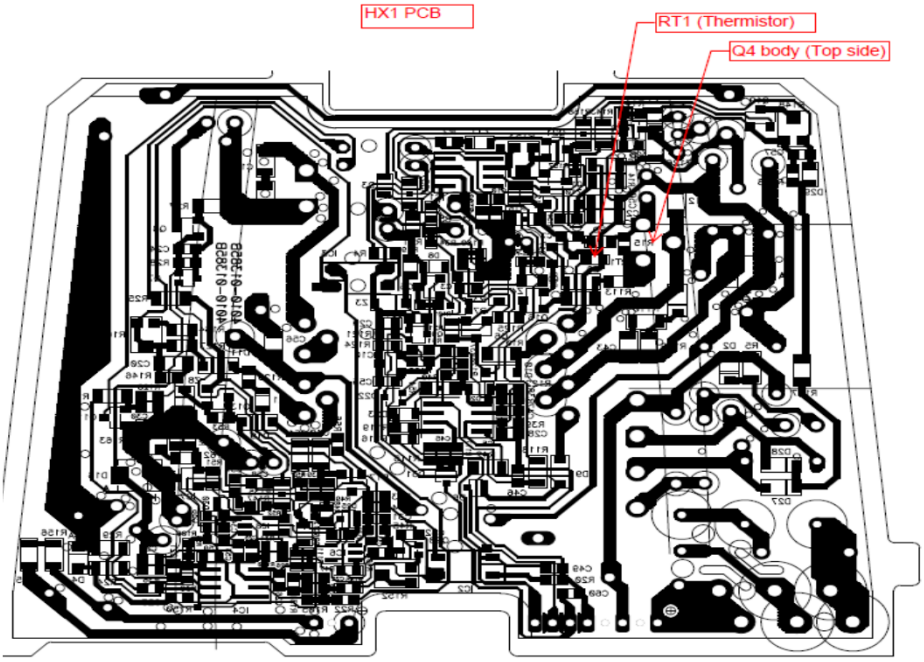
In-Situ Picture – Ts location



RESULTS:

In-Situ Maximum Measured Driver Source Temperature

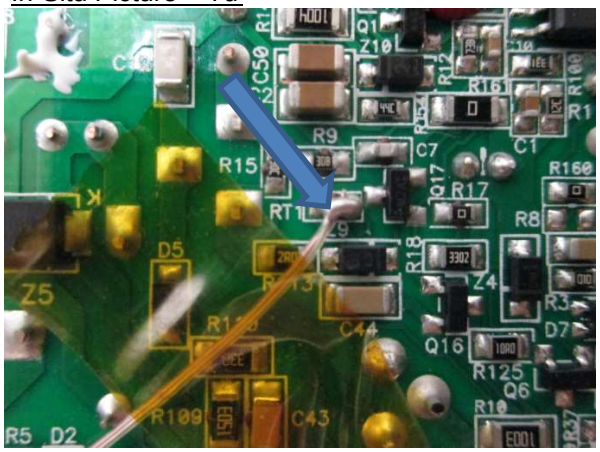
Driver Source Manufacturer's Spec
Driver Identified As:

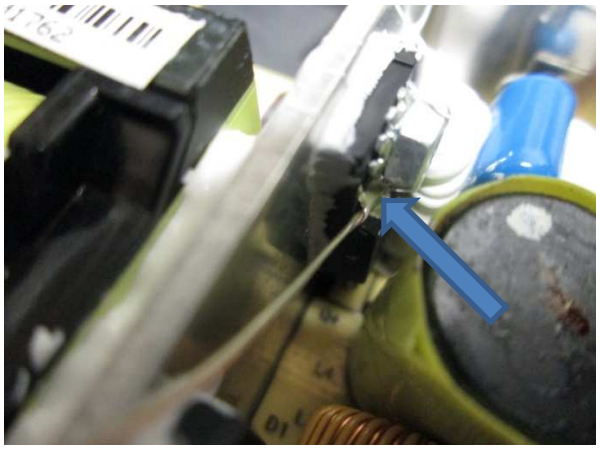



Maximum Measured Manufacturer Designated Driver Source Temperature

| Intertek Sample No. | Maximum Measured Driver Temperature ℃ | Location of Thermocouple | Maximum Rated Driver Temperature ℃ |
|---|---|---|---|
| CRT1602050826-001 | 64.1 | Q4 | 130.0 |
| | 75.4 | RT11 | 130.0 |

In-Situ Picture – Td



PRODUCT PICTURE:

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Gerald Gray
Associate Engineer
Lighting Division

Report Reviewed By:

Jeffrey Davis
Engineering Supervisor
Lighting Division

Attachments: IES File - CRT1602050826-001

# REPORT

**545 E. Algonquin Rd., Arlington Heights, IL 60005**

Project No. G102482931 Date: March 9, 2016

REPORT NO. 102482931CHI-001

TEST OF ONE LED LUMINAIRE RETROFIT

MODEL NO. SOURCE4WRD GALLERY
LED MODEL NO. CREE – HXCAWT-90
DRIVER MODEL NO. ERP – HX1

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC
3031 PLEASANT VIEW ROAD
MIDDLETON WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number QU-00675911.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number Source4wrd Gallery. The sample was received by Intertek on February 24, 2016, in undamaged condition and one sample was tested as received. The sample designation was AH02242016031959.

DATES OF TESTS: March 5, 2016 through March 9, 2016.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | Source4wrd Gallery |
| Description: | LED Luminaire Retrofit |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 6424 | 6278 |
| Total Power (W) | 157.4 | 157.4 |
| Luminaire Efficacy (LPW) | 40.81 | 39.89 |

| Criteria | Result |
|---|---|
| Power Factor | 0.992 |
| Current ATHD % | 10.44 |
| Correlated Color Temperature (CCT - K) | 3024 |
| Color Rendering Index (CRI - Ra) | 90.2 |
| Color Rendering Index (CRI - R9) | 50.8 |
| DUV | 0.001 |
| Chromaticity Coordinate (x) | 0.434 |
| Chromaticity Coordinate (y) | 0.401 |
| Chromaticity Coordinate (u') | 0.250 |
| Chromaticity Coordinate (v') | 0.520 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date | Date Used |
|---|---|---|---|---|---|
| Yokogawa Power Meter | WT210 | 146919 | 07/14/15 | 07/14/16 | 03/05/16 |
| Omega Newport Thermometer | DPI8-C24 | 146920 | 10/09/15 | 10/09/16 | 03/05/16 |
| LSI High Speed Mirror Goniometer | 6440T | 146928 | VBU | VBU | 03/05/16 |
| Newport Thermohygrometer | iServer | 146956 | 01/04/16 | 01/04/17 | 03/05/16 |
| Pacific, AC power supply | 118-ACX | CHI0358 | VBU | VBU | 03/05/16 |
| Extech temperature logger | SD700 | 146965 | 01/04/16 | 01/04/17 | 03/09/16 |
| Labsphere Spectroradiometer | CDS1100 | CHI0091 | VBU | VBU | 03/08/16 |
| 3 Meter Sphere | SPR600 | CHI0088 | VBU | VBU | 03/08/16 |
| Elgar AC Power Supply | CW1251M | 146112 | VBU | VBU | 03/08/16 |
| Sorenson DC Power Supply | XFR150-8 | 146846 | VBU | VBU | 03/08/16 |
| Newport Humidity Recorder | iTHX-SD | 146382 | 07/09/15 | 07/09/16 | 03/08/16 |
| Yokogawa Power Meter | WT1600 | 146768 | 01/14/16 | 01/14/17 | 03/08/16 |
| Omega Temperature Meter | MDSi8 | 146139 | 04/03/15 | 04/03/16 | 03/08/16 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

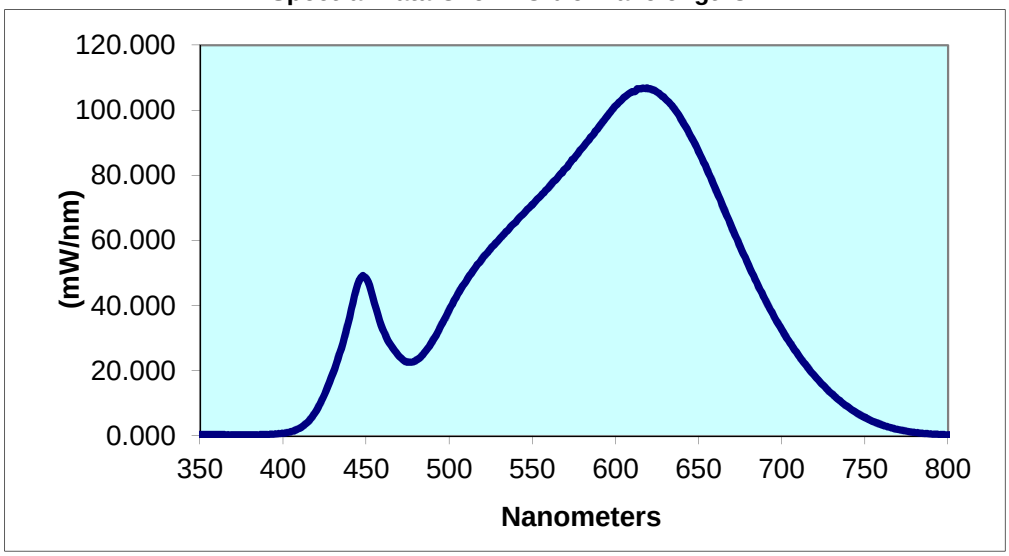
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| AH02242016031959 | Up | 120.0 | 1321 | 157.4 | 0.992 | 10.44 | 6424 | 40.81 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3024 | 90.2 | 50.8 | 0.001 | 0.434 | 0.401 | 0.250 | 0.520 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.408 | 440 | 36.51 | 530 | 60.36 | 620 | 106.6 | 710 | 24.82 |
| 355 | 0.396 | 445 | 46.63 | 535 | 63.06 | 625 | 105.6 | 715 | 21.41 |
| 360 | 0.398 | 450 | 48.47 | 540 | 65.71 | 630 | 103.6 | 720 | 18.29 |
| 365 | 0.361 | 455 | 40.81 | 545 | 68.45 | 635 | 100.7 | 725 | 15.53 |
| 370 | 0.353 | 460 | 32.77 | 550 | 71.13 | 640 | 96.88 | 730 | 13.03 |
| 375 | 0.335 | 465 | 27.80 | 555 | 73.77 | 645 | 92.36 | 735 | 10.79 |
| 380 | 0.339 | 470 | 24.36 | 560 | 76.47 | 650 | 87.32 | 740 | 8.831 |
| 385 | 0.350 | 475 | 22.61 | 565 | 79.26 | 655 | 81.93 | 745 | 7.123 |
| 390 | 0.432 | 480 | 23.22 | 570 | 82.22 | 660 | 76.24 | 750 | 5.666 |
| 395 | 0.576 | 485 | 25.55 | 575 | 85.22 | 665 | 70.03 | 755 | 4.436 |
| 400 | 0.825 | 490 | 29.16 | 580 | 88.48 | 670 | 64.07 | 760 | 3.449 |
| 405 | 1.365 | 495 | 33.84 | 585 | 91.80 | 675 | 58.13 | 765 | 2.628 |
| 410 | 2.413 | 500 | 38.75 | 590 | 94.95 | 680 | 52.48 | 770 | 1.960 |
| 415 | 4.366 | 505 | 43.42 | 595 | 98.21 | 685 | 47.05 | 775 | 1.464 |
| 420 | 7.755 | 510 | 47.44 | 600 | 101.3 | 690 | 41.91 | 780 | 1.070 |
| 425 | 12.85 | 515 | 51.25 | 605 | 103.9 | 695 | 37.08 | | |
| 430 | 19.24 | 520 | 54.59 | 610 | 105.7 | 700 | 32.65 | | |
| 435 | 26.89 | 525 | 57.57 | 615 | 106.5 | 705 | 28.56 | | |

**Spectral Data Over Visible Wavelengths**



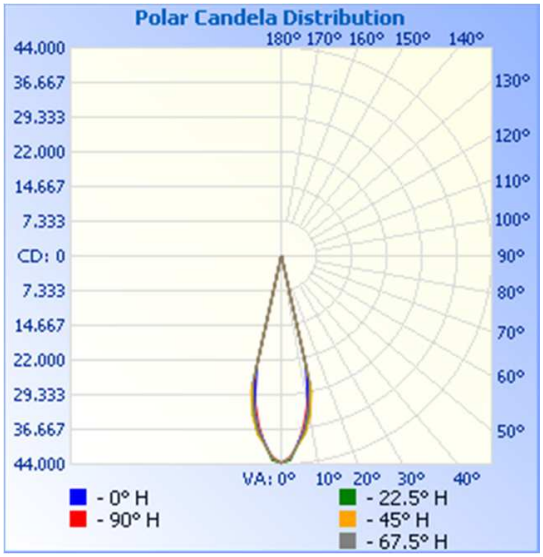
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|
| AH02242016031959 | Up | 120.1 | 1320 | 157.4 | 0.993 | 6278 | 39.89 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 43639 | 43639 | 43639 | 43639 | 43639 |
| 5 | 39602 | 40025 | 39974 | 39767 | 39878 |
| 10 | 32641 | 34040 | 34343 | 32612 | 31454 |
| 15 | 683 | 692 | 698 | 663 | 645 |
| 20 | 264 | 264 | 267 | 265 | 263 |
| 25 | 146 | 146 | 145 | 146 | 146 |
| 30 | 87 | 87 | 87 | 87 | 87 |
| 35 | 53 | 53 | 53 | 53 | 53 |
| 40 | 36 | 36 | 36 | 36 | 36 |
| 45 | 24 | 25 | 25 | 25 | 25 |
| 50 | 15 | 17 | 18 | 17 | 17 |
| 55 | 9 | 10 | 12 | 12 | 12 |
| 60 | 3 | 5 | 7 | 7 | 7 |
| 65 | 0 | 0 | 0 | 1 | 0 |
| 70 | 0 | 0 | 0 | 0 | 0 |
| 75 | 0 | 0 | 0 | 0 | 0 |
| 80 | 0 | 0 | 0 | 0 | 0 |
| 85 | 0 | 0 | 0 | 0 | 0 |
| 90 | 0 | 0 | 0 | 0 | 0 |

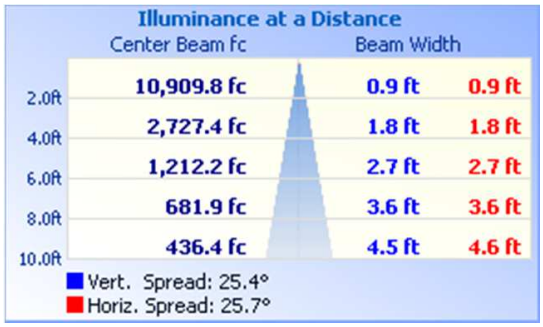


RESULTS OF TEST (cont'd)

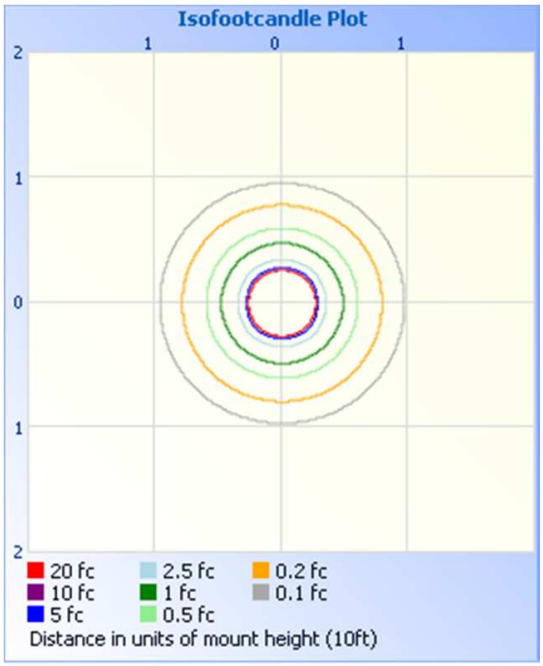
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 6212 | 99.0 |
| 0-40 | 6247 | 99.5 |
| 0-60 | 6276 | 100.0 |
| 60-90 | 1.4 | 0.0 |
| 0-90 | 6278 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 6278 | 100.0 |

Zonal Lumens and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 3606 | 57.4 |
| 10-20 | 2536 | 40.4 |
| 20-30 | 69.9 | 1.1 |
| 30-40 | 34.7 | 0.6 |
| 40-50 | 19.6 | 0.3 |
| 50-60 | 10.1 | 0.2 |
| 60-70 | 1.4 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

PICTURES (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Vladimir Kozak
Senior Associate Engineer
Lighting Division

Report Reviewed By:

Timothy Quigley
Engineer
Lighting Division

Attachment: None

# SECTION 9 - Irideon

FPZ

# REPORT

**545 E. Algonquin Rd., Arlington Heights, IL 60005**

Project No. G102413460 Date: January 6, 2016

REPORT NO. 102413460CHI-001

TEST OF ONE LED TRACK LUMINAIRE

MODEL NO. IRIDEON FPZ
LED MODEL NO. CREE MT-G2
DRIVER MODEL NO. ELECTRONIC THEATRE CONTROLS IRIDEON FPZ

RENDERED TO

ELECTRONIC THEATRE CONTROLS INC
3031 PLEASANT VIEW RD
MIDDLETON WI 53562

TEST: Electrical and Photometric tests as required to the IESNA test standard.

STATEMENT OF LIMITATION: This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the federal government.

AUTHORIZATION: The testing performed was authorized by signed quote number Qu-00663064-1.

STANDARDS USED: The following American National Standards or Illuminating Engineering Society of North America Test Guides were used in part or totally to test each specimen:

IESNA LM-79 - 2008: Electrical and Photometric Measurements of Solid State Lighting

ANSI NEMA ANSLG C78.377: 2012: Specifications of the Chromaticity of Solid State Lighting Products

UL 1598-2009: Underwriters Laboratories Inc. Standard for Safety - Luminaires

DESCRIPTION OF SAMPLE: The client submitted one production sample of model number Irideon FPZ. The sample was received by Intertek on December 23, 2015, in undamaged condition and one sample was tested as received. The sample designation was AH12232015113941.

DATES OF TESTS: December 30, 2015 through January 6, 2016.

This report is for the exclusive use of Intertek's Client and is provided pursuant to the agreement between Intertek and its Client. Intertek's responsibility and liability are limited to the terms and conditions of the agreement. Intertek assumes no liability to any party, other than to the Client in accordance with the agreement, for any loss, expense or damage occasioned by the use of this report. Only the Client is authorized to copy or distribute this report and then only in its entirety. Any use of the Intertek name or one of its marks for the sale or advertisement of the tested material, product or service must first be approved in writing by Intertek. The observations and test results in this report are relevant only to the sample tested. This report by itself does not imply that the material, product, or service is or has ever been under an Intertek certification program.

SUMMARY

| | |
|---|---|
| Model No.: | Irideon FPZ |
| Description: | LED Track Luminaire |

| | Result | |
|---|---|---|
| Criteria | Sphere | Goniometer |
| Total Lumen Output (Lumens) | 811.9 | 797.9 |
| Total Power (W) | 19.28 | 19.33 |
| Luminaire Efficacy (LPW) | 42.11 | 41.28 |

| Criteria | Result |
|---|---|
| Power Factor | 0.995 |
| Current ATHD % | 5.54 |
| Correlated Color Temperature (CCT - K) | 3086 |
| Color Rendering Index (CRI - Ra) | 81.0 |
| Color Rendering Index (CRI - R9) | 6.0 |
| DUV | 0.002 |
| Chromaticity Coordinate (x) | 0.429 |
| Chromaticity Coordinate (y) | 0.398 |
| Chromaticity Coordinate (u') | 0.248 |
| Chromaticity Coordinate (v') | 0.518 |
| Maximum In-Situ Source Temperature Point (°C) | 70.9 |
| Maximum In-Situ Driver Case Temperature (°C) | 47.1 |

EQUIPMENT LIST

| Equipment Used | Model Number | Control Number | Last Date Calibrated | Calibration Due Date | Date Used |
|---|---|---|---|---|---|
| Yokogawa Power Meter | WT210 | 146919 | 07/14/15 | 07/14/16 | 01/04/16 |
| Omega Newport Thermometer | DPI8-C24 | 146920 | 10/09/15 | 10/09/16 | 01/04/16 |
| LSI High Speed Mirror Goniometer | 6440T | 146928 | VBU | VBU | 01/04/16 |
| Newport Thermohygrometer | iServer | 146956 | 01/06/15 | 01/06/16 | 01/04/16 |
| Pacific, AC power supply | 118-ACX | CHI0358 | VBU | VBU | 01/04/16 |
| Extech temperature logger | SD700 | CHI0279 | 04/03/15 | 04/03/16 | 01/06/16 |
| Yokogawa power meter | WT210 | 146453 | 01/14/15 | 01/14/16 | 01/06/16 |
| Agilent data logger | 34970A | 146441 | 01/13/15 | 01/13/16 | 01/06/16 |
| Staco Energy variac | 3PN1510B | 146399 | VBU | VBU | 01/06/16 |
| Labsphere Spectroradiometer | CDS1100 | CHI0091 | VBU | VBU | 12/30/15 |
| 3 Meter Sphere | SPR600 | CHI0088 | VBU | VBU | 12/30/15 |
| Elgar AC Power Supply | CW1251M | 146112 | VBU | VBU | 12/30/15 |
| Sorenson DC Power Supply | XFR150-8 | 146846 | VBU | VBU | 12/30/15 |
| Newport Humidity Recorder | iTHX-SD | 146382 | 07/09/15 | 07/09/16 | 12/30/15 |
| Yokogawa Power Meter | WT1600 | 146768 | 01/15/15 | 01/15/16 | 12/30/15 |
| Omega Temperature Meter | MDSi8 | 146139 | 04/03/15 | 04/03/16 | 12/30/15 |

TEST METHODS

Seasoning in Sample Orientation – LED Products

No seasoning was performed in accordance with IESNA LM-79.

Photometric and Electrical Measurements – Integrating Sphere Method

A Labsphere Model CDS 1100 CCD Array Spectroradiometer and Two Meter or Ten Foot Sphere was used to measure correlated color temperature, chromaticity coordinates, and the color rendering index for each SSL unit.

Ambient temperature was measured at a position inside the sphere. Each SSL unit was operated on the client provided driver at the rated input voltage in its designated orientation. Each SSL unit was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

The calibration of the sphere photometer-spectroradiometer system is traceable to the National Institute of Standards and Technology.

Photometric and Electrical Measurements – Distribution Method

A LSI Type C High Speed Model 6440 Mirror Goniometer was used to measure the intensity (candelas) at each angle of distribution for each sample.

Ambient temperature was measured equal to the height of the sample mounted on the Goniometer equipment. Each sample was operated at input rated voltage in its designated orientation. Each sample was allowed to stabilize for at least thirty minutes before measurements were made. Electrical measurements including voltage, current, and power were measured using the Xitron or Yokogawa Power Analyzer.

Some graphics were created with Photometrics Plus software.

In-Situ Maximum Measured Power Supply Case and LED Source Point Temperature

Power supply case and/or LED source operating temperature measurements were taken on one test sample per model with a thermocouple and Fluke 87 temperature meter. The SSL sample was allowed to reach thermal equilibrium for seven and a half hours before measurements were taken. Power supply or source temperature measurements were measured at the TMPPS or TS point as indicated by the included diagram in accordance with manufacturers declared hot spot location, or at a hot spot location found with a thermal camera when no diagram from the manufacturer is given. The maximum temperature was recorded for the sample. A simulated ceiling or other enclosure may be used in accordance to UL 1598 or UL 153 as applicable.

RESULTS OF TEST

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) - Integrating Sphere Method

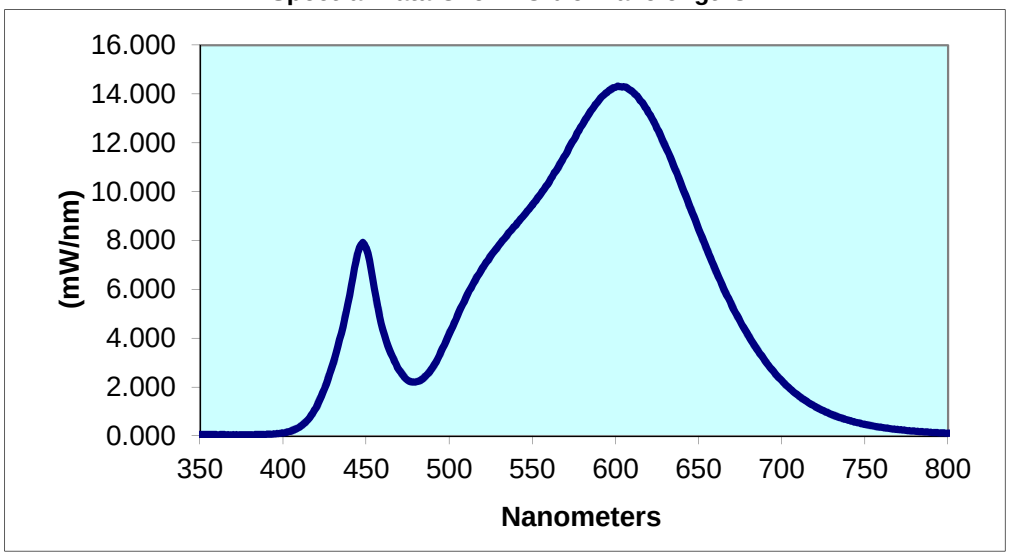
| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Current ATHD (%) | Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|---|
| AH12232015113941 | Up | 120.0 | 161.5 | 19.28 | 0.995 | 5.54 | 811.9 | 42.11 |

| Correlated Color Temperature (K) | CRI -Ra | CRI -R9 | DUV | CIE 31' Chromaticity Coordinate (x) | CIE 31' Chromaticity Coordinate (y) | CIE 76' Chromaticity Coordinate (u') | CIE 76' Chromaticity Coordinate (v') |
|---|---|---|---|---|---|---|---|
| 3086 | 81.0 | 6.0 | 0.002 | 0.429 | 0.398 | 0.248 | 0.518 |

Spectral Distribution over Visible Wavelengths

| nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm | nm | mW/nm |
|---|---|---|---|---|---|---|---|---|---|
| 350 | 0.058 | 440 | 5.761 | 530 | 7.821 | 620 | 13.23 | 710 | 1.647 |
| 355 | 0.049 | 445 | 7.479 | 535 | 8.241 | 625 | 12.60 | 715 | 1.411 |
| 360 | 0.049 | 450 | 7.707 | 540 | 8.631 | 630 | 11.87 | 720 | 1.205 |
| 365 | 0.049 | 455 | 6.029 | 545 | 9.043 | 635 | 11.07 | 725 | 1.033 |
| 370 | 0.052 | 460 | 4.363 | 550 | 9.460 | 640 | 10.21 | 730 | 0.880 |
| 375 | 0.043 | 465 | 3.365 | 555 | 9.909 | 645 | 9.342 | 735 | 0.753 |
| 380 | 0.047 | 470 | 2.682 | 560 | 10.40 | 650 | 8.473 | 740 | 0.644 |
| 385 | 0.051 | 475 | 2.278 | 565 | 10.94 | 655 | 7.639 | 745 | 0.552 |
| 390 | 0.061 | 480 | 2.218 | 570 | 11.53 | 660 | 6.837 | 750 | 0.473 |
| 395 | 0.084 | 485 | 2.399 | 575 | 12.14 | 665 | 6.059 | 755 | 0.407 |
| 400 | 0.125 | 490 | 2.809 | 580 | 12.74 | 670 | 5.347 | 760 | 0.352 |
| 405 | 0.208 | 495 | 3.430 | 585 | 13.31 | 675 | 4.697 | 765 | 0.302 |
| 410 | 0.372 | 500 | 4.173 | 590 | 13.76 | 680 | 4.099 | 770 | 0.260 |
| 415 | 0.673 | 505 | 4.945 | 595 | 14.08 | 685 | 3.558 | 775 | 0.224 |
| 420 | 1.174 | 510 | 5.649 | 600 | 14.27 | 690 | 3.073 | 780 | 0.192 |
| 425 | 1.954 | 515 | 6.306 | 605 | 14.29 | 695 | 2.643 | | |
| 430 | 2.962 | 520 | 6.879 | 610 | 14.11 | 700 | 2.265 | | |
| 435 | 4.200 | 525 | 7.377 | 615 | 13.72 | 705 | 1.933 | | |

**Spectral Data Over Visible Wavelengths**



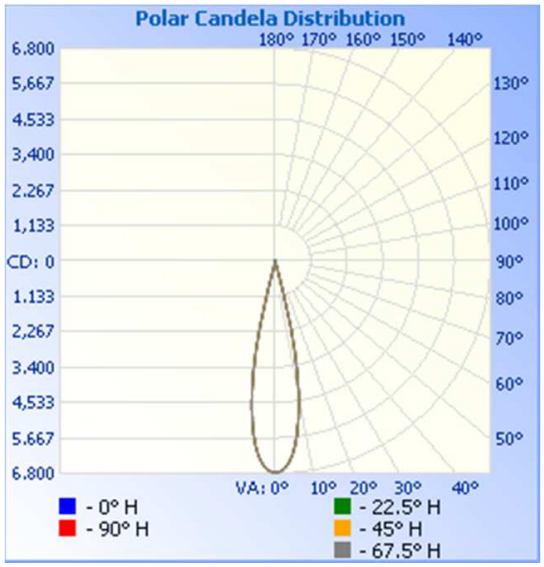
RESULTS OF TEST (cont'd)

Photometric and Electrical Measurements at Ambient Temperature (25°C +/- 1°C) – Distribution Method

| Intertek Sample No. | Base Orientation | Input Voltage {Vac} | Input Current (mA) | Input Power (Watts) | Input Power Factor | Absolute Luminous Flux (Lumens) | Lumen Efficacy (LPW) |
|---|---|---|---|---|---|---|---|
| AH12232015113941 | Up | 119.9 | 162.1 | 19.33 | 0.996 | 797.9 | 41.28 |

Intensity (Candlepower) Summary at 25°C - Candelas

| Angle | 0 | 22.5 | 45 | 67.5 | 90 |
|---|---|---|---|---|---|
| 0 | 6754 | 6754 | 6754 | 6754 | 6754 |
| 1 | 6743 | 6743 | 6743 | 6743 | 6743 |
| 2 | 6696 | 6696 | 6696 | 6696 | 6696 |
| 3 | 6589 | 6589 | 6589 | 6589 | 6589 |
| 4 | 6430 | 6430 | 6430 | 6430 | 6430 |
| 5 | 6230 | 6230 | 6230 | 6230 | 6230 |
| 6 | 5943 | 5943 | 5943 | 5943 | 5943 |
| 7 | 5602 | 5602 | 5602 | 5602 | 5602 |
| 8 | 5202 | 5202 | 5202 | 5202 | 5202 |
| 9 | 4745 | 4745 | 4745 | 4745 | 4745 |
| 10 | 4238 | 4238 | 4238 | 4238 | 4238 |
| 11 | 3649 | 3649 | 3649 | 3649 | 3649 |
| 12 | 2951 | 2951 | 2951 | 2951 | 2951 |
| 13 | 2162 | 2162 | 2162 | 2162 | 2162 |
| 14 | 1002 | 1002 | 1002 | 1002 | 1002 |
| 15 | 44 | 44 | 44 | 44 | 44 |
| 16 | 8 | 8 | 8 | 8 | 8 |
| 17 | 7 | 7 | 7 | 7 | 7 |
| 18 | 6 | 6 | 6 | 6 | 6 |
| 19 | 5 | 5 | 5 | 5 | 5 |
| 20 | 5 | 5 | 5 | 5 | 5 |
| 21 | 4 | 4 | 4 | 4 | 4 |
| 22 | 4 | 4 | 4 | 4 | 4 |
| 23 | 3 | 3 | 3 | 3 | 3 |
| 24 | 3 | 3 | 3 | 3 | 3 |
| 25 | 2 | 2 | 2 | 2 | 2 |
| 26 | 2 | 2 | 2 | 2 | 2 |
| 27 | 1 | 1 | 1 | 1 | 1 |
| 28 | 1 | 1 | 1 | 1 | 1 |
| 29 | 0 | 0 | 0 | 0 | 0 |
| 30 | 0 | 0 | 0 | 0 | 0 |

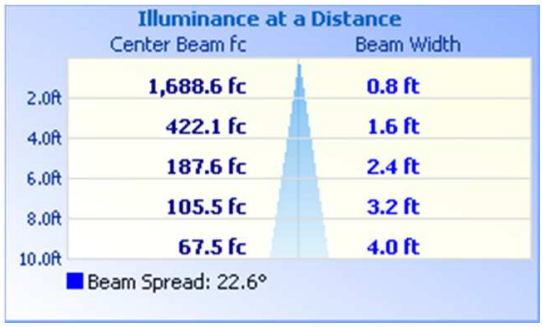


RESULTS OF TEST (cont'd)

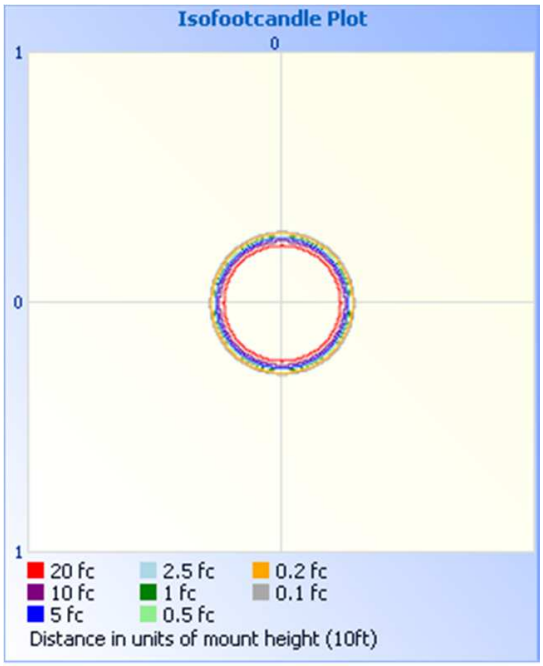
Illumination Plots

Mounting Height: 10 ft.

Illuminance - Cone of Light




Isoillumination Plot




Zonal Lumen Summary and Percentages at 25°C

| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-30 | 797.9 | 100.0 |
| 0-40 | 797.9 | 100.0 |
| 0-60 | 797.9 | 100.0 |
| 60-90 | 0.0 | 0.0 |
| 0-90 | 797.9 | 100.0 |
| 90-180 | 0.0 | 0.0 |
| 0-180 | 797.9 | 100.0 |

Zonal Lumens and Percentages at 25°C

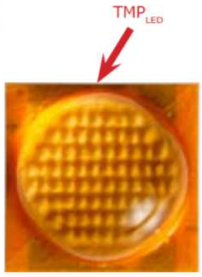
| Zone | Lumens | % Luminaire |
|---|---|---|
| 0-10 | 529.8 | 66.4 |
| 10-20 | 267.1 | 33.5 |
| 20-30 | 0.9 | 0.1 |
| 30-40 | 0.0 | 0.0 |
| 40-50 | 0.0 | 0.0 |
| 50-60 | 0.0 | 0.0 |
| 60-70 | 0.0 | 0.0 |
| 70-80 | 0.0 | 0.0 |
| 80-90 | 0.0 | 0.0 |

RESULTS OF TEST (cont'd)

In-Situ Maximum Measured LED Source Temperature

Manufacturer Supplied Documentation:

| Characteristics | Unit | Minimum | Typical | Maximum |
|---|---|---|---|---|
| Viewing Angle (FWHM) | degrees | | 115 | |
| ESD Classification (HBM per Mil-Std-883D) | | | Class 2 | |
| Effective Thermal Resistance, Junction to Solder Point | °C/W | | 1.5 | |
| LED Junction Temperature | °C | | | 150 |
| DC Forward Current (6 V) | mA | | | 3000 |
| DC Forward Current (9 V) | mA | | | 2000 |
| DC Forward Current (36 V) | mA | | | 500 |
| Forward Voltage (6 V, 1100 mA, 85 °C) | V | | 5.7 | |
| Forward Voltage (6 V, 1100 mA, 25 °C) | V | | | 7 |
| Forward Voltage (9 V, 735 mA, 85 °C) | V | | 8.55 | |
| Forward Voltage (9 V, 735 mA, 25 °C) | V | | | 10.5 |
| Forward Voltage (36 V, 185 mA, 85 °C) | V | | 34.2 | |
| Forward Voltage (36 V, 185 mA, 25 °C) | V | | | 42 |

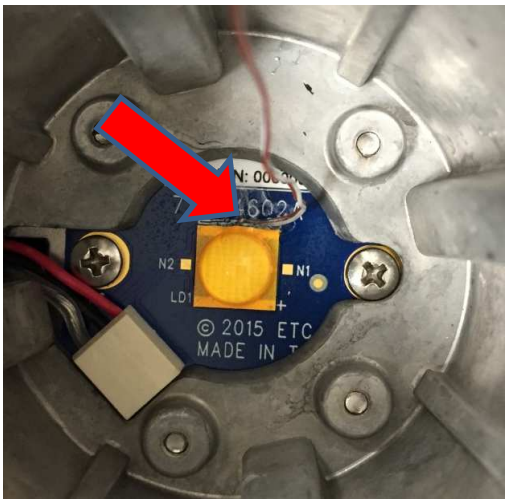



Maximum Junction Temperature from LED specification (Tj) = 150°C
Thermal Resistance Formula from LED specification = 1.5°C/W
Maximum Forward Voltage (Vf) from LED specification = 42V
Measured LED Current = 441.5mA
Calculated LED Wattage = Vf x Measured LED Current = 18.543W
Maximum Source Temperature (Ts) = Tj – (LED Wattage x Thermal Resistance) = 122.2°C

Maximum Measured Manufacturer Designated Source Temperature

| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| AH12232015113941 | 70.9 | Per specs above | 122.2 |

LED In-Situ Picture – Ts

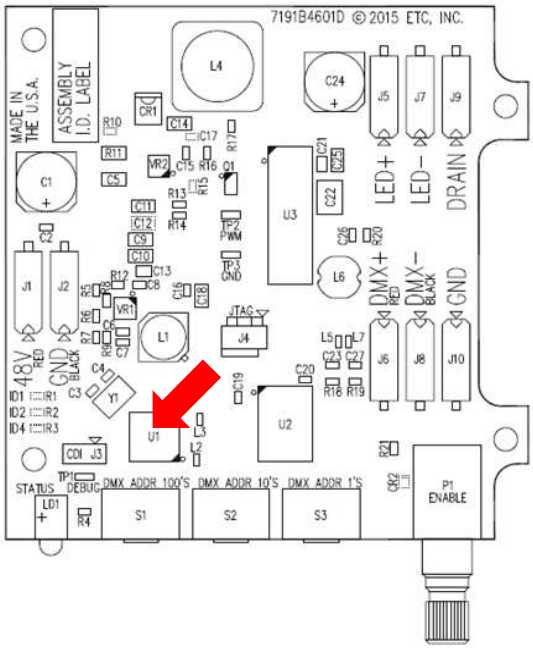


RESULTS OF TEST (cont'd)

In-Situ Maximum Measured Power Supply Case Temperature

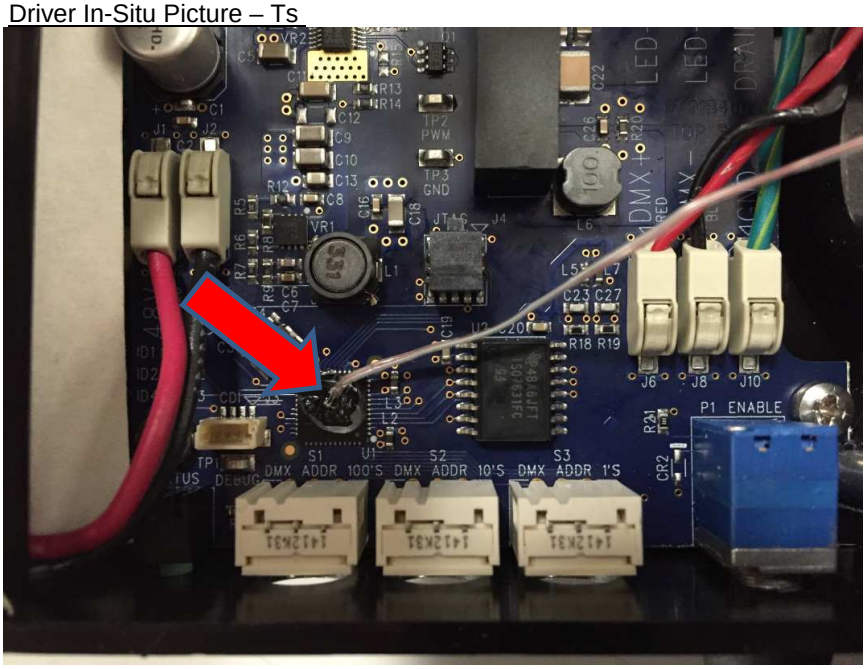
Manufacturer Supplied Documentation:

Please measure the driver temperature at U1. The limit is 100C.




| Sample No. | Maximum Measured Source Temperature (°C) | Location | Maximum Rated Source Temperature (°C) |
|---|---|---|---|
| AH12232015113941 | 47.1 | Per specs above | 100.0 |

Driver In-Situ Picture – Ts



PICTURE (not to scale)

CONCLUSION

The results tabulated in this report are representative of the actual test samples submitted for this report only. The data is provided to the client for further evaluation. Compliance to the referenced specification requirements was not determined in this report.

In Charge Of Tests:

Vladimir Kozak
Senior Associate Engineer
Lighting Division

Report Reviewed By:

Timothy Quigley
Engineer
Lighting Division

Attachment: None